

The Pennsylvania State University

The Graduate School

**A VISION TOWARDS DEVELOPING A SPATIALLY AND  
TEMPORALLY ROBUST DATA-DRIVEN MODELING  
FRAMEWORK TO PREDICT CONTINUOUS STREAM NITRATE  
CONCENTRATION AT DATA-SCARCE LOCATIONS**

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by

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## ABSTRACT

Stream nitrate concentration provides critical insights into nutrient dynamics and can help improve the effectiveness of sustainable ecosystem management decisions. Conventional stream nitrate monitoring is conducted through lab analysis using in-situ water samples, typically at coarse temporal resolution. In the United States, federal agencies and a few state and local agencies started collecting high-frequency (5-60 min intervals) nitrate data using optical sensors in the last decade. These sensor-based high-frequency stream nitrate concentrations at multiple stream locations in a region provide valuable information on the dynamics of nitrate transport, including the timing, magnitude, and sources of nitrate loading in the environmental system. This study hypothesized that the nitrate dynamics information available in a region's high-frequency stream nitrate monitoring sites could be used to estimate spatially and temporally continuous nitrate concentration at other low-frequency monitoring locations. Deep learning (DL) models could use to extract the complex nutrient dynamics from the high-frequency sites and transferred the information to low-frequency monitoring sites. The primary goal of this study was to develop a spatially and temporally robust data-driven modeling framework to predict continuous stream nitrate concentration at data-limited sites in a region using high-frequency nitrate data and biophysical attributes of the region. The objectives of this study are to (1) develop a deep learning (DL)-based modeling approach to generate continuous daily stream nitrate concentration for nitrate data-sparse watersheds and compare the performance of the developed modeling approach with other statistical models; (2) analyze the deep learning approach-based modeling framework to understand the critical environmental drivers of predicting daily nitrate concentrations, and; (3) explore the potential of extending the developed modeling framework to stream discharge unavailable locations for

making the framework more spatially robust. This study used a DL model called Long Short-Term Memory (LSTM) to estimate continuous daily stream nitrate. The DL model received climate, land use, fertilization, topography, and soil characteristics data as inputs during the model development. The DL model was trained comprehensively by using four hyperparameters (batch size, hidden layer size, time window, and epoch) and isolating the best combination of hyperparameters based on five performance metrics, including RMSE, bias, correlation, NSE, and KGE. The hypothesis was tested with Iowa, USA, as a case study region because the state had more high-frequency nitrate monitoring sites with long-term data.

In the first objective, a DL model-based stream nitrate estimation framework was trained and tested for thirty-four (34) high-frequency and eight (8) low-frequency nitrate monitoring sites. DL model demonstrated median test-period Nash-Sutcliffe efficiency (NSE) = 0.75 to estimate continuous daily stream nitrate concentration, which is unprecedented performance. Twenty-one sites (50% of all nitrate monitoring sites) and thirty-four sites (76%) demonstrated NSE greater than 0.75 and 0.50, respectively. The concentration (c) - discharge (Q) relationship analysis showed that the study watersheds had four dominant nitrate transport patterns from landscapes to streams with increasing discharge, including (i) flushing, (ii) flushing during low Q ( $Q < \text{median } Q, Q_{50}$ ) and chemostatic during high Q ( $Q > Q_{50}$ ), (iii) flushing during low Q and dilution during high Q and (iv) chemodynamic. The flushing pattern was identified as the most dominant c-Q relationship pattern.

The developed DL modeling framework's performance was compared with other widely used statistical models, including Weighted Regressions on Time, Discharge, and Season (WRTDS) and Load Estimator (LOADEST), for the case study region. The study assumed that each basin had only one high-frequency monitoring site within the region. In this study, high-frequency



nitrate data from each site were randomly subsampled to biweekly data and used as a pseudo-low-frequency site to train the DL model. These sites' remaining daily nitrate data were used for the developed DL model's performance evaluation. DL and WRTDS models performed similarly in most low-frequency nitrate monitoring sites. However, the developed DL model performed better at a few low-frequency nitrate monitoring sites. The DL and WRTDS models' performance in continuous nitrate estimation was better than the LOADEST model.

Though the developed DL model performed exceptionally well in many sites, the performance was low for a few nitrate monitoring sites. The second objective identified the critical environmental drivers that impacted the data-driven modeling approach in estimating continuous stream nitrate concentrations. DL models were developed to predict daily stream nitrate concentrations at locations lacking continuous data. The sensitivity of daily varying environmental variables, including high-frequency nitrate and stream discharge and day length (representing seasonality) data, was identified in estimating continuous nitrate concentrations. The similarity between the sensitive variables was determined using Dynamic Time Warping (DTW) method to assess the contribution of biophysical similarity on the DL model performance. The DL model learned and transferred nitrate dynamics to sites within a radius of 300 km of a high-frequency nitrate monitoring site. The dominant annual c-Q relationship plots inferred that the DL model potentially learned nitrate dynamics from the data of a high-frequency nitrate monitoring site and predicted continuous nitrate concentrations accurately at low-frequency nitrate sites. DTW analysis indicated a similarity in nitrate concentration, stream discharge, and day length between a low-frequency and a high-frequency nitrate monitoring site, contributing to the accurate daily stream nitrate prediction.

The third objective aimed to expand the data-driven modeling framework to those stream

locations where stream discharge information is unavailable and make the framework more spatially robust. This study evaluated the potential of precipitation or simulated stream discharge data to be used as a surrogate of observed stream discharge data for the data-driven model development. Five cases were developed based on the DL modeling framework using the combination of precipitation, observed, and simulated stream discharge values to identify the most significant variable for estimating stream nitrate concentrations. The DL model, developed with observed stream discharge as a crucial environmental characteristic, demonstrated the best performance. The DL model performed similarly with simulated stream discharge as the critical variable, indicating that simulated discharge can be a potential surrogate. The cross-correlation analysis showed that precipitation, stream discharge, and temperature influence the winter periods' nitrate dynamics. For the other three seasons (e.g., Summer, Fall, and Spring) except winter, stream discharge was the most significant environmental for stream nitrate variabilities.

This comprehensive study offers crucial insights into developing a spatially and temporally robust data-driven modeling framework for continuous stream nitrate estimation. The study results will help understand nitrate dynamics at data-limited locations, isolate the period for water withdrawal, optimize the new nitrate sensors installing locations, and design appropriate conservation practices to restrict landscape nitrate transport.

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for their endless love, support, and encouragement.

# **CHAPTER 1**

# INTRODUCTION

## 1.1 Introduction

Nitrogen is a crucial nutrient for crop production. However, nitrate, one of the derivatives of field-applied nitrogen fertilizer, affects aquatic ecosystems and human health when nitrate concentrates excessively in water bodies and is present in drinking water sources. Nitrate originates primarily from agricultural activities, urban runoff, atmospheric deposition, and wastewater. Thorough investigations portrayed nitrate contamination as worsening steadily, and about 21 million people were exposed to nitrate contamination across the vast Midwest, Southwest, Atlantic Coast, and California (Schechinger, 2020). Several Midwestern states, including Iowa, Illinois, Minnesota, and Missouri, within the Mississippi River basin have experienced persistent challenges with stream nitrate pollution. In the Mid-Atlantic region, total maximum daily load (TMDL) goals have not been achieved to attenuate the high volumes of nitrate transport to the Chesapeake Bay estuary (Boesch et al., 2010). On the US west coast, the Central Valley in California also faces higher stream nitrate concentration due to nitrate runoff from farmlands (Tariqi & Naughton, 2021). Intensive agricultural practices, including extensive inorganic and organic fertilizers and livestock farming, contributed to elevated stream nitrate levels within several US basins. As a result, eutrophication, and frequent formation of hypoxic zones in many estuaries, including the Gulf of Mexico and the Chesapeake Bay, are evident. To minimize nitrate transport from agricultural and urban landscapes, US federal, state, and local agencies implemented water quality management programs, regulations, and voluntary initiatives (Mississippi River/Gulf of Mexico Watershed Nutrient Task Force, 2008; Shortle et al., 2012; Van Meter et al., 2018; Ritter, 2019; Chesapeake Bay Progress, 2020). Major initiatives included the US Department of Agriculture

(USDA) Farm Bill Programs (Claassen and Horan, 2001) and the US Environmental Protection Agency (USEPA) Clean Water Act (CWA) (Dowd et al., 2008). These initiatives set water quality standards, promoted conservation practices, implemented riparian buffer zones, and suggested adopting new wastewater treatment technologies. However, the success and progress of these nitrate load reduction initiatives require further evaluation (USEPA, 2015; Rabotyagov et al., 2014; Jones et al., 2018).

Nitrate concentration data is essential to reduce the nitrate from reaching streams and plan appropriate best management practices (BMPs). Nitrate data provides critical insights into stream nitrate dynamics, transport mechanisms, transformation processes, and source characteristics (Duncan et al., 2015; Duncan, Welty, et al., 2017; Gorski & Zimmer, 2021; Meulemans et al., 2020). Nitrate data is also crucial for ensuring compliance monitoring with water quality regulations and assessing the ecological impacts severity and impact of climate change. In addition, nitrate data contributes to developing integrated decision support systems with other environmental and watershed characteristics data (G. K. Saha et al., 2021, 2022), helps raise public awareness, and promotes environmental stewardship (Singh et al., 2022). About 400 federal, state, and local agencies, including the USEPA, the United States Geological Survey (USGS), and the USDA, collect water samples from approximately 1.5 million locations in all fifty (50) US states to get different types of water quality data, including physical, chemical, biological, habitat, metrics, and indices (NWQMC, 2021). Often these agencies collect water samples at periodical scales (biweekly, monthly, seasonal) and analyze samples at labs due to the cost of sample collection and analysis. The water quality portal (<https://www.waterqualitydata.us/>) currently has 380 million data throughout the US. However, only about 800 US locations have consistent periodically collected nitrate data

between 2000 and 2020 (Sadayappan et al., 2022).

A crucial issue of these periodically collected lab-analyzed nitrate data is their insufficiency to portray accurate nitrate dynamics. Stream nitrate concentration varies substantially due to several environmental factors, including seasonal changes, agricultural activities, the geological formation of soils, precipitation, and runoff events (Burns et al., 2019a). Events like heavy rainfall, floods, or wastewater discharge incidents, could also lead to significant spikes in stream nitrate loads (Moatar et al., 2017; Zhi and Li, 2020). The periodically collected low-frequency (data collection frequency < rate of change in nitrate concentration) nitrate data miss these short-term fluctuations and fail to capture the full range of stream nitrate dynamics. In addition, nitrate concentration varies considerably across different locations in a region due to the variabilities in land use, proximity to streams, and hydrologic conditions. Intermittent data collection, especially in a limited number of stream monitoring locations, may partially capture spatial heterogeneity in nitrate concentrations. The spatially and temporally scarce nitrate data limits the ability to identify localized hotspots and the implementation of targeted interventions.

Statistical models generally fill the temporal gaps in nitrate data at low-frequency monitoring sites to estimate seasonal and annual nitrate loads. The existing statistical models include simple aggregations or interpolations, ratio estimators (Cochran, 1977), and regression-based models (Ferguson, 1986, 1987). The most recent regression-based models apply the flexible-functional-form parametric methods to a broader range of environmental conditions (Hirsch et al., 2010). A few of these statistical models, including Weighted Regressions on Time, Discharge, and Season (WRTDS; (Hirsch et al., 2010) and Load Estimator (LOADEST; (Runkel et al., 2004), provide reasonable estimates of monthly or annual nitrate loads and have

limited focus on estimating temporal variability of stream nitrate concentrations. These models utilize a combination of the three core explanatory variables, discharge, time, and seasonal variabilities, to estimate stream nitrate concentrations (Lee et al., 2016). Therefore, the respective site's nitrate concentration frequency and the nitrate concentration (c) - stream discharge (Q) relationship information impacted the accuracy of these statistical models' nitrate estimation (Hirsch, 2014; Lee et al., 2016; Maestre et al., 2014). Since stream discharge influences the variability in stream nitrate concentrations in a watershed, these statistical tools utilize stream discharge as the dominant environmental variable to estimate nitrate concentrations, ignoring other relevant environmental processes, including fertilizer application, land cover changes, changes in soil conditions, and landscape runoff. Moreover, the statistical models produce bias estimates with complex nitrate c-Q relationships. Another crucial aspect of these statistical models is that these statistical models learn nitrate dynamics from the nitrate data of one stream location and estimate nitrate concentrations for the respective location for another period. Though Spatially Referenced Regressions On Watershed-attributes (SPARROW; (Smith et al., 1997), a regression-based model, simulates spatial patterns in water quality data across large basins in a region, the SPARROW model was designed to estimate annual nutrient loads depending on annual hydrologic variability, with a narrow focus on persistent temporal variability. Process-based ecohydrological models are also used to estimate nitrate loads. Process-based models simulate the movement of water and the transformation of nitrate using climate and watershed biophysical characteristics data and require intensive calibration and validation to provide a comprehensive understanding of nitrate dynamics.

In the last decade, USGS and many state and local agencies, including Iowa Water Quality Information System (IWQIS), Monterey Bay Aquarium Research Institute (MBARI), California, Gulf of Mexico Coastal Ocean Observing System (GCOOS), Minnesota Pollution Control Agency, Deschutes River Continuous Nitrate Monitoring by Department of Ecology, State of Washington, and St. Joseph River Watershed Initiative (SJRWI) and the City of Fort Wayne, started placing optical nitrate sensors throughout the USA. These nitrate sensors can measure concentrations at 15- to 60-minute intervals (USGS, 2023b). This high-frequency (data collection frequency  $>$  rate of change in nitrate concentration) nitrate data has dense nitrate variability information, which can be utilized to understand regional nitrate dynamics and make informed management decisions. Though the sensor-based high-frequency nitrate data has more nitrate dynamics information, the number of monitoring sensors with long-term data is minimal. The nitrate sensors are also expensive and require a fair amount of logistic support. Therefore, collecting high-frequency nitrate data in an adequate number of stream locations requires considerable time and effort.

The available high-frequency nitrate data of multiple stream locations contains detailed information on a region's stream nitrate transport dynamics (such as source characteristics, magnitude, and timing of nitrate loading) and nitrate c-Q relationships. The nitrate dynamics information in the high-frequency stream nitrate data in the region could be used to estimate continuous nitrate concentrations at other data-limited locations. Data-driven models, especially deep learning (DL) models, can extract the complex nitrate dynamics from high-frequency nitrate data and transfer the information to estimate continuous nitrate at low-frequency nitrate monitoring sites in the region. Therefore, this study hypothesized that the DL model could learn regional stream nitrate dynamics from nitrate concentration, stream

discharge, and other climatic and watershed biophysical characteristics of multiple nitrate monitoring locations and transfer the learned information to estimate continuous nitrate data at other data-limited stream nitrate monitoring locations within the region.

DL modeling techniques for stream nitrate estimation will deploy a holistic approach to understand the complex interactions better between integrated environmental variables that impacted stream nitrate concentrations variability. DL models have the potential to learn and transfer regional dynamics, dominant patterns, non-linear and complex c-Q relationships, and dependencies among influential environmental variables from the available nitrate data of high-frequency nitrate monitoring sites to other low-frequency sites. Besides, the DL modeling techniques have been used in several other hydrological and eco-hydrological research (C. Shen, 2018). Researchers have used DL models, such as boosted regression tree (BRT), random forest (RF), support vector machine (SVM), and k-nearest neighbor (k-NN), to estimate groundwater nutrient concentrations and to analyze nutrient load estimation uncertainty (Rahmati et al., 2019; Ransom et al., 2017). DL models also have the potential to understand the in-stream nutrient dynamics and learn the nutrient's unknown c-Q relationships or c-Q relationships continuously vary depending on the watershed characteristics, fertilization, cropping patterns, and in-stream processes (Zhi et al., 2020; Zhi & Li, 2020). Existing nitrate concentration studies used machine learning models to estimate groundwater nitrate loads (He et al., 2019; Rahmati et al., 2019; Ransom et al., 2017; Wheeler et al., 2015). A primary reason behind the limited number of stream nitrate estimation studies is the unavailability of enough nitrate information to represent accurate temporal variability and land management information, including fertilization (rate, type, timing), land preparation (method and timing), planting timing, and artificial drainage system information. The recently available



high-frequency stream nitrate data from nitrate sensors portrayed the higher temporal nitrate variability in stream nitrate concentrations, which expanded the opportunity of utilizing DL models to learn stream nitrate dynamics and predict continuous stream nitrate at required locations. Therefore, developing a new DL model-based continuous stream nitrate modeling framework will address the multi-directional challenges, including cost and temporal delay in manual nitrate data collection and lab analysis, and contribute to making a spatially and temporally robust nitrate monitoring network for the USA.

## **1.2 Research goal and objectives**

With a vision to develop a temporally and spatially robust stream nitrate modeling framework, Iowa was selected as a case study region because Iowa has a higher number of high-frequency nitrate monitoring sites. The overarching goal of this study is to develop a data-driven modeling framework to estimate temporally and spatially continuous nitrate concentration data for data-limited locations in a region using high frequency nitrate data and biophysical characteristics of the region. The study objectives include -

- (1) Developing a deep learning-based modeling approach to generate continuous daily stream nitrate concentration for nitrate data-sparse watersheds and comparing the performance of the developed modeling approach with other statistical models.
- (2) Analyzing the deep learning approach-based modeling framework to understand the critical environmental drivers of predicting daily nitrate concentrations.
- (3) Exploring the potential of extending the data-driven stream nitrate concentration modeling framework to stream discharge unavailable locations for making the framework more spatially robust.

A DL model called Long Short-Term Memory (LSTM) was used in this study to estimate

continuous stream nitrate concentrations. Four hyperparameters (batch size, hidden layer size, time window, and epoch) and five performance metrics (RMSE, bias, correlation, NSE, and KGE) were used to train and evaluate the DL model for a case study region, Iowa, USA.

### **1.3 Dissertation Organization**

The entire dissertation has been crafted into six chapters. Chapter 1, Introduction chapter, introduces current stream nitrate status, the importance of tracking nitrate, existing nitrate data issues, strategies to estimate stream nitrate concentrations, the applicability of data-driven modeling to estimate nitrate concentrations, opportunities to understand regional nitrate dynamics by utilizing high-frequency nitrate data, developing a robust data-driven modeling framework for nitrate prediction, and goal and research objectives of the study. Chapter 2 explains the data-driven modeling approach of learning stream nitrate dynamics from the nitrate and biophysical characteristics data of multiple high-frequency sensor-based and periodically collected low-frequency nitrate monitoring sites and predicting continuous nitrate data for data-unavailable periods and data-scarce locations. Chapter 3 focuses on developing a data-driven modeling framework to fill the temporal gaps in the low-frequency nitrate data at a data-scarce region using high-frequency nitrate data from one nitrate monitoring location and comparing the developed framework with other widely used statistical models for nitrate estimation. Chapter 4 describes the analysis of the data-driven modeling framework, which was developed to fill the temporal nitrate data gaps to identify the critical environmental drivers influencing stream nitrate variability. Chapter 5 discusses the potential of extending the developed modeling framework to stream discharge data unavailable locations for building a more spatially robust data-driven modeling framework. Chapters 2, 3, 4, and 5 are formatted as manuscripts and have their individual supplemental information in the Appendices. Chapter

6 summarizes the overall study outcome, limitations, and future scope of expanding this research.

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## **CHAPTER 2**

# A DEEP LEARNING-BASED NOVEL APPROACH TO GENERATE CONTINUOUS DAILY STREAM NITRATE CONCENTRATION FOR NITRATE DATA-SPARSE WATERSHEDS.

## 2.1 Introduction

Nitrate contamination of streams, lakes, and estuaries is a pressing issue in many agricultural watersheds (Kuwayama et al., 2020; Liang et al., 2020). Excessive nitrate concentration in streams accelerates eutrophication and alters the physical characteristics of aquatic flora and fauna (Blaen et al., 2017; Pellerin et al., 2014). Nitrate reaches streams from natural and anthropogenic sources, including fertilizer-derived agricultural runoff, sewage and septic tank leakages, industrial activities, and atmospheric deposition (Nijesh et al., 2021; A. Saha et al., 2022; G. Saha et al., 2018; G. K. Saha et al., 2021, 2022). Stream nitrate concentration data is crucial for evaluating stream health and planning and implementing conservation practices. The stream nitrate concentration is generally measured through the lab analysis of grab-sample data. These data are collected bi-weekly, monthly, or seasonally depending on their specific purposes. In the United States (US), about 400 agencies (state, federal, tribal, and local), including the United States Geological Survey (USGS), United States Environmental Protection Agency (USEPA), and United States Department of Agriculture (USDA), collect stream nitrate concentration data (Read et al., 2017; Sprague et al., 2017). According to the Water Quality Portal database (Read et al., 2017), about 5,500 sites in the US have a minimum of 20 nitrate measurements over 20 years. However, measuring stream health with these intermittently collected discrete nitrate data is challenging since the data may not portray complex nitrate concentration trends and seasonal variabilities.

In many previous studies, researchers used statistical methods to generate continuous nitrate

time-series data by mapping intermittently collected nitrate data onto known temporal and seasonal streamflow patterns (Alam et al., 2020; Y. Jiang et al., 2014a, 2014b; Li et al., 2019; Mize et al., 2018; Wu et al., 2022; Q. Zhang et al., 2021). Nitrate concentration and load estimation methods include linear interpolation between measured data (Le Moal et al., 2019) as well as regression-based tools such as Load Estimator (LOADEST) (Runkel et al., 2004) and Weighted Regressions of Time, Discharge, and Season (WRTDS) (Hirsch et al., 2010). The LOADEST and WRTDS tools utilize discharge, time, and seasonal information to determine the temporal variability in nitrate concentration. Though these regression-based tools are widely used, they could not consider all relevant environmental processes, such as fertilizer application rate and landscape characteristics. The accuracy of statistical estimation methods depends on the availability of nitrate data and the complexity of the nitrate concentration-discharge (c-Q) relationship (Hirsch, 2014; Lee et al., 2016; Maestre et al., 2014). The nitrate c-Q relationship varies considerably in managed catchments with spatially and temporally varying nitrogen sources (e.g., fertilization, manure application, atmospheric deposition, and N fixation), nitrogen sinks (e.g., plant uptake and transformations), and dominant flow characteristics (e.g., surface runoff, groundwater flow, tile flow, and lateral flow). Thus, the accuracy of these models is not optimal.

In-situ nitrate sensors have grown popular in the last decade. In-situ sensors can collect high-frequency (sampling rate > rate of change) nitrate data at 5- to 60-minute intervals. In the US, the USGS has 167 active nitrate gauges reporting nitrate data at 15-minute intervals (USGS, 2023b), with records spanning 1-10 years. High-frequency nitrate sensor data contains nitrate trend, variability, and c-Q relationship information that can improve understanding of the hydrologic and biochemical processes underlying episodic, diel, and long-term stream nitrate



dynamics. Furthermore, high-frequency nitrate data can provide new knowledge about the influence of autotrophic uptake, denitrification, riparian evapotranspiration, temperature-driven diel patterns, and climate change on stream nitrate concentration (Burns et al., 2019a; Duncan, Band, et al., 2017; Duncan et al., 2015; Duncan, Welty, et al., 2017). The study hypothesizes that deep learning (DL) models can extract the trend and temporal variability information available in high-frequency stream nitrate sensor data and transfer the information to predict continuous daily nitrate for unavailable data periods and data-limited locations in a region.

DL techniques allow for learning complex temporal trends and transferring the learned information to new situations (S. Jiang et al., 2020; Ma et al., 2021; C. Shen, 2018). DL techniques have been used with high-frequency time-series data to predict streamflow, soil moisture, water temperature, and dissolved oxygen concentration at neighboring locations (Fang et al., 2017a; Feng et al., 2020; Rahmani, Shen, et al., 2021; Zhi et al., 2021). DL techniques have also been used to predict groundwater nitrate concentrations (Anning et al., 2012; Ransom et al., 2017; Wheeler et al., 2015). However, due to the complexity of nitrate c-Q relationships, DL techniques have yet to be used to predict surface water nitrate concentrations at large spatial scales. Recently, Long Short-Term Memory (LSTM), a deep recurrent neural network, has been used widely in hydrology and water quality modeling studies involving complex environmental processes. The LSTM model can learn from the temporal variability of high-frequency data and predict data for other periods and locations (Kratzert et al., 2018; C. Shen, 2018; Zhi et al., 2021).

In this study, a novel data-driven modeling framework was developed for using high-frequency nitrate data and biophysical attributes of the landscape to estimate continuous daily stream

nitrate concentration. Specific objectives were to (i) develop a data-driven modeling framework to estimate continuous daily stream nitrate concentration for unavailable data periods at existing nitrate monitoring sites, focusing primarily on data-limited sites; and (ii) study the nitrate concentration-discharge (c-Q) relationships to assess model's capability to represent the nutrient dynamics in the study region.

## **2.2 Methods**

Nitrate loss from landscapes to streams depends on several biophysical characteristics, human-induced activities, and climatic processes, including land cover, soil properties, slope condition, fertilizer application rate, type and timing, and weather pattern. The exploratory study was conducted in Iowa, USA since the state has many sensor-based nitrate monitoring sites with long-term data. The datasets critical for nitrate prediction were collected, processed, and divided into training and testing period data for the DL model development. The LSTM was used as the DL model to estimate continuous daily nitrate at gauged stream locations in Iowa. Five cases were formulated to explore the robustness of data-driven nitrate modeling and evaluate the significance of precipitation, discharge, and regional and local nitrate information on stream nitrate concentration estimation. Each case was simulated with multiple hyperparameters (epochs, batch size, hidden layer size, time window) combinations to select the best LSTM model. The developed LSTM models were evaluated based on statistical metrics (Bias, RMSE, NSE, Correlation, KGE) and used to model continuous daily stream nitrate concentration.

### **2.2.1 Study area**

Iowa is an agriculture-dominated state; about 90% of the state is under row crops (60%) or grasslands (30%) (USDA-NASS, 2020). Most of the farmers grow corn and soybean in the

row cropping areas. In addition, the state has rolling hills and landscapes with natural lakes, which facilitate agricultural nutrient transport from lands to streams. The state's water quality is degrading because of the increasing number of high-density livestock facilities and fertilizer and pesticide runoff from crop production (Babcock et al., 2003; The Iowa Journal, 2009). The state has 34 sensor-based continuous nitrate monitoring sites. In addition, the USGS and USEPA collected nitrate samples periodically from more than 5000 sites in Iowa. Since high-frequency nitrate data used in the study is available from 2012 and onwards, a strong focus of this study was to identify intermittently collected nitrate samples for 2012-2019 from the intermittently collected nitrate monitoring sites. Of these nitrate monitoring sites, eight (8) nitrate monitoring sites have more than 25 nitrate samples collected between 2012 and 2019 (Fig. 2.1).

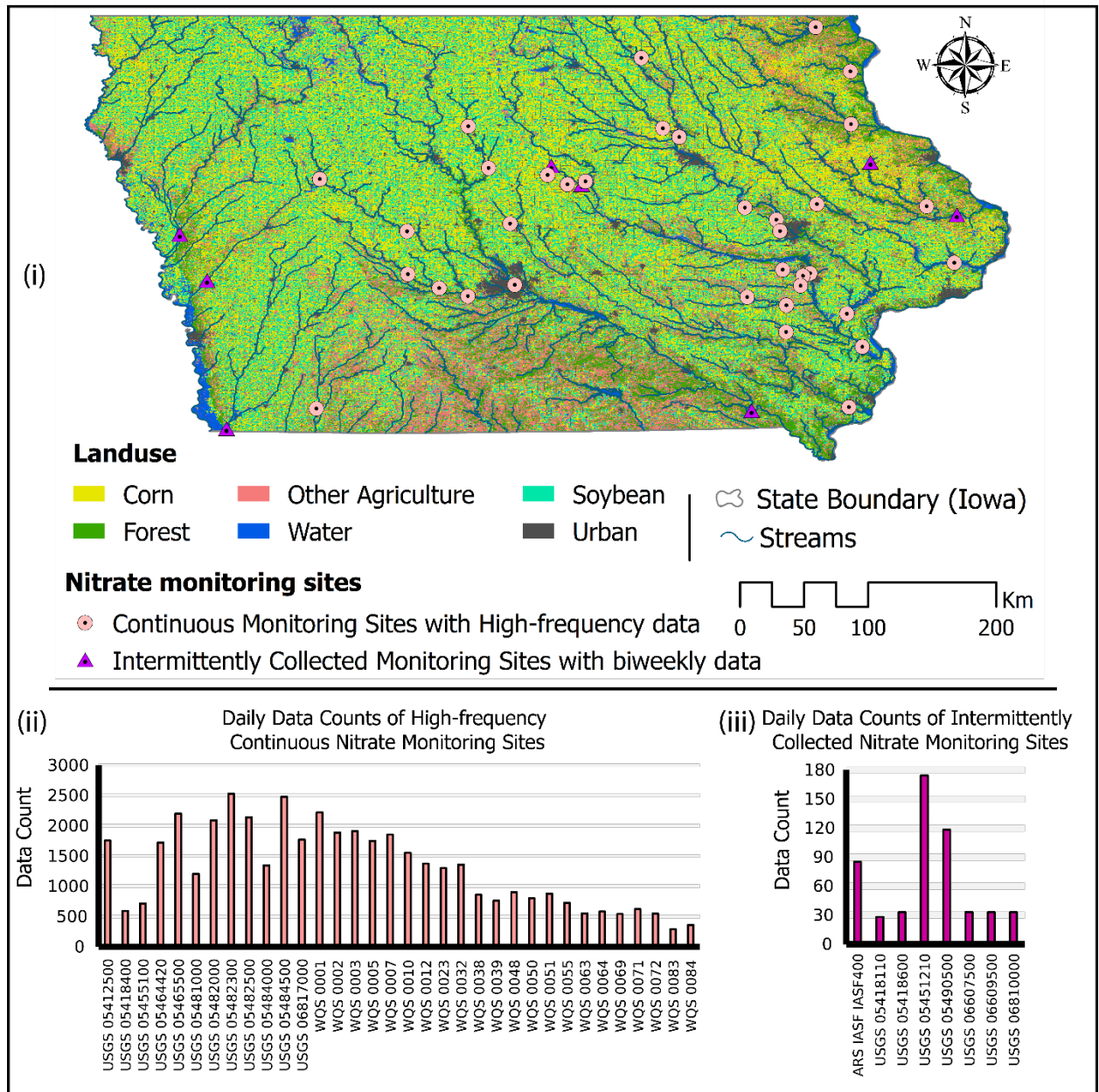


Figure 2.1 Nitrate monitoring site locations, nitrate data count, and land use categories in Iowa, USA: (i) 42 nitrate monitoring site's locations, including major land use categories; (ii) 34 high-frequency nitrate monitoring sites; and (iii) 8 intermittently collected nitrate monitoring sites.

## 2.2.2 Data sources and preparation.

Biophysical, hydrologic, and climatic information is critical for explaining the stream nitrate dynamics. Considerable effort was put into accumulating and processing the collected data

from several secondary data sources and made the data consistent for the model input. The data for each selected nitrate monitoring site in Iowa were processed based on the site's contributing watershed area. Land cover, topography, soil, precipitation, temperature, day length, discharge, and fertilization data were collected for the contributing area of each monitoring site. The biophysical data sources include the USDA-NASS geo-database from CropScape – NASS CDL Program (USDA-NASS, 2020) for land cover; the USGS National Geospatial Program (USGS, 2020c) for elevation and hydrography data; USGS Water Data (USGS, 2020a) for discharge data; NuGIS (The Fertilizer Institute, 2020) for annual fertilizer N and manure N input data, and USDA's SSURGO database (USDA-NRCS, 2020) for soil characteristics. Precipitation, temperature, and day length data were obtained from the DAYMET website (Thornton et al., 2020). Sensor-based continuous nitrate data were gathered from the websites of USGS WaterWatch (USGS, 2020b) and the IWQIS (The University of Iowa, 2022). Intermittently collected nitrate data were downloaded from the Water Quality Portal (NWQMC, 2021). Nitrate data collection at most sensor-based monitoring sites started in 2012. Sensor-based continuous nitrate data were obtained from 2012 (the earliest year for sensor-based data availability) to 2019. Intermittent nitrate data were obtained for the same period. Precipitation, temperature, and discharge data are available at daily timesteps. Land-use data (corn, soybean, urban, and forest areas) and fertilization data (fertilizer N and manure N applications data) are available at the annual time step. Annual data were converted to a daily time step by assuming a constant value for the entire year. The remaining datasets, including slope, shallow soil area, drainage class-based soil areas, hydrologic soil group-based soil areas, erodibility factors for soil, and areas of dominant soil textures, are watershed-specific attributes. The county-level inorganic fertilizer N and manure N inputs data were downscaled

to contributing head-watershed-level data. Downscaling was conducted based on the area proportion of respective counties and head-watershed. The DAYMET website provided gridded daily precipitation, temperature (both maximum and minimum), and day length data. The daily gridded values of each contributing watershed were averaged to produce daily time series for each contributing head-watershed. The LSTM model requires all the input data to be continuous without any data gaps. The missing discharge data (10%) were filled using simple linear regression and three-year moving average extrapolation for fertilization data, available only until 2016.

The average soil depth, soil drainage class, hydrologic soil group, soil erodibility factor, and soil texture data were processed for each site's contributing area from the SSURGO database. Shallow and deep soil areas were calculated using the soil depth data since the depth data provided us with insights into soil infiltration and runoff characteristics. Soil drainage class information was categorized into three broader classes: excessively drained, well-drained, and poorly drained areas. The soil area of each contributing watershed was divided into four hydrologic soil groups: A, B, C, and D. The dominant soil erodibility factor, dominant and second most dominant soil texture, and related area coverages were estimated.

Among thirty-four sensor-based continuous nitrate monitoring sites in Iowa, IWQIS operates twenty-two, and USGS operates twelve monitoring sites, respectively. Each of these sites has nitrate information at 15-minute intervals. Since we are interested in daily predictions in this work, we aggregated sensor data into daily average concentration data. Two nitrate time series, termed regional nitrate average (RegionNO<sub>3</sub>) and local nitrate average (LocalNO<sub>3</sub>), were prepared using the nitrate data of existing USGS and IWQIS monitoring sites. A site's regional nitrate average (mg/L) is the mean of all nitrate data from other considered sites in a study

region, without including the nitrate data for the prediction site. In this study, the entire Iowa was considered the study region. The local nitrate average (mg/L) for a site is the mean of all nitrate data of neighboring sites of adjacent watersheds within 50 km Euclidean distance, excluding the nitrate data for the prediction site. Table S2.1 in Appendix A lists all forcings and attributes data, descriptions, and corresponding measuring units.

### **2.2.3 Description of Deep Learning technique: LSTM Network.**

The LSTM model algorithm used in the study was previously developed and employed for soil moisture estimation (Fang et al., 2017a, 2019), streamflow forecasting (Feng et al., 2020), stream water temperature prediction (Rahmani, Lawson, et al., 2021a; Rahmani, Shen, et al., 2021), and dissolved oxygen (DO) in stream estimation (Zhi et al., 2021). The algorithm uses several units called memory cells and gates to learn and store information for long periods. Memory cells store the information, and different gates decide the fate of the information. The model has an input gate, an output gate, and a forget gate. During the model training, the gates simultaneously receive all the input information and predict the target variable. This workflow is end-to-end training that avoids external decision inputs. The architectural design and LSTM equations are demonstrated in the SI document (Fig. S2.1 in Appendix A). More details on the LSTM model's structure refer to Hochreiter and Schmidhuber (1997) and Fang et al. (2019, 2017) several units called memory cells and gates to learn and store information for long periods. Memory cells store the information, and different gates decide the fate of the information. The model has an input gate, an output gate, and a forget gate. During the model training, the gates simultaneously receive all the input information and predict the target variable. This workflow is end-to-end training that avoids external decision inputs. The architectural design and LSTM equations are demonstrated in the SI document (Fig. S2.1 in

Appendix A). More details on the LSTM model's structure refer to Hochreiter and Schmidhuber (1997) and Fang et al. (2019, 2017).

In our LSTM implementation, the user can specify different timelines for model training and testing for each monitoring site (Table S2.2 in Appendix A). The updated model algorithm selects data based on user inputs and simulates accordingly. This improvement gives the LSTM model more flexibility for users to apply for discrete and data-limited target variables. The developed LSTM model to predict daily stream nitrate in this study is available at GitHub ([https://github.com/gourabsaha010/LSTM\\_Nitrate](https://github.com/gourabsaha010/LSTM_Nitrate)).

#### **2.2.4 Simulation cases to evaluate sensitivity of input data.**

Five different simulation cases were developed with different forcing combinations to evaluate the sensitivity of the selected forcing inputs on the nitrate estimation and spatial robustness (Table S2.3 in Appendix A). In case 1 (baseline case), the LSTM model was developed using all forcings and attributes except RegionNO<sub>3</sub> and LocalNO<sub>3</sub> to investigate the model performance using only the climatic and watershed characteristics. The following three cases (Case 2, 3, and 4) were developed to isolate different variables (RegionNO<sub>3</sub>, LocalNO<sub>3</sub>, discharge) and investigate their impacts on the model performance. In case 2, RegionNO<sub>3</sub> and LocalNO<sub>3</sub> were included along with the inputs used in Case 1 (all inputs listed in Table S2.1 in Appendix A were used). In Case 3, the LSTM model was developed with all input data as in Case 2, excluding RegionNO<sub>3</sub>. Case 4 excludes both discharge and RegionNO<sub>3</sub> to identify the impact of discharge on continuous daily nitrate estimation. Case 5 was developed by generating pseudo intermittent nitrate monitoring sites, randomly selected 10% of the continuous nitrate monitoring sites (4 sites) were subsampled biweekly. These four sites' input data, training, and testing periods for were the same as in Case 2. The measured nitrate



concentration data from these pseudo intermittent nitrate monitoring sites were compared with the LSTM estimated concentration data to evaluate the robustness of the model in predicting the continuous nitrate concentration data in intermittently collected nitrate monitoring sites. The LSTM model was trained, including all the sites at once. The developed LSTM models for all 5 cases were trained with different hyperparameter combinations (minibatch, epoch, hidden layer, and time window) and random seeds. The final hyperparameters-combination was selected based on the simulated outputs for the testing periods. The SI document shows the hyperparameter-selection details (Section: Setting of hyperparameters for each simulation case).

#### **2.2.5 Evaluation matrices.**

The LSTM model was developed, minimizing the root-mean-square error (RMSE) as the loss function. Statistical scores of RMSE, bias, correlation, Nash–Sutcliffe efficiency coefficient (NSE) (Nash & Sutcliffe, 1970), and Kling Gupta Efficiency (KGE) (Gupta et al., 2009) (Details available in the SI document) were calculated for the testing period data of all 42 stream nitrate monitoring sites. The optimal LSTM model was selected for each case based on the median values of the five statistical scores for all 42 nitrate monitoring sites.

### **2.3 Results**

The data-driven model used in the study learned temporal variability in stream nitrate concentration and predicted continuous daily nitrate for data-unavailable periods and at data-limited locations within a region. The model showed unprecedented performance of  $NSE = 0.75$  and  $KGE = 0.83$  in daily stream nitrate prediction. The nitrate data from neighboring nitrate monitoring sites helps the model to learn temporal and spatial variability in daily nitrate concentration throughout the region and make a general model to estimate the daily nitrate

concentration. The developed data-driven model performed better for the summer and fall seasons. The estimated continuous daily nitrate portrayed similar c-Q relationships as the observed nitrate, which indicated the model's robustness in daily nitrate prediction.

### 2.3.1 LSTM based nitrate modeling framework.

The simulated LSTM models demonstrated strong performance in estimating stream nitrate concentration for the respective testing periods of all monitoring sites (Table 2.1). The Case 2 framework with all inputs provided the best model performance with the median test-period NSE, KGE, correlation, RMSE, and bias were 0.75, 0.83, 0.88, 1.53 mg/L, and 0.15 mg/L, respectively (Fig. 2.2). Twenty-one (21) sites (50% of all selected nitrate monitoring sites) had test-period NSEs greater than 0.75 individually. Hyperparameter combination of batch size 42, hidden layer size 50, and a time window of 150 days (Case 2 and Case 3) or 180 days (Case 1 and Case 4) provided the best results for the testing period. The SI document provides boxplots of all simulation cases and median test-period evaluation metrics in figures S2.2 - S2.6 and Table S2.4 in Appendix A.

Table 2.1 Median scores of evaluation matrices for all five cases' best outputs

| Case No. | Testing |      |      |             |      | Training |      |      |             |       |
|----------|---------|------|------|-------------|------|----------|------|------|-------------|-------|
|          | NSE     | KGE  | RMSE | Correlation | Bias | NSE      | KGE  | RMSE | Correlation | Bias  |
| 1        | 0.63    | 0.77 | 2.09 | 0.83        | 0.54 | 0.75     | 0.78 | 1.55 | 0.91        | -0.06 |
| 2        | 0.75    | 0.83 | 1.53 | 0.88        | 0.15 | 0.89     | 0.90 | 1.19 | 0.96        | -0.03 |
| 3        | 0.72    | 0.82 | 1.52 | 0.87        | 0.19 | 0.87     | 0.86 | 1.33 | 0.94        | -0.05 |
| 4        | 0.70    | 0.78 | 1.74 | 0.86        | 0.25 | 0.85     | 0.82 | 1.39 | 0.93        | 0.06  |
| 5        | 0.74    | 0.84 | 1.57 | 0.88        | 0.26 | 0.88     | 0.88 | 1.29 | 0.95        | 0.03  |

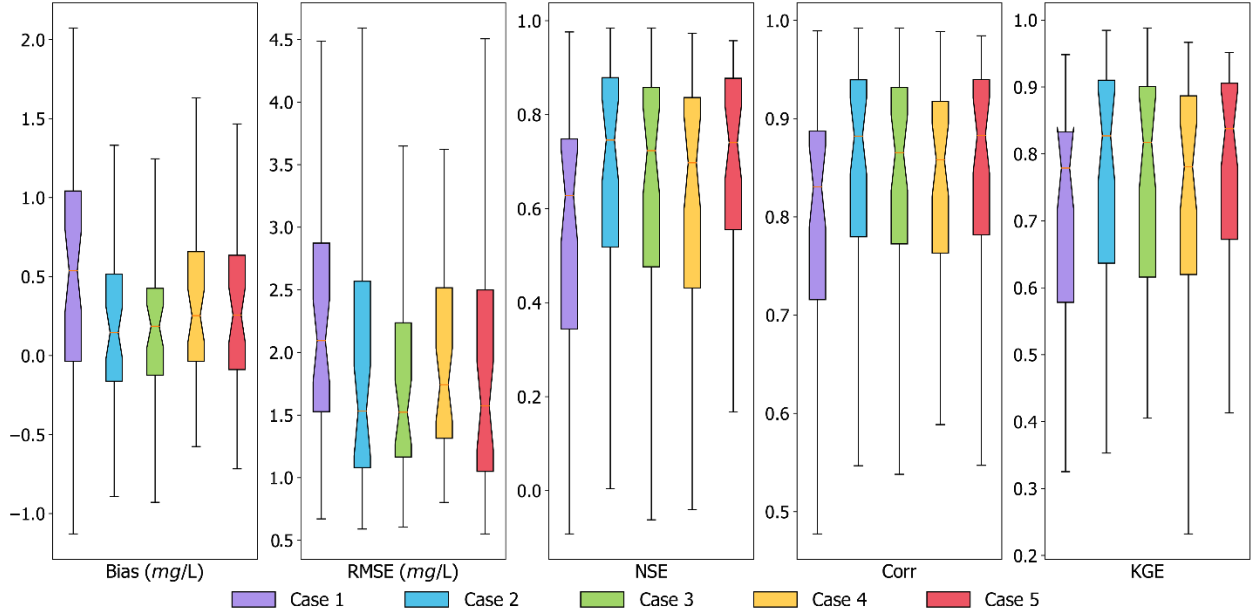


Figure 2.2 Comparison among boxplots of the best simulated outputs from all five cases

### 2.3.2 Comparison among models based on the change in dominant dynamic inputs.

Comparing the baseline case (Case 1) with other test cases (Case 2-5) indicated that the inclusion of local and regional averages of nitrate (LocalNO<sub>3</sub> and RegionNO<sub>3</sub>) improved the model's ability to estimate continuous daily stream nitrate concentration in the case study region (Fig. 2.2). Cases 2 to 4 comparison indicates that LocalNO<sub>3</sub> is more critical in nitrate concentration prediction in the study region among the three variables (Q, RegionNO<sub>3</sub>). Evaluation scores (Table 2.1) were reduced slightly for the case without discharge (Q) as input (Case 4). Case 2 and Case 5 simulation results showed similar values of evaluation metrics (median test-period NSE, KGE, RMSE, correlation, and bias values). The four pseudo-intermittent nitrate monitoring sites in Case 5 (USGS 05481000, USGS 05482500, WQS 0005, and WQS 0012) had test-period RMSE scores of 1.61, 1.20, 2.38, and 1.90, NSE scores 0.96, 0.95, 0.58, and 0.59, and KGE scores 0.92, 0.90, 0.63, and 0.78, respectively, which were very similar to Case 2 (Table S2.5 in Appendix A).

The Case 5 model-based evaluation metrics for the four selected pseudo intermittent nitrate

monitoring sites were promising while slightly lower than Case 2, which was trained with complete data (Table S2.5 in Appendix A). For USGS 05481000, USGS 05482500, WQS 0005, and WQS 0012, RMSEs were 2.41, 1.50, 2.20, 2.09, NSEs were 0.90, 0.92, 0.55, and 0.53, and KGEs were 0.88, 0.90, 0.68, and 0.75 (based on Case 5). The load estimations based on the Case 2 model from these four sensor-based continuous nitrate monitoring sites were within the 10% error limit for all four pseudo-intermittent nitrate monitoring sites (Table S2.6 in Appendix A).

### **2.3.3 Concentration time series analysis of DL model**

The Case 2 LSTM model, which performed best considering median test-period evaluation metrics, was used to evaluate the continuous daily stream-nitrate concentration time series for all nitrate monitoring sites. All intermittently collected nitrate monitoring sites performed well for their respective testing periods. Among eight (8) intermittently collected nitrate monitoring sites, five (5) sites demonstrated NSE greater than 0.75, two (2) sites greater than 0.7, and one (1) site greater than 0.50. The estimated test-period NSEs for intermittently collected nitrate monitoring sites support the research hypothesis of using the temporal variability information in the high-frequency nitrate data to model daily nitrate data at other gauged stream locations (Fig. 2.3). The time series comparison of pseudo intermittent nitrate monitoring sites in Case 5 with Case 2 indicated similar nitrate concentration predictions in both Cases (Fig. S2.7 in Appendix A).

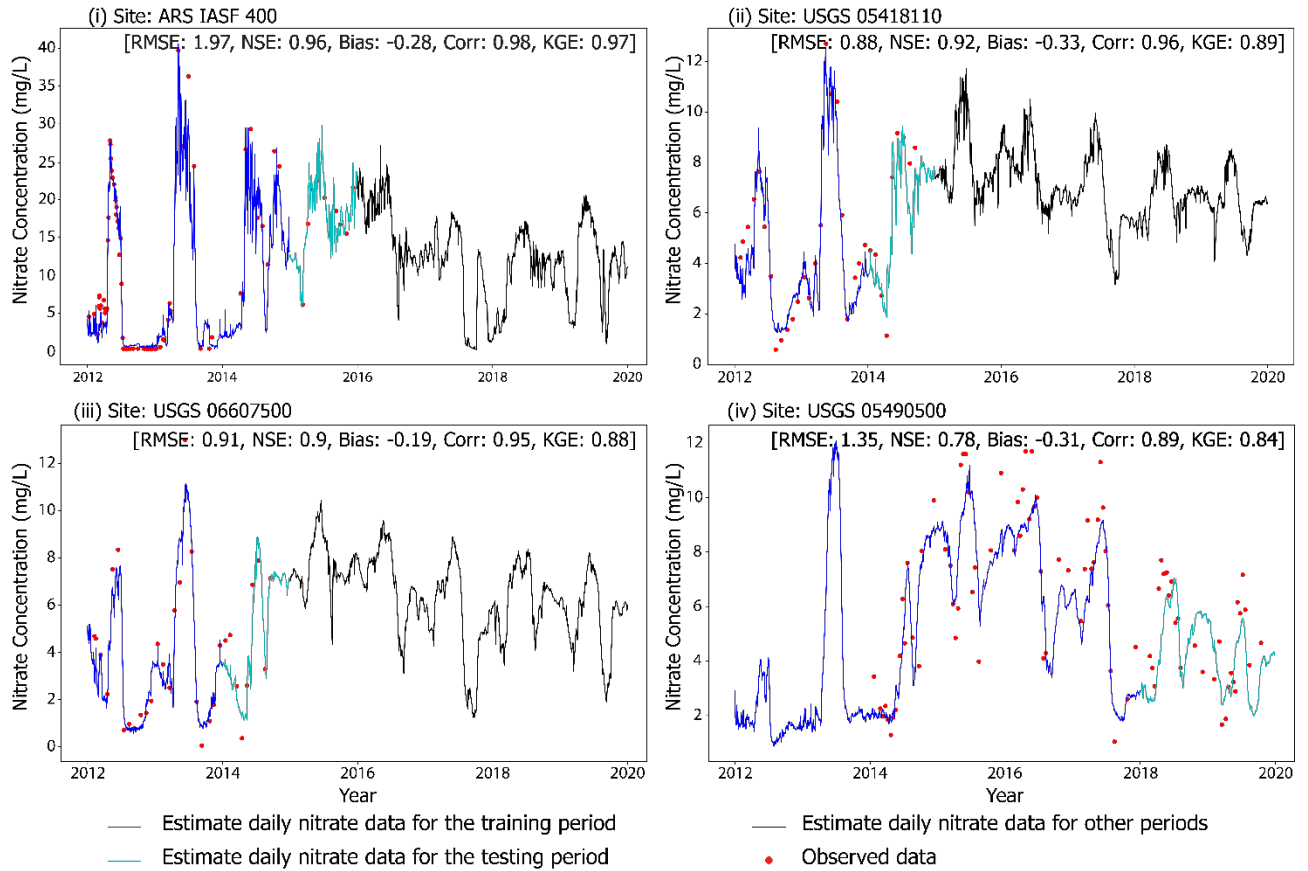


Figure 2.3 Time series plots of continuous daily nitrate concentration for four intermittently collected nitrate monitoring sites for period 2012-2019. These four sites perform best for their respective testing periods. The test-period model performances for four intermittently collected nitrate monitoring sites: (5i) Site ARS IASF 400 [RMSE: 1.97, NSE: 0.96, Bias: -0.28, Correlation: 0.98, and KGE: 0.97], (5ii) Site USGS 05418110 [RMSE: 0.88, NSE: 0.92, Bias: -0.33, Correlation: 0.96, and KGE: 0.89], (5iii) Site USGS 06607500 [RMSE: 0.91, NSE: 0.90, Bias: -0.19, Correlation: 0.95, and KGE: 0.88], (5iv) Site USGS 05490500 [RMSE: 1.35, NSE: 0.78, Bias: -0.31, Correlation: 0.89, and KGE: 0.84].

### 2.3.4 Seasonal performance of the model.

The Case 2 result was divided into four seasons, including Winter (December – February), Spring (March – May), Summer (June – August), and Fall (September – November), to evaluate the estimation performance of the model in these seasons. The quantified median test-period NSE values in the winter, spring, summer, and fall seasons were -1.11, 0.13, 0.65, and

0.45 (Fig. S2.8 and Table S2.7 in Appendix A). In addition, the model showed comparatively lower performance for most nitrate monitoring sites during the winter than summer and fall seasons. The model performance was comparatively less effective for the winter and spring seasons (Fig. 2.4).

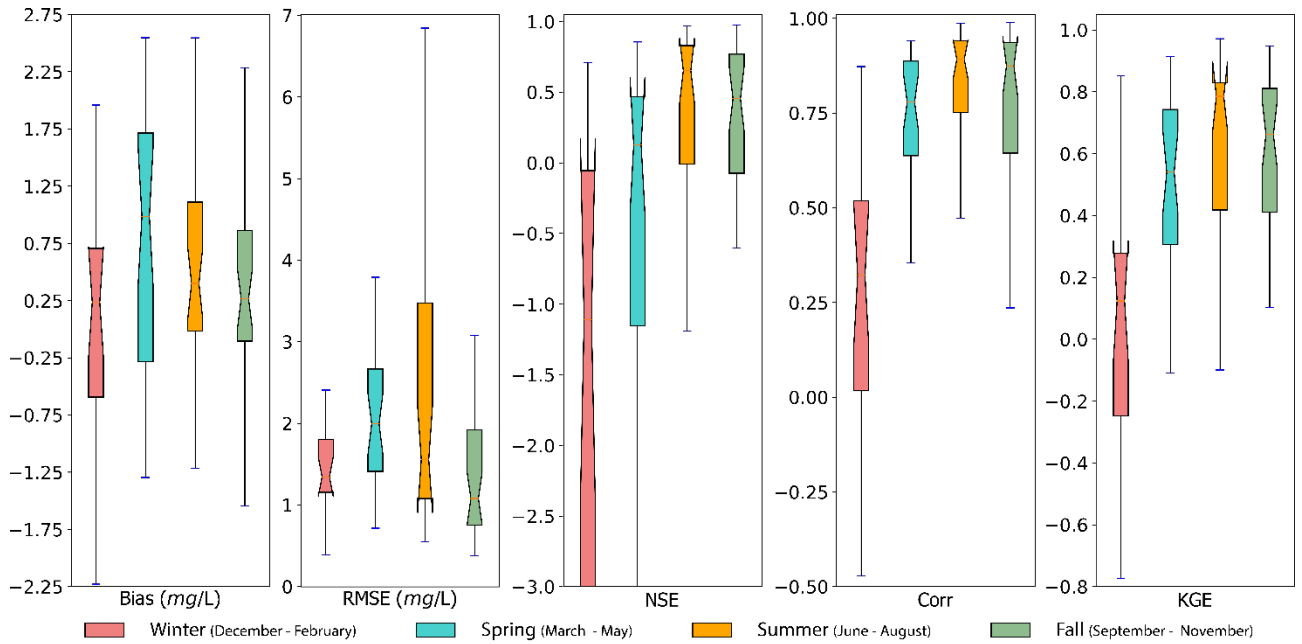


Figure 2.4 Boxplot comparison of estimated daily stream nitrate concentrations among four seasons based on the simulated output of Case 2 model.

### 2.3.5 Concentration-Discharge (c-Q) analysis to assess the model's capability representing nitrate dynamics.

Based on the estimated daily nitrate concentration (c) and stream discharge (Q), logarithmic nitrate concentration versus logarithmic discharge values were plotted for each site annually (2012-2019). The calculated slope (b) and correlation (r) values from  $\log_{10}(c)$  versus  $\log_{10}(Q)$  plots range from -0.2 to 2.10 and -0.40 to 0.90, respectively (Fig. S2.9 in Appendix A). The 50% nitrate monitoring sites (21 sites) had slope (b) values higher than the calculated mean b value ( $b_{\text{mean}} = 0.35$ ), which indicated the dominance of agricultural watersheds in the study

region. Twenty-seven (27) sites (64%) showed a correlation greater than 0.60 between concentration and discharge in most of the years between 2012 and 2019. The calculated  $b$  (range: - 0.60 – 1.82) and  $r$  (range: -0.90 – 0.92) values of observed  $\log_{10}(c)$  versus  $\log_{10}(Q)$  were identified as similar to the  $b$  and  $r$  values calculated for the estimated nitrate concentration. The observed ranges of slope and correlation values indicated the interannual variabilities in the  $c$ - $Q$  relationships of each site. The results portrayed the agreement between nitrate concentration and stream discharge, a common feature of agricultural watersheds.

The  $\log(c)$  versus  $\log(Q)$  plot for each site for each year showed four dominant patterns with increasing discharge: (i) flushing pattern, (ii) flushing before median discharge ( $Q_{50}$ ) and chemostatic after  $Q_{50}$  pattern, (iii) flushing before  $Q_{50}$  and dilution after  $Q_{50}$  pattern, and (iv) chemodynamic pattern were identified (Musolff et al., 2015) (Fig. 2.5). Twenty-seven (64%) sites showed a flushing pattern for 2-6 years from 2012 to 2019. About 38% of sites (16 sites) demonstrated a flushing pattern until  $Q_{50}$  and chemostatic at higher discharge for 2-4 years between 2012 and 2019. Nine monitoring sites (21%) presented a flushing pattern before  $Q_{50}$  and a dilution pattern after  $Q_{50}$  for 2-3 years. Fifteen sites (36%) demonstrated a chemodynamic pattern for 2-3 years from 2012 to 2019. Flushing and flushing with chemostatic patterns dominate most agricultural watersheds, portraying stream nitrate concentrations sensitive to significant hydrologic events such as floods (Moatar et al., 2017; Zhi & Li, 2020). Only two sites demonstrated a chemodynamic pattern consistently for six years between 2012-2019. About 87% of contributing watershed areas of these two monitoring sites include soil hydrologic groups C and D, indicating potential tile drain-based nitrate loading in these areas.

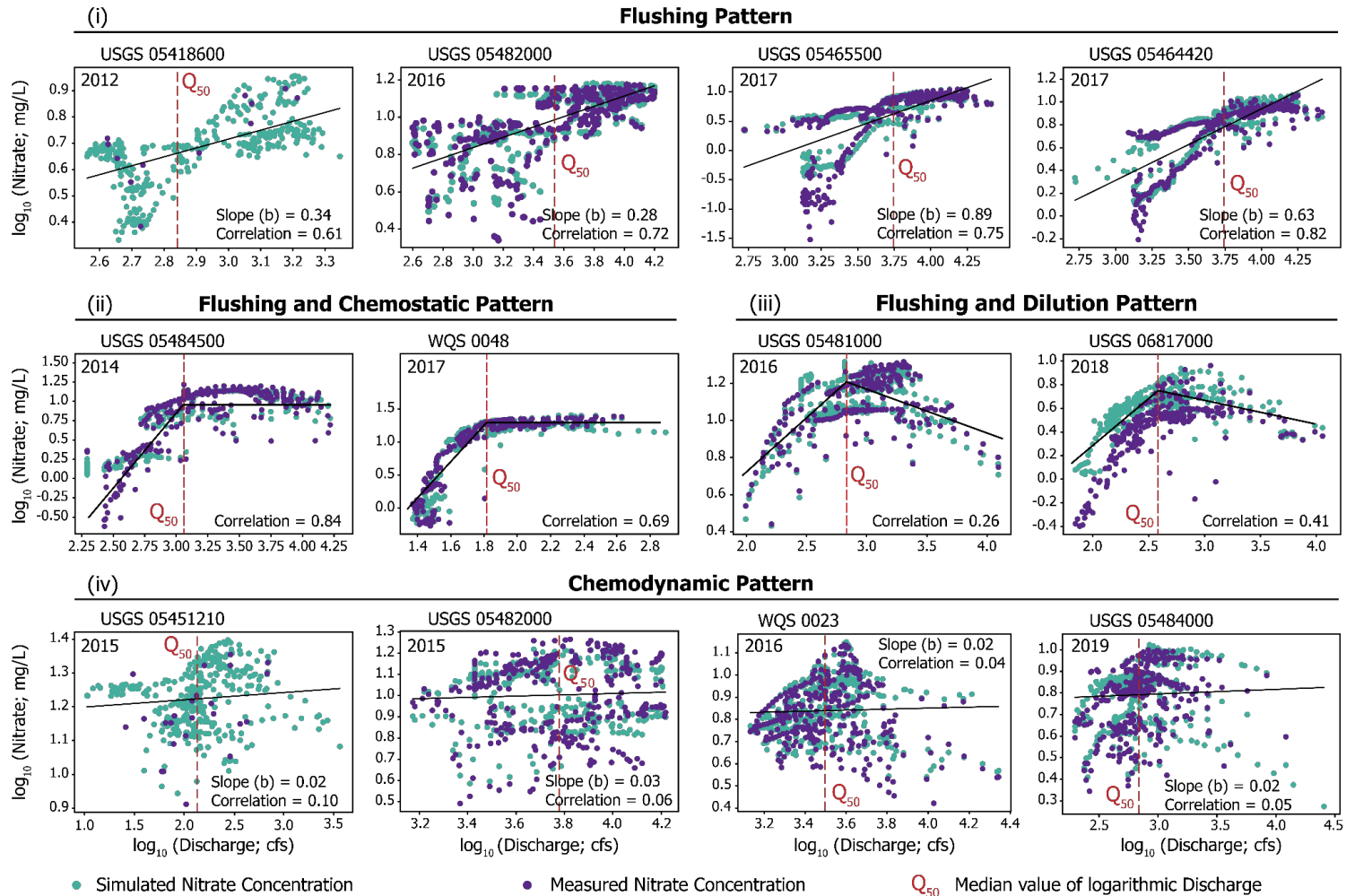


Figure 2.5 Scatter plots of logarithmic estimated nitrate concentration and logarithmic stream discharge values for different years. Estimated daily nitrate concentration and discharge values were plotted on annual basis. Four most dominant patterns, including (i) flushing pattern, (ii) flushing before median discharge ( $Q_{50}$ ) and chemostatic after  $Q_{50}$  pattern, (iii) flushing before  $Q_{50}$  and dilution after  $Q_{50}$  pattern, and (iv) chemodynamic pattern were identified.



## 2.4 Discussion

### 2.4.1 Overall Model performance in context to the existing literature

The daily nitrate estimation performance of the developed data-driven models (LSTM models) performance is encouraging. Unlike the metrics computed by other hydrologic and statistical models during stream nitrate concentration estimation in the existing literature, daily NSE was reported to be greater than 0.50 for the first time for all the cases in this study (Aouissi et al., 2014; Epelde et al., 2015; Jha et al., 2007; Moriasi et al., 2007; Noori et al., 2020; Nossent & Bauwens, 2012; van Liew et al., 2005; Verma et al., 2012). To give some context, Li et al. (2019) applied coefficient-of-variance (CV) as an evaluation metric to estimate annual total N concentration and reported a high CV (35.2%). Moatar and Meybeck (2005) identified linear interpolation as the best method among seven considered annual load estimation methods based on relative error variability as a metric. Jiang et al. (2014a) reported a one (1) mg/L mean absolute error while predicting monthly flow-weighted mean nitrate-N concentration from non-point agricultural sources in Indiana without reporting daily metrics. Jha et al. (2007) reported NSE scores of 0.73 and 0.78 for SWAT-simulated monthly and annual nitrate load prediction, respectively, but did not provide any daily metrics. In contrast, developed LSTM models performed much better estimating in-stream nitrate concentration for Iowa. Typically, daily predictions are more complex than monthly or annual predictions.

In general, precipitation and stream discharge are highly correlated (Breinl et al., 2021; Rohith et al., 2021). The minimal reduction in model performance (Table 2.1) for the case without discharge (Q) as input (Case 4) indicates precipitation as a good surrogate for discharge in modeling continuous daily stream nitrate concentration. Stream discharge is a critical component of the c-Q relationship and can be estimated at ungauged locations using

established methods (Marti et al., 2022). The potential of precipitation data as a surrogate provides an opportunity to develop more spatially explicit nitrate prediction models even at locations where streamflow data is unavailable.

#### **2.4.2 Model's robustness assessment to fill temporal gaps in daily nitrate concentration.**

The similarity in evaluation metrics (Table 2.1) among these four sensor-based continuous nitrate monitoring sites and their pseudo intermittent surrogates (Case 2 versus Case 5 results) implied pseudo intermittent nitrate monitoring sites, together with the remaining sensor-based continuous nitrate monitoring sites, had enough nitrate information to capture the trend and temporal variability in nitrate. This result also reflects the potential for spatial robustness of the developed LSTM models and indicates the possibility of making robust DL models with fewer continuous sensor-based gauges. However, a more detailed analysis is required to identify a minimum number of sensor-based nitrate monitoring sites. The pseudo intermittent nitrate monitoring sites' daily estimates were also compared with the remaining observed daily nitrate data to evaluate the effectiveness of the DL model in filling data gaps in intermittently measured nitrate sites. The evaluation metrics for the two USGS sites with larger contributing watershed areas were high. The high values of the evaluation metrics for the remaining daily nitrate data periods for the pseudo intermittent nitrate monitoring sites represented the potential of the developed LSTM model to utilize it for filling the data for unavailable periods. In addition, the time-series comparison plots for pseudo nitrate monitoring sites based on Case 5 and Case 2 indicated the similarity in trends and temporal variabilities of the two sites, which reinforced the hypothesis of using the developed LSTM model to fill the temporal data gaps for the unavailable period and treating the models robust spatially.

### **2.4.3 Comparison between LSTM-based nitrate estimates and reported statewide observed nitrate.**

The estimated mean nitrate concentrations of twenty-eight (28) best-performed and six (6) poorly performed sensor-based continuous nitrate monitoring sites were 6.90 and 8.70 mg/L, with standard deviations of 4.6 and 6.0 mg/L, respectively. These statistics portray that the variability in estimated nitrate concentration is higher for the poorly performed sites. The estimated mean nitrate concentration and standard deviations of these eight intermittently collected nitrate monitoring sites were 5.0-11.80 mg/L and 1.7 – 7.8 mg/L, respectively, based on Case 2. The estimated mean nitrate concentration is close to Iowa's reported statewide observed nitrate concentration mean (8.70 mg/L) (Jones et al., 2018). The calculated statistics implied that the developed LSTM model learned from the nitrate monitoring sites' information and transferred the info temporally and spatially to predict the continuous daily nitrate concentration of neighboring intermittently collected nitrate monitoring sites.

### **2.4.4 Seasonal performance evaluation of the developed LSTM model.**

The limited number of inputs on other environmental processes (e.g., base flow, tiled drains) beyond agricultural practices in the model could be the primary reason behind the low performance during the winter season. Previous studies also discussed the limited prediction capabilities of statistical models during the winter season (Rahmani, Lawson, et al., 2021a; Rahmani, Shen, et al., 2021). Besides, the Case 2 model performed well during the summer and fall seasons (Correlation: 0.89 and 0.87; KGE: 0.79 and 0.66), indicating that the model efficiently captures nitrate loss to streams due to surface runoff.

#### **2.4.5. Identifying potential reasons that could affect the developed stream nitrate concentration estimation model.**

The developed LSTM model also performed exceptionally well for sensor-based continuous nitrate monitoring sites. The study model performance was better than the recommended threshold model performance value ( $NSE > 0.35$ ) for daily, monthly, or annual nitrate estimation (Moriassi et al., 2007, 2012, 2015). During monthly and annual nitrate estimation for an individual site, many researchers reported  $NSE > 0.65$  as a cognizable model evaluation metric (Arnold et al., 2012; Daggupati et al., 2015; Wang et al., 2022). Other studies used model evaluation metrics, including percentage change to baseline, mean absolute error, and nitrogen loss score, to report the model performance statistically (F. Jiang et al., 2021; Mason et al., 2020). About 14% of monitoring sites (6 sites) performed poorly with the selected LSTM model. The time series of poorly performed sensor-based continuous nitrate monitoring sites for 2012-2019 showed that although the developed model could not predict concentration for those sites but could capture the trends and temporal variability effectively (Fig. S10 in Appendix A). Further investigation indicated that stream discharge and precipitation triggered the impulses in the estimated nitrate concentration time series of well-performed nitrate monitoring sites (Fig. 2.6). Temperature and day length also influenced the concentration's seasonal variability. Poor-performed sites do not agree with stream discharge and precipitation for several years between 2012 and 2019.

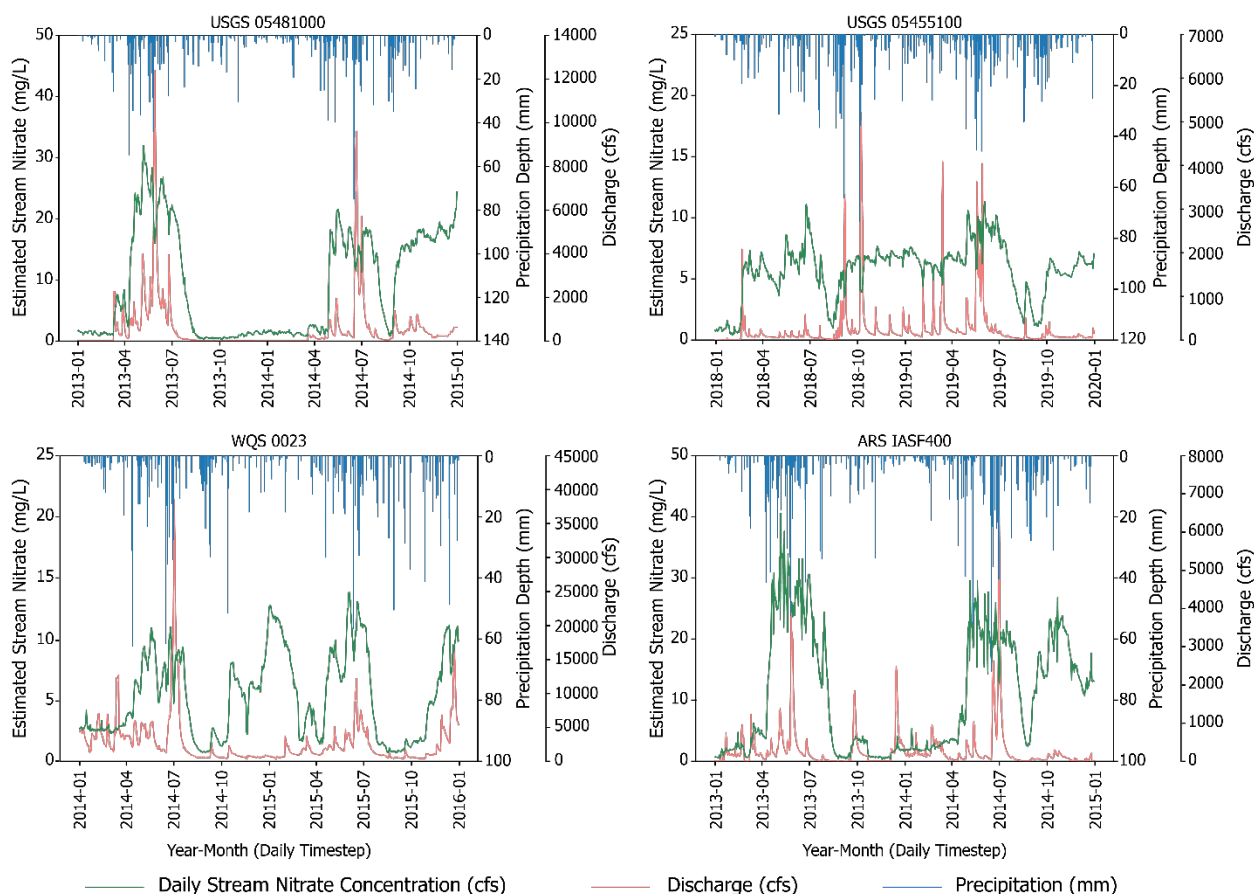


Figure 2.6 The variability of estimated daily nitrate concentration time series for nitrate monitoring sites with the variabilities of discharge and precipitation: Sites USGS 05481000 (period: 2013-2015), USGS 05455100 (period: 2018-2019), WQS 0023 (period: 2014-2016), ARS IASF400 (period: 2013-2015).

The spatial distribution of site-specific test-period NSE was mapped to identify spatial patterns of model performance in poorly performing sites (Fig. S2.11). The nitrate monitoring sites that performed poorly (sites with low test-period NSE) did not have high-frequency sensor-based nitrate monitoring sites within 50 km to calculate LocalNO<sub>3</sub> input. The nearest continuous sensor gauge was more than 300 km from each site. The low-NSE sites also had comparatively small contributing areas. Though the model input data were standardized to remove the effects of differing sizes and wetness in the contributing head watershed, the input data uncertainty

could be high from downscaling in small catchments. The Fertilizer-N and Manure-N data were downscaled from the county level to the level of contributing watershed. The downscaled data for smaller contributing head watersheds might inaccurately represent conditions at the downstream nitrate monitoring sites. The nitrate monitoring sites with low test-period NSE had fewer nitrate data for the training periods. Inadequate data in the training period for any site restricts the model from learning about the mechanisms happening in the respective site's contributing head watershed. Sites with a comparatively high nitrate data count for the training periods had high test-period NSE and vice versa (Fig. 2.7).

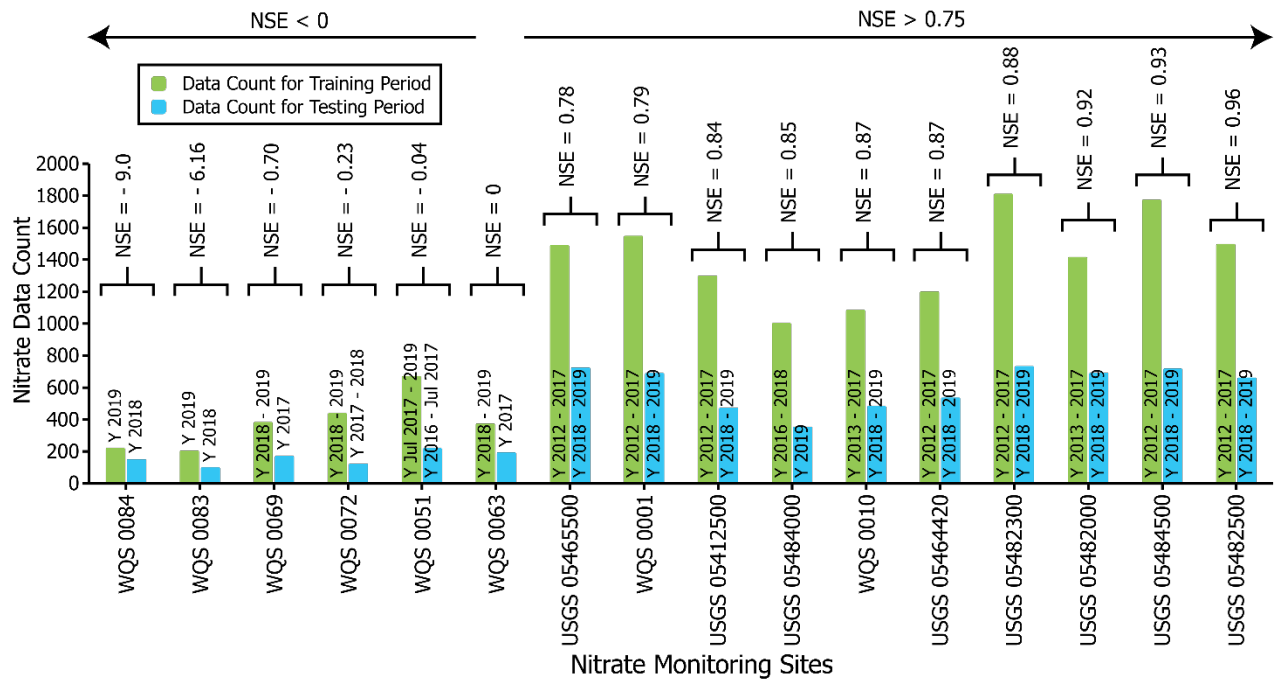


Figure 2.7 Bar graphs of nitrate counts of nitrate monitoring sites based on their high and low test-period NSEs and corresponding training and testing periods Impacts of these results.

The developed LSTM model can model continuous daily nitrate time series and fill the data gaps for chosen 42 nitrate monitoring sites in Iowa. The estimated continuous daily nitrate concentration provides a reasonable estimate of nitrate trends and seasonal variabilities. The

simulated LSTM model could not perform better for a few nitrate-monitoring sites having inadequate nitrate information. Many of these poorly performing nitrate monitoring sites have been established recently. Still, the estimated nitrate concentration for the poorly performed sites can be used as an initial measure for those sites' nitrate trends and seasonal variability. The developed LSTM models would perform better when the poorly performed nitrate monitoring sites have more high-frequency nitrate concentration data in the future.

#### **2.4.6 Capabilities of the developed stream nitrate concentration estimation model.**

The developed LSTM model's excellent performance indicates the potential for regional transfer of temporal and trend information among nitrate monitoring sites. Temporal and spatial transfer of the temporal variability and trend information from one gauged location to another using the LSTM model is one novel outcome of this study. Another crucial contribution of this study is the DL model's ability to handle more than one nitrate monitoring site's data and provide good model performance. No other existing nitrate estimation tools [e.g., LOADEST (Runkel et al., 2004) and WRTDS (Hirsch, 2014; Hirsch et al., 2010)] can extract information from more than one nitrate monitoring site and estimate continuous daily stream nitrate concentration. Because the developed DL model can handle data from multiple nitrate monitoring sites with relevant watersheds' climatic and biophysical attributes as inputs, the developed data-driven model provides the opportunity to expand to estimate surface water nitrate at regional and continental scales.

Estimating continuous daily stream nitrate concentration accurately in each location could be challenging for a large region like Iowa. However, accurate estimation of stream nitrate concentration at critical areas (e.g., water withdrawal sites) can benefit numerous stakeholders, including local watershed managers, water supply providers, and policymakers. For instance,

the U.S. water supply providers must meet the USEPA maximum contamination level (MCL) of 10 mg/L for nitrate. Observed nitrate data from 42 monitoring sites in Iowa showed nitrate exceeded the MCL 210 days (22% of the observed nitrate data) on an average between 2012 and 2019. The DL model estimated nitrate data exceeded USEPA MCL 230 days (25% of the days when observed nitrate data is available) on an average for all monitoring sites during 2012-2019. The average number of observed USEPA MCL exceeding days for the eight (8) intermittent sites was 21 (day range: 1-100 days). The average number of USEPA MCL exceeding days for the same sites based on DL model prediction was 22 (day range: 1 – 116 days). The similarity in observed and predicted nitrate values exceeding the USEPA MCL indicates the utility of the developed data-driven model in the drinking water supply. The developed DL model performed exceptionally well (median test-period NSE 0.75) and can estimate continuous daily nitrate for both intermittently collected and sensor-based nitrate monitoring sites. Therefore, regions with sensor-based monitoring sites can use this modeling technique to estimate nitrate concentration in nearby areas. Currently, Midwest, Florida, Louisiana, California, and Northeast in the USA have most of the sensor-based nitrate monitoring sites. The study could be extended to other states in the USA or even expanded to a continental scale as more sensor-based continuous nitrate data becomes available. The study results are promising; they support the hypothesis and encourage the development of more DL models for estimating continuous stream nitrate concentration. This study provides opportunities to explore the temporal and spatial transfer of trend and seasonal variability information to predict stream nitrate concentration at other data-limited locations using the LSTM model. A detailed investigation could provide insights into the dynamics of c-Q relationships, identify critical parameters, and choose the best DL model. Due to limited data



availability, this study did not include land management and crop information, such as the timing of season-based fertilization or the crop vegetative index. More sensor-based nitrate data can also offer opportunities for an improved understanding of watershed dynamics and a mechanistic understanding of in-stream biogeochemical processes (Burns et al., 2019a).

## **2.5 Conclusions**

This exploratory research successfully tested the hypothesis that the data-driven model learned temporally varying regional nitrate concentration relationship information available in sensor-based high-frequency nitrate data and generated continuous nitrate data for unavailable data periods and at other data-limited stream locations. The study results are encouraging and reinforce the research hypothesis. Thirty-four (34) sites (76% of all the nitrate monitoring sites) provided satisfactory evaluation metrics ( $NSE > 0.50$ ), which is state-of-art in surface water nitrate estimation. Among the sites, 21 sites (50%) showed test-period NSEs of more than 0.75 individually. Among eight (8) intermittently collected nitrate monitoring sites, five (5) sites demonstrated NSE greater than 0.75, two (2) sites greater than 0.7, and one (1) site greater than 0.50. The study reported that average nitrate information from neighboring monitoring sites (local nitrate,  $LocalNO_3$ ) influenced continuous daily nitrate estimation. Ten percent (10%) of the sites' nitrate information was subsampled to check the temporal robustness of the developed model. The simulated LSTM model with subsampled nitrate data was robust (median test-period NSE 0.74). These results indicated that it is potentially possible to hindcast nitrate data to fill the data gaps that exist at grab-sample nitrate monitoring locations. Further investigation showed that temporal impulses in the estimated daily nitrate concentration were influenced primarily by precipitation and discharge. Seasonal performance evaluation of the model showed that the model performed effectively during the summer and fall seasons. Since model

performances for the winter and spring seasons are comparatively lower than the summer and fall seasons due to insufficient representation of other environmental processes (e.g., base flow, tiled drains) as inputs beyond agricultural practices. Watershed managers and policymakers should consider this information while utilizing the estimated data for decision-making. Nitrate concentration-discharge analysis showed four dominant nitrate transport patterns from landscapes to streams with increasing discharge, including (i) flushing pattern, (ii) flushing before median discharge ( $Q_{50}$ ) and chemostatic after  $Q_{50}$  pattern, (iii) flushing before  $Q_{50}$  and dilution after  $Q_{50}$  pattern, and (iv) chemodynamic pattern. Most of the contributing watersheds in Iowa lost nitrate to streams following the flushing pattern of nitrate concentration increasing with stream discharge. About 64% of the monitoring sites (26 sites) showed correlation values  $> 0.60$  between estimated nitrate concentration and discharge. The results also described Iowa's local and regional nitrate status, which might help watershed managers to make informed nutrient management decisions. The modeled nitrate data can provide meaningful insights into maintaining stream health, understanding ecosystem functions, and managing water resources. The deep learning model-based modeling framework developed here can estimate daily stream nitrate concentration. Detailed studies in the future would be beneficial to evaluate the framework's potential to quantify nitrate status in regions with limited sensor-based nitrate monitoring sites.

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## **CHAPTER 3**

# A DATA-DRIVEN MODELING FRAMEWORK TO FILL STREAM NITRATE DATA GAPS AT LOW-FREQUENCY NITRATE MONITORING BASINS.

## 3.1 Introduction

Continuous stream nitrate concentration is critical for accurate stream health monitoring (G. K. Saha et al., 2021, 2022). Nitrate concentration monitoring is traditionally conducted as lab analysis with grab water samples. Outside of long-term and experimental watersheds, most of these data are collected biweekly or monthly timestep depending on their specific purposes, and often the monitoring locations are rotated in the stream to get a more spatial representation of nitrate data (USEPA, 2013). In the United States (US), about 400 federal, state, local, and tribal agencies, including the United States Geological Survey (USGS), United States Environmental Protection Agency (USEPA), and United States Department of Agriculture (USDA), collect stream nitrate concentration data periodically at more than 5000 locations in the US (Read et al., 2017; Sprague et al., 2017). The periodically collected nitrate data is used to estimate seasonal or annual nitrate load using statistical models. The statistical models work on the principles of learning the nitrate concentration to discharge (c-Q) relationship from periodically collected nitrate data and continuously monitored discharge data and fill the concentration data gaps using the retained c-Q relationship.

The existing statistical methods to estimate water quality constituents have three general types: simple aggregation or interpolation techniques (Ferguson, 1986, 1987; Larson et al., 1995; Le Moal et al., 2019; Likens, 1977), ratio estimators (Cochran, 1977), and regression-based methods (Hirsch, 2014; Hirsch et al., 2010; Runkel et al., 2004). In addition, Aulenbach and his fellow researchers used a composite method (combination of simple aggregation or

interpolation- and regression-based methods) to estimate several constituents, including nitrate, at multiple watersheds in the USA (Aulenbach et al., 2016; Aulenbach & Hooper, 2006). However, selecting the best constituent estimation method depends on the constituent type, data availability, and concentration dynamics (Aulenbach et al., 2016; Hirsch, 2014). The most updated statistical methods, including the Load Estimator (LOADEST) (Runkel et al., 2004) and Weighted Regressions of Time, Discharge, and Season (WRTDS) (Hirsch et al., 2010) are widely used regression methods for estimating seasonal or annual nitrate loads from periodically collected nitrate data. An important advantage of these regression methods is their use of the most significant explanatory variables as inputs to estimate nitrate loads and variability. These tools utilize a stream nitrate monitoring site's discharge, time, and seasonal information to calculate nitrate concentrations and loads. The accuracy of these statistical tools depends on the respective site's nitrate concentration frequency and the nitrate concentration-stream discharge (c-Q) relationship information (Hirsch, 2014; Lee et al., 2016; Maestre et al., 2014). Though Hirsch (2014) pointed out WRTDS as a valuable tool to identify nitrate trends and had limited use for accurate load calculation over a few years, the recently available WRTDS tool with Kalman smoothing improved the tool's concentration and load estimation capability (Lee et al., 2019; Zhang & Hirsch, 2019). Moreover, these tools depend heavily on stream-discharge and do not consider other relevant environmental processes, such as fertilizer application rate, land cover changes, and landscape runoff.

Nitrate sensors that collect continuous (15 – 60 minutes intervals) concentration data are getting more common in the last decade. The United States Geological Survey (USGS) has about 175 nitrate monitoring sensors around the US (Pellerin et al., 2013; USGS, 2023a). Different state agencies, including the Iowa Water Quality Information System (IWQIS), also

install nitrate sensors to collect high-frequency (sampling rate  $>$  rate of change) nitrate data to understand nitrate dynamics within the states. These sensor-based high-frequency nitrate data can provide crucial insights into the region's temporal variability of nitrate data. A data-driven modeling framework was developed to estimate daily nitrate concentration data at any gauged stream locations (Saha et al., 2023). The developed data-driven model learned stream nitrate dynamics from environmental and biophysical attributes and temporal variabilities in nitrate concentration data from sensor-based high-frequency and low-frequency (sampling rate  $<$  rate of change) nitrate monitoring sites in a region (G. K. Saha, Rahmani, et al., 2023). This study hypothesized that the data-driven modeling framework would learn the nitrate c-Q information from high-frequency monitoring sites and fill nitrate data gaps in low-frequency nitrate monitoring sites. The specific study objectives were (1) to develop a data-driven modeling framework that can predict daily nitrate at low-frequency nitrate monitoring sites to fill the nitrate data gaps and (2) to compare the performance of the developed data-driven model with other widely used statistical models, including WRTDS and LOADEST.

### **3.2 Methods**

This study utilized a previously developed long short-term memory (LSTM) model-based deep learning (DL) modeling framework (Fang et al., 2017; Saha et al., 2023; Shen, 2018; Shen et al., 2018) for multiple large basins with a dominant number of low-frequency stream nitrate measuring sites (Chapter 2). In this study, the developed DL model predicted daily nitrate concentration at these low-frequency stream nitrate measuring sites using the high-frequency nitrate and stream discharge information of a high-frequency nitrate monitoring site as inputs. Twenty-four high-frequency nitrate monitoring sites within three river basins in Iowa, USA, were considered for testing the hypothesis in this study (Fig. 3.1). Nine (9) DL models were

developed and simulated using the high-frequency nitrate and stream discharge data of nine different high-frequency sites within these three river basins. Daily nitrate data from each high-frequency monitoring site were subsampled randomly into biweekly nitrate data to create pseudo-low-frequency nitrate data. The subsampled biweekly data was used for model development, and the remaining nitrate data were used for performance evaluation. The WRTDS and LOADEST models were also developed with the subsampled nitrate data of the same pseudo-low-frequency nitrate monitoring site used in DL model development. The daily nitrate concentrations estimated by the DL model, WRTDS-Kalman, and 7-parameter LOADEST were compared by calculating five evaluation metrics, including bias, root means square error (RMSE), Nash-Sutcliffe efficiency (NSE) (Nash & Sutcliffe, 1970), correlation, and Kling-Gupta efficiency (KGE) (Gupta et al., 2009). Finally, the estimated nitrate loads by these three models were compared with the observed nitrate loads to assess the robustness of the DL modeling framework.

### **3.2.1 Study Area**

According to the sixth annual Feeding the Economy report, Iowa has an \$88.3 billion agricultural industry with its traditional commodities, including corn, soybeans, oats, hogs, cattle, eggs, and dairy products (Lamm, 2022). The IOWATER program identified increasing nitrate concentration levels in several parts of Iowa (Izaak Walton League of America, 2022). Three (3) large basins were selected in this study, including Racoon and Des Moines, Iowa, and Cedar rivers, to develop data-driven models (Fig. 3.1). Twenty-four (24) high-frequency nitrate monitoring sites were identified within the three selected basins (Fig. S3.1 in Appendix B). Des Moines River Basin combined the Racoon River and Des Moines River networks with eight (8) high-frequency nitrate monitoring sites. The area around the Iowa River network with



ten (10) high-frequency nitrate monitoring sites was called Iowa River Basin. Cedar River Basin considered seven (7) high-frequency nitrate monitoring sites within the Cedar River network. Daily nitrate concentration data of each of the selected twenty-four sites was highly autocorrelated. The autocorrelation metric dropped from 1.0 to 0.80 for sixteen (16) sites (67% of all sites) with lag days of two to three months and for eight (8) sites (33% of all sites) with lag days of one month.

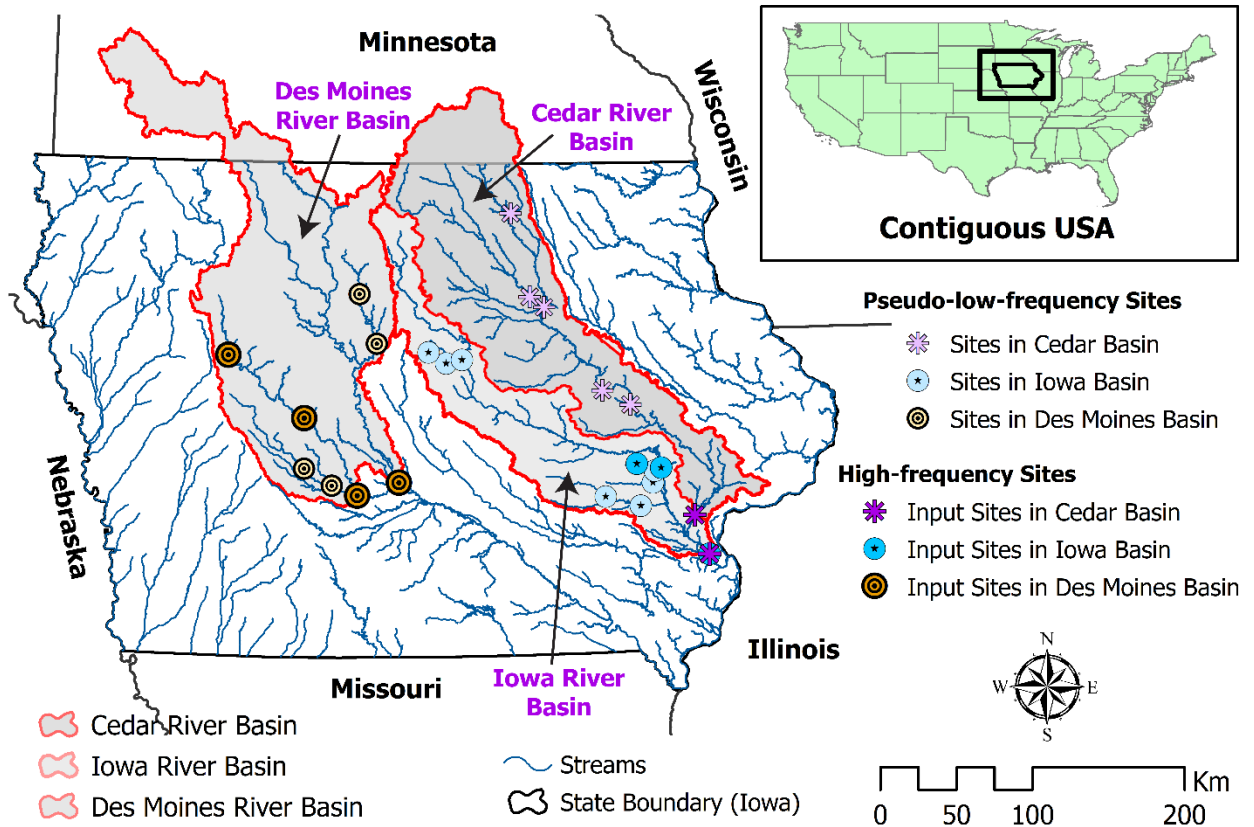


Figure 3.1 Three (3) selected large basins (Des Moines River Basin, Iowa River Basin, and Cedar River Basin), nitrate monitoring sites, and land use categories in Iowa, USA. The Des Moines, Iowa, and Cedar River basins have 8, 10, and 7 high-frequency nitrate monitoring sites, respectively. Among the nitrate monitoring sites, 4, 3, and 2 high-frequency nitrate monitoring sites were used from Des Moines River Basin, Iowa River Basin, and Cedar River Basin for DL model development. Daily nitrate concentration ( $c$ ) and stream discharge ( $Q$ ) of

high-frequency nitrate monitoring sites were used to develop a DL model development. The remaining high-frequency nitrate monitoring sites were considered pseudo-low-frequency nitrate monitoring sites. A high-frequency nitrate monitoring site was common in Iowa and Cedar River Basins.

### **3.2.2 Model selection**

DL models such as recurrent neural networks (RNN) and Gated Recurrent Units (GRU) can be used to develop the data-driven modeling framework for time series prediction. An RNN model, such as LSTM, can capture data's long-term dependencies by addressing the vanishing gradients problem (Gers, 2001; Pascanu et al., 2013). The LSTM can selectively remember or forget information, filtering out noise and irrelevant information in the data. Many hydrologic studies used the LSTM algorithm in soil moisture estimation (Fang et al., 2017b, 2019; Liu et al., 2022), streamflow forecasting (Feng et al., 2020), stream water temperature prediction (Rahmani, Lawson, et al., 2021b; Rahmani, Shen, et al., 2021), and dissolved oxygen (DO) in stream estimation (Zhi et al., 2021). This study developed an LSTM-based modeling framework and utilized the framework for filling nitrate concentration gaps at a daily timestep for low-frequency nitrate measuring sites. The construction of the developed DL modeling framework is detailed in Chapter 2 and Appendix A. The previously developed DL modeling framework was designed to learn regional nitrate dynamics using relevant input environmental drivers (Chapter 2 and Saha et al., 2023). The developed DL model simulated and captured nitrate relationships from high-frequency stream nitrate concentration for a certain period and estimated daily stream nitrate concentration for data-limited locations. The previously developed framework primarily applies to regions with a reasonably high number of high-frequency stream nitrate monitoring sites. However, most US regions have a few high-

frequency nitrate monitoring sites and many low-frequency sites. A DL modeling framework that fills the temporal gaps in the low-frequency nitrate data using the available nitrate information of these few high-frequency nitrate monitoring sites would be crucial to creating a high-frequency nitrate database, which can be used to identify accurate nitrate dynamics of a nitrate data-limited region and design appropriate conservation practices. In this study, the developed DL framework was applied to three large river basins in Iowa, USA, to test the validity of the developed framework in filling temporal gaps in the low-frequency nitrate data.

### **3.2.3 Conceptual study design**

The focus of this study was to develop an LSTM-based modeling framework for filling daily nitrate concentration gaps at low-frequency nitrate monitoring sites with the help of temporal variabilities in the nitrate concentration ( $c$ ) and stream discharge ( $Q$ ) of a neighboring high-frequency nitrate monitoring site as inputs within a basin. Therefore, daily nitrate and stream discharge data of a high-frequency nitrate monitoring site within the basin were used as model inputs. The study was conducted for eight years (2012- 2019). Daily nitrate data from remaining monitoring sites within the basin were subsampled randomly into biweekly nitrate data to create pseudo-low-frequency nitrate monitoring sites to develop a DL model. For instance, daily nitrate data of seven (7) among eight (8) high-frequency nitrate monitoring sites in the Des Moines River Basin were subsampled into pseudo-low-frequency nitrate monitoring sites. The remaining high-frequency nitrate monitoring site's daily nitrate and stream discharge data were used as inputs to develop the DL model for Des Moines River Basin. The nitrate data of pseudo-low-frequency nitrate monitoring sites were represented as low-frequency data for the DL model development. The remaining daily nitrate data of the seven (7) pseudo-low-frequency nitrate monitoring sites in the Des Moines Basin were used for evaluating the

performance of the developed DL model.

### **3.2.4 Input data to the DL model**

The input data for model development was divided into forcings (time-series data) and attributes (watershed-based constant information). Forcings include stream discharge data of respective nitrate monitoring sites, precipitation, maximum and minimum temperatures, day length, and daily nitrate and stream discharge data of a neighboring high-frequency nitrate monitoring site. Annually changing land use data, including corn, soybean, forest, urban, and other agriculture areas, and fertilization data, including fertilizer N input and manure N input, were also considered forcings for model development. Attributes include drainage area, average precipitation, areas of shallow and deep soils, areas of excessively drained, well-drained, and poorly drained soils, areas with hydrologic soil groups A, B, C, and D, dominant soil erodibility factors, the area covered by dominant soil erodibility factors, dominant soil textures and its respective areas for each monitoring site. A detailed explanation, measuring units, and preparation of these forcings and attributes for model development are available in Chapter 2 and Saha et al. (2023) manuscript.

### **3.2.5 Data-driven model development**

The developed DL framework was tested in three major river basins, including Des Moines River Basin, Iowa River Basin, and Cedar River Basin, in Iowa. In this study, nine (9) DL models were developed using nine (9) different high-frequency nitrate monitoring sites daily c and Q data as inputs for three (3) selected river basins: four DL models for Des Moines River Basin, three DL models for Iowa River Basins, and two DL models for Cedar River Basin. In Des Moines River Basin, daily nitrate concentration and steam discharge data of sites USGS 05482300, USGS 05484500, USGS 05482500, and USGS 05482000 were used as model

inputs in Model 1, Model 2, Model 5, and Model 6, respectively (Table S3.1 in Appendix B). Daily nitrate concentration and stream discharge data of sites WQS0001, USGS 05465500, and WQS0003 were used in Model 3, Model 4, and Model 7 as inputs for Iowa River Basin. Two models (Model 8 and Model 9) were developed for the Cedar River Basin with daily nitrate and stream discharge data of WQS007 and USGS 05465500 as model inputs. Since both Iowa and Cedar Rivers go through the location of site USGS 05465500, the daily nitrate concentration and stream discharge information of site USGS 05465500 were used as model inputs in Model 4 and Model 9 within Iowa River Basin and Cedar River Basin, respectively. The data of the input high-frequency nitrate monitoring site were eliminated from each DL model to prevent information leakage in the DL model during model training.

Multiple combinations of five hyperparameters, including minibatch, epoch, hidden layer, time window, and random seeds, were utilized during each model training to identify the best combination for each model. The best combination of hyperparameters was selected based on the model performance for each developed model (Table S3.2 of Appendix B). A detailed explanation of selected hyperparameters and their definitions are available in Appendix A of Chapter 2.

### **3.2.6 WRTDS and LOADEST model development**

Pseudo-low-frequency nitrate monitoring sites' data were simulated using WRTDS (Hirsch et al., 2010) and LOADEST (Runkel et al., 2004) tools to estimate daily nitrate concentrations and annual loads. EGRETS (Hirsch & Cicco, 2015) and EGRETS-Kalman (R packages developed for WRTDS models) (Hirsch & Cicco, 2015; Q. Zhang & Hirsch, 2019) were used to simulate the nitrate concentrations and loads. The utilized WRTDS-Kalman method (EGRETS-Kalman) works in three steps: (i) the method develops a regular WRTDS model

using observed nitrate concentration data; (ii) the method determines the standard residual of the observed nitrate concentration data, (iii) the method assumes that the standard residuals have a first-order autoregressive serial correlation structure with a positive correlation coefficient between 0 and 1 (Q. Zhang & Hirsch, 2019). The WRTDS-based simulation requires nine (9) parameters, including half window width for the time weighting (WindowY), half window width for the weighting in terms of  $\ln(Q)$  (WindowQ), half window width for the seasonal weighting (WindowS), the minimum numbers of data with nonzero weights (minNumObs), the minimum number of uncensored observations with nonzero weight (minNumUncen), logical variable (edgeAdjust), rho value, number of iteration (niter), and random seed. Initially, WRTDS parameters, including WindowY 2 – 4 years, WindowQ 1.5 – 2.0 m<sup>3</sup>/s, WindowS 0.30 – 0.75 years, and minNumObs 45 – 60, count were used to estimate nitrate concentration for twenty-four sites. Initial parameter ranges were used based on the parameter selection process recommended in literature and minimizing the RMSE between estimated and observed nitrate concentrations. Later, the Kalman smoothing filter (WRTDS-Kalman model) with rho 0.95 and 500 iterations (niter) was applied to the nitrate data to get the best nitrate concentration estimates for each monitoring site. The best combination of parameters to estimate daily nitrate concentrations and loads by WRTDS for each monitoring site is illustrated in Table S3.3 of Appendix B. Average annual nitrate load and respective standard error were estimated for each nitrate monitoring site using Web-based Load Calculation using LOADEST (Park et al., 2015). LOADEST model requires nitrate concentration (mg/L), missing values confirmation in the dataset, stream discharge (cms), output load unit (ton per day), and model number (0-9) as inputs. This study used the 7-parameter LOADEST model number 9 to estimate nitrate load and concentration.

### **3.3 Results and Discussion**

#### **3.3.1 Performance evaluation of the data-driven models**

The DL models for all three river basins were excellent, with test-period metrics ranging KGEs 0.67 - 0.94, correlations 0.66 - 0.95, biases -0.16 - 2.90, and RMSEs 1.50 - 2.95 (Table S3.4 in Appendix B). The best model performance for each nitrate monitoring site within each basin was identified from the performance metrics of all DL models developed for three river basins in Iowa. The DL model-based performance evaluation metrics for all the nitrate monitoring sites within the Des Moines River, Iowa River, and Cedar River Basins demonstrated the best NSE and KGE values ranging from 0.19 to 0.95 and 0.61 to 0.97 in Iowa (Fig. 3.2, Table S3.5 in Appendix B). Individual nitrate monitoring sites' evaluation metrics showed that 18 monitoring sites (75% of all) had NSE greater than 0.50, and all 24 nitrate monitoring sites (100%) had KGE greater than 0.60. About 83% of all nitrate monitoring sites (20 sites) demonstrated a minimal bias range within  $\pm 0.50$ . The DL model estimated daily nitrate concentration showed high correlations (correlation  $\geq 0.70$ ) with the observed nitrate concentration for all nitrate monitoring sites. The overall DL model performance indicated that the developed DL models learned the trends and temporal variabilities from the input environmental and biophysical information of respective low-frequency nitrate monitoring sites and nitrate and stream discharge (primarily c-Q relationship) information of a neighboring high-frequency nitrate monitoring site within the basin. The developed DL models predict unknown daily nitrate concentrations using the learned information at low-frequency (biweekly) nitrate monitoring locations.

The spatial distribution of the NSE values for the Des Moines River Basin, Iowa River Basin, and Cedar River Basin ranged from 0.51 to 0.95, 0.20 to 0.89, and 0.38 to 0.80, respectively

(Fig. 3.2). Eighty-nine percent (89%, 7 out of 8 monitoring sites), sixty percent (60%, 6 out of 10 monitoring sites), and forty-three percent (43%, 3 out of 7 monitoring sites) nitrate monitoring sites had NSE values greater than 0.70 in the Des Moines River Basin, Iowa River Basin, and Cedar River Basin, respectively (Table S3.5 in Appendix B). DL model performed better in Des Moines River Basin compared to Iowa and Cedar River Basins. The DL model performance for six (6) nitrate monitoring sites (3 sites in Iowa River Basin and three (3) nitrate monitoring sites in Cedar River Basin) was less than 0.50. The stream distance between the target low-performing pseudo-low-frequency nitrate monitoring sites and respective input high-frequency nitrate monitoring sites was higher for these sites (Fig. 3.2 and Table S3.6 in Appendix B). Due to data availability for Iowa River Basin and Cedar River Basin, the selected high-frequency sites were mostly near the basin outlets. The calculated stream distances between the pseudo-low-frequency nitrate monitoring sites and their respective high-frequency nitrate monitoring sites ranged from 25 to 315 km for Des Moines River Basin, 17 to 349 km for Iowa River Basin, and 41 to 433 km for Cedar River Basin (Table S3.6 in Appendix B).



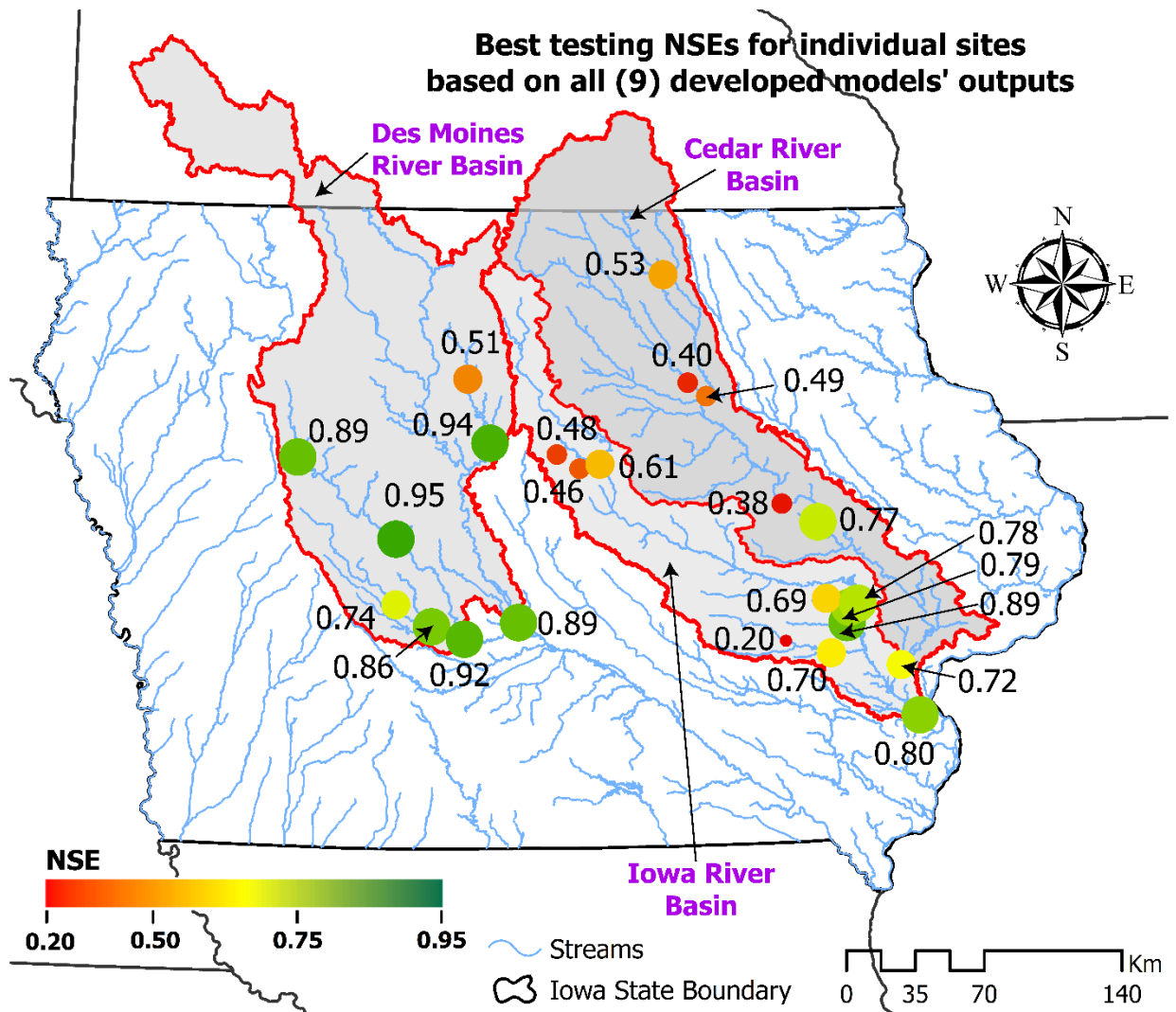


Figure 3.2 Spatial distribution of the best model performance (NSE) values calculated during the performance metrics calculation for nine (9) developed data-driven models for all 24 nitrate monitoring sites within selected three study basins.

In most cases, the DL model's performance was higher ( $NSE > 0.70$ ) when the stream distance between the pseudo-low-frequency nitrate monitoring sites and their respective input high-frequency nitrate monitoring sites was low (Fig. S3.2ii and S3.2iii in Appendix B). In a few cases, the pseudo-low-frequency sites and their input high-frequency nitrate monitoring sites were situated in two rivers within the river basin. For instance, the pseudo-low-frequency

nitrate monitoring site, USGS 05481000, and corresponding input high-frequency nitrate monitoring site, USGS 05482500, were on Racoon and Des Moines Rivers, respectively, within the Des Moines River Basin. Though the stream distance between these sites was 315 km, the Euclidean distance between them was lower (Fig. S3.2i in Appendix B). The results indicated either the similar dynamic or biophysical characteristics between the target pseudo-low-frequency and input high-frequency nitrate monitoring sites or temporal variability information transfer within a certain stream distance from the input high-frequency to low-frequency through stream network contributed to better daily nitrate prediction by the developed DL model. In addition, the stream distances for three (3) low-performed pseudo-low-frequency nitrate monitoring sites (WQS 0039, WQS 0071, and WQS0072) were comparatively lower with their input high-frequency nitrate monitoring sites (Fig. S3.3 in Appendix B). Analysis showed that all these low-performed nitrate monitoring sites had less than 50 data points for a two-years of time span in the subsampled pseudo-low-frequency site data used for DL model training, and the DL model experienced difficulties learning the trends and temporal variabilities from this limited input information.

### **3.3.2 Performance comparison of the DL, WRTDS, and LOADEST model-based nitrate estimates**

The daily nitrate concentration prediction performance for the DL and the WRTDS were similar in most gauging stations, and they performed better than LOADEST (Fig. 3.3 and Table S3.6 in Appendix B). The average NSE value difference between DL and WRTDS models was within 20% for eighteen (18) out of twenty-four (24) nitrate monitoring sites (75% monitoring sites) (Table S3.6 in Appendix B). The WRTDS model performed better in the Iowa River and Cedar River basins, where the high-frequency nitrate monitoring sites were much farther than

the pseudo-low-frequency sites. Nitrate concentration estimates based on the WRTDS model showed more abrupt fluctuations on many days within the estimated period. The primary reason behind these abrupt fluctuations was large temporal gaps and magnitude variability between two consecutive nitrate data of a monitoring site used for WRTDS model simulation (Fig. 3.3iv and 3.3v). The utilized WRTDS-Kalman model replaced the initially estimated nitrate concentrations with observed data for sampling days and adjusted the standard residuals from nitrate estimates for days without a nitrate record. The nitrate data of pseudo-low-frequency nitrate monitoring sites that held the assumption of positive correlation among standard residuals demonstrated better model performance than those pseudo-low-frequency nitrate monitoring sites that did not have positive correlations among standard residuals. LOADEST model consistently underpredicted the daily nitrate concentration for most of the nitrate monitoring sites within the selected three (3) river basins (Fig. 3 and Table S3.6 in Appendix B). LOADEST model calculates nitrate concentration and loads based on eleven predetermined equations. There is no guarantee that these equations will estimate daily nitrate concentration with accuracy and certainty using predefined equations. In addition, the LOADEST model assumes that high fitness exists between the log of nitrate concentration and the log of stream discharge, and residuals are stationary (Hirsch, 2014; Lee et al., 2016; Runkel et al., 2004). These assumptions in LOADEST invalidate when seasonal variation dominates the concentration and stream discharge relationship substantially.

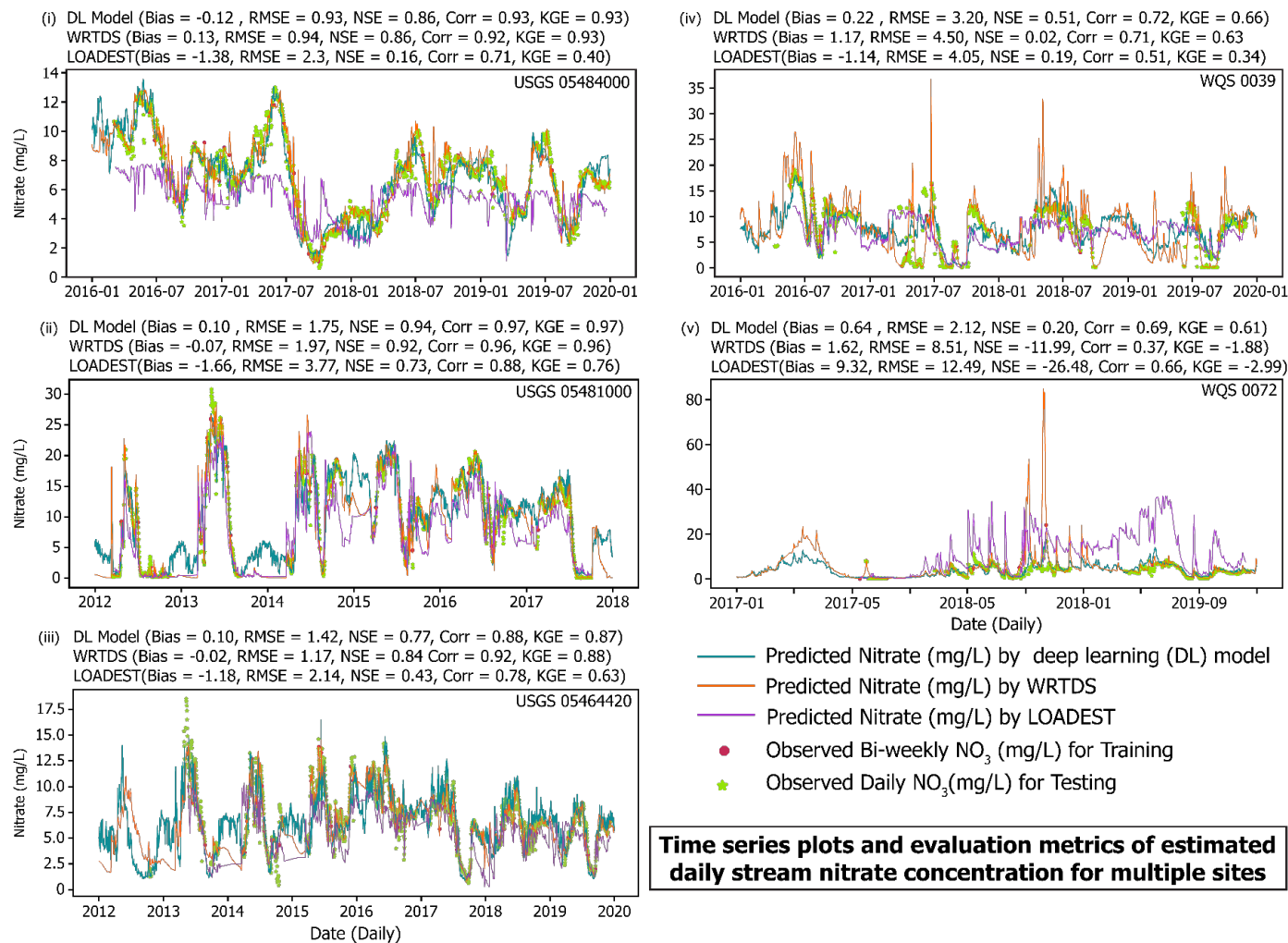


Figure 3.3 Time series plots of five nitrate monitoring sites with their evaluation metrics. The DL model performed similarly to WRTDS for the most pseudo-low-frequency nitrate monitoring sites (Fig 3.3i, 3.3ii, and 3.3iii). WRTDS and LOADEST models did not perform well for a few pseudo-low-frequency nitrate monitoring sites (Fig. 3.3iv and 3.3v).

The DL, WRTDS, and LOADEST models based on daily nitrate estimates showed different nitrate concentration variability patterns during the winter seasons. Between 2012 – 2019, all twenty-four (24) nitrate monitoring sites had a total of 48 winter seasons that had nitrate information. The DL model predicted the nitrate concentration better in 60% of the winter seasons than the WRTDS model (Fig. 3.4i and 3.4ii). In 33% of winter seasons, the WRTDS model followed the observed winter season's nitrate variability more comprehensively than the DL model (Fig. 3.4iii and 3.4iv). All three models failed to predict the observed nitrate variability in the remaining 7% of winter seasons. LOADEST demonstrated limited capabilities in predicting daily nitrate concentrations during the winter season. LOADEST model-based nitrate estimates mostly followed the discharge trend (Fig. S3.4 in Appendix B). In contrast, the DL and WRTDS models can capture the nitrate dilution trend due to comparatively high snowmelt or streamflow (Fig. S3.4i and S3.4ii in Appendix B). However, the WRTDS model demonstrated an excessive dilution pattern during high flood events than the observed nitrate dilution (Fig. S3.4i and S3.4ii in Appendix B). In addition, WRTDS could not estimate nitrate concentration accurately for nitrate data-unavailable days using less than 50 observed nitrate samples and demonstrated excessive fluctuations in estimated daily concentrations. Hirsch (2014) discussed this limitation of the WRTDS to compute nitrate concentrations and loads thoroughly.

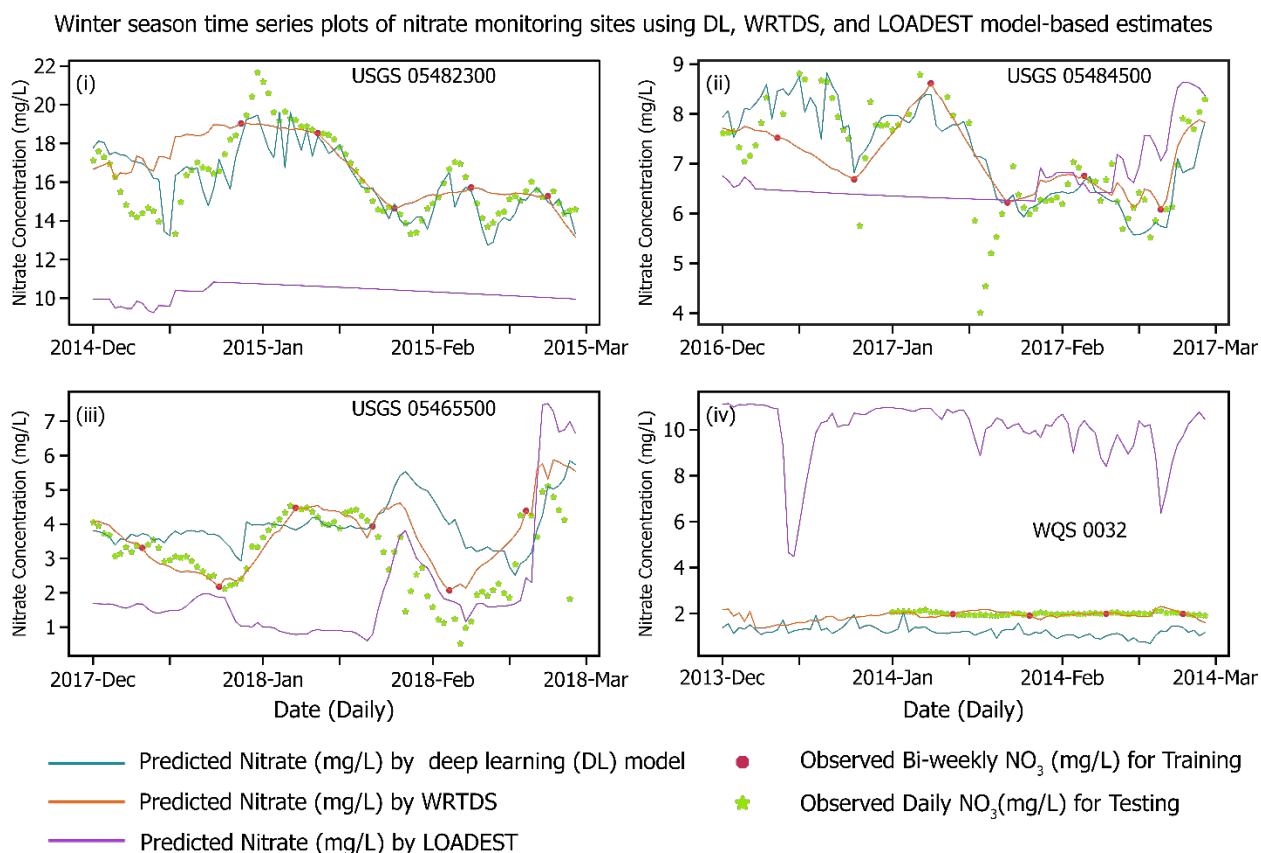


Figure 3.4 Time series of DL, WRTDS, and LOADEST models-based nitrate estimates during the winter season and their variability compared to observed nitrate concentrations: (i) and (ii) DL model-based nitrate estimates followed the observed nitrate concentrations better than WRTDS and LOADEST models-based nitrate estimates; (iii) and (iv) WRTDS model-based nitrate estimates demonstrated similar variability as observed nitrate estimates compared to the nitrate estimates computed by DL and LOADEST models.

The nitrate concentration estimations were also compared with the USEPA drinking water standard for nitrate concentration of 10 mg/L, called maximum contamination level (MCL). The daily nitrate measured concentration data exceeded 25% (0-80% for different sites) of the total number of observed nitrate concentrations. The average number of days measured nitrate concentrations and DL, WRTDS, and LOADEST model-based nitrate estimates exceeded

USEPA MCL was 354, 353, 362, and 165, respectively (Table S3.7 in Appendix B). The DL model was superior in predicting exceedance over MCL, the WRTDS slightly over predicted while the LOADEST underpredicted the trends significantly.

### **3.3.3 DL model, WRTDS, and LOADEST estimated load comparison.**

The DL model-based load estimates were similar to the observed nitrate-based load estimates (Fig. 3.5). Calculated average annual loads by the DL model were analogous to WRTDS-based nitrate load estimates since daily nitrate concentration estimates by the DL model were comparable to WRTDS-based nitrate estimates (Fig. S3.5 and Table S3.8 in Appendix B). However, WRTDS- and LOADEST-based nitrate load estimates were extremely high in a few cases (Fig. 3.5). Two nitrate monitoring sites (WQS 0012 and WQS0055) were placed to track nitrate near drainage tiles and an inlet to the wetlands designed for nitrate treatment. Because of the significant sub-surface nitrate contribution, the contributing watershed area of these two nitrate monitoring sites may not represent its hydrologic characteristics. The nitrate load (kg/ha/year) for these two sites was not calculated. LOADEST computed comparatively higher nitrate loads for 16 among 24 nitrate monitoring sites (67% monitoring sites) (Table S3.9 in Appendix B).

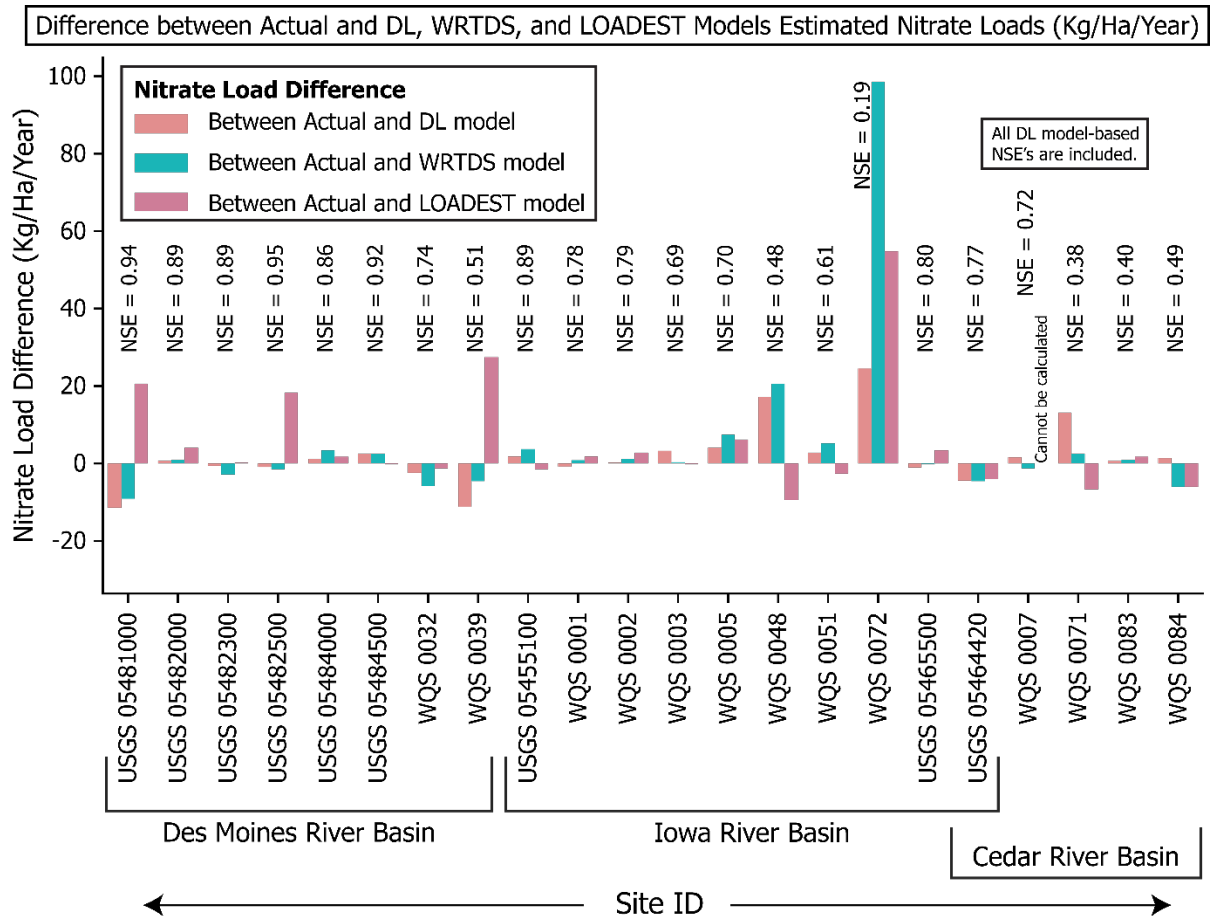


Figure 3.5 A different of estimated daily nitrate loads (kg per ha per day) calculated based on DL, WRTDS, and LOADEST models from observed daily nitrate load for each nitrate monitoring site within Des Moines, Iowa, and Cedar River basins.

### 3.3.4 Strengths and opportunities of the data-driven model

The similar prediction performance by DL and already established WRTDS models indicated that the DL model captured the trends and temporal variabilities of stream nitrate concentrations from respective low-frequency nitrate monitoring site's biweekly nitrate data, and environmental and biophysical attributes and a daily nitrate and stream discharge data of a neighboring high-frequency nitrate monitoring. The DL model predicted daily nitrate concentrations better at USGS sites than at the IWQIS sites. USGS placed most high-frequency



nitrate sensors at more significant streams with higher contributing areas. In comparison, IWQIS installed many nitrate sensors at smaller watershed streams where potential water quality issues were identified. The nitrate dynamics of these smaller watersheds are more challenging to estimate due to higher temporal variability from the timing of fertilization, planting, harvesting, and other management practices based on land use. The DL model inputs on fertilization and land management practices were developed for contributing watersheds using county-scale data due to data availability. More detailed data availability could improve DL models' ability to represent smaller watersheds better. However, the developed data-driven modeling framework can fill the nitrate data gaps at low-frequency nitrate monitoring sites in a river basin.

The performance of the DL model was encouraging in estimating daily nitrate concentrations using biweekly nitrate concentration data. The data-driven model performed comparatively well for nitrate monitoring sites with short-term data (1-3 years data). Since the data-driven algorithm learns information from the data of multiple low-frequency nitrate monitoring sites at once in a basin, the DL model offers better daily nitrate estimates compared to other statistical models. Unlike other statistical methods, the DL model learned information from a site's biophysical and climate characteristics and a neighboring high-frequency nitrate monitoring site's nitrate and stream discharge data. The DL model considers hydrologic and biophysical attributes of the watersheds; therefore, the model can mimic the trends and temporal variabilities of observed nitrate concentrations better than other statistical models.

The local watershed managers can utilize the DL model or the model-based daily nitrate estimates to schedule the water withdrawal at their required locations. They can identify the days the stream nitrate concentration meets the USEPA MCL (10 mg/L). The policymakers

can isolate the time window in a year when nitrate can exceed the permissible level and plan their specific conservation practices accordingly. The DL model can be expanded to other regions in the USA, mode developed with more high frequency sites with larger geophysical characteristic variability could improve the model's ability to estimate nitrate concentration in low-frequency measurement sites and other ungauged watersheds. The continuous nitrate concentration data could also be utilized for calibration and validation of the process-based environmental system models. The process-based model predictions could be used as an input to the DL model or an initial estimate of the DL model to improve the DL training.

More studies are required to improve the capabilities of the DL models in estimating stream nitrate concentrations. In many cases, stream nitrate concentrations heavily depend on the stream discharge and demonstrate extreme values. The developed DL model could not capture all the extreme events because the model has limited capabilities to prioritize any specific variable automatically. The developed DL model accounts for multiple environmental and biophysical attributes of the respective watersheds to predict stream nitrate concentration in a study region. While the biweekly nitrate concentration data were used in this study to create the DL model, more detailed analysis with different sampling intervals and strategic sampling time selection could help design future monitoring strategies in low-frequency sites.

### **3.4 Conclusions**

Availability of continuous nitrate data helps analyzing the nitrate trends and decision making. The developed DL framework utilizes high frequency nitrate data from the region, environmental and biophysical characteristics, and available nitrate data in low-frequency site to fill data gaps and generate continuous nitrate data. In general, the DL model performed well in filling data gaps in pseudo-low-frequency sites in the study watersheds. The DL model

performance was relatively lower if the nearby high-frequency nitrate monitoring site is farther than 150 km. The nitrate concentration and load estimates by the DL model and WRTDS-Kalman were comparable and better than 7-parameter LOADEST estimations. However, the DL model performed comparatively better for the nitrate monitoring sites having fewer nitrate data (less than 50 data points) within two-years of time span. The DL model was also better in winter nitrate concentration gap filling and the USEPA MCL exceedance days. The DL model for nitrate data gap filling in low-frequency monitoring sites learning trends from nearby high frequency monitoring sites provides new opportunities in designing nitrate monitoring networks to generate more spatially and temporally continuous monitoring network.

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## **CHAPTER 4**

# ANALYZING THE DEEP LEARNING APPROACH-BASED MODELING FRAMEWORK TO UNDERSTAND THE CRITICAL ENVIRONMENTAL DRIVERS OF PREDICTING DAILY NITRATE CONCENTRATIONS.

## 4.1 Introduction

Excessive stream nitrate concentrations can cause human health and environmental issues, including methemoglobinemia, cancer, algal blooms, eutrophication, and hypoxia (Burns et al., 2019b; Doney, 2010; Kuwayama et al., 2020; Liang et al., 2020; Vorosmarty et al., 2010). Accurate estimation of stream nitrate concentrations would be beneficial for developing and implementing effective management strategies to minimize nitrate reaching streams (G. K. Saha et al., 2022). In the United States (US), federal, state, and many local agencies, including the United States Geological Survey (USGS), Iowa Water Quality Information System (IWQIS), St. Joseph River Watershed Initiative (SJRWI) and the City of Fort Wayne, Monterey Bay Aquarium Research Institute (MBARI), California, Gulf of Mexico Coastal Ocean Observing System (GCOOS), Minnesota Pollution Control Agency, and Deschutes River Continuous Nitrate Monitoring by Department of Ecology, State of Washington, placed high-frequency (concentration changing rate < data collection frequency) instream nitrate monitoring sensors to understand the nitrate dynamics better. The deployment of sensor-based monitoring sites still needs to be improved due to high capital investment requirements. Nitrate concentrations are generally measured periodically (biweekly, monthly, or seasonally) using grab water samples analyzed in labs. These grab-sample stream nitrate concentrations are collected by 400 federal, state, local, and tribal agencies throughout the US (Read et al., 2017; Sprague et al., 2017). A deep learning (DL) approach-based modeling framework was



developed to predict daily stream nitrate concentration at low-frequency nitrate monitoring locations utilizing nitrate concentration and stream discharge inputs of a high-frequency nitrate monitoring site within a river basin (Chapters 2 and 3).

The developed DL model-based stream nitrate prediction framework successfully predicted daily nitrate concentrations at low-frequency monitoring sites (periodically measured sites) within a river basin to fill the existing temporal gaps in the nitrate data sequence (G. K. Saha, Cibi, et al., 2023). However, the DL model performance varied spatially and temporally across different sites. Understanding the environmental drivers influencing daily stream nitrate concentration estimation from a DL framework perspective could give critical insights into the transferability of the model to other regions and focus on efficient data collection efforts.

Analyzing the contributing watershed's environmental information is crucial to identify the governing environmental drivers influencing stream nitrate concentration variability. The primary purpose of these critical environmental drivers' identification is to isolate region-specific predictor variables and watershed model parameters for accurate stream nitrate estimation and to make the study adaptive to changing environmental conditions. Nitrate dynamics and nitrate c-Q relationships continuously change depending on the dominance of multiple runoff event characteristics (Winter et al., 2022); therefore, isolating critical environmental drivers influencing stream nitrate variabilities becomes challenging. Many researchers identified environmental drivers, including stream flow, hydro-climatic characteristics, watershed biophysical attributes, and soils conditions, as critical factors for the stream nitrate concentrations variability (Gascuel-Oudoux et al., 2010; Poor & McDonnell, 2007; Schilling & Spooner, 2006; L. Q. Shen et al., 2020; Wherry et al., 2021). Streamflow and components of streamflow (e.g., baseflow, stormflow) were identified as the crucial

predictor variables for estimating daily nitrate load in many regions (Ayers et al., 2021; Wherry et al., 2021). Precipitation was also identified as one of the dominant drivers of nitrogen load in several US watersheds (L. Q. Shen et al., 2020; Wolf et al., 2020). Villarini et al. (2016) developed statistical models to identify critical drivers influencing stream nitrate concentration using eight environmental drivers, representing monthly climate, monthly hydrology, and annual cropping practices. They identified baseflow and the previous year's soybean as the two most important drivers. In another study, changes in trend detection analysis were conducted on high-frequency stream nitrate concentration ( $c$ ) and stream discharge ( $Q$ ) relationships to locate the source of nitrate export (Jones et al., 2017). However, identifying the critical environmental factors for the developed DL modeling framework is essential since the complex DL model could extract information from input environmental variables differently than traditional process-based approaches. A practical approach to identifying influential drivers would be analyzing the data-driven models' performance, identifying the input environmental drivers' variability patterns, and connecting the variability patterns with the DL model performance (Wherry et al., 2021). The identified most influential drivers would help estimate stream nitrate concentrations at other significant areas and facilitate effective water quality management.

The overarching goal of this study was to identify the environmental drivers that influence the accuracy of the DL model performance in predicting daily stream nitrate concentration. The specific study objectives included (i) identifying the critical environmental drivers influencing the nitrate concentration prediction based on the data-driven modeling framework and (ii) assessing the performance of the data-driven modeling approach in predicting seasonal stream nitrate concentration.

## **4.2 Methods**

This study evaluated the critical environmental drivers that impacted predicting daily nitrate concentrations using a DL modeling framework. The model learned nitrate dynamics from biophysical and climate information and high-frequency nitrate data available within the region and predicted daily nitrate concentration at low-frequency nitrate monitoring locations. The DL modeling framework was developed with the Long Short-Term Memory (LSTM) model, and a case study was conducted in Iowa, which had a rich nitrate monitoring dataset. The high-frequency sites data were randomly sub-sampled to biweekly data to create pseudo-low-frequency nitrate monitoring sites. The pseudo-low-frequency sites data and one high frequency site data, along with other environmental data from the region was used to develop the DL model. The developed DL model performance was evaluated with remaining nitrate concentration data from pseudo-low-frequency sites. The environmental drivers were evaluated using time series similarity analysis of drivers and prediction variables using the dynamic time warping (DTW) method, the distance between sites, and sensitivity analysis of DL model inputs. The seasonal performance of the developed DL models was evaluated for four seasons, including winter, spring, summer, and fall. The nitrate concentration (c) – stream discharge (Q) relationships were compared for growing and harvesting seasons based on the DL model-estimated and observed nitrate concentrations.

### **4.2.1 Case Study Area - Iowa.**

Iowa is one of the US's major corn- and soybean-producing states (Official State of Iowa Website, 2022). According to the Iowa Nutrient Reduction Strategy, 92% of total nitrogen (TN) enters Iowa's stream annually from non-point sources (Crespi & Chen, 2019; Libra et al., 2004). Since 60% of Iowa's farmlands have 2-5% slopes and annual average precipitation

between 28- and 40-inch, intensive agricultural activities are the dominant reason behind non-point source pollution (Fenton & Miller, 2016; Iowa State University Extension and Outreach, 2012). The statewide mean nitrate concentration is 8.70 mg/L, close to the USEPA maximum contamination level (MCL) of 10 mg/L (Jones et al., 2018). The state requires precise nitrate information to make informed decisions and plan accurate conservation practices. Towards this vision, federal and state agencies, including USGS and IWQIS, placed more than 100 sensors to track water quality status (The University of Iowa, 2022). Most of these nitrate measuring sensors were placed recently and had limited nitrate information. This study utilized thirty-four (34) high-frequency nitrate monitoring sensors data with at least two years of high-frequency nitrate data and eight (8) low-frequency nitrate measuring locations for the DL model development (Fig. 4.1 and Fig. S4.1 in Appendix C). These eight (8) low-frequency monitoring sites were only used for DL model training, and not used for performance evaluation due to limited nitrate data.

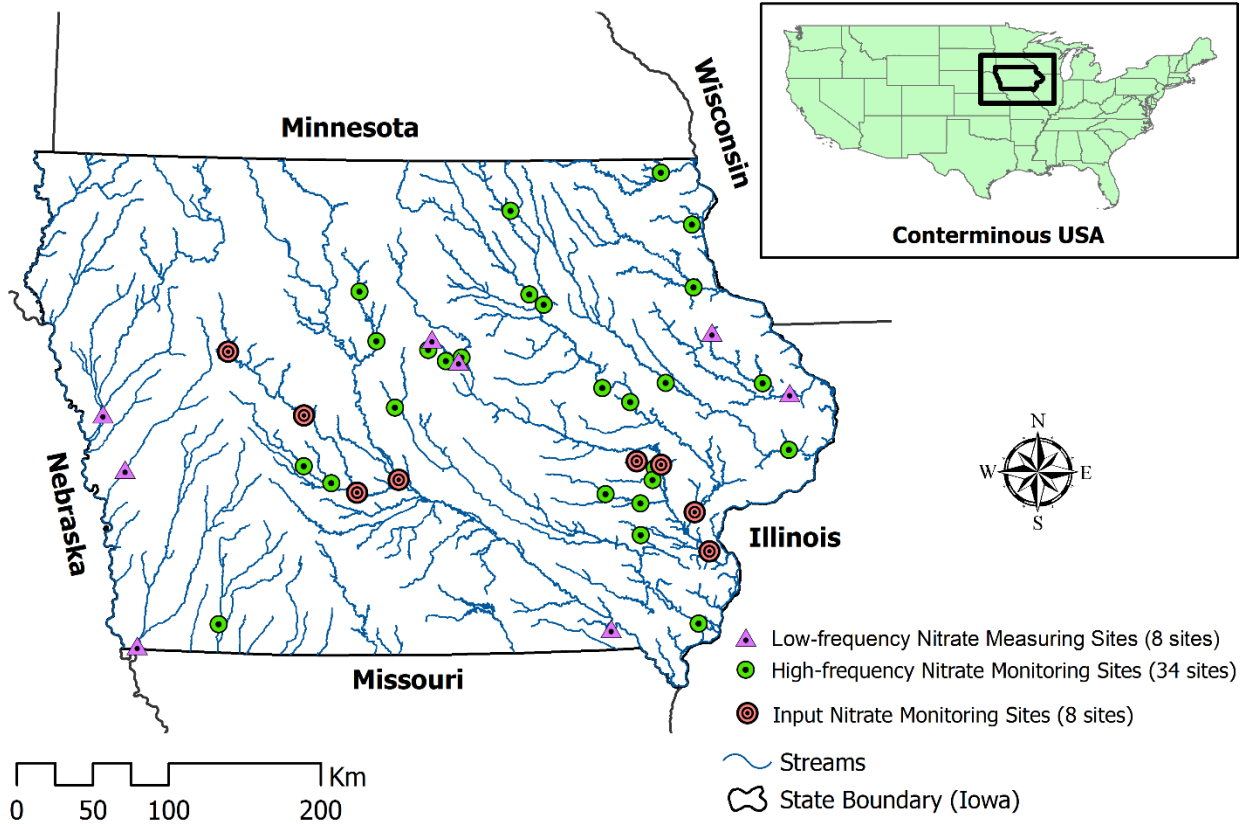


Figure 4.1 Forty-two (42) Nitrate monitoring site locations in Iowa, USA. Among thirty-four (34) selected high-frequency nitrate monitoring sites, eight (8) high-frequency sensor-based nitrate monitoring sites' nitrate, and stream discharge data were used as inputs to the models.

#### 4.2.2 DL Model Development.

The DL modeling framework was able to regionalize the nitrate concentration dynamics of a region from high-frequency monitoring sites and transfer the information to low-frequency nitrate monitoring sites to generate continuous daily scale nitrate concentration data (G. K. Saha, Rahmani, et al., 2023). The DL model was developed using the nitrate concentration and stream discharge data from one high-frequency monitoring site, nitrate concentration and stream discharge from all low-frequency sites, climate, and landscape biophysical characteristics (soil, land use, and fertilization) of the contributing catchment area of individual sites. Eight DL models were created, each corresponding to one of the selected high-frequency

nitrate monitoring sites (Fig. 4.1). The remaining high-frequency nitrate monitoring sites were utilized as pseudo-low-frequency sites, where their daily nitrate data were randomly subsampled into biweekly data and used for training the DL model. The remaining daily nitrate data from these pseudo-low-frequency sites were then employed to evaluate the performance of the trained DL model.

Two modeling scenarios were developed to assess the potential of the DL model for temporal gap filling and forecasting: (Scenario 1) predicting daily nitrate concentration to fill the temporal gaps between two biweekly nitrate data for the pseudo-low-frequency sites and (Scenario 2) forecasting or hindcasting the daily nitrate concentration beyond the available nitrate data period. Scenario 2 would help understand the forecasting or hindcasting potential of the developed DL model-based stream nitrate prediction framework with the forecasted or hindcasted daily nitrate of at least one high-frequency monitoring site, stream discharge, precipitation, temperature, and day length data. All the prepared biophysical and climatic characteristics data of each selected nitrate monitoring site were split following two periods based on the data availability (Table S4.1 in Appendix C). For both scenarios, the DL model was trained with pseudo-low-frequency site nitrate data of the same period. However, the developed DL model's performance was evaluated with daily nitrate data of different periods according to the scenario (Table S4.1 in Appendix C). The eight (8) high-frequency nitrate monitoring sites used for eight DL model development were listed in Table S4.2 in Appendix C.

#### **4.2.2.1 Input data for DL model development.**

The biophysical and climatic characteristics used to develop the DL models were divided into two categories: forcings (time-series data) and attributes (constant information for a

watershed). Daily varying forcings, including stream discharge, maximum and minimum temperatures, precipitation, day length (a climate characteristic representing seasonality), and daily nitrate c and Q data, were prepared for each nitrate monitoring site and used for the DL model development. Among annually changing forcings, such as land use, including corn, soybean, forest, urban, other agriculture areas, and fertilization data, including fertilizer and manure N inputs, were also used to develop the DL models. In addition, the DL models used numerous attributes, including watershed drainage area, average precipitation, shallow and deep soil areas, areas with hydrologic soil groups A, B, C, and D, poorly drained, well-drained, and excessively drained soil areas, dominant soil textures and its respective areas for each monitoring site, dominant soil erodibility factors, and the area covered by dominant soil erodibility factors (Chapter 2 and Saha et al., 2023).

The DL model development used multiple combinations of five hyperparameters (epoch, minibatch, hidden layer, time window, and random seeds) to train and the best hyperparameter combination model was selected for detailed analysis (Table S4.3 in Appendix C). Five performance evaluation metrics including bias, root mean squared error (RMSE), Nash–Sutcliffe efficiency coefficient (NSE) (Nash & Sutcliffe, 1970), correlation, and Kling Gupta Efficiency (KGE) (Gupta et al., 2009) were used to assess model performance. A detailed description of data preparation, hyperparameters and evaluation metrics used for DL model development is available in Chapter 2 and Saha et al. (2023).

#### **4.2.3 Sensitivity analysis of daily varying environmental variables.**

The sensitivity of seven environmental varies (stream discharge, precipitation, maximum temperature, minimum temperature, and day length) were evaluated using one variable at a time sensitivity analysis. Each DL model was simulated to estimate the nitrate concentration

of all sites for the testing period, with the daily values of one environmental variable changed by 25% and 50% consecutively kept all other environmental variables constant. Later, RMSE was calculated using the estimated nitrate and observed data for the testing period (J. Liu et al., 2023). Simulations of each DL model with 25% and 50% increased values of seven (7) environmental variables provided fourteen (14) sets of estimated daily nitrate time series of all sites. Fourteen (14) sets of RMSE values for all sites were calculated using each DL model's estimated and observed nitrate time series. RMSE values calculated for the default DL model were compared with these fourteen (14) sets of RMSE values. The procedure was followed for all eight (8) developed DL models. A higher RMSE value represents a higher error in nitrate estimates. The environmental variables for which the DL model showed comparatively higher RMSE values could be treated as more sensitive for daily nitrate prediction.

After identifying the most sensitive environmental variables used as model inputs, the similarity of the environmental variable for each of the thirty-four (34) pseudo-low-frequency nitrate monitoring sites and the input high-frequency nitrate site was calculated. The Dynamic Time Warping (DTW), which calculates the similarity between two-time sequences (Dupas et al., 2015; Samara et al., 2022; Vintsyuk, 1968), was used to identify the capability of the developed DL modeling framework to transfer information from a high-frequency nitrate monitoring site to a low-frequency site of similar environmental characteristics. The DTW calculates the cumulative distance between two continuous time series as an index similarity. A higher distance index in DTW indicates lower similarity between two-time sequences and vice versa. A more detailed discussion is provided in Appendix C (Fig. S2 in Appendix C).

#### **4.2.4 Seasonal performance of the DL models.**

Among the eight (8) developed DL models, the best model was identified for each pseudo-



low-frequency nitrate monitoring site based on the developed DL models' prediction performance. Seasonal performance analysis for each site was conducted to understand the significance of seasonal data using the estimates of the site's best model. Four (4) seasons included winter (December – February), spring (March – May), summer (June – August), and fall (September – November). The model performance for each season was estimated based on bias, RMSE, NSE, correlation, and KGE. Later, concentration-stream discharge (c-Q) relationship plots were analyzed for growing (April-July) and harvesting periods (October-November) of dominant annual row crops (corn and soybean) for each year for each high-frequency nitrate monitoring site. The estimated c-Q and observed c-Q relationships were compared to check the developed DL models' consistency in retaining the c-Q trends.

#### **4.3 Results and Discussion**

##### **4.3.1 The performance of the DL model predicting daily nitrate at each site.**

The best performing model for each nitrate monitoring site was determined from the eight DL models developed based on different high frequency nitrate sites based on testing period evaluation metrics. In Scenario 1, thirty-four (34) sites showed an NSE range of -2.16 – 0.96, KGE range of 0.28 – 0.97, correlation range of 0.55 – 0.98, RMSE range of 0.96 – 4.57, and bias range of -1.23 – 2.53 (Table S4.4 in Appendix C). About 70% (24 out of 34 sites) of the sites were simulated with greater than 0.75 NSE and 18% (6 sites out of 34 sites) of sites showed NSE less than 0.25. The results indicated that the DL model could learn the nitrate trends and variabilities from input high-frequency nitrate monitoring sites and transfer to pseudo-low-frequency nitrate monitoring sites. Model 1 with input high-frequency nitrate monitoring site USGS 05482300 and Model 2 with input high-frequency nitrate monitoring site USGS 05484500 contributed to the best daily nitrate prediction performance for nine (9)

and fifteen (15) pseudo-low-frequency nitrate monitoring sites (Table S4.5 in Appendix C). The spatial distribution of the NSE values showed that most low-performed nitrate monitoring sites, including WQS 0051, WQS 0055, WQS 0063, WQS 0072, WQS 0083, and WQS 0084, were at the upstream locations in Iowa and Cedar River basins (Fig. 4.2a and Table S4.5 in Appendix C). These nitrate monitoring sites were considerably far from their input high-frequency nitrate monitoring sites: input high-frequency nitrate monitoring sites were located far downstream of the same river basin or at different river basins. The difference in trends and variability of nitrate, stream discharge, and other hydrologic characteristics of pseudo-low-frequency and input high-frequency nitrate monitoring sites might be reflected by the geographical distance between sites. Another influencing factor on the accuracy of the DL model performance is the similarity between the biophysical and climatic characteristics of the predicted pseudo-low-frequency nitrate monitoring site and input high-frequency nitrate monitoring site.

Calculated evaluation metrics in the forecasting or hindcasting scenario (Scenario 2) included NSE range -2.18 – 0.99, KGE range - 0.08 – 0.99, correlation range 0.41 – 0.99, RMSE range 0.23 – 7.69, and bias -0.68 – 5.52 (Table S4.5 in Appendix C). Only 26% (9 out of 34 sites) of the nitrate sites demonstrated NSE greater than 0.75. Low-performed NSE values for Scenario 2 were spatially distributed, indicating that the biweekly measured nitrate data used for DL model training is insufficient to predict the daily nitrate concentration beyond the available nitrate data period. In addition, daily nitrate forecast of biweekly measured nitrate site will require the respective site's estimated or forecasted stream discharge, precipitation, maximum and minimum temperature, and predicted nitrate and stream discharge information of input high-frequency nitrate monitoring site as inputs. In the case of daily nitrate hindcast at low-

frequency nitrate monitoring locations for a historic period, the user needs a continuous high-frequency nitrate time series as an input to the DL model for the required hindcasting period. Since getting predicted values of these required inputs for daily nitrate prediction are difficult and involves substantial uncertainty, forecasting daily nitrate concentration seems unrealistic compared to its use case. Therefore, the investigation continued further with Scenario 1, which represents biweekly measured nitrate sites requiring daily nitrate data gaps.

Spatial distribution of the best test-period Nash-Sutcliffe Efficiency (NSE) values for each nitrate monitoring site based on eight (8) developed models

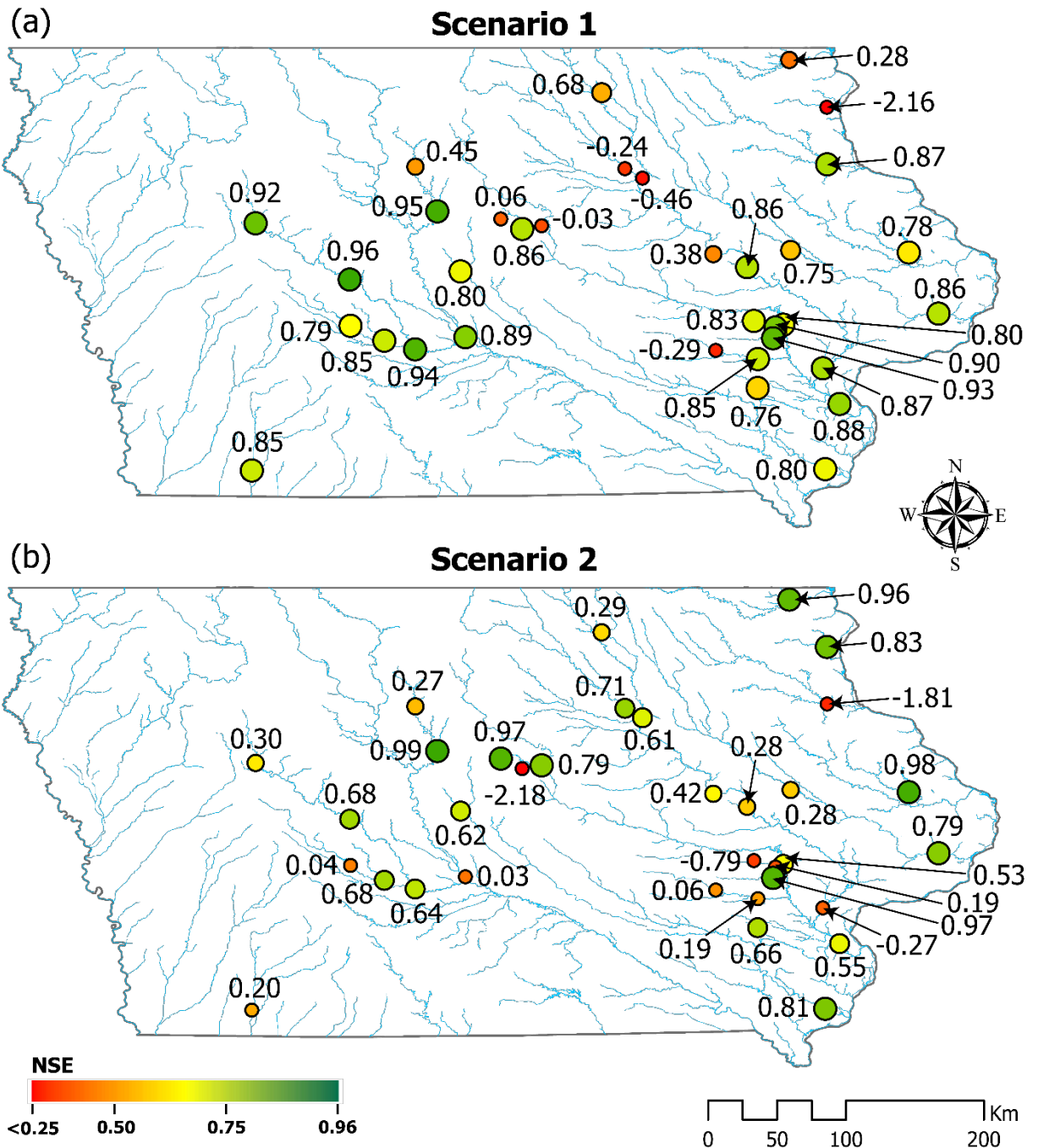


Figure 4.2 Spatial distribution of the best NSE performances at thirty-four (34) nitrate monitoring sites in Iowa based on eight (8) model outputs for Scenario 1 and Scenario 2. NSE ranges for Scenario 1 and Scenario 2 was -2.16 – 0.96 and -2.18 – 0.99, respectively.

#### 4.3.2 Analysis of nitrate concentration and stream discharge information transferability

In general, simulation results indicate that the input high-frequency nitrate monitoring site need not be in the same river basin as the low frequency site. The similarity of c-Q trend between the high-frequency nitrate monitoring site and the low frequency site seems to be more critical. Most input high-frequency nitrate monitoring sites that contributed to the best prediction at respective pseudo-low-frequency nitrate monitoring sites were at different river basins (Fig. S4.3 in Appendix C). The Euclidean distance between each pseudo nitrate monitoring site and the respective input high-frequency nitrate monitoring site ranged from 18 to 374 km, with a median stream distance of 189 km (Fig. 4.3 and Table S4.6 in Appendix C). The results support initial hypothesis that variability information of biophysically similar (similar seasonal, stream discharge, and nitrate variability) high-frequency nitrate monitoring helps to predict daily nitrate at low-frequency nitrate monitoring sites. In addition, distance between the biophysically similar input high-frequency and predicted low-frequency nitrate monitoring sites inferred the potential of extending the DL modeling framework for filling the stream nitrate data gaps in low-frequency nitrate monitoring sites in other regions.

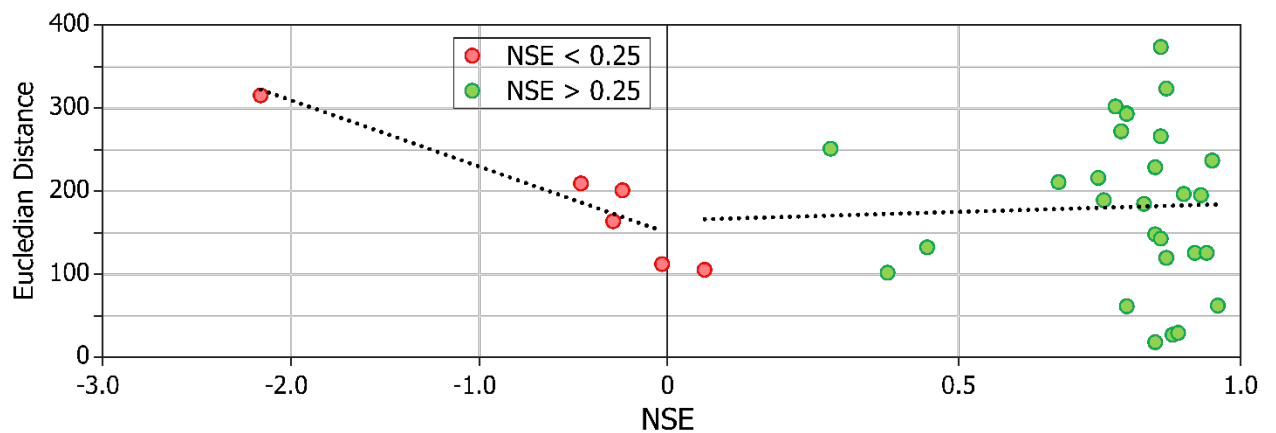


Figure 4.3 Euclidean distances between each of thirty-four (34) pseudo-low-frequency and respective input high-frequency nitrate monitoring sites versus DL model's NSE metrics for those pseudo-low-frequency nitrate monitoring sites.

Annual logarithmic nitrate concentration ( $c$ ) versus logarithmic stream discharge ( $Q$ ) were plotted using estimated nitrate  $c$  and  $Q$  of each pseudo-low-frequency and its respective model-input high-frequency nitrate monitoring site to identify the dominant  $c$ - $Q$  patterns for each pseudo-low-frequency nitrate monitoring site and understand the DL model's ability of learning and transferring  $c$ - $Q$  information. The annual  $c$ - $Q$  patterns of all thirty-four (34) pseudo-low-frequency nitrate monitoring sites for eight years (2012 – 2019), 272  $c$ - $Q$  relationship plots, demonstrated nine (9)  $c$ - $Q$  relationship patterns; three of the identified  $c$ - $Q$  relationship patterns were dominant (Table 1). The patterns included (i) flushing pattern (99 of 272 plots, 36.4%), (ii) flushing during low  $Q$  ( $Q < \text{median } Q, Q_{50}$ ), and dilution during high  $Q$  ( $Q > Q_{50}$ ) pattern (71 of 272 plots, 26.1%), and (iii) flushing during low  $Q$  and chemostatic during high  $Q$  (47 of 272 plots, 17.3%). Two (2) other  $c$ - $Q$  patterns were also identified for several years among the remaining six identified annual  $c$ - $Q$  relationship patterns: (iv) chemostatic during low  $Q$  and flushing during high  $Q$  (22 of 272 plots, 8.1%) and (v) chemodynamic patterns (16 of 272 plots, 5.9%). Flushing pattern, flushing during low  $Q$  and chemostatic during high  $Q$  pattern, and flushing during low  $Q$  and dilution during high  $Q$  pattern are predominant patterns in most of the US agricultural watersheds (Moatar et al., 2017; Zhi & Li, 2020) and identified dominant  $c$ - $Q$  relationship patterns for the agriculturally intensive case study area based on the developed DL model-based stream nitrate concentration estimates were similar (Fig. S4.4 in Appendix C). The results indicated that the DL model-based nitrate estimates can be used to understand nitrate  $c$ - $Q$  relationship patterns at data-

limited locations.

About 82% (28 out of 34 sites) and 65% (22 out of 34 sites) of pseudo-low-frequency sites demonstrated a flushing pattern and flushing during low Q and dilution during high Q pattern, respectively, for 1-5 years between 2012 and 2019 (Table 1). Twenty-seven (27) pseudo-low-frequency nitrate monitoring sites (79%) presented a flushing pattern during low Q and a chemostatic pattern during high Q for 1-4 years. In addition, analyses of c-Q relationship patterns of input high-frequency nitrate monitoring sites showed that all the high-frequency nitrate monitoring sites that contributed to accurate nitrate estimation at pseudo-low-frequency sites had all the dominant three c-Q relationship patterns. High-frequency nitrate monitoring sites, including USGS 05484500 used for Model 2, USGS 05482300 used for Model 1, and USGS 05482500 used for Model 5, demonstrated the dominant three c-Q relationship patterns for 1-3 years during the eight years of the study period (2012 – 2019) (Fig. S4.4 in Appendix C). The results indicated that the DL models learned the c-Q relationship patterns from daily nitrate concentrations and stream discharge data of input high-frequency nitrate monitoring sites and transferred the information accordingly for estimating daily nitrate concentration accurately at pseudo-low-frequency sites.

Table 4.1 Counts of year-wise estimated daily nitrate concentration (c) and stream discharge (Q) relationship patterns for thirty-four (34) pseudo-low-frequency and their input high-frequency nitrate monitoring sites.

| Gauge ID      | Year-wise estimated nitrate concentration (c) - stream discharge (Q) relationship patterns for pseudo-low-frequency nitrate monitoring sites |                        |                        |                              |                        |                        |                   |                        |                        | No. of annual c-Q relationship patterns for input high-frequency nitrate monitoring sites           |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|------------------------------|------------------------|------------------------|-------------------|------------------------|------------------------|-----------------------------------------------------------------------------------------------------|
|               | Flsh <sup>1</sup>                                                                                                                            | Flsh&Chst <sup>1</sup> | Flsh&Dilt <sup>1</sup> | Chdm <sup>1</sup>            | Chst&Flsh <sup>1</sup> | Chst&Dilt <sup>1</sup> | Dilt <sup>1</sup> | Dilt&Flsh <sup>1</sup> | Dilt&Chst <sup>1</sup> |                                                                                                     |
| USGS 05464420 | 2013, 2015, 2017                                                                                                                             | 2019                   | 2016, 2018             |                              | 2012, 2014             |                        |                   |                        |                        | <b>USGS 05482300</b><br><br>Flsh&Dilt = 3<br>Chst&Flsh = 2<br>Flsh&Chst = 1<br>Flsh = 1<br>Dilt = 1 |
| USGS 05484500 | 2012, 2013, 2017                                                                                                                             | 2014, 2018, 2019       | 2015, 2016             |                              |                        |                        |                   |                        |                        |                                                                                                     |
| WQS 0001      | 2012, 2013, 2015, 2018, 2019                                                                                                                 | 2014, 2017             | 2016                   |                              |                        |                        |                   |                        |                        |                                                                                                     |
| WQS 0007      | 2014, 2015, 2017                                                                                                                             |                        | 2016, 2018, 2019       |                              | 2013                   |                        |                   | 2012                   |                        |                                                                                                     |
| WQS 0023      | 2013, 2014, 2015, 2016                                                                                                                       | 2017, 2019             | 2018                   |                              |                        | 2012                   |                   |                        |                        |                                                                                                     |
| WQS 0048      | 2012, 2015, 2017                                                                                                                             | 2013, 2016, 2019       | 2018                   | 2014                         |                        |                        |                   |                        |                        |                                                                                                     |
| WQS 0063      | 2013, 2014, 2015, 2016, 2017, 2018, 2019                                                                                                     |                        |                        |                              |                        |                        |                   | 2012                   |                        |                                                                                                     |
| WQS 0083      | 2012, 2013, 2017, 2018                                                                                                                       | 2014, 2016, 2019       |                        | 2015                         |                        |                        |                   |                        |                        |                                                                                                     |
| WQS 0084      | 2012, 2013                                                                                                                                   | 2017, 2019             | 2018                   | 2014                         |                        |                        | 2015, 2016        |                        |                        |                                                                                                     |
| USGS 05455100 | 2012, 2014, 2017                                                                                                                             | 2019                   | 2016, 2018             | 2015                         | 2013                   |                        |                   |                        |                        | <b>USGS 05484500</b><br><br>Flsh&Dilt = 3<br>Flsh&Chst = 2<br>Flsh = 2<br>Chst&Flsh = 1             |
| USGS 05482000 | 2012, 2013, 2016                                                                                                                             | 2014, 2018             |                        | 2015, 2019                   |                        |                        |                   |                        | 2017                   |                                                                                                     |
| USGS 05482300 | 2012, 2013, 2017                                                                                                                             |                        | 2015, 2016, 2019       |                              |                        |                        | 2014, 2018        |                        |                        |                                                                                                     |
| USGS 05482500 | 2012, 2017                                                                                                                                   | 2019                   | 2015, 2016, 2018       | 2014                         | 2013                   |                        |                   |                        |                        |                                                                                                     |
| USGS 05484000 | 2013, 2017                                                                                                                                   | 2014, 2016             | 2015, 2018, 2019       |                              | 2102                   |                        |                   |                        |                        |                                                                                                     |
| WQS 0002      | 2012, 2013, 2015, 2017                                                                                                                       | 2018                   | 2014, 2019             |                              |                        |                        | 2016              |                        |                        |                                                                                                     |
| WQS 0003      | 2012, 2013, 2017, 2018                                                                                                                       | 2014, 2019             | 2016                   |                              | 2015                   |                        |                   |                        |                        |                                                                                                     |
| WQS 0012      | 2017                                                                                                                                         | 2012                   | 2013, 2016, 2018, 2019 | 2014, 2015                   |                        |                        |                   |                        |                        |                                                                                                     |
| WQS 0038      | 2017, 2018, 2019                                                                                                                             | 2014                   | 2015, 2016             |                              | 2012, 2013             |                        |                   |                        |                        |                                                                                                     |
| WQS 0039      | 2012, 2015                                                                                                                                   | 2013, 2017, 2018       | 2014, 2016, 2019       |                              |                        |                        |                   |                        |                        |                                                                                                     |
| WQS 0050      | 2012, 2013, 2016, 2017                                                                                                                       | 2019                   | 2018                   |                              | 2015                   |                        |                   | 2014                   |                        |                                                                                                     |
| WQS 0051      | 2017                                                                                                                                         | 2015, 2016             |                        | 2018                         | 2012, 2013             | 2014, 2019             |                   |                        |                        |                                                                                                     |
| WQS 0055      |                                                                                                                                              | 2019                   |                        | 2012, 2013, 2014, 2017, 2018 |                        | 2015, 2016             |                   |                        |                        |                                                                                                     |
| WQS 0069      | 2012, 2013, 2015                                                                                                                             | 2017                   | 2016, 2018, 2019       | 2014                         |                        |                        |                   |                        |                        |                                                                                                     |
| WQS 0072      | 2012, 2013, 2017, 2019                                                                                                                       |                        | 2014, 2015, 2016, 2018 |                              |                        |                        |                   |                        |                        |                                                                                                     |
| WQS 0032      | 2012, 2013, 2017                                                                                                                             | 2014, 2016, 2018, 2019 | 2015                   |                              |                        |                        |                   |                        |                        | <b>USGS 05465500</b><br>Flsh = 3, Flsh&Dilt = 2<br>Chst&Flsh = 1, Chst&Dilt = 1                     |
| WQS 0064      | 2012, 2014, 2016, 2017                                                                                                                       | 2018                   | 2013, 2015, 2019       |                              |                        |                        |                   |                        |                        |                                                                                                     |

<sup>1</sup>. Median stream discharge = Q<sub>50</sub>; low Q = Q < Q<sub>50</sub>; high Q = Q > Q<sub>50</sub>. Flsh = Flushing; Flsh&Chst = Flushing during low Q and Chemostatic during high Q; Flsh&Dilt = Flushing during low Q and Dilution during high Q; Chdm = Chemodynamic; Chst&Flsh = Chemostatic during low Q and Flushing during high Q; Chst&Dilt = Chemostatic during low Q and Dilution during high Q; Dilt = Dilution; Dilt&Flsh = Dilution during low Q and Flushing during high Q; Dilt&Chst = Dilution during low Q and Chemostatic during high Q.



| Gauge ID      | Year-wise estimated nitrate concentration (c) - stream discharge (Q) relationship patterns for pseudo-low-frequency nitrate monitoring sites |                        |                              |                   |                        |                        |                   |                        |                        | No. of annual c-Q relationship patterns for input high-frequency nitrate monitoring sites |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------------|-------------------|------------------------|------------------------|-------------------|------------------------|------------------------|-------------------------------------------------------------------------------------------|
|               | Flsh <sup>1</sup>                                                                                                                            | Flsh&Chst <sup>1</sup> | Flsh&Dilt <sup>1</sup>       | Chdm <sup>1</sup> | Chst&Flsh <sup>1</sup> | Chst&Dilt <sup>1</sup> | Dilt <sup>1</sup> | Dilt&Flsh <sup>1</sup> | Dilt&Chst <sup>1</sup> |                                                                                           |
| USGS 05418400 | 2014                                                                                                                                         |                        | 2013, 2016, 2017, 2018, 2019 |                   | 2015                   |                        |                   | 2012                   |                        | <b>USGS 05482500</b><br>Flsh&Dilt = 3, Flsh = 3<br>Flsh&Chst = 1, Chst&Flsh = 1           |
| USGS 06817000 | 2016, 2017                                                                                                                                   | 2019                   | 2014, 2015, 2018             |                   | 2012, 2013             |                        |                   |                        |                        |                                                                                           |
| WQS 0005      | 2012, 2013, 2014, 2017, 2019                                                                                                                 | 2016                   | 2015, 2018                   |                   |                        |                        |                   |                        |                        |                                                                                           |
| WQS 0010      | 2012, 2017                                                                                                                                   | 2016, 2019             | 2014, 2015, 2018             |                   |                        |                        |                   | 2013                   |                        |                                                                                           |
| USGS 05412500 | 2012, 2013, 2014, 2015                                                                                                                       |                        | 2016, 2017, 2018, 2019       |                   |                        |                        |                   |                        |                        | <b>WQS 0007</b><br>Chdm = 3, Flsh&Dilt = 2, Flsh = 1<br>Flsh&Chst = 1, Chst&Flsh = 1      |
| USGS 05465500 | 2014, 2017                                                                                                                                   | 2019                   | 2016, 2018                   |                   | 2012, 2013, 2015       |                        |                   |                        |                        |                                                                                           |
| USGS 05481000 | 2013, 2017                                                                                                                                   | 2014, 2018             | 2015, 2016, 2019             |                   | 2012                   |                        |                   |                        |                        |                                                                                           |
| WQS 0071      | 2014                                                                                                                                         |                        | 2015, 2018, 2019             |                   | 2012, 2013, 2017       | 2016                   |                   |                        |                        |                                                                                           |

<sup>1</sup>: Median stream discharge = Q<sub>50</sub>; low Q = Q < Q<sub>50</sub>; high Q = Q > Q<sub>50</sub>. Flsh = Flushing; Flsh&Chst = Flushing during low Q and Chemostatic during high Q; Flsh&Dilt = Flushing during low Q and Dilution during high Q; Chdm = Chemodynamic; Chst&Flsh = Chemostatic during low Q and Flushing during high Q; Chst&Dilt = Chemostatic during low Q and Dilution during high Q; Dilt = Dilution; Dilt&Flsh = Dilution during low Q and Flushing during high Q; Dilt&Chst = Dilution during low Q and Chemostatic during high Q.

### **4.3.3 Identifying the sensitive environmental variables.**

Each developed DL model was simulated fourteen (14) times by changing an environmental variable once a time, keeping the remaining environmental variables fixed. The simulated results of the eight (8) developed DL models showed day length as the most sensitive parameter influencing daily nitrate prediction (Fig. 4.4 and Fig. S4.5-S6 in Appendix C). The day length is a good representative of seasonality in the region. Other sensitive variables were the nitrate concentration and stream discharge of input high-frequency nitrate monitoring sites and nitrate and stream discharge of pseudo-low-frequency nitrate monitoring sites. Precipitation was also a sensitive environmental variable for a few models (Models 1, 2, 3, and 7; Fig. 4.4 and Fig. S4.5-S4.6 in Appendix C). Even though the temperature is highly correlated with seasonality, the temperature was less sensitive in the DL model. Less sensitivity could be attributed to the higher variability in daily temperature (maximum and minimum temperature) within seasons, while day length was gradually changed within seasons.

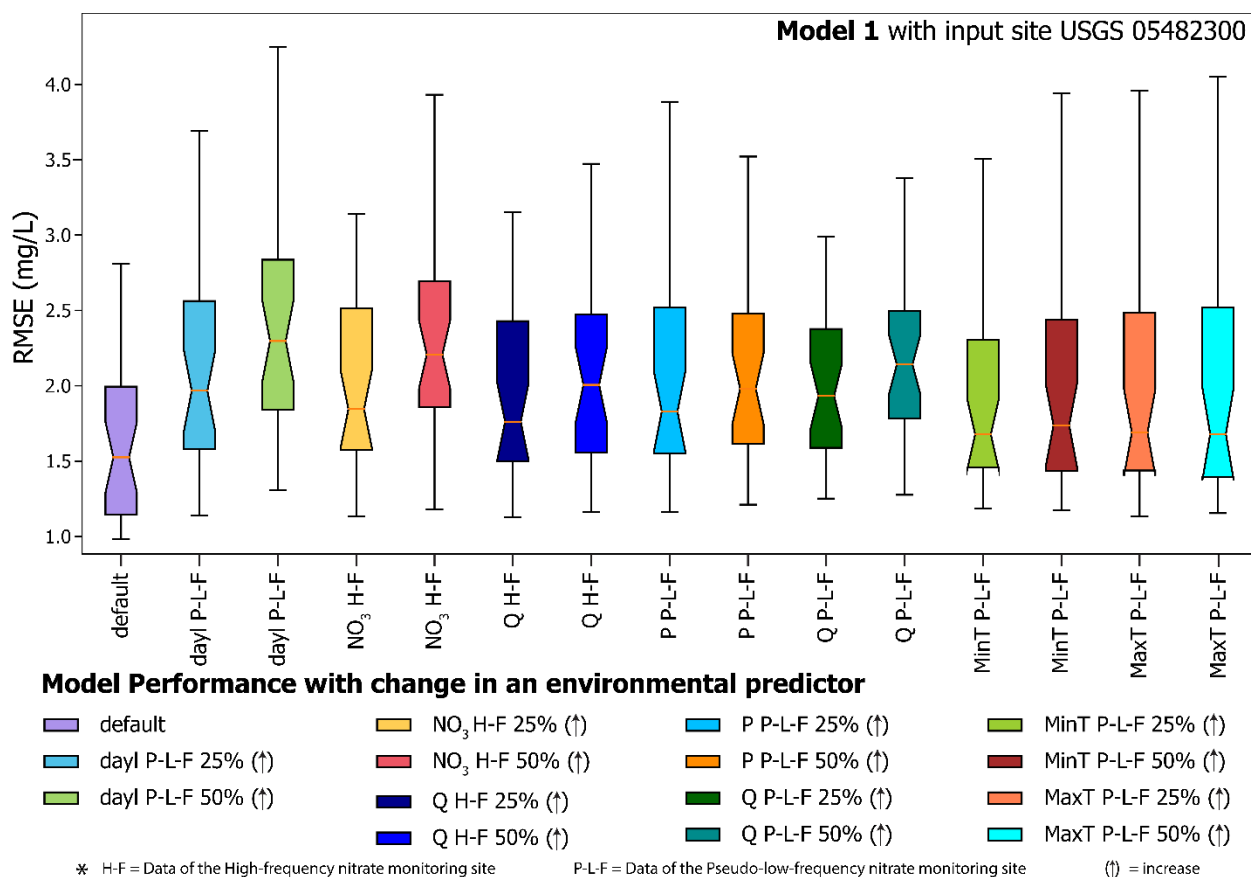


Figure 4.4 Boxplots of RMSEs calculated for Model 1 after increasing (directional sign) 25% and 50% of the magnitude of each daily varying environmental characteristics, including day length (dayl), stream discharge (Q), precipitation (P), maximum temperature (MaxT), minimum temperature (MinT) of pseudo-low-frequency (P-L-F) nitrate monitoring sites, and daily nitrate concentrations (NO<sub>3</sub>) and stream discharge (Q) of respective input high-frequency nitrate monitoring (H-F) site.

The comparative evaluation among time series of observed and estimated nitrate concentrations and sensitive environmental variables, including day length and stream discharge of a pseudo-low-frequency nitrate monitoring site and daily nitrate concentration and stream discharge of the input high-frequency nitrate monitoring site, supports the hypothesis of biophysical similarity in the environmental variables of low-frequency and input high-frequency nitrate monitoring sites contributed to more accurate continuous daily nitrate

concentration estimation. Time series plots of good-performing pseudo-low-frequency nitrate monitoring sites demonstrated DL model estimated daily nitrate concentration varied similarly with the site's day length. This similarity represented the influence of seasonal variability on accurate daily nitrate prediction at low-frequency nitrate monitoring sites. Similar variabilities in nitrate concentration and stream discharge time series between pseudo-low-frequency nitrate and input high-frequency nitrate monitoring sites also contributed to accurate daily nitrate prediction (Fig. 4.5i). For poorly performed pseudo-low-frequency nitrate monitoring sites, nitrate concentration and stream discharge variability patterns were identified different between a pseudo-low-frequency nitrate monitoring sites and their respective input high-frequency nitrate monitoring sites (Fig. 4.5ii).

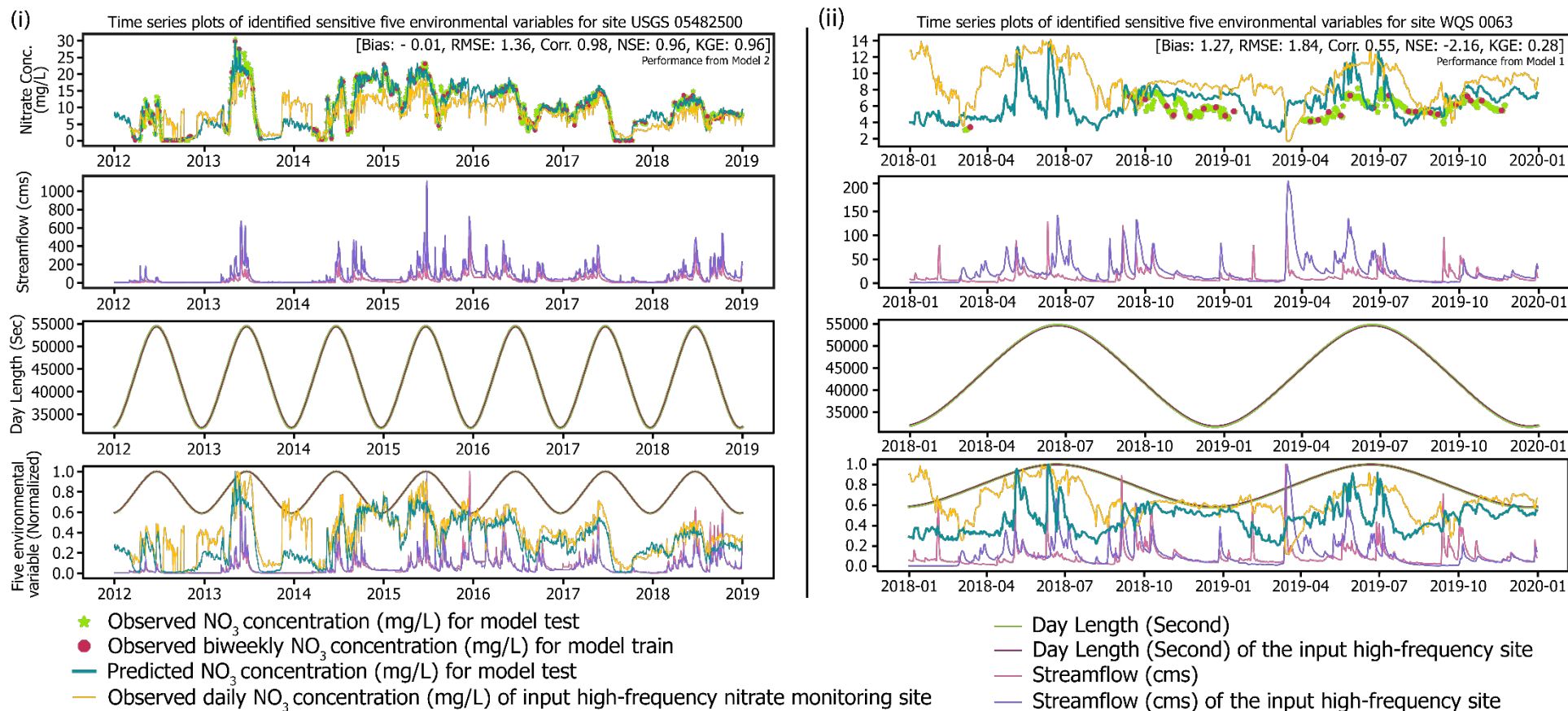


Figure 4.5 Time series plots of five environmental variables, including observed and estimated daily nitrate concentrations, stream discharge, and day length of pseudo-low-frequency and input high-frequency nitrate monitoring sites for training and testing periods: (i) USGS 05482500 (a well-performed site), and (ii) WQS 0063 (a poorly performed site).

#### **4.3.4 Evaluating environmental variable's similarity contribution.**

The similarity index values for day length and stream discharge between the pseudo-low-frequency site and the high-frequency site used for training ranged from 0 to 3067 and 146568 to 3444490, respectively (Table S4.7 in Appendix C). The index values were normalized from 0 to 1 to compare between thirty-four nitrate monitoring sites. The results indicated that the combination of similarity index values of day length (which is a representative variable of seasonality) and stream discharge contributed more to daily nitrate prediction at pseudo-low-frequency nitrate monitoring sites (Fig. 4.5). Though some pseudo-low-frequency nitrate monitoring sites, including WQS 0051, WQS 0055, WQS 0063, WQS 0072, WQS 0083, and WQS 0084, had similar day length and stream discharge (Fig. 4.5), the developed DL models could not learn the nitrate variability due to the smaller number of nitrate information (less than 50 data points) for the DL model training (Fig. 4.6).

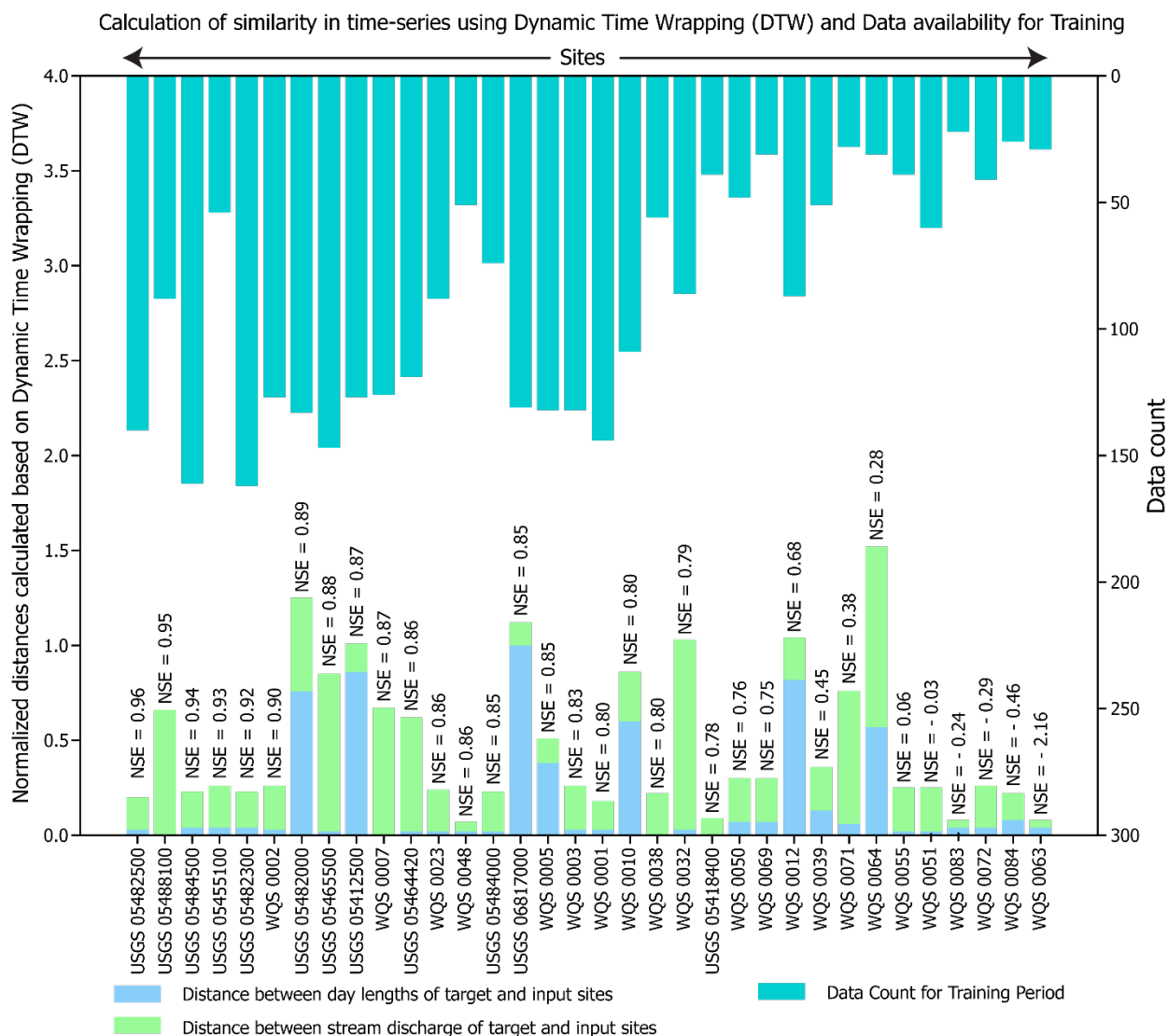


Figure 4.6 Site-wise normalized similarity indices calculated for day length and stream discharge (normal axis) and respective data counts used for DL model training (inverted axis)

The DL model framework excellently filled the temporal data gaps in the low-frequency nitrate data. The similarity in stream discharge, seasonality, and nitrate trends variability between low-frequency and respective input high-frequency nitrate monitoring sites contributed to the nitrate concentration prediction ability of the DL model. Data availability to train the DL model was also crucial; the models' performance was low when the low-frequency monitoring site had less than 50 counts (two years of biweekly data) of measured nitrate data. The DL model

demonstrated limited ability to predict daily nitrate concentrations beyond the available nitrate data period at low-frequency monitoring sites. The trained DL model can predict nitrate concentrations appropriately within the input high-frequency nitrate data available period since high-frequency nitrate concentration data is an input to the DL model. More detailed analysis is required to explore the potential of using the DL modeling framework for hindcasting to periods where the high frequency data is not available. The extrapolated or filled nitrate data can be used as an input to forecast daily nitrate at other low-frequency nitrate monitoring locations. More accurate forecasts of stream discharge, precipitation, maximum and minimum temperatures, day length, and land uses could improve model predictions in the forecast period.

#### **4.3.5 Seasonal performance of the DL models.**

The DL models performed well in predicting daily nitrate concentration for all four seasons at most pseudo-low-frequency nitrate monitoring sites (Fig. 4.7i). About 68% (23 out of 34 sites) and 56% (19 out of 34 sites) showed NSEs greater than 0.75 for the summer and fall seasons, respectively (Table S4.8 in Appendix C). The DL model performance for the spring and winter seasons was comparatively lower than the summer and fall seasons. About 50% (17 out of 34 sites) of pseudo-low-frequency nitrate monitoring sites performed well for spring with NSE values of 0.62 (Table S4.8 in Appendix C). Among the twenty-seven (27) pseudo-low-frequency nitrate monitoring sites with nitrate data in winter, thirteen (13) sites performed well with NSE greater than 0.75. The DL model predicted poorly for 6 to 11 sites for all four seasons. In a few cases, the DL model cannot predict the comparatively high nitrate concentration values in winter seasons. For instance, the DL model underestimated the daily nitrate concentration of the 2013, 2014, and 2018 winter seasons at the nitrate monitoring site USGS 06817000 (Fig. 4.7ii). The biophysical characteristics and climate input data used to



develop DL models may not contain adequate information on these specific processes, leading to inaccurate daily nitrate concentration predictions during winter. Moreover, the DL model overestimated daily nitrate concentration from October 15 to March 30, during the late fall (October 15 – November), winter (December – February), the early spring (March), seasons at the poorly performed nitrate monitoring sites (Fig. 4.7iii).

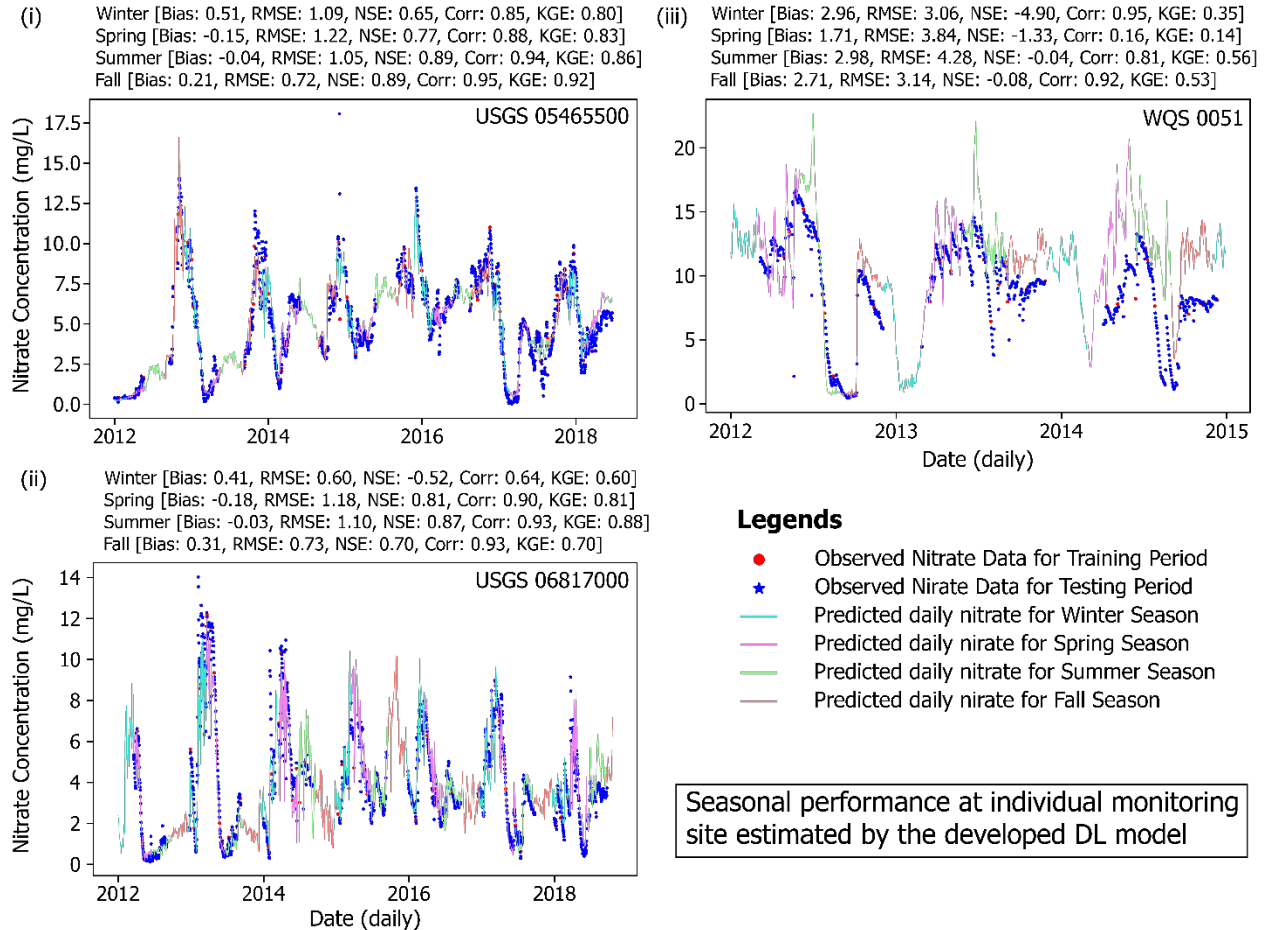


Figure 4.7 Estimated daily nitrate concentration time series at three nitrate monitoring sites for four seasons: (i) the DL model predicted daily nitrate concentration well for four seasons at USGS 05465500; (ii) the DL model performed well to predict daily nitrate concentrations for three seasons except for winter at USGS 06817000; and (iii) the DL model overestimated daily nitrate concentration for four seasons at WQS 0051.

The c-Q relationships during each year's growing (April – August) and harvesting (October – November) seasons were identified using the DL model estimated daily nitrate concentrations and compared with the observed nitrate data. The analysis identified seven (7) c-Q relationship patterns based on DL-model estimated daily nitrate concentrations for growing and harvesting seasons. The patterns included (i) flushing, (ii) flushing during low Q and chemostatic during the high Q, (iii) flushing during low Q and dilution during the high Q, (iv) chemostatic during low Q and flushing during the high Q, (v) chemostatic during low Q and dilution during the high Q, (vi) dilution, and (viii) chemodynamic. During growing seasons, three c-Q relationship patterns, including flushing (42%) (Fig. 4.8i), flushing during low Q and dilution during the high Q (29%) (Fig. 4.8iii), and flushing during low Q and chemostatic during the high Q (16%) (Fig. 4.8ii) were identified dominant. The growing season includes the late spring and entire summer seasons, and the model performed well for the summer season and the c-Q relationship patterns were identified similarly. During the harvesting seasons, other three c-Q relationship patterns, including flushing during low Q and dilution during the high Q (45%) (Fig. 4.8v), dilution (21%) (Fig. 4.8vi), and flushing (20%) (Fig. 4.8iv). Agricultural lands generally contain lower soil nitrogen levels near harvesting period compared to early growing periods since fertilizers are applied near planting periods. A comparatively lower nutrient availability and less intensive precipitation might show more flushing during low Q and dilution during the high Q pattern, followed by dilution patterns for harvesting periods (Burns et al., 2019a).

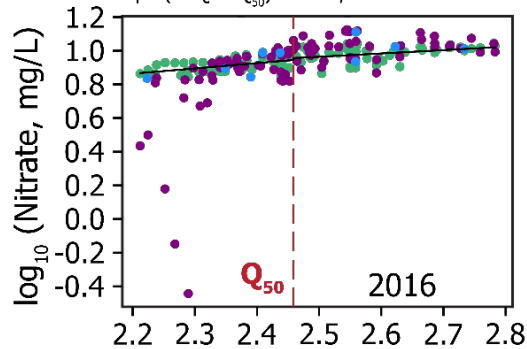
## Growing Season

(April - August)

### (i) Flushing Pattern

WQS 0002

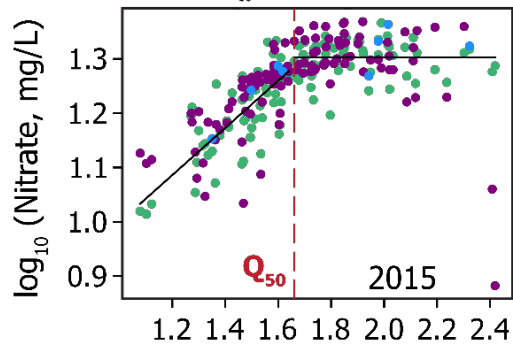
Slope (at  $Q < Q_{50}$ ) = 0.93; Correlation = 0.54  
Slope (at  $Q > Q_{50}$ ) = 0.77; Correlation = 0.39



### (ii) Flushing and Chemostatic Pattern

USGS 05482500

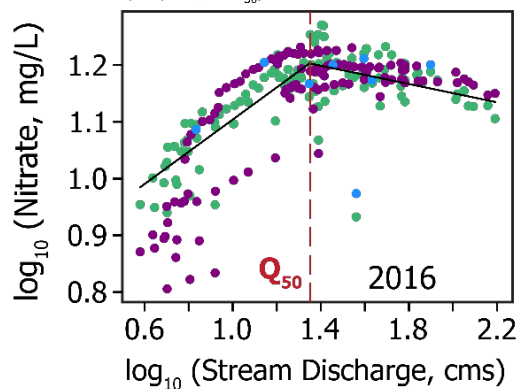
Slope (at  $Q < Q_{50}$ ) = 1.39; Correlation = 0.78  
Slope (at  $Q > Q_{50}$ ) = -0.01; Correlation = -0.002



### (iii) Flushing and Dilution Pattern

USGS 05482300

Slope (at  $Q < Q_{50}$ ) = 2.71; Correlation = 0.88  
Slope (at  $Q > Q_{50}$ ) = -1.56; Correlation = -0.36



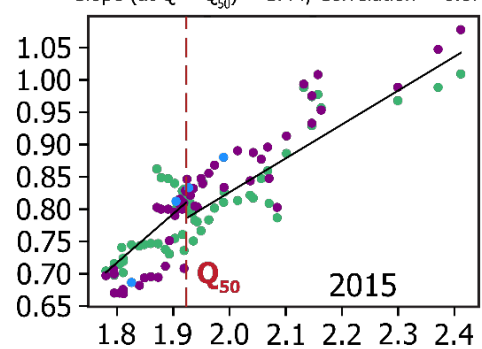
## Harvesting Season

(October - November)

### (iv) Flushing Pattern

USGS 05464420

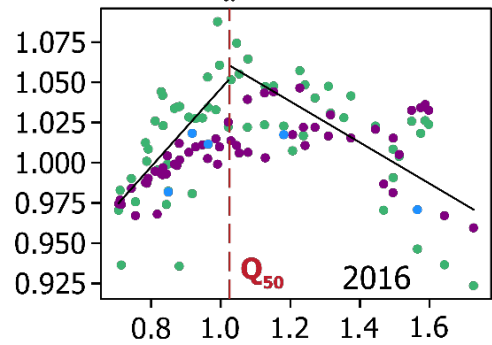
Slope (at  $Q < Q_{50}$ ) = 0.68; Correlation = 0.72  
Slope (at  $Q > Q_{50}$ ) = 1.44; Correlation = 0.87



### (v) Flushing and Dilution Pattern

WQS 0039

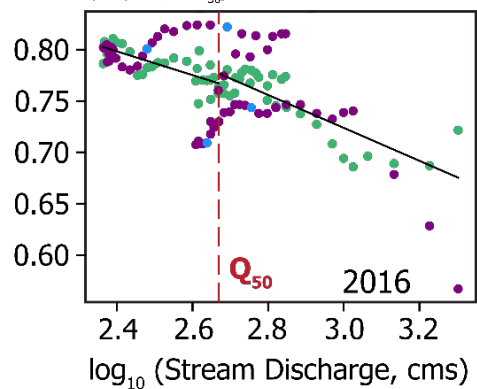
Slope (at  $Q < Q_{50}$ ) = 1.60; Correlation = 0.62  
Slope (at  $Q > Q_{50}$ ) = -4.11; Correlation = -0.73



### (vi) Dilution Pattern

USGS 05465500

Slope (at  $Q < Q_{50}$ ) = -4.92; Correlation = -0.75  
Slope (at  $Q > Q_{50}$ ) = -4.33; Correlation = -0.83



- Predicted Daily Nitrate Concentration (mg/L)
- Observed Daily Nitrate Concentration for testing period (mg/L)
- Observed Daily Nitrate Concentration for training period (mg/L)

Figure 4.8 Three (3) dominant c-Q relationship patterns based on estimated daily nitrate concentration for growing and harvesting seasons. For growing season, the patterns were (i) flushing, (ii) flushing during low Q and chemostatic during the high Q, and (iii) flushing during low Q and dilution during the high Q. For harvesting season, were (iv) flushing, (v) flushing during low Q and dilution during the high Q, and (vi) dilution.

Depending upon the availability of observed stream nitrate concentration data, only 157 nitrate c-Q relationship plots developed for DL model-estimated nitrate and observed nitrate concentrations could be compared. The nitrate c-Q relationship plots of all thirty-four nitrate monitoring sites were comparable to observed nitrate c-Q relationship plots produced for growing and harvesting seasons between 2012 – 2019. About 89% of estimated nitrate c-Q relationship plots were analogous with observed nitrate c-Q relationship plots for the growing season. The remaining 11% estimated nitrate concentration-based c-Q relationships did not follow the observed c-Q relationship because of two reasons: (i) unavailability or limited availability of observed nitrate data at high stream discharge (  $Q > \text{median } Q$  ) and (ii) unavailability or limited availability of observed nitrate data at low stream discharge (  $Q < \text{median } Q$  ) (Fig. S4.7i in Appendix C). The estimated c-Q relationship plots are similar to the observed c-Q relationship plots for the harvesting season of each year for 83% of the 157 c-Q plots. Among the remaining 17% of estimated c-Q relationship plots, about 9% of the estimated c-Q relationship plots showed identical patterns as observed but with a consistent bias (overestimated or underestimated) daily nitrate concentration values (Fig. S4.7iv in Appendix C). The rest 8% of estimated c-Q relationship plots showed different c-Q relationship patterns than observed (Fig. S4.7iii in Appendix C). Since the DL models overestimated or underestimated during the late fall season, getting different c-Q relationship patterns during the harvesting season is possible.

#### **4.3.6 Applicability and future scope of the DL model-based daily stream nitrate prediction framework.**

The study shows potential of the DL model for filling the nitrate data gaps in low-frequency nitrate monitoring sites for regions which have a high frequency nitrate monitoring available in the region or a biophysically similar region. The study shows potential to develop a spatially and temporally continuous nitrate data network for the US with more emerging high frequency nitrate sensor network. Currently, the USGS have about 160 sensor-based nitrate monitoring sites in 27 US states, and the agency continuously is expanding their sensor-based nitrate monitoring network throughout the USA (USGS, 2023b). The modeling framework could also be expanded to sub-daily temporal scale, the high frequency data is available at 15-minute interval. The modeling framework could also be used to identify strategic new nitrate monitoring locations based on models' inability to estimate specific regions. Respective agencies can also prioritize the low-frequency nitrate monitoring locations to convert them into high-frequency nitrate monitoring sites based on the modeling framework. The modeling framework also expands the potential to develop more accurate historic nitrate data for low-frequency nitrate monitoring sites and better understand nitrate c-Q dynamics at data-scarce watersheds. Though the DL model-based daily stream nitrate prediction framework performed well in the case study region, more studies are required to extend the framework to a national scale. The framework performed excellently for an agriculture-dominated region; however, the framework's applicability to diverse land-use practices needs further investigation. The developed DL models demonstrated comparatively poor performance in the winter season, emphasizing the need for more data collection efforts in the winter months.

#### 4.4 Conclusions

The developed DL framework indicated a high potential for developing more temporally continuous daily nitrate data in low-frequency nitrate sites. The DL model accurately predicted daily stream nitrate concentration at 70% (24 out of 34 sites) pseudo-low-frequency nitrate monitoring site with NSE greater than 0.75. The analysis suggests that the DL models require high-frequency nitrate data as the input data and the hindcasting or forecasting could be challenging, and more studies are required to explore this. The DL model can learn temporal nitrate variability from a high-frequency monitoring site within a 300 km radius. Three (3) dominant nitrate c-Q relationship patterns were identified by analyzing the estimated nitrate concentrations of the pseudo-low-frequency nitrate monitoring sites: (i) flushing, (ii) flushing during low Q ( $Q < \text{median } Q, Q_{50}$ ) and dilution during high Q ( $Q > Q_{50}$ ), and (iii) flushing during low Q and chemostatic during high Q. Corresponding input high-frequency sites also demonstrated these dominant c-Q relationship patterns, which indicated the DL models ability to learn and transfer nitrate dynamics from the high-frequency nitrate concentrations to predict continuous nitrate accurately at low-frequency nitrate sites. Further analysis indicated that similarity in day length (a representative variable of seasonality), stream discharge, and daily nitrate concentration contributed more to the accurate daily nitrate prediction at pseudo-low-frequency nitrate monitoring sites in Iowa. The DL models require biweekly nitrate data (50 data counts) of more than two years to learn the nitrate dynamics of any low-frequency nitrate monitoring site. The DL model performed well for the summer and fall seasons. The DL model could not predict extremely high nitrate concentrations during winter. The DL model predicted accurate c-Q relationships for growing and harvesting seasons at 89% and 83%, respectively. Six dominant c-Q relationship patterns were identified based on the DL model estimated daily

nitrate concentration for growing and harvesting seasons: for growing seasons, (i) flushing, (ii) flushing during low Q, and dilution during the high Q, (iii) flushing during low Q and chemostatic during the high Q; for harvesting seasons (iv) flushing during low Q and dilution during the high Q, (v) dilution, and (vi) flushing. The DL modeling framework-derived insights will expand the framework's applicability to other nitrate data-scarce regions and widen the opportunity of conducting more water quality research in the US.

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## **CHAPTER 5**

# EXPLORING THE POTENTIAL OF EXTENDING THE DATA-DRIVEN STREAM NITRATE CONCENTRATION MODELING FRAMEWORK TO STREAM DISCHARGE DATA UNAVAILABLE LOCATIONS FOR MAKING THE FRAMEWORK MORE SPATIALLY ROBUST.

## 5.1 Introduction

Stream nitrate concentration is a crucial indicator of water quality and the ecological health of aquatic ecosystems. Therefore, identifying the potential sources and understanding the trends, seasonal variabilities, and influential environmental drivers of stream nitrate concentrations become critical to plan appropriate conservation practices. Stream nitrate concentrations heavily depend on the concentration (c) - stream discharge (Q) relationship (Duncan, Band, et al., 2017; Duncan, Welty, et al., 2017; X. Zhang et al., 2020). The stream nitrate's c-Q relationships provide valuable insights into the sources and pathways of nitrate inputs to streams, including agricultural runoff, groundwater inputs, geological contributions, and atmospheric deposition (Bowes et al., 2014; Husic et al., 2020; W. Liu et al., 2022; Saavedra et al., 2022). The c-Q relationship patterns also offer information on hydrological connectivity, the lag time between nitrate sources to their delivery locations within a watershed, the role of subsurface flow, nitrate storage in soils, and nitrate runoff (Kaandorp et al., 2021). Since c-Q relationships are non-stationary over time and space due to the influence of land use, climate, and land management practices (Bieroza et al., 2018; Q. Zhang, 2018), tracking and estimating stream nitrate and discharge at multiple locations is essential.

Among numerous ways of tracking or estimating stream nitrate concentration, the most effective way can be the frequent collection (sampling rate > rate of change) of stream nitrate

concentration in multiple locations and consequent data analysis. Towards the effort, a federal agency, and a few state and local agencies, such as the United States Geological Survey (USGS), Iowa Water Quality Information System (IWQIS), and Minnesota Pollution Control Agency, collect high-frequency stream nitrate data using nitrate sensors for the last decade. However, a crucial issue in many nitrate monitoring locations is the unavailability of stream discharge data due to needing more information demand and adequate financial and logistic support. The unavailability of stream discharge data at a nitrate monitoring site limits understanding the contributing watershed's c-Q relationship patterns and stream nitrate dynamics.

In a previous study, Saha et al. (2023) developed a deep learning (DL)-based modeling framework (Chapters 2 and 3) that learned nitrate concentration dynamics from the periodical nitrate measurements and a few available high-frequency nitrate monitoring sites in a region and utilized the learned information generating continuous daily stream nitrate concentration at those periodically nitrate measured locations (called low-frequency nitrate monitoring sites where sampling rates are less than nitrate change rate). DL model development also requires information on nitrate monitoring sites' biophysical, climatic, and watershed characteristics to predict stream nitrate concentrations.

DL modeling analysis demonstrated stream discharge and seasonality (day length) and precipitation (in a few cases) as the crucial environmental characteristics to estimate stream nitrate concentrations accurately at data-scarce locations in a region (Chapter 3). However, the unavailability of stream discharge in many nitrate data measuring locations creates a significant challenge to making a more spatially robust data-driven modeling framework for filling nitrate gaps. Therefore, one potential solution can be utilizing precipitation data as the

crucial predictor of stream nitrate concentration because of the wide availability of precipitation data across the US. Another practical approach can be using simulated stream discharge stream locations where stream discharge is not measured. Incorporating precipitation or simulated stream discharge data or both to develop deep learning (DL) models can enhance the potential of building a more spatially and temporally robust data-driven modeling framework for filling the temporal nitrate data gaps in the data-limited regions.

Precipitation data at the stream nitrate monitoring locations can be used as a surrogate, which may provide essential insights into the c-Q relationship dynamics, enabling the accurate estimation of stream nitrate concentrations. Precipitation represents the primary source of water input to streams and contributes to the nitrate transport from the land surface to water bodies. Precipitation frequency, duration, and variability in intensity affect the nitrate transport rate and timing (Wolf et al., 2020). The influence of precipitation on nitrate transport from landscape to stream makes it a critical factor to consider when predicting nitrate concentrations. An advantage of precipitation data is its availability throughout the US (Thornton et al., 2020). Assessing the potential of precipitation data of high-temporal resolution as a critical predictor of stream nitrate variability enhances the opportunity of expanding the developed data-driven modeling framework (Chapter 2 and Saha et al. 2023) to stream discharge data-limited locations and make the modeling framework more spatially robust in estimating stream nitrate concentrations.

Another potential option for understanding the nitrate c-Q relationship dynamics could be using simulated stream discharge instead of unavailable observed stream discharge data. Simulated stream discharge derived from hydrological or statistical models can represent the flow variability where no stream discharge is measured. One valuable source of obtaining

simulated stream discharge could be StreamStats (<https://streamstats.usgs.gov/ss/>), a web-based Geographic Information System (GIS) application created by the USGS. StreamStats offers simulated stream discharge at any stream location in the US using the Flow-Duration Curve Transfer Method (FDCTM) (Marti et al., 2022). the simulated discharge can be an appropriate surrogate that portrays the hydrological connectivity and nitrate transport pathways within the watershed and provide valuable information on stream discharge variability.

The study aims to make the developed DL modeling framework more spatially robust in predicting daily nitrate concentration. The objectives of this study were (i) to evaluate the potential of precipitation and stream discharge data as the critical environmental characteristics in predicting continuous nitrate concentrations for filling the data gaps at data-limited locations using the data-driven modeling framework and (ii) to analyze the developed DL models' capability in predicting nitrate concentrations for different seasons and days of precipitation.

## **5.2 Methods**

This study assessed the potential of precipitation and stream discharge to expand the DL modeling framework to nitrate measuring locations where stream discharge measurement is unavailable and make the DL modeling framework more spatially robust. In this regard, a Long Short-Term Memory (LSTM)-based DL model was developed to estimate stream nitrate concentrations. The developed DL model learned nitrate dynamics from climate and watershed's biophysical information and predicted continuous nitrate concentrations at data-limited stream locations. The case study was conducted in Iowa because of the state's higher stream nitrate data availability. In this study, thirty-two (32) among selected thirty-four (34) high-frequency nitrate monitoring sites were randomly subsampled to biweekly data and used for DL model development. These 32 sites with biweekly nitrate data were called pseudo-low-

frequency sites. These pseudo-low-frequency sites' remaining daily nitrate concentrations were used to evaluate the developed DL model. The daily nitrate concentrations and stream discharge data of the remaining two high-frequency nitrate monitoring sites were used as DL model inputs. Five DL modeling cases were formulated considering the combinations of precipitation, observed stream discharge, and simulated stream discharge as the most significant variables. The DL model simulated results were analyzed to identify the crucial environmental variable for spatial expansion of the DL modeling framework. The cross-correlation among estimated nitrate concentrations, precipitation, and stream discharge was calculated to evaluate the impacts on stream nitrate variability. The DL model performance was also assessed for precipitation and no-precipitation days to understand the contribution of the precipitation data in estimating stream nitrate concentrations as the most significant environmental variable.

### **5.2.1 Case Study area – Iowa.**

Iowa was selected as the case study region since the state monitored stream nitrate status using many high-frequency nitrate monitoring sensors. These nitrate sensors can measure stream nitrate in 15- to 60-minute intervals (Pellerin et al., 2013; USGS, 2023a). In addition, Iowa is a leading agriculture-dominated state in the USA and one of the most significant N contributors to the Mississippi River (Wolf et al., 2020). Iowa has primarily flat terrain with a slope of less than 5%. Average precipitation ranges from 28- to 40 inches annually. Multiple studies identified that rainfall, streamflow, land cover, soybean production, and N fertilization impacted Iowa's stream nitrate concentration (Villarini et al., 2016; Wolf et al., 2020). The Iowa Nutrient Reduction Strategy wants to reduce the N loads by 45% annually to improve the water quality of Iowa's streams, rivers, and the Gulf of Mexico (Jones et al., 2019). Recent

analyses showed that manure N contributes to nitrate loss since livestock concentration is increasing in many watersheds in Iowa (Jones et al., 2019). A few federal and state agencies, including the United States Geological Survey (USGS) and the Iowa Water Quality Information System (IWQIS), placed more than a hundred high-frequency nitrate sensors on many of Iowa's rivers and creeks to track stream nitrate concentrations. However, seventy percent (70%) of these high-frequency nitrate sensors were recently placed and have limited high-frequency nitrate information. The high-frequency stream nitrate concentrations help identify the most accurate stream nitrate dynamics and make informed management decisions. This study used nitrate data from thirty-four (34) high-frequency nitrate sensors considering at least two years of data availability (Fig. 5.1 and Fig. S5.1 in Appendix D).

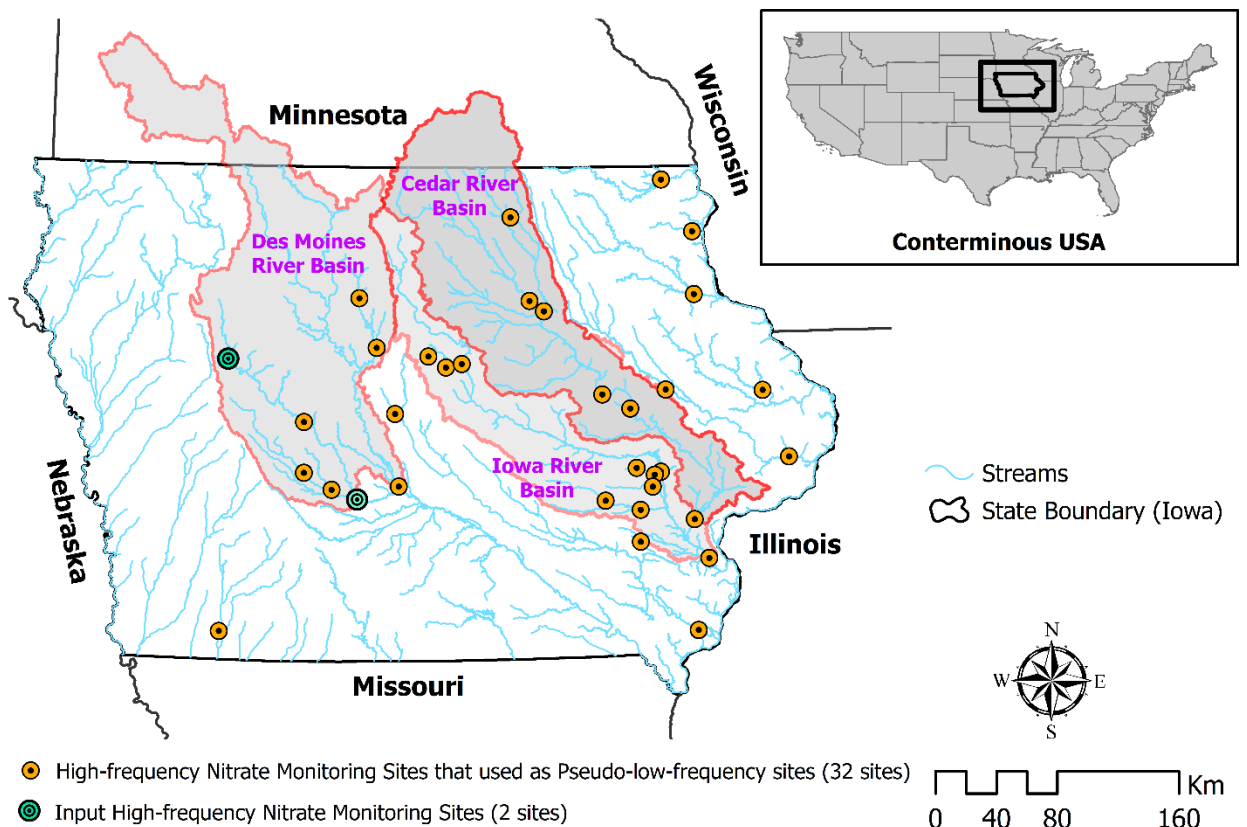




Figure 5.1 Nitrate monitoring site locations in Iowa, USA: (i) thirty-two (32) high-frequency nitrate monitoring sites used as pseudo-low-frequency nitrate sites; (ii) high-frequency nitrate data of two (2) nitrate monitoring sites used as model inputs. Most of these high-frequency nitrate monitoring sites were placed in three significant basins: Des Moines, Iowa, and Cedar River Basins.

### **5.2.2 DL modeling framework**

This study analyzed a developed data-driven modeling framework further to make the framework more spatially and temporally robust for filling the stream nitrate data gaps in the data-scarce region. The data-driven modeling framework used Long Short-Term Memory (LSTM) model as the deep learning (DL) model to predict continuous stream nitrate concentration at any gauged location (Chapter 3). DL model can learn stream nitrate dynamics from the variability of nitrate concentration, stream discharge, climate, land use, soil characteristics, fertilization of multiple low-frequency and a few neighboring high-frequency nitrate monitoring sites and predict continuous nitrate concentrations low-frequency sites in a data-scarce region (G. K. Saha, Rahmani, et al., 2023). In previous studies, continuous nitrate and stream discharge data of a high-frequency nitrate monitoring site were used as inputs to develop DL models assuming a nitrate data-scarce region had at least one high-frequency nitrate monitoring site (G. K. Saha, Rahmani, et al., 2023). Practically, the nitrate data-scarce areas may have multiple high-frequency nitrate monitoring sites. Therefore, data from various high-frequency nitrate monitoring locations in a data-scarce region could be used as inputs for DL model development. A previous study demonstrated continuous nitrate concentration and stream discharge data of two high-frequency nitrate monitoring sites (Site USGS 05482300 and USGS 05484500) helped the DL models predict daily nitrate concentrations at 22 out of 34 (70%) of Iowa's low-frequency nitrate monitoring sites (Chapter 4). This study used

continuous stream nitrate concentration and stream discharge data from Iowa's two high-frequency nitrate monitoring sites for the DL model development. The remaining thirty-two (32) high-frequency nitrate monitoring sites were utilized as pseudo-low-frequency sites, where their daily nitrate data were randomly subsampled into biweekly data and used for training the DL model. The remaining daily nitrate data from these pseudo-low-frequency sites were then employed to evaluate the performance of the trained DL model. The data availability, training, and testing periods for each of the pseudo-low-frequency nitrate monitoring sites used for the DL model development are available in Table S5.1 in Appendix D.

DL model development required the watershed's biophysical and climatic information, including forcings (time-series information) and attributes (watershed characteristics information). Among daily varying forcings, a DL model used precipitation, observed or simulated stream discharge, maximum and minimum temperatures, and day length data of all low-frequency nitrate monitoring sites and continuous nitrate and observed stream discharge data of available high-frequency nitrate monitoring sites as inputs. Annually varying forcings include land use (corn, soybean, forest, urban, and other agriculture areas) and fertilization data (fertilizer and manure N applications). The DL model also required attributes, such as drainage area, mean precipitation, shallow and deep soils, soils with hydrologic soil groups A, B, C, and D, poorly drained, well-drained, and excessively drained soils, dominant soil textures, dominant soil erodibility factors and its area coverage of each nitrate monitoring site's contributing watersheds.

USGS only measures stream discharge at a few nitrate monitoring locations in the USA. In Iowa, the Iowa Institute of Hydraulic Research (IIHR)-Hydroscience & Engineering also collects stream discharge at other specific locations to better understand Iowa's flooding and

water quality status. Since observed stream discharge is a critical environmental characteristic for estimating stream nitrate concentrations, simulated stream discharge for observed stream discharge data unavailable locations can be a good surrogate for the DL model development. Recently, USGS provided simulated stream discharge data through StreamStats (<https://streamstats.usgs.gov/ss/>). StreamStats is a web application that delineates drainage-basin boundaries, basin characteristics, and estimates of streamflow statistics using USGS-collected hydrologic and hydraulic information throughout the USA. StreamStats used the study area and a nitrate monitoring site's coordinate to delineate the site's contributing basin. The web application also calculated many basin characteristics, including basin area, average slope, mean precipitation, base flow index, basin shape factor, longest flow path slope, and hydrologic soil groups. StreamStats also offers eight different regression-based scenarios, including Peak-Flow Statistics, Low-Flow Statistics, Flow-Duration Statistics, Flow-Duration Curve Transfer Method (FDCTM), Seasonal Flow Statistics, General Flow Statistics, Bankfull Statistics, and Maximum Probable Flood Statistics, to provide daily stream discharge for the selected stream location (for the chosen pseudo-low-frequency nitrate monitoring site's location). The study used FDCTM to estimate simulated stream discharge for the thirty-two (32) pseudo-low-frequency nitrate monitoring sites. The FDCTM method required a neighboring USGS stream discharge measuring location to calculate the simulated stream discharge for the required location. The web application provides options to select neighboring USGS stream discharge measuring gauge locations based on stream proximity or the best correlation with the required stream location. The simulated stream discharge data was calculated for all thirty-two (32) pseudo-low-frequency nitrate monitoring locations from January 01, 2012, to December 31, 2019.

During each DL model training, several combinations of five hyperparameters, including epoch, minibatch, hidden layer, time window, and random seeds, were used to identify the best DL model for result analysis (Table S5.2 in Appendix D). The best DL model was determined by five performance evaluation metrics: bias, root mean squared error (RMSE), Nash–Sutcliffe efficiency coefficient (NSE) (Nash & Sutcliffe, 1970), correlation, and Kling Gupta Efficiency (KGE) (Gupta et al., 2009). Data preparation and details about hyperparameters and performance evaluation metrics were described in Chapter 2 and Saha, Rahmani, Shen, et al. (2023).

### **5.3 Five DL Model Simulation Case Development**

Five DL model-based simulation cases were developed with different combinations of precipitation, observed stream discharge, and simulated stream discharge with a vision to estimate continuous nitrate concentration at any stream location within a region and make a more spatially and temporally robust data-driven modeling framework. In Case 1, the DL model was developed using all forcings and attributes as DL model inputs except observed and simulated stream discharge. The case represented the influence of precipitation data to predict daily stream nitrate concentrations at pseudo-low-frequency nitrate monitoring sites. Case 2 and Case 3 were developed with observed stream discharge and simulated stream discharge, respectively, to evaluate the importance of observed and simulated stream discharge data in predicting continuous stream nitrate concentrations at a data-scarce region. Precipitation data was not used in these two simulation cases (Cases 2 and 3). Between the remaining two cases, Case 4 was developed by combining precipitation and simulated stream discharge, and Case 5 was set up with precipitation and observed stream discharge to identify the combined impacts of these environmental characteristics on daily stream nitrate prediction at pseudo-low-

frequency sites. Other forcings and attributes were kept the same to develop DL models for all five cases except for the combinations of precipitation, observed stream discharge, and simulated stream discharge. The best DL model with specific hyperparameters-combination was identified using testing period performance metrics for each simulation case.

## **5.3 Results and Discussion**

### **5.3.1 The best identified DL models for all five cases**

Case 2 DL model provided the best model performance with the median NSE 0.77, KGE 0.85, RMSE 1.41 mg/L, correlation 0.88, and bias 0.02 mg/L for the testing period (Table 5.1 and Fig. S5.2 in Appendix D). The DL model developed with precipitation as the significant environmental characteristic (Case 1) performed 15% less than the Case 2 model (Table 1 and Table S3 in Appendix D). The DL models set for Case 3, Case 4, and Case 5, showed similar estimation performance as the Case 2 model performance. The results indicated that simulated stream discharge was the most significant environmental characteristic that helped the DL model estimate continuous daily nitrate concentration at the pseudo-low-frequency nitrate monitoring locations. Case 3 model performance demonstrated that simulated stream discharge would be an excellent surrogate to predict stream nitrate concentration at low-frequency nitrate monitoring locations where observed stream discharge data is unavailable. Precipitation data could also be a potential surrogate of stream discharge, except for the probability of having less nitrate estimation accuracy. According to Cases 4 and 5 simulation results, utilizing both precipitation and stream discharge data for DL model development included limited additional information to the DL model for accurate nitrate estimation. One of them could be enough as the most significant environmental characteristic to estimate stream nitrate concentrations at data-limited locations.

Table 5.1 Median scores of evaluation metrics for all five DL model simulation cases

| Case No | Major Variables | Testing Period |      |      |             |      | Training Period |      |      |             |      |
|---------|-----------------|----------------|------|------|-------------|------|-----------------|------|------|-------------|------|
|         |                 | NSE            | KGE  | RMSE | Correlation | Bias | NSE             | KGE  | RMSE | Correlation | Bias |
| 1       | P               | 0.67           | 0.79 | 1.73 | 0.82        | 0.06 | 0.89            | 0.90 | 1.08 | 0.96        | 0.10 |
| 2       | Q               | 0.77           | 0.85 | 1.41 | 0.88        | 0.02 | 0.93            | 0.92 | 0.85 | 0.96        | 0.06 |
| 3       | SimQ            | 0.76           | 0.83 | 1.45 | 0.88        | 0.08 | 0.90            | 0.90 | 1.00 | 0.96        | 0.14 |
| 4       | P & SimQ        | 0.75           | 0.81 | 1.48 | 0.87        | 0.20 | 0.88            | 0.89 | 1.01 | 0.96        | 0.20 |
| 5       | P & Q           | 0.75           | 0.84 | 1.49 | 0.87        | 0.15 | 0.90            | 0.84 | 1.08 | 0.95        | 0.16 |

P = Precipitation, Q = Observed Stream Discharge, SimQ = Simulated Stream Discharge.

### 5.3.2 DL model's capabilities to predict nitrate concentrations at numerous locations in Iowa.

Among the performance metrics of thirty-two (32) pseudo-low-frequency nitrate sites by the Case2 model, about 62.5% of the sites showed NSE performances of 0.75 and more (Table S4 in Appendix). The Case 2 DL model also predicts continuous daily nitrate concentration well at seven other (7) pseudo-nitrate monitoring sites (22% of all thirty-two) with NSE 0.50 or higher. Though the developed DL models for all five cases demonstrated good prediction performance for the USGS and early IWQIS-installed (sites that have data from 2012 and onwards) nitrate monitoring sites, the model performance for sites having nitrate data from 2016 and later was identified as comparatively poor ( $NSE < 0.50$ ) (Fig. 5.2 and Fig. S5.3 in Appendix D). For a few pseudo-low-frequency nitrate monitoring sites, the Case 1 DL model (Case 1) failed to estimate continuous daily nitrate concentrations (WQS 0063, WQS 0071, and WQS 0083). The result indicated that precipitation did not influence the nitrate variability of these pseudo-low-frequency sites. Observed stream discharge contributed to good performance in predicting continuous stream nitrate at these nitrate measuring sites (Table S5.4 and Figure S5.3 in Appendix D).

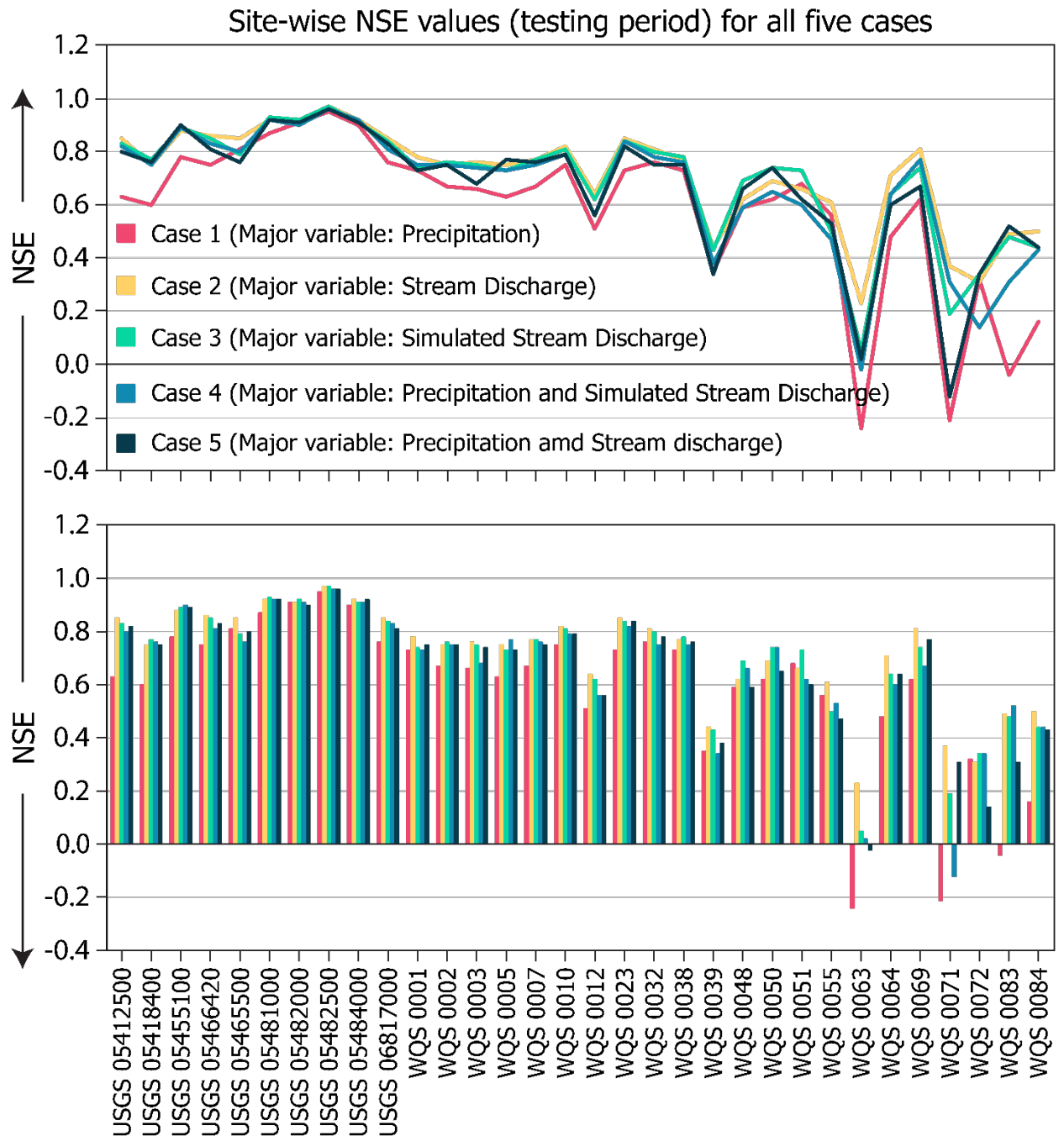


Figure 5.2 Site-wise NSE trends of thirty-two pseudo-low-frequency nitrate monitoring sites for all five cases: (i) overall progression of NSE values among five cases and (ii) bar plots of NSE values of all five cases for each pseudo-low-frequency site.

Regional NSE distribution throughout Iowa showed that DL models for all five cases predicted continuous daily stream nitrate concentrations well for filling the temporal data gaps at pseudo-

low-frequency sites at Des Moines and Iowa River Basins (Fig. 5.3 and Fig. S5.4 in Appendix D). Wolf et al. (2020) also identified the impacts of current and previous year precipitation on stream nitrate variability in Des Moines and Iowa River Basins. They pointed out that excessive availability of surplus or lack of fillable soil pore space regulated the overland flow, infiltration, and percolation processes in these basins and resulted in precipitation's governance on stream nitrate concentration. The DL model performance was reduced for a few pseudo-low-frequency nitrate monitoring sites placed in the middle of the Cedar River Basin and the outlet of the Yellow River Basin (YRB; at the northeast corner of Iowa). Meulemans et al. (2020) identified extensive agricultural practice and, consequently, higher stream nitrate delivery from the tributaries of the Middle Cedar River as the root cause of erratic and excessive stream nitrate concentrations in the middle part of the Cedar River Basin. The developed DL model, with less information on agricultural management practices, could not explain the nitrate dynamics of pseudo-low-frequency nitrate monitoring sites at the mid-part of Cedar River Basin. All five DL models developed for five simulation cases failed to estimate continuous stream nitrate concentration at the pseudo-low-frequency nitrate monitoring site (WQS0063) located at the outlet of the YRB (Fig. S5.4 in Appendix D). One reason was the limited nitrate data availability for DL model training at that location. IWQIS placed the nitrate sensor in mid of 2017. In addition, the northeastern region of Iowa, including the Yellow River Basin, has different landform regions than other parts of Iowa. YRB is situated in Paleozoic Plateau, with unglaciated sedimentary rock-formed farmed uplands above narrow valleys (Jones, Davis et al., 2018). Since karst and sinkholes are common in the YRB, hydrological processes related to stream nitrate transport from uplands to the Yellow River can differ from other Iowa places. The developed DL models identified dominant trends and variabilities in



the stream nitrate concentrations explained by the input environmental characteristics. The developed DL model might need more information about the region to determine karst and sinkholes' impacts on the variability of stream nitrate concentrations at the YRB outlet.

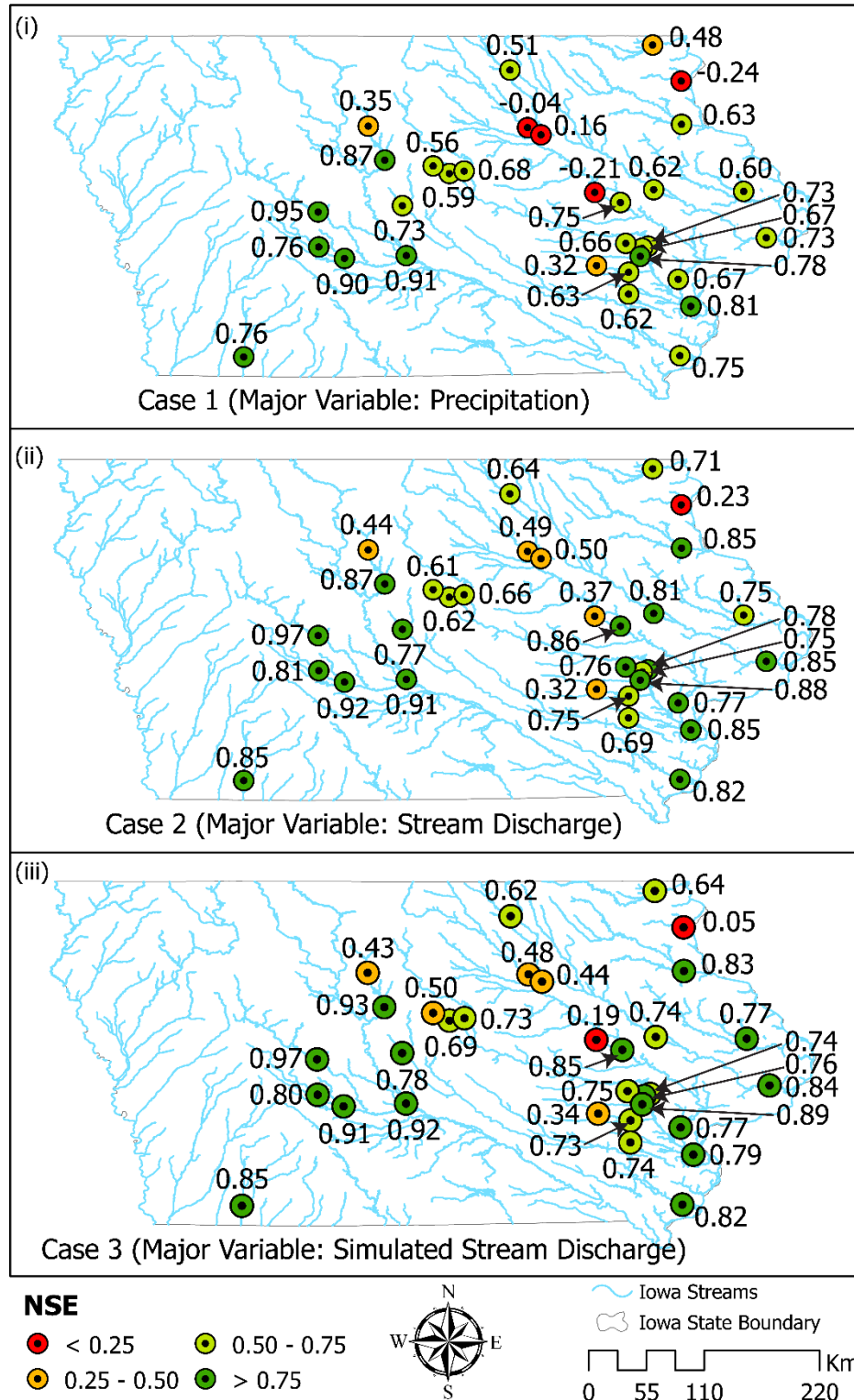


Figure 5.3 Regional NSE distribution throughout Iowa: NSE from (i) Case 1, (ii) Case 2, and (iii) Case 3 models.

### **5.3.3 Environmental factors influencing the estimated stream nitrate variability.**

Time series plots showed that daily stream nitrate concentrations estimated for five cases varied similarly with the precipitation variation and observed and simulated stream discharge. In general, nitrate concentrations increased in streams during or after precipitation and increased stream discharge (Fig. 5.4 and Fig. S5.5 in Appendix D). In addition, contributing watershed area versus NSE (Case 2) plot for all thirty-two (32) pseudo-low-frequency nitrate monitoring demonstrated that stream discharge was the crucial predictor of stream nitrate concentrations at the pseudo-low-frequency nitrate monitoring sites that had comparatively large contributing basins (Fig. S5.6i in Appendix D). Time series plots indicated that the variability of observed stream nitrate concentrations was substantially different from the variability in the DL model estimated nitrate concentrations for certain periods for several pseudo-low-frequency sites (Fig. 5.4). Precipitation or stream discharge might not be the dominant environmental factor for stream nitrate variability during these periods. Cross-correlation between observed and simulated stream discharge values for each pseudo-low-frequency site versus the DL model's performance metrics plot showed a positive trendline (Fig. S5.6ii in Appendix D). This positive trend inferred that simulated stream discharge, having similar variability as observed stream discharge, contributed considerably to accurate stream nitrate estimation. Therefore, if simulated stream discharge misrepresents the dynamics of any nitrate monitoring site's contributing basin, the DL model developed with simulated stream discharge may inaccurately predict stream nitrate concentrations.

The cross-correlations between Case 2 DL model-based estimated nitrate concentrations and observed stream discharge ranged from 0.09 to 0.57 with a median of 0.26 (Fig. S5.7i in Appendix D). The cross-correlation between Case 3 DL model-based estimated nitrate

concentrations and simulated stream discharge was similar to the correlation range of Case 2 model estimated nitrate concentrations and observed stream discharge (cross-correlation value range 0.14 – 0.46 with a median 0.28). However, the cross-correlation between Case 1 model-based estimated nitrate concentrations and precipitation were calculated as low (cross-correlation values  $< 0.1$  for thirty-two pseudo-low-frequency nitrate sites). The results indicated that stream nitrate concentration variability was not directly correlated with precipitation. Instead, observed stream discharge information or simulated stream discharge (where observed stream discharge was unavailable) could be the crucial environmental characteristics that helped the model predict continuous stream nitrate concentrations. Further analysis of Case 1 DL model's results demonstrated that day length had higher cross-correlation with the estimated stream nitrate concentrations (Fig. S5.7ii in Appendix D), which inferred that seasonality contributed more than precipitation to accurate estimation of DL model-based stream nitrate concentrations.

Since the variability in stream nitrate concentration was substantially correlated to the variability of stream discharge, the accurate prediction of stream nitrate concentrations depended on the accuracy of stream discharge measurements. USGS ensures the data quality of stream discharge by visiting each stream gage location every 6-8 weeks to measure stream discharge manually (Turnipseed & Sauer, 2010; USGS, 2018). The agency continuously updates the stage-discharge relations (rating curve) and calculates the stream discharge using the rating curve. The agency's stage measurement accuracy lies between -0.1 and 0.1 foot or 0.2% of the effective stage (Sauer & Turnipseed, 2010). USGS always tries to maintain the accuracy of simulated stream discharge for ungaged stream locations by ensuring the quality of observed stream stage and watershed characteristics. Since StreamStats utilizes FDCTM (a

parameter-based regression model) to develop a continuous daily flow-duration curve (the relation between exceedance probability and streamflow for each day of observed streamflow) at the ungagged stream location, and the curve is used to estimate daily mean stream discharge quantiles, the accuracy of the regression model was crucial to minimize the uncertainty in the FDCTM and get accurate simulated stream discharge. In addition, the representativeness of the gaged site for ungagged sites is crucial. This study prioritized two factors while obtaining simulated stream discharge using StreamStats: (i) the gaged site situated on the same river system and proximate to the ungagged site and (ii) the gaged site at a different river network that had a higher correlation with the ungagged site based on contributing basin characteristics. The analysis of the developed DL model results validated the simulated stream discharge estimation priorities used in this study.

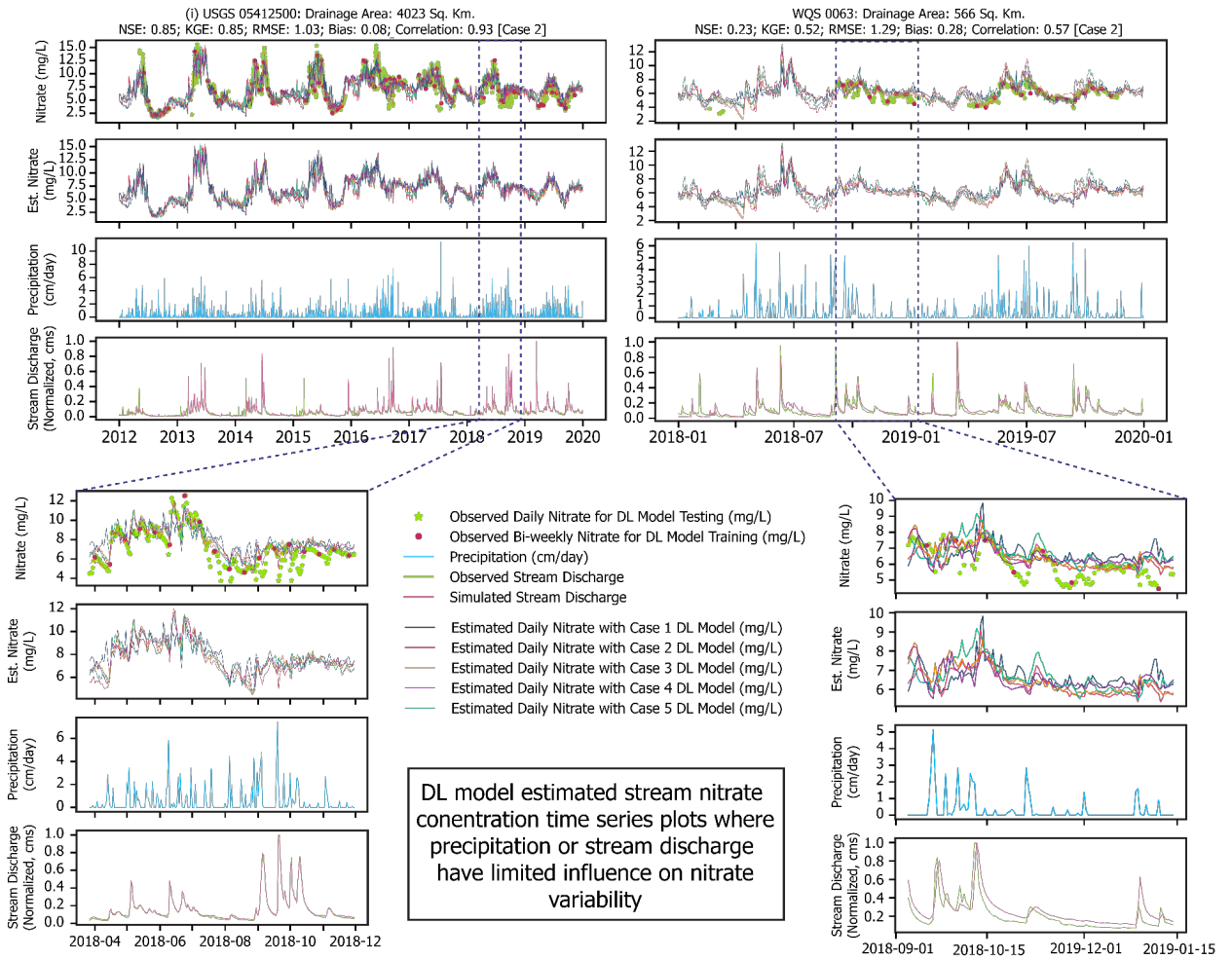


Figure 5.4 Time series plots of nitrate concentrations, including precipitation, observed and simulated stream discharge of pseudo-low-frequency sites for training and testing periods: (i) DL model predicted well for USGS 05412500; however, the model cannot predict for a certain period, and (ii) DL model performed poorly at WQS 0063; trend in the observed nitrate data was different and had less influence of stream discharge and precipitation.

#### **5.3.4 The DL models' robustness in predicting nitrate concentrations for no-precipitation and precipitation days.**

Based on the DL model performance comparison between no-precipitation and precipitation days, Case 1 and Case 2 models performed primarily well for no-precipitation days with median NSE 0.69 and 0.78, respectively (Fig. 5.5). The results indicated that precipitation data added insufficient information to the DL models for learning stream nitrate trends and variabilities during the DL model development. The DL models' nitrate estimation performance was further analyzed for no-precipitation and precipitation days during growing and non-growing seasons and identified inconclusive results. Later, the DL models' (Case 1 and Case 2) prediction performance was analyzed for no-precipitation and precipitation days during all seasons, including winter, spring, summer, and fall. Analysis indicated that the developed DL models predicted inaccurate daily stream nitrate concentrations for precipitation days during the winter season (Fig. S5.8 in Appendix D). Hydrologic processes that govern nitrate transport from lands to streams in agricultural watersheds during the winter seasons were substantially different from other seasons. Snow accumulation and snow melting, depending on the change in atmospheric temperature, impact the nitrate dynamics during winter seasons. The developed DL models might need more information to learn the nitrate dynamics of winter seasons and contribute to poor daily stream nitrate estimation at precipitation days during winter seasons.

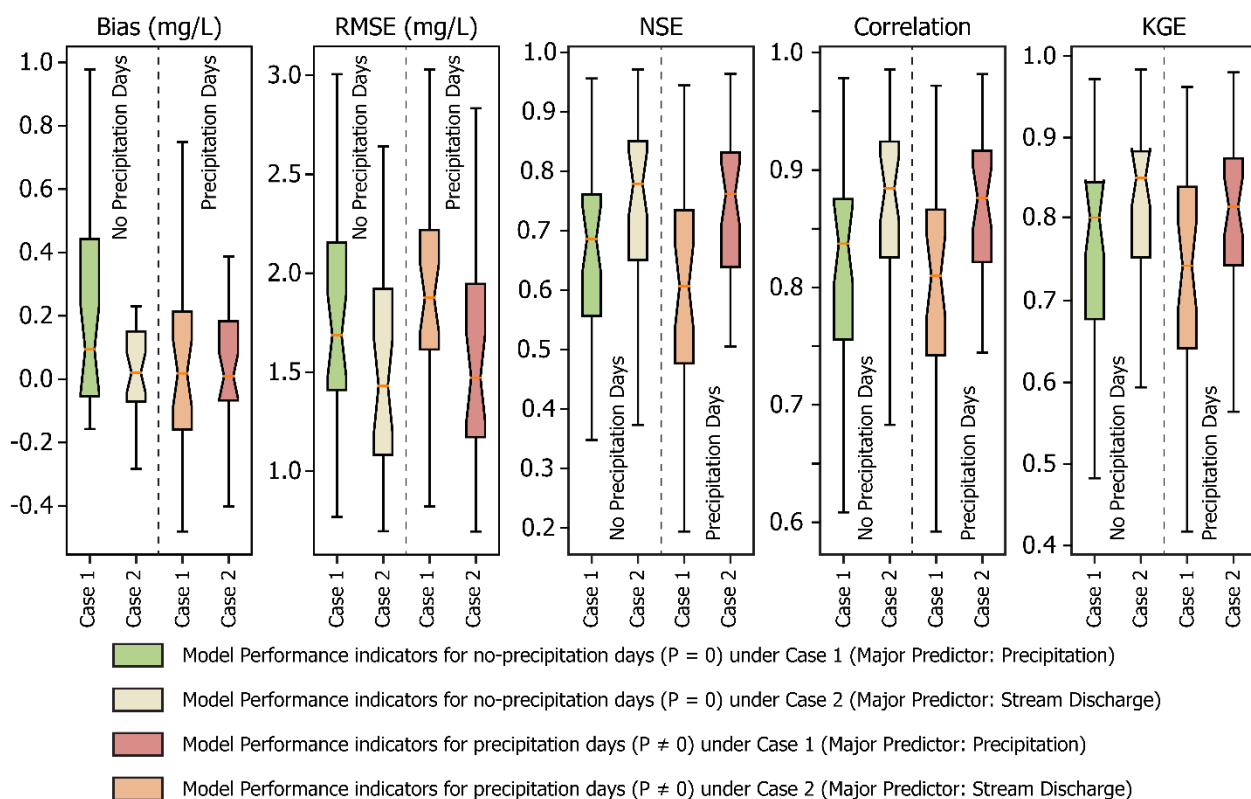


Figure 5.5 Boxplots comparison between the Case 1 and Case 2 DL models' performance for no-precipitation and precipitation days.

Many pseudo-low-frequency nitrate monitoring sites did not have enough nitrate data during winter seasons (Fig. S5.9 in Appendix D). Winter season's NSE distribution demonstrated poor model performance of nine (9) pseudo-low-frequency nitrate monitoring sites located primarily in Iowa and Cedar River Basins and the eastern part of Iowa (Fig. S5.9 in Appendix D). Cross-correlation between the DL model estimated stream nitrate concentrations for precipitation days of four seasons and four input environmental characteristics, including stream discharge, precipitation, and maximum and minimum temperature, was determined for those nine pseudo-low-frequency nitrate monitoring sites to evaluate the influence of these environmental characteristics on the precipitation-days stream nitrate concentration variability. Here, estimated stream nitrate concentrations by the Case 2 DL model were used for the



analyses. Six (6) among nine (9) low-performed nitrate monitoring sites showed positive correlations between estimated stream nitrate concentration and the mentioned four environmental characteristics for precipitation days during most of the winter seasons (Fig. 5.6). The results indicated the DL model required more nitrate data during winter seasons to learn winter stream nitrate dynamics appropriately. Limited nitrate data availability during winter seasons affected the learning of the DL models and resulted in inaccurate stream nitrate prediction during the precipitation days in winter. In addition, the cross-correlations of estimated nitrate concentrations for precipitation days during winter seasons with these four environmental variables were not consistent over the years (Fig. 5.6). These results indicated that the stream nitrate variability could be influenced by different combinations of environmental variables depending upon the occurrence of runoff events. The cross-correlation between estimated daily stream nitrate concentrations and stream discharge was identified as dominant for precipitation days during the summer and fall seasons. The result indicated that different seasons had different stream nitrate dynamics. The developed DL models might learn the dominant relationships between stream nitrate and input environmental characteristics and predict poorly during winter seasons.

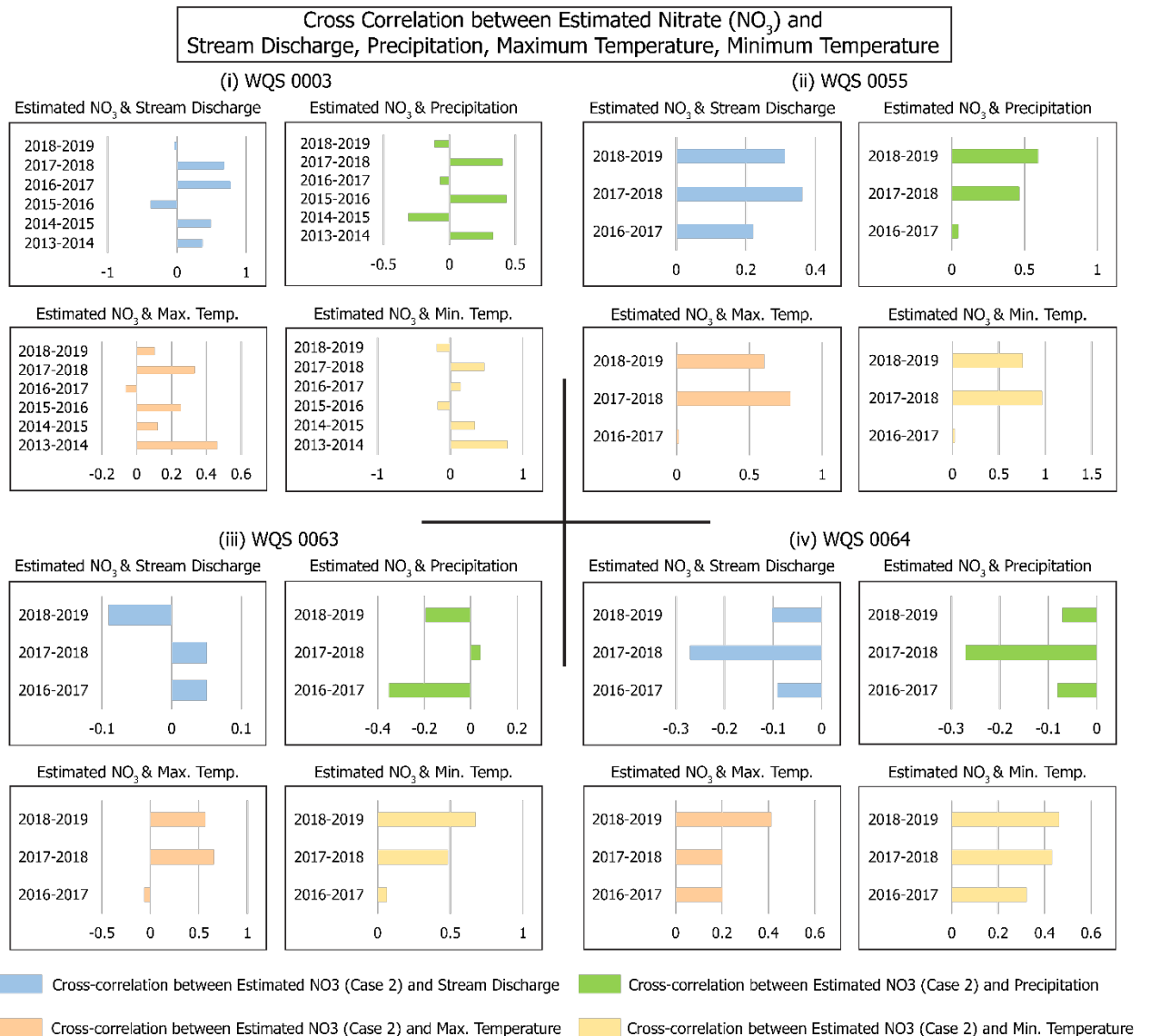


Figure 5.6 Cross correlation between estimated nitrate concentration (based on Case 2 Model) and

four environmental characteristics, including stream discharge, precipitation, maximum and minimum temperatures. Cross correlation values were calculated for precipitation days during winter season of each year. Sites include (i) WQS 0003, (ii) WQS 0055, (iii) WQS 0063, and (iv) WQS 0064.

### 5.3.5 Future potential of making more spatially and temporally robust framework.

Analyses showed the significant applicability and future potential of making the developed data-driven modeling framework more spatially and temporally robust for filling the nitrate data gaps at periodically monitored stream locations using simulated stream discharge or

precipitation data. This approach addresses the challenge of limited stream discharge measurements at nitrate monitoring sites. The framework also offers a viable solution for understanding stream nitrate dynamics, facilitating effective water quality management. The modeling framework with precipitation and/or simulated streamflow will expand the spatial coverage of stream nitrate predictions. More accurate simulated stream discharge or precipitation enables precise stream nitrate estimations at low-frequency nitrate measuring locations. With ongoing technological advancements and high-frequency nitrate data availability, the framework holds promising future research and application prospects.

## **5.4 Conclusion**

Understanding the potential of precipitation and stream discharge information in estimating accurate stream nitrate concentrations is critical because this information will help us expand the DL modeling framework to stream discharge unavailable locations and build a more spatially robust data-driven modeling framework. Five cases were developed with different combinations of precipitation, observed, and simulated stream discharge data in this study. Among the five cases, the DL model developed with stream discharge as the critical environmental characteristic (Case 2) performed well with a median NSE of 0.77 in filling the nitrate data gaps of the low-frequency sites. The DL model with simulated stream discharge as the dominant environmental driver performed similarly to the DL model developed with observed discharge as the crucial environmental driver, inferring the potential of simulated stream discharge to be used for nitrate monitoring sites that lack stream discharge measurements. The DL model developed with precipitation as the dominant variable performed 15% less than the DL model developed with stream discharge. The developed DL model could have been better estimated for nitrate monitoring sites with less than two years of

data. DL model predicted well with precipitation at low-frequency nitrate monitoring sites in the Des Moines and Iowa River Basins, indicating the influence of precipitation in stream nitrate variability. Literature illustrated that the nitrate contribution of mid-tributaries to the Cedar River was higher due to extensive agricultural activities. The developed DL model had limited capability to learn environmental processes that led to excessive nitrate transport with user input information, resulting in inaccurate nitrate predictions at sites in the middle of the Cedar River Basin. Estimated nitrate time series plots with precipitation, observed, and simulated stream discharge indicated stream discharge as the primary contributor to stream nitrate variability for large river basins. In addition, the developed DL model offered inaccurate nitrate estimations when simulated discharge was not representative of the respective basin and stream nitrate was not impacted by stream discharge. The DL model predicted nitrate concentration well for no-precipitation days compared to precipitation days. Estimated nitrates for precipitation days during winter seasons by the DL model were identified as less accurate than the estimated nitrates for precipitation days of the other seasons. Cross-correlation analysis showed stream discharge, precipitation, and maximum and minimum temperatures were correlated with estimated nitrate concentrations during winter seasons. However, nitrate concentrations were highly correlated with stream discharge during other seasons. The developed DL model captured the dominant trends and variabilities of the input data to predict nitrate variability. This modeling study will help researchers and stakeholders build a spatially and temporally robust data-driven modeling framework to predict stream nitrate at low-frequency monitoring locations accurately.

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## **CHAPTER 6**

## SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS FOR THE FUTURE RESEARCH

### 6.1 Research Overview

Stream nitrate concentration is generally collected as grab samples for lab analysis at weekly/biweekly/monthly scales in most monitoring sites in the United States. Evaluating the nitrate trends and variability with these periodically collected stream nitrate data is often challenging. Sensor-based high-resolution (15-minute to 1-hour intervals) nitrate data collection is getting popular in the last decade in the US. While these sensors-based continuous nitrate data provide critical information on nitrate trends and variability, the number of stations with more than five (5) years of data is limited. In addition, understanding more about the historical nitrate trends is critical to evaluate the changing nitrate patterns. The current study hypothesized that high-frequency nitrate sensors could provide critical nitrate dynamics of the region, and data-driven models could extract the nitrate dynamics information and estimate continuous nitrate concentrations at low-frequency sites to generate more spatially and temporally continuous nitrate data in the US. With this hypothesis, the primary goal of this study was to develop an integrated and spatially and temporally robust data-driven modeling framework to predict continuous stream nitrate concentration. The study developed a comprehensive data-driven modeling framework that incorporates nitrate concentrations of multiple locations, influences climatic and biophysical attributes of contributing watersheds, and predicts continuous stream nitrate at low-frequency nitrate monitoring sites. A long short-term memory (LSTM) networks-based deep learning (DL) modeling framework was developed where the DL model received climate, land use, fertilization, topography, and soil characteristics data as inputs and was trained comprehensively by using four hyperparameters

(batch size, hidden layer size, time window, and epoch) and isolating the best combination of hyperparameters based on five performance metrics, including RMSE, bias, correlation, NSE, and KGE. The DL model was developed for a case study region – Iowa since the state has many nitrate monitoring sites.

The objectives of the data-driven modeling study were -

- (1) Developing a deep learning-based novel approach to generate continuous daily stream nitrate concentration for nitrate data-sparse watersheds and comparing the performance of the developed modeling approach with other statistical models.
- (2) Analyzing the deep learning approach-based modeling framework to understand the critical environmental drivers of predicting daily nitrate concentrations.
- (3) Exploring the potential of extending the data-driven stream nitrate concentration modeling framework to stream discharge unavailable locations for making the framework more spatially robust.

In the first objective (Chapters 2 and 3), a DL model-based continuous stream nitrate estimation framework was developed for the data-limited locations and compared the performance of the developed framework with other widely used statistical models, including WRTDS and LOADEST. The study used daily stream nitrate data from thirty-four (34) high-frequency sites and biweekly stream nitrate data from eight (8) low-frequency sites to train and test the DL model. The trained DL model predicted continuous daily stream nitrate concentrations for unavailable periods at all forty-two (42) sites. The seasonal performance of the developed DL model and consistency comparison between annual c-Q relationships of all sites were conducted to evaluate the capability of the developed modeling framework. The developed data-driven modeling framework's performance was compared with other statistic



tools, including WRTDS and LOADEST, for nitrate data-scarce regions, assuming each basin had only one high-frequency monitoring site within the region (Chapter 3). The DL model was developed with twenty-four (24) high-frequency nitrate monitoring sites in Iowa's three large river basins: Des Moines River Basin, Iowa River Basin, and Cedar River Basin. Here, high-frequency nitrate data of each site were randomly subsampled to biweekly data and used to train the DL model. These sites were called pseudo-low-frequency nitrate monitoring sites. The remaining daily data of these pseudo-low-frequency sites were used to determine the DL model's performance. Later, biweekly nitrate data of pseudo-low-frequency sites were used to develop WRTDS and LOADEST models, and the remaining daily data were used to assess these models' performance.

The second objective (Chapter 4) analyzed the developed DL models to identify critical environmental drivers influencing stream nitrate variability in the case study area. The DL model was developed with biweekly nitrate data from thirty (34) pseudo-low-frequency and eight (8) low-frequency nitrate monitoring sites. Initially, the most sensitive daily varying and, later, crucial environmental drivers that influence stream nitrate variability were identified utilizing the results of thirty-four (34) pseudo-low-frequency nitrate monitoring sites for Iowa. The transferability of nitrate dynamics from high-frequency concentrations to continuous nitrate prediction at the low-frequency monitoring sites was also evaluated. The DL model performance for four seasons (winter, spring, summer, and fall) and growing and harvesting seasons were assessed to understand the DL modeling framework's robustness. The third objective (Chapter 5) assessed the potential of expanding the framework to locations where stream discharge data is not measured to develop more spatially continuous nitrate data. The study explored the potential of precipitation and simulated streamflow discharge data for DL

model development. In this study, the developed DL modeling framework was tested for thirty-four (34) pseudo-low-frequency sites with precipitation and simulated stream discharge as critical environmental variables, which could be used as a surrogate of stream discharge in estimating spatially continuous nitrate concentration. The cross-correlation of precipitation and stream discharge data with estimated nitrate concentrations was calculated to understand the variability of precipitation and stream discharge data with nitrate variability. In addition, the DL model's performance for precipitation days and no-precipitation days was evaluated, which portrayed the DL model's capability and opportunity towards developing a more spatially and temporally robust data-driven modeling framework.

## **6.2 Major Research Findings.**

Objective 1 (Chapter 2): Data-driven modeling approach development to estimate continuous stream nitrate at data-limited locations.

1. The hypothesis that the DL model can learn from regional nitrate relationship information available in the high-frequency stream nitrate data and generate continuous daily nitrate concentration for data unavailable periods and at data-scarce nitrate measuring stream locations (at low-frequency nitrate monitoring sites) was tested successfully (with median test-period NSE 0.75).
2. The developed DL model predicted continuous nitrate concentrations satisfactorily with  $NSE > 0.50$  at 76% (34 sites) of the total number (42 sites) of nitrate monitoring sites. The DL model performance was exceptionally well ( $NSE > 0.75$ ) for 21 sites (50% of all 42 sites). The developed DL model predicted continuous daily stream nitrate concentration with  $NSE > 0.75$  at five (5),  $NSE > 0.70$  at two (2), and  $NSE > 0.50$  at one (1) low-frequency nitrate monitoring sites.

3. DL model results analysis portrayed average nitrate information from neighboring monitoring sites (local nitrate, LocalNO<sub>3</sub>) impacted continuous daily nitrate estimation. Further study demonstrated that precipitation and stream discharge primarily influenced temporal impulses in the estimated daily nitrate concentrations. The DL model's estimation performance is comparatively better in the summer and fall seasons.
4. The DL model estimated nitrate concentration-based c-Q relationship patterns were identified as analogous to observed nitrate c-Q relationship patterns. The analysis demonstrated four dominant nitrate c-Q relationship patterns, including (i) flushing pattern, (ii) flushing during low Q ( $Q < \text{median } Q, Q_{50}$ ) and chemostatic after high Q ( $Q > Q_{50}$ ) pattern, (iii) flushing during low Q and dilution during high Q pattern and (iv) chemodynamic pattern. Twenty-six (26) monitoring sites (64%) had a high correlation (correlation > 0.60) between estimated nitrate concentration and stream discharge.

Objective 1 (Chapter 3): Performance evaluation of DL models with LOADEST and WRTDS.

5. This study hypothesized that the DL model utilizes high-frequency nitrate information from one high-frequency nitrate monitoring site in a region as input, along with environmental and biophysical characteristics and available nitrate data in low-frequency sites to generate continuous nitrate data and fill temporal data gaps at the low-frequency sites. The developed DL modeling framework successfully filled the temporal nitrate data gaps for three large river basins: Des Moines, Iowa, and Cedar River Basins.

6. The DL model generated continuous nitrate concentrations at 67% and 79% pseudo-low-frequency nitrate monitoring sites with  $NSE > 0.70$  and  $KGE > 0.70$ , respectively. The model performance was reduced when the distance between pseudo-low-frequency sites and their respective input high-frequency nitrate monitoring sites was more than 150 km.
7. DL and WRTDS estimated stream nitrate concentrations were similar in most gauges. However, the DL model performed better in gauges with fewer observed nitrate data. The WRTDS model demonstrated unrealistic fluctuations in the estimated daily nitrate time series observed nitrate data points were less than 50. Both DL and WRTDS model's nitrate concentration estimations were much better than the LOADEST estimations. The DL model represented the exceedance days from the US Environmental Protection Agency (USEPA) maximum contamination level (MCL) better than WRTDS and LOADEST.

Objective 2 (Chapter 4): Identification of critical environmental drivers by analyzing the developed data-driven modeling framework.

8. DL model performance was exceptionally well because the developed DL models predicted continuous stream nitrate concentrations at 70% of the pseudo-low-frequency nitrate monitoring sites with  $NSE > 0.75$ .
9. The developed DL models could learn temporal nitrate variability from high-frequency nitrate data of a high-frequency nitrate sensor and predict continuous nitrate concentrations well at biophysically similar pseudo-low-frequency sites situated within a 300 km radius from the high-frequency nitrate sensor.

10. Annual  $c$ - $Q$  relationship plots for pseudo-low-frequency sites and their input high-frequency nitrate monitoring sites demonstrated three dominant patterns. These patterns included: (i) flushing, (ii) flushing during low  $Q$  and dilution during high  $Q$ , and (iii) flushing during low  $Q$  and chemostatic during high  $Q$ . Pseudo-low-frequency sites showed these  $c$ - $Q$  relationship patterns for 1-5 years, and their input high-frequency nitrate monitoring sites, which contributed to accurate nitrate estimates, had the same dominant pattern for 1-3 years. The results inferred that the DL model learned  $c$ - $Q$  relationship patterns and accurately predicted continuous nitrate at pseudo-low-frequency nitrate monitoring sites with similar biophysical characteristics.
11. According to the Dynamic Time Warping (DTW) analysis, the DL model learned the similarity in nitrate concentration, stream discharge, and day length (representing seasonality) between a low-frequency and a high-frequency nitrate monitoring site contributed to the accurate estimation of stream nitrate concentrations.
12. The DL model estimated continuous stream nitrate concentration well at pseudo-low-frequency sites having 50 counts of nitrate data over two years. Seasonal DL model performance analysis indicated a better continuous nitrate estimation performance during the summer and fall seasons. Analyzing growing (April – August) and harvesting (October – November) seasons'  $c$ - $Q$  relationship patterns demonstrated that about 89% and 83% of estimated nitrate  $c$ - $Q$  relationship plots were similar to observed nitrate  $c$ - $Q$  relationship plots for growing and harvesting seasons.

Objective 3 (Chapter 5): Importance of precipitation and stream discharge information to build a temporally and spatially robust data-driven nitrate modeling framework.

13. This DL modeling study identified observed stream discharge as the best environmental variable among three crucial environmental variables, precipitation, observed stream discharge, and simulated stream discharge, to estimate continuous stream nitrate concentrations at low-frequency nitrate monitoring sites. Simulated stream discharge can be used as a surrogate of observed discharge for the DL model development for those nitrate monitoring locations where stream discharge measurements are unavailable. This result helps to develop a more spatially robust continuous nitrate estimation framework.
14. The DL model performed poorly for those pseudo-low-frequency nitrate monitoring sites with less than two years of nitrate data. The DL model developed with precipitation data as the dominant environmental variable predicted well for pseudo-low-frequency nitrate monitoring sites in the Des Moines and Iowa River Basins. This result indicated the influence of precipitation in stream nitrate variability in the Des Moines and Iowa River Basins.
15. The developed DL model needed more strength in learning environmental processes (e.g., land management) that contributed to substantial nitrate transport to streams. Due to extensive agricultural activities, the middle part of the Cedar River Basin received higher nitrate contributions from its mid-tributaries. With minimal land management input information, the DL model made inaccurate predictions at the pseudo-low-frequency nitrate monitoring sites near the mid-part of Cedar River Basin.
16. The analysis also indicated that the developed DL model estimated nitrate concentrations inaccurately for those pseudo-low-frequency sites when simulated

discharge data was not representative of the respective basin or did not influence the variability of stream nitrate concentrations.

17. The DL model performance was calculated better for no-precipitation days than for precipitation. The analysis also demonstrated that the DL model-estimated nitrate concentrations for precipitation days during winter seasons contributed to lower modeling performance for precipitation days overall. Analyzing the cross-correlation between stream discharge, precipitation, and maximum and minimum temperatures indicated the influence of all these four environmental variables on estimated nitrate concentrations for precipitation days during the winter seasons. However, the influence of stream discharge on nitrate estimation was dominant for precipitation days during the other three seasons (summer, fall, and spring).

### **6.3 Limitation of Current Study and Recommendations for Future Research.**

1. The current study was developed for an agriculture-dominated region, like Iowa, where corn and soybeans were the primary land uses. The robustness of the developed data-driven modeling framework should be evaluated for the region with multiple land use practices to improve the developed framework.
2. The current data-driven modeling framework-based DL models were simulated for the data available from 2012 or later. The primary reason behind this initiative was the availability of high-frequency nitrate data at most monitoring locations. Since low-frequency nitrate data has been available in many US stream locations for the last four decades, accurately estimating continuous stream nitrate concentrations at those low-frequency nitrate monitoring locations by improving the data-driven modeling framework would improve the framework's application.

3. The DL modeling framework can be deployed to hindcast continuous stream nitrate concentrations at low-frequency nitrate monitoring locations for a historical period with regional biophysical and climatic information and continuous nitrate concentration of one monitoring site within the region for the respective historical period.
4. Due to data limitations, the study included limited information on cropping patterns and land management, including the timing of season-based fertilization or the crop vegetative index. Incorporating this information (if available) can improve the model's performance in predicting stream nitrate concentrations.
5. The study developed DL models using soil information, including hydrologic soil group, soil drainage class, soil erodibility factors, soil texture, and soil depths. Depending upon the study locations, the modeling framework can adopt other soil information, such as soil organic matter, hydraulic conductivity, runoff potential, leaching potential, Depth to the water table, flood frequency class, pond frequency class, and soil with frost-free days.
6. The study identified different stream nitrate dynamics for winter seasons than other seasons. The possibility of developing two separate DL models for winter and non-winter seasons or a DL model incorporating the environmental characteristics information for all seasons can be evaluated.
7. This study included the most relevant climatic and biophysical characteristics of non-point source stream nitrate pollution. More information should be included for the regions with point-source nitrate contamination to make the modeling framework effective for those regions.



8. The current study developed the DL models with biweekly stream nitrate concentrations and compared the model performance with WRTDS and LOADEST models. Future research can assess the modeling framework's capability to predict continuous nitrate concentrations at low-frequency monitoring sites with nitrate data collected in 3-day, 5-day, 7-day, or 10-day intervals and compare the DL model performance with the WRTDS and LOADEST performance. The results will demonstrate the modeling framework's estimation capability with nitrate data of different collection frequencies and indicate the optimum water quality sampling frequency at the low-frequency nitrate monitoring sites.
9. Coupling ecohydrological models with the DL modeling framework can be another exciting area of research, where ecohydrological models will provide initial nitrate estimates, and the DL model can improve the estimations based on observed nitrate data.
10. DL model-based daily stream nitrate prediction framework also had the opportunity to extend the framework to a national scale.

#### **6.4 Concluding remarks.**

Except for long-term and experimental watersheds, most US watersheds have periodically collected low-frequency nitrate data for shorter periods. Designing appropriate conservation practices with low-frequency nitrate data is difficult. The study successfully tested the hypothesis and developed a DL modeling framework that accurately estimated continuous stream nitrate concentrations at data-scarce watersheds in an agriculture-dominated region. The DL framework could also be used to optimize the new nitrate sensor placement locations and identify critical periods which the nitrate concentration is greater than drinking water

standards for water withdrawal. The developed DL framework has the potential to be expanded or transferred to other US regions for estimating spatially and temporally continuous stream nitrate concentrations, especially to those regions that have a high-frequency stream nitrate monitoring site. The DL framework's robustness can also be explored for regions with changing hydrology, land use, and geology, for hindcasting continuous stream nitrate at historical periods, and for coupling with other eco-hydrological models. Overall, this data-driven modeling study validated the regional transfer potential of nitrate information from biophysically similar high-frequency monitoring sites to low-frequency sites and widened the potential of creating an integrated continuous stream nitrate monitoring network for the future.

## Appendix A

### Data processing

The input data were available in time series and spatial formats. The contributing watershed area was delineated for all 42 nitrate monitoring sites using Hydrology Toolbox in ArcGIS Desktop and the digital elevation model (DEM). An overview of the delineation process is available at the ESRI Hydrology toolset overview (<https://desktop.arcgis.com/en/arcmap/latest/tools/spatial-analyst-toolbox/an-overview-of-the-hydrology-tools.htm>). In this study, land cover data of 2012-2019 was classified into five (5) major land covers: Corn, Soybean, Other agriculture, Forest, and Urban areas. Deciduous, evergreen, and mixed forests were combined and considered Forest. Urban areas include developed open spaces and low-, medium-, and high-intensity developed areas. All row croplands and forage lands, excluding the Corn and Soybean, were lumped under the Other Agriculture. Areas of these five major land covers were calculated using the 'Calculate Geometry' function calculation in ArcGIS Desktop. Fertilizer nitrogen (N) and manure N information are available from 1987 to 2016 on the NuGIS website ([https://nugis.tfi.org/nugis\\_map](https://nugis.tfi.org/nugis_map)) in both the county and HUC8 watershed scales. The data is free to download; however, users should register on this website. County-scale fertilizer N and manure N data were accumulated based on the contributing watershed boundaries. Fertilizer N and manure N information for each year was calculated based on the area proportion of respective counties and watershed boundaries. Since fertilizer N and manure N data were unavailable for 2017-2019, the 3-year moving average was used to extrapolate the data for 2017-2019. Mean daily streamflow/discharge data has several missing values during 2012-2019. The linear interpolation method was used to fill the streamflow data gaps. Climate data,

including precipitation, maximum temperature, minimum temperature, and day length data, were collected using the Single Pixel Extraction Tool available on the Daymet website. The tool requires the site's location and date range information to extract the climate information. More detailed information is available at <https://daymet.ornl.gov/single-pixel/>. The location of each nitrate monitoring site was used to get the climate data.

## Input and Target variables

Table S2.1 Summary of the forcing, attribute, and target variables used as inputs to the LSTM model.

| Data type                                                          | Input variable name                     | Description                                                             | Unit            |
|--------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------|-----------------|
| Forcings                                                           | Q                                       | Mean daily streamflow/discharge                                         | cfs             |
|                                                                    | PRCP                                    | Precipitation                                                           | mm/day          |
|                                                                    | Dayl                                    | Day length                                                              | s               |
|                                                                    | T <sub>max</sub>                        | Maximum temperature                                                     | °C              |
|                                                                    | T <sub>min</sub>                        | Minimum temperature                                                     | °C              |
|                                                                    | Corn                                    | Corn area                                                               | ha              |
|                                                                    | Soybean                                 | Soybean area                                                            | ha              |
|                                                                    | Other_Ag                                | Remaining agricultural area                                             | ha              |
|                                                                    | Forest                                  | Forest area                                                             | ha              |
|                                                                    | Urban                                   | Developed area                                                          | ha              |
|                                                                    | Fertilizer_N                            | Nitrogen input from inorganic fertilizer                                | ton             |
|                                                                    | Manure_N                                | Nitrogen input from manure                                              | ton             |
|                                                                    | RegionNO <sub>3</sub>                   | Daily mean nitrate of all sites within the study region                 | mg/L            |
|                                                                    | LocalNO <sub>3</sub>                    | Daily mean nitrate of neighboring sites                                 | mg/L            |
| Attributes                                                         | PPTAvg_basin                            | Mean precipitation of a basin                                           | mm/day          |
|                                                                    | Drain_area                              | Contributing watershed area                                             | Km <sup>2</sup> |
|                                                                    | Shallow_soil                            | Area of the shallow soils in a watershed                                | Km <sup>2</sup> |
|                                                                    | Deep_soil                               | Area of the deep soils in a watershed                                   | Km <sup>2</sup> |
|                                                                    | Excess_Drained_Area                     | Area of the excessively drained soils in a watershed                    | Km <sup>2</sup> |
|                                                                    | Well_Drained_Area                       | Area of the well-drained soils in a watershed                           | Km <sup>2</sup> |
|                                                                    | Poor_Drained_Area                       | Area of the poorly drained soils in a watershed                         | Km <sup>2</sup> |
|                                                                    | HydGrpA                                 | Area of the soils with hydrologic drainage class A                      | Km <sup>2</sup> |
|                                                                    | HydGrpB                                 | Area of the soils with hydrologic drainage class B                      | Km <sup>2</sup> |
|                                                                    | HydGrpC                                 | Area of the soils with hydrologic drainage class C                      | Km <sup>2</sup> |
|                                                                    | HydGrpD                                 | Area of the soils with hydrologic drainage class D                      | Km <sup>2</sup> |
|                                                                    | Dominant_K_factor                       | Most dominant soil erodibility factor of a watershed.                   | -               |
|                                                                    | Dominant_K_factor_Area                  | Area that has the most dominant soil erodibility factor in a watershed. | Km <sup>2</sup> |
|                                                                    | 1 <sup>st</sup> _dominant_soil_tex      | Dominant soil texture in a watershed                                    | -               |
|                                                                    | 1 <sup>st</sup> _dominant_soil_tex_area | Area of the most dominant soil texture in a watershed                   | Km <sup>2</sup> |
|                                                                    | 2 <sup>nd</sup> _dominant_soil_tex      | Second most dominant soil texture in a watershed                        | -               |
|                                                                    | 2 <sup>nd</sup> _dominant_soil_tex_area | Area of the second most dominant soil texture in a watershed            | Km <sup>2</sup> |
| Target                                                             | NO <sub>3</sub>                         | Surface water nitrate                                                   | mg/L            |
| Forcings – time series data with daily timesteps                   |                                         |                                                                         |                 |
| Attributes – constant values of respective contributing watersheds |                                         |                                                                         |                 |

## Data Availability, training, and testing periods

Table S2.2 Data availability, and training and testing periods for sensor-based continuous and intermittently collected discrete nitrate monitoring sites in Iowa.

| Data Type  | Site No.         | Data Availability Period | Training Period | Testing Periods |
|------------|------------------|--------------------------|-----------------|-----------------|
| High-freq. | USGS 05412500    | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | USGS 05418400    | 2012-2014                | 2012-2013       | 2014            |
| High-freq. | USGS 05455100    | 2012-2015                | 2012-2014       | 2014-2015       |
| High-freq. | USGS 05464420    | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | USGS 05465500    | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | USGS 05481000    | 2014-2017                | 2014-2015       | 2016-2017       |
| High-freq. | USGS 05482000    | 2013-2019                | 2013-2017       | 2018-2019       |
| High-freq. | USGS 05482300    | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | USGS 05482500    | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | USGS 05484000    | 2016-2019                | 2016-2018       | 2019            |
| High-freq. | USGS 05484500    | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | USGS 06817000    | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | WQS 0001         | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | WQS 0002         | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | WQS 0003         | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | WQS 0005         | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | WQS 0007         | 2012-2019                | 2012-2017       | 2018-2019       |
| High-freq. | WQS 0010         | 2013-2019                | 2013-2017       | 2018-2019       |
| High-freq. | WQS 0012         | 2014-2019                | 2014-2017       | 2018-2019       |
| High-freq. | WQS 0023         | 2014-2019                | 2014-2017       | 2018-2019       |
| High-freq. | WQS 0032         | 2014-2019                | 2014-2017       | 2018-2019       |
| High-freq. | WQS 0038         | 2016-2019                | 2016-2018       | 2019            |
| High-freq. | WQS 0039         | 2016-2019                | 2016-2018       | 2019            |
| High-freq. | WQS 0048         | 2016-2019                | 2016-2018       | 2019            |
| High-freq. | WQS 0050         | 2016-2019                | 2016-2018       | 2019            |
| High-freq. | WQS 0051         | 2016-2019                | 2017-2019       | 2016-2017       |
| High-freq. | WQS 0055         | 2017-2019                | 2018-2019       | 2017            |
| High-freq. | WQS 0063         | 2017-2019                | 2018-2019       | 2017            |
| High-freq. | WQS 0064         | 2017-2019                | 2018-2019       | 2017            |
| High-freq. | WQS 0069         | 2017-2019                | 2017-2018       | 2019            |
| High-freq. | WQS 0071         | 2017-2019                | 2017-2018       | 2019            |
| High-freq. | WQS 0072         | 2017-2019                | 2018-2019       | 2017-2018       |
| High-freq. | WQS 0083         | 2018-2019                | 2019            | 2018            |
| High-freq. | WQS 0084         | 2018-2019                | 2019            | 2018            |
| Low-freq.  | ARS IASF IASF400 | 2102-2015                | 2012-2014       | 2015            |
| Low-freq.  | USGS 05418110    | 2012-2014                | 2012-2013       | 2014            |
| Low-freq.  | USGS 05418600    | 2012-2014                | 2012-2013       | 2014            |
| Low-freq.  | USGS 05451210    | 2012-2019                | 2012-2017       | 2018-2019       |
| Low-freq.  | USGS 05490500    | 2012-2019                | 2012-2017       | 2018-2019       |
| Low-freq.  | USGS 06607500    | 2012-2014                | 2012-2013       | 2014            |
| Low-freq.  | USGS 06609500    | 2012-2014                | 2012-2013       | 2014            |
| Low-freq.  | USGS 06810000    | 2012-2014                | 2012-2013       | 2014            |

### **Data standardization process and Architecture of the used LSTM model.**

In the data preprocessing step, the values of all inputs and target variables were standardized. Initially, the LSTM model divided the input discharge data by the respective contributing head-watershed area and mean annual precipitation to produce a dimensionless discharge dataset. Later, the model transformed the factorized discharge data into normally distributed data (Feng et al., 2020). The LSTM model applied the following equation for this data transformation.

$$v^* = \log_{10} (\sqrt{v} + 0.1) \quad (1)$$

Here  $v^*$  and  $v$  are the variables after and before transformation, respectively.

The transformed discharge data and all other forcings and attributes data were standardized using the following formula.

$$x_{i,new} = \frac{(x_i - \bar{x})}{o'} \quad (2)$$

In equation (2),  $x_{i,new}$ ,  $x_i$ ,  $\bar{x}$ , and  $o'$  are the standardized value, raw value, variable mean, and standard deviation of the variable, respectively. This step is necessary because the LSTM model performs the gradient descent function better when the model gets the standardized input data. The standardization of the input data also helps the model treat all basins equally, irrespective of their size and wetness (Feng et al., 2020).

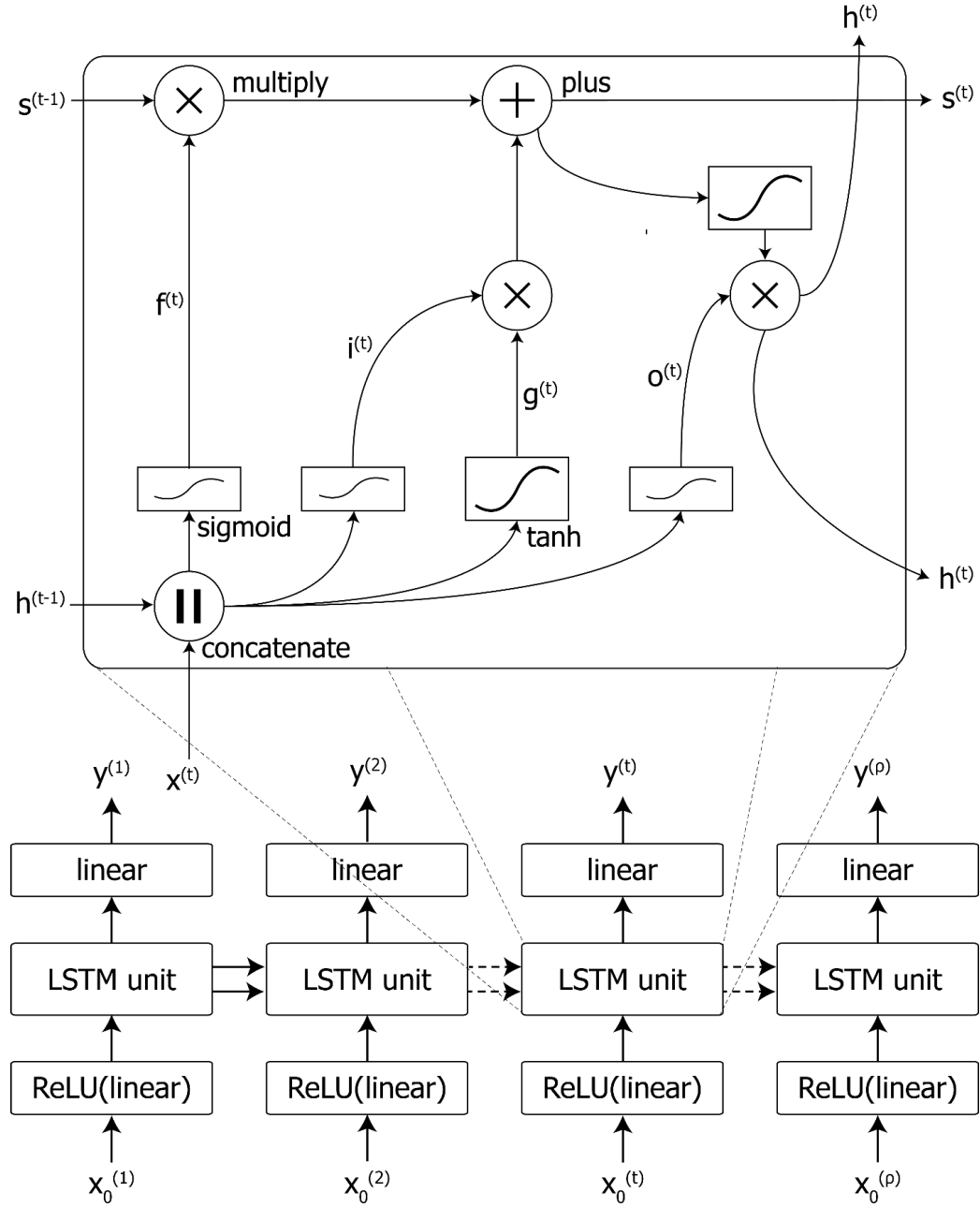


Figure S2.1 Different components of the LSTM model structure.  $x_0^{(t)}$  is the vector of raw inputs for the time step  $t$ ,  $\rho$  is the time sequence length within the training period.  $f^{(t)}$ ,  $i^{(t)}$ , and  $o^{(t)}$  are the forget, input and output gates, respectively.  $x^{(t)}$  is the input vector to the LSTM cell and  $g^{(t)}$  is the output of the input node. In addition,  $s^{(t)}$  represents the states of memory cells and  $h^{(t)}$  represents hidden states.  $y^{(t)}$  is the predicted output that is compared to daily stream nitrate observations. ReLU(linear) is the rectified linear unit.



### Equations used by the LSTM model.

$$\text{Input transformation:} \quad x^t = \text{ReLU}(W_I I^t + b_I) \quad (1)$$

$$\text{Input node:} \quad g^t = \tanh(\mathcal{D}(W_{gx} x^t) + \mathcal{D}(W_{gh} h^{t-1}) + b_g) \quad (2)$$

$$\text{Input gate:} \quad i^t = \sigma(\mathcal{D}(W_{ix} x^t) + \mathcal{D}(W_{ih} h^{t-1}) + b_i) \quad (3)$$

$$\text{Forget gate:} \quad f^t = \sigma(\mathcal{D}(W_{fx} x^t) + \mathcal{D}(W_{fh} h^{t-1}) + b_f) \quad (4)$$

$$\text{Output gate:} \quad o^t = \sigma(\mathcal{D}(W_{ox} x^t) + \mathcal{D}(W_{oh} h^{t-1}) + b_o) \quad (5)$$

$$\text{Cell state:} \quad s^t = g^t \odot i^t + s^{t-1} \odot f^t \quad (6)$$

$$\text{Hidden state:} \quad h^t = \tanh(s^t) \odot o^t \quad (7)$$

$$\text{Output:} \quad y^t = W_{hy} h^t + b_y \quad (8)$$

where each superscript containing  $t$  represents the temporal step for time-dependent variables;  $I^t$  is the raw inputs to the model (Equation 1);  $x^t$  represents the input vector to the LSTM cell (Equations 1-2); the dropout operator is denoted by  $\mathcal{D}$  (Equations 2-5); all  $W$ 's and  $b$ 's with different subscripts denote the gate-specific network weights and bias parameters (Equations 1-6 and 8), respectively;  $\sigma$  is the sigmoidal function (Equations 3-6);  $\odot$  is the element-wise multiplication operator (Equations 6-7);  $g^t$  is the output of the input node (Equations 2 and 8);  $i^t, f^t, o^t$  are the input, forget, and output gates, respectively (Equations 3-5, 7); the hidden states are represented by  $h^t$  (Equations 7-8);  $s^t$  is the memory cell states (Equations 6-7); and  $y^t$  represents the predicted output at time step  $t$  (Equation 8). In this study, a fast and flexible LSTM code has been implemented. This code can utilize the highly optimized NVIDIA CUDA® Deep Neural Network (cuDNN) library from the PyTorch platform.  $\mathcal{D}$  applies constant dropout masks to reduce overfitting (Boolean matrices to set some weights to 0.5). The theory of dropout rate was explained in Gal and Ghahramani (2016).

### Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) represents the difference between the predictions and measured values using Euclidean distance. The metric is used widely to evaluate the quality of predictions. The metric is mostly applied in supervised learning applications to minimize model error.

$$RMSE = \sqrt{\frac{\sum_{i=1}^N |y_i - \hat{y}_i|^2}{N}}$$

Here,  $N$  is the number of measured values,  $y_i$  is the  $i$ -th measurement, and  $\hat{y}_i$  is the corresponding prediction of the  $i$ -th measurement.

RMSE is an effective metric to evaluate machine learning model performance. The metric can be applied during model training, cross-validation, and monitoring for deployed models. The metric is well-established because of its intuitive nature to understand and compatible with the most common statistical assumptions. In addition, RMSE is not scaled invariant; therefore, comparison among different models developed by numerous scaled data using RMSE should be affected by the scale factor. RMSE is mostly used in climatology, forecasting, and regression analysis to verify experimental results.

### Nash–Sutcliffe efficiency (NSE)

Nash–Sutcliffe efficiency (NSE) (Nash & Sutcliffe, 1970) is a metric often used during calibrating and evaluating the hydrological models. The metric provides information on the similarity between observed and simulated discharge (Gupta et al., 2009). NSE normalizes model performance into an interpretable scale.

$$NSE = 1 - \frac{\sum_{t=1}^{t=T} (Q_{sim}(t) - Q_{obs}(t))^2}{\sum_{t=1}^{t=T} (Q_{obs}(t) - \bar{Q}_{obs})^2}$$

where  $T$  is the total number of time steps,  $Q_{sim}(t)$  the simulated discharge at time  $t$ ,  $Q_{obs}(t)$  the

observed discharge at time  $t$ , and  $\overline{Q_{obs}}$  the mean observed discharge.

$NSE = 1$  indicates no error and perfect correspondence between simulated and observed data. In addition,  $NSE = 0$  represents that the simulated outputs of the model have the same explanatory power as the mean of the observed data. Schaefli and Gupta (2007) interpreted  $NSE < 0$  as a worse predictor than the mean of the observed data. In several research,  $NSE = 0$  was used as a benchmark to separate ‘good’ and ‘bad’ models (Houska et al., 2014; Moriasi et al., 2007; Schaefli & Gupta, 2007; Seibert, 2001). However, understanding the study-specific hydrologic processes is crucial to selecting the benchmark (Moriasi et al., 2007). One  $NSE$  value is not a representative benchmark for all flow regimes. Especially, mean observed flow is not an appropriate approximation for the flow regime having a strong seasonality component (Knoben et al., 2019; Schaefli & Gupta, 2007). Many studies made an arbitrary choice in defining the  $NSE$  threshold based on different qualitative levels of model performance (Moriasi et al., 2007). Different  $NSE$  thresholds influence the prediction uncertainty bounds of the simulated results (Freer et al., 1996). However, thresholds or a benchmark can provide a context for evaluating the model performance (Schaefli & Gupta, 2007).

### **Kling–Gupta efficiency (KGE)**

Gupta et al. (2009) developed Kling–Gupta efficiency (KGE) to calibrate and evaluate the model performance. The KGE can address the incapability of  $NSE$  in evaluating model performance. The KGE has three constitutive components: correlation, variability bias, and mean bias.

$$KGE = 1 - \sqrt{(r - 1)^2 + (\alpha - 1)^2 + (\beta - 1)^2}$$

Where  $r$  is the linear correlation between simulated and observed data,  $\alpha$  is a measure of the flow variability error, and  $\beta$  is a bias term. The KGE equation can be decomposed into the

following equation.

$$KGE = 1 - \sqrt{(r - 1)^2 + \left(\frac{\sigma_{sim}}{\sigma_{obs}} - 1\right)^2 + \left(\frac{\mu_{sim}}{\mu_{obs}} - 1\right)^2}$$

where  $\sigma_{obs}$  is the standard deviation in observed data,  $\sigma_{sim}$  the standard deviation in simulated data,  $\mu_{sim}$  the mean of the simulated data, and  $\mu_{obs}$  mean of the observed data.  $KGE = 1$  indicates the perfect agreement between simulated and observed data like NSE. Most of the existing literature treated KGE as analogous to NSE and stated positive and negative KGE values as indicatives of ‘good’ and ‘bad’ model simulations, respectively (Andersson et al., 2017; Castaneda-Gonzalez et al., 2018; Koskinen et al., 2017; Rogelis et al., 2016; Schönfelder et al., 2017; Sutanudjaja et al., 2018). Knoben et al. (2019) challenged the analogous interpretation of KGE and NSE value ranges. The researchers proved the KGE range  $-0.41 > KGE > 1$  compared the NSE range  $(0 > NSE > 1)$ . There is no direct relationship between KGE and NSE. KGE metric does not have any inherent benchmark. Modelers can interpret KGE values between  $-0.41 > KGE > 1$ . KGE values make modelers better understand the model errors, but interpretation based on the evaluation metric should be purpose driven. In addition, Kling et al. (2012) and Pool et al. (2018) developed a modified and a non-parametric version of KGE, respectively.

### **Setting of hyperparameters for each simulation case.**

The developed LSTM models for all 5 cases were trained using different hyperparameter combinations (minibatch, epoch, hidden layer, and time window) and random seeds. Minibatch size determines the number of samples given to the DL model to calculate loss before updating weights in a training step. One epoch represents the use of all training data once for a simulation. The layer located between the input and output layers is called the intermediate layer. The time window portrays the maximum number of days the model used for

backpropagation steps for each training sample. Minibatch was selected based on the number of nitrate monitoring sites considered for simulation. For 42 nitrate monitoring sites, 42 and 21 were used as three minibatch sizes. The hidden layer depends on the minibatch size and the model performance. If the model performed well for the training and poor for testing, then the hidden layer size should be reduced and vice versa. Different values of the hidden layer sizes, including 45, 50, and 60, were used for model simulation. Since nitrate data showed seasonal cycles, the time window was defined as four months (120 days), five months (150 days), and six months (180 days), considering the overlap among the seasons. The initial epoch was considered 4000. Finally, the epoch was reduced to 2500 because model performances were comparable from epochs 2500 to 4000. The dropout rate (the probability that some weights will be set randomly to 0 during model training, which helps prevent model overfitting) was set to 0.50 based on the existing hydrology and machine learning literature (Rahmani et al., 2021b, 2021a). The model was simulated with multiple hyperparameter combinations to identify the best combination for each case. The final hyperparameters-combination was selected based on the simulated outputs for the testing periods.

### Table describing the combinations of Forcings and Hyper-parameters.

Table S2.3 Five (5) developed cases based on different combinations of forcings and hyper-parameters.

| Case No. | No. of nitrate monitoring sites  | Forc1 | Forc2 | Forc3                 | Forc4                | Batch size | Hidden layer size | Time window   | No. of epoch     | Random seeds     |
|----------|----------------------------------|-------|-------|-----------------------|----------------------|------------|-------------------|---------------|------------------|------------------|
| 1        | 34 continuous and 8 intermittent | PRCP  | Q     |                       |                      |            |                   |               |                  |                  |
| 2        |                                  | PRCP  | Q     | RegionNO <sub>3</sub> | LocalNO <sub>3</sub> |            |                   |               |                  |                  |
| 3        |                                  | PRCP  | Q     |                       | LocalNO <sub>3</sub> |            |                   |               |                  |                  |
| 4        |                                  | PRCP  |       |                       | LocalNO <sub>3</sub> |            |                   |               |                  |                  |
|          | 30 continuous,                   |       |       |                       |                      | 21, 42     | 45, 50, 60        | 120, 150, 180 | 2500, 3000, 4000 | 0, 1, 3, 5, 7, 9 |
| 5        | 8 intermittent, and 4 subsampled | PRCP  | Q     | RegionNO <sub>3</sub> | LocalNO <sub>3</sub> |            |                   |               |                  |                  |

\* All other forcings (Forc) and attributes information were used as LSTM inputs for all 5 cases.

### Evaluation metrics calculation and boxplot generation methods

The test-period observed daily nitrate data of each nitrate monitoring station were identified. The DL-model estimated daily nitrate data of each respective station were also selected for the dates where the observed daily nitrate data was available. After choosing the observed and estimated nitrate data for each nitrate monitoring station, five evaluation metrics, including bias, RMSE, NSE, correlation, and KGE, were calculated. Multiple packages and programming libraries are available to calculate the evaluation metrics. 'statsmodels' (a Python module) was used to calculate the evaluation metrics in this study (Seabold & Josef, 2010). Calculated evaluation metrics for all forty-two nitrate monitoring stations for all five cases based on the hyperparameter combinations were plotted as boxplots (Fig. S2.2-S2.6). For each case, four boxplots for four hyperparameter combinations of five evaluation metrics (bias, RMSE, NSE, and KGE) were plotted using Matplotlib (a plotting module of Python). The Case 2 model with Epochs 2500, mini-batch size 42, hidden layer size 50, and a time window of 150 days was identified as the best model with NSE 0.75 (Table S2.4).

## Boxplots for Case 1-5

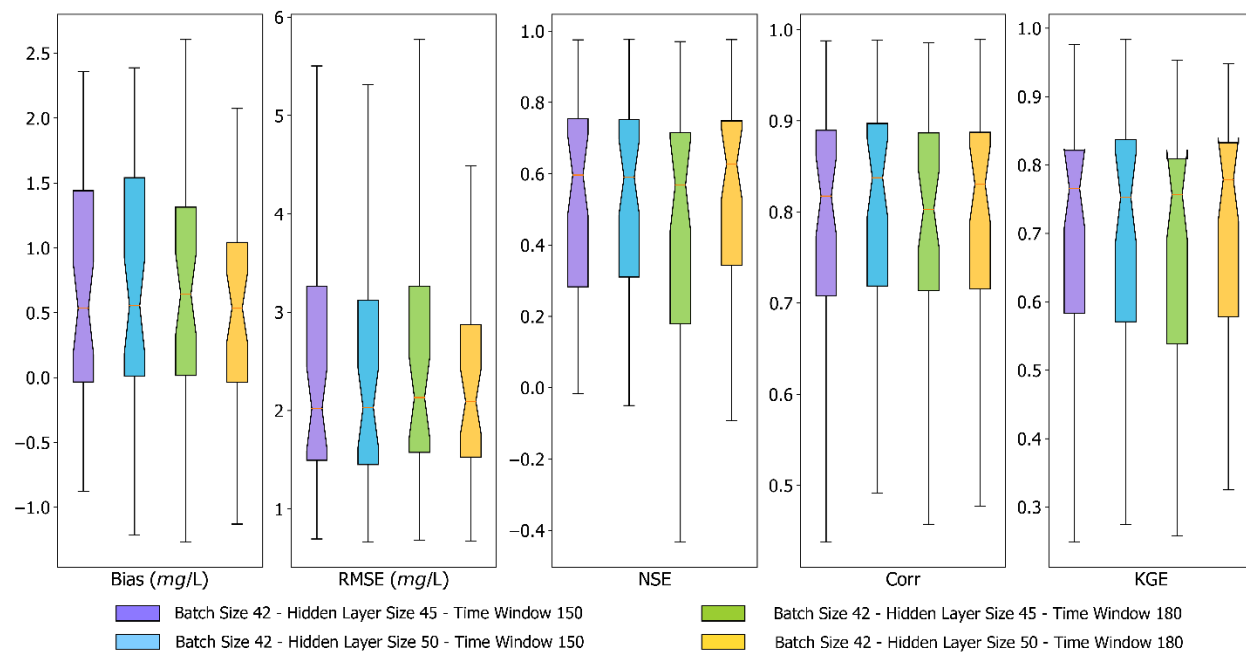


Figure S2.2 Boxplots of LSTM model outputs simulated with four different hyperparameters combinations for Case 1.

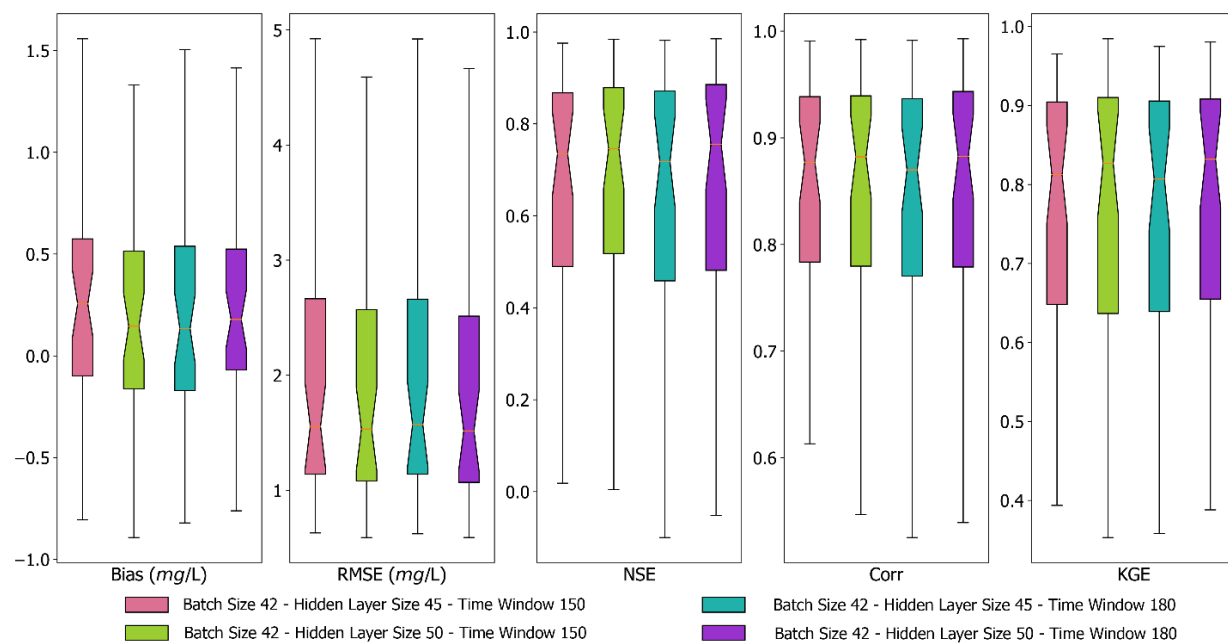


Figure S2.3 Boxplots of LSTM model outputs simulated with four different hyperparameters combinations for Case 2.

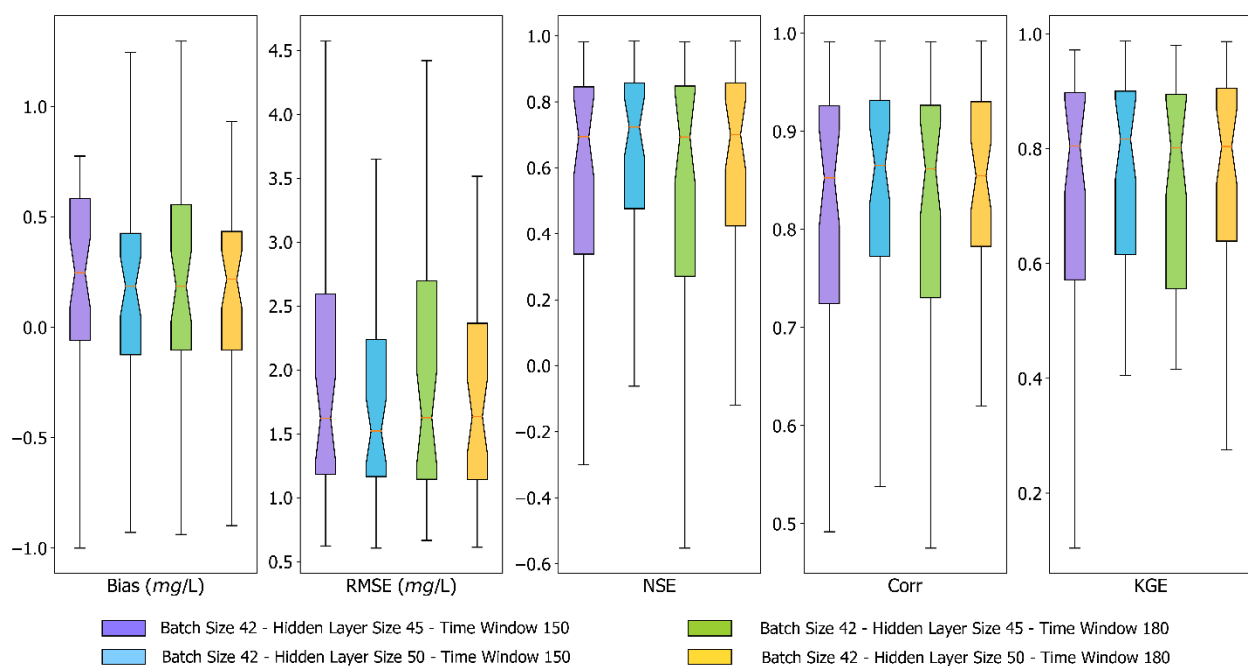


Figure S2.4 Boxplots of LSTM model outputs simulated with four different hyperparameters combinations for Case 3.

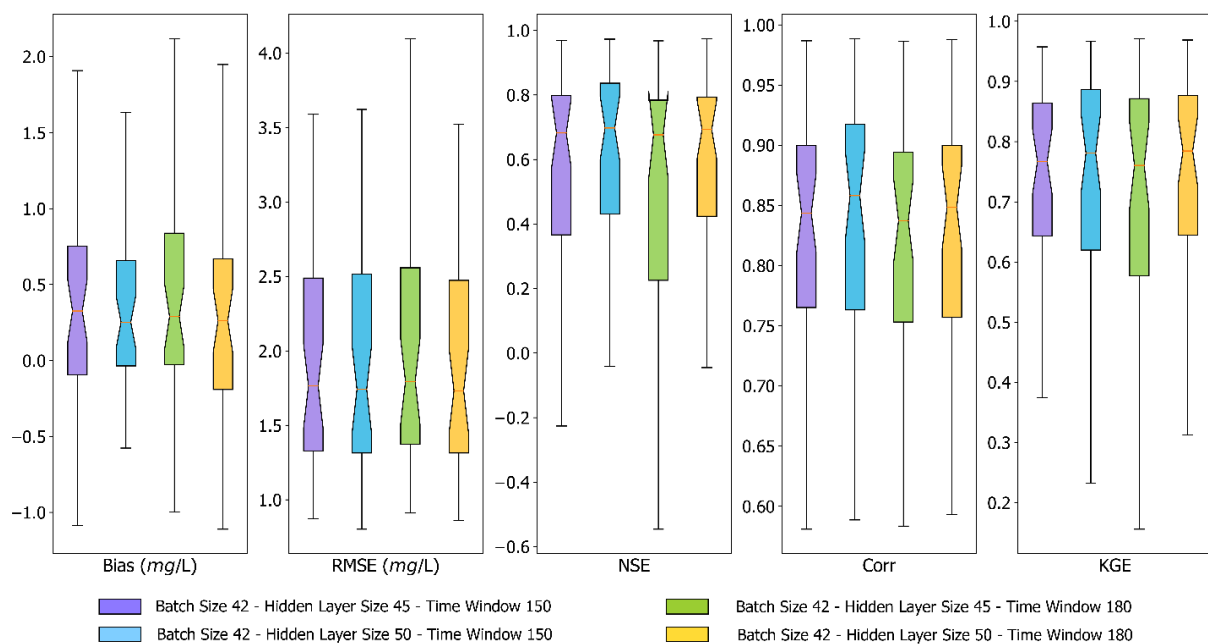


Figure S2.5 Boxplots of LSTM model outputs simulated with four different hyperparameters combinations for Case 4.



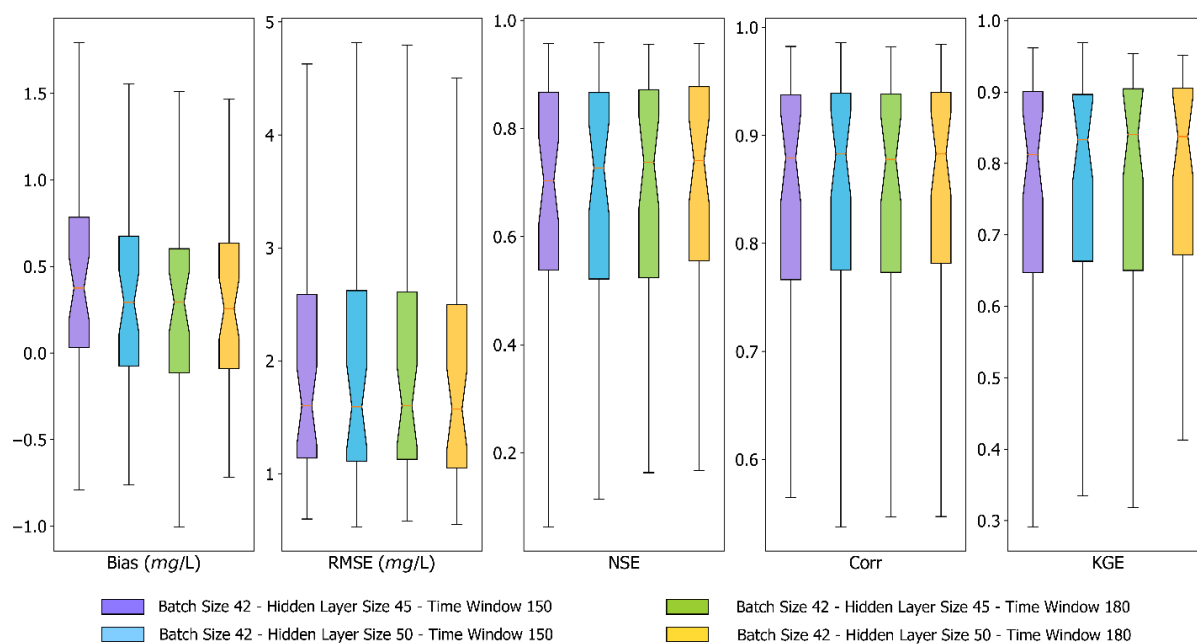


Figure S2.6 Boxplots of LSTM model outputs simulated with four different hyperparameters combinations for Case 5.

Table S2.4 Median evaluation metric values for each combination of hyperparameters for all five cases considering the test periods of all the nitrate monitoring sites.

| Case No. | Epoch | Batch size | Hidden Layer Size | Time Window | Evaluation Metrics for Testing Periods |      |      |             |      |
|----------|-------|------------|-------------------|-------------|----------------------------------------|------|------|-------------|------|
|          |       |            |                   |             | Bias                                   | RMSE | NSE  | Correlation | KGE  |
| Case1    | 2500  | 42         | 45                | 150         | 0.53                                   | 2.02 | 0.60 | 0.82        | 0.77 |
|          |       |            | 50                |             | 0.55                                   | 2.03 | 0.59 | 0.84        | 0.75 |
|          |       |            | 45                | 180         | 0.64                                   | 2.13 | 0.57 | 0.80        | 0.76 |
|          |       |            | 50                |             | 0.54                                   | 2.09 | 0.63 | 0.83        | 0.78 |
| Case2    | 2500  | 42         | 45                | 150         | 0.26                                   | 1.56 | 0.73 | 0.88        | 0.81 |
|          |       |            | 50                |             | 0.15                                   | 1.53 | 0.75 | 0.88        | 0.83 |
|          |       |            | 45                | 180         | 0.13                                   | 1.57 | 0.72 | 0.87        | 0.81 |
|          |       |            | 50                |             | 0.18                                   | 1.52 | 0.75 | 0.88        | 0.83 |
| Case3    | 2500  | 42         | 45                | 150         | 0.25                                   | 1.62 | 0.69 | 0.85        | 0.81 |
|          |       |            | 50                |             | 0.19                                   | 1.52 | 0.72 | 0.87        | 0.82 |
|          |       |            | 45                | 180         | 0.19                                   | 1.63 | 0.69 | 0.86        | 0.80 |
|          |       |            | 50                |             | 0.22                                   | 1.64 | 0.70 | 0.85        | 0.80 |
| Case4    | 2500  | 42         | 45                | 150         | 0.33                                   | 1.77 | 0.68 | 0.84        | 0.77 |
|          |       |            | 50                |             | 0.25                                   | 1.74 | 0.70 | 0.86        | 0.78 |
|          |       |            | 45                | 180         | 0.29                                   | 1.79 | 0.68 | 0.84        | 0.76 |
|          |       |            | 50                |             | 0.26                                   | 1.73 | 0.69 | 0.85        | 0.78 |
| Case5    | 2500  | 42         | 45                | 150         | 0.38                                   | 1.60 | 0.70 | 0.88        | 0.81 |
|          |       |            | 50                |             | 0.29                                   | 1.59 | 0.73 | 0.88        | 0.83 |
|          |       |            | 45                | 180         | 0.29                                   | 1.60 | 0.74 | 0.88        | 0.84 |
|          |       |            | 50                |             | 0.26                                   | 1.57 | 0.74 | 0.88        | 0.84 |

**Comparison between two sensor-based continuous nitrate monitoring sites and their pseudo intermittent nitrate monitoring sites according to the Case 2 and Case 5 results**

Table S2.5 Evaluation matrices scores of randomly selected four sensor-based continuous nitrate monitoring sites (10% of all continuous monitoring sites), which were considered as pseudo intermittent nitrate monitoring sites with biweekly nitrate data.

| Case No.                                                                                                                                | Site ID       | Evaluation Metrics for training and testing periods of 4 sites based on Case 2 model |      |             |       |      |                |      |             |       |      |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------|------|-------------|-------|------|----------------|------|-------------|-------|------|
|                                                                                                                                         |               | Training Period                                                                      |      |             |       |      | Testing Period |      |             |       |      |
|                                                                                                                                         |               | NSE                                                                                  | RMSE | Correlation | Bias  | KGE  | NSE            | RMSE | Correlation | Bias  | KGE  |
| Case 2 Model                                                                                                                            | USGS 05481000 | 0.98                                                                                 | 1.05 | 0.99        | 0.05  | 0.98 | 0.98           | 0.96 | 0.99        | 0.11  | 0.98 |
|                                                                                                                                         | USGS 05482500 | 0.99                                                                                 | 0.73 | 0.99        | 0.10  | 0.98 | 0.96           | 1.11 | 0.98        | 0.19  | 0.94 |
|                                                                                                                                         | WQS 0005      | 0.97                                                                                 | 0.51 | 0.99        | 0.05  | 0.96 | 0.78           | 1.57 | 0.89        | 0.28  | 0.79 |
|                                                                                                                                         | WQS 0012      | 0.57                                                                                 | 1.55 | 0.79        | -0.59 | 0.73 | 0.70           | 1.66 | 0.84        | -0.22 | 0.79 |
| Evaluation Metrics of same 4 pseudo intermittent nitrate monitoring sites based on Case 5 model                                         |               |                                                                                      |      |             |       |      |                |      |             |       |      |
| Case 5 Model                                                                                                                            |               | NSE                                                                                  | RMSE | Correlation | Bias  | KGE  | NSE            | RMSE | Correlation | Bias  | KGE  |
|                                                                                                                                         | USGS 05481000 | 0.96                                                                                 | 1.58 | 0.98        | 0.45  | 0.95 | 0.96           | 1.61 | 0.98        | 0.69  | 0.92 |
|                                                                                                                                         | USGS 05482500 | 0.98                                                                                 | 1.02 | 0.99        | 0.17  | 0.93 | 0.95           | 1.20 | 0.98        | 0.33  | 0.90 |
|                                                                                                                                         | WQS 0005      | 0.95                                                                                 | 0.68 | 0.98        | 0.21  | 0.85 | 0.58           | 2.38 | 0.77        | 0.4   | 0.63 |
|                                                                                                                                         | WQS 0012      | 0.39                                                                                 | 1.77 | 0.78        | -1.05 | 0.76 | 0.59           | 1.90 | 0.79        | -0.43 | 0.78 |
| Evaluation Metrics using remaining continuous nitrate data of same 4 pseudo intermittent nitrate monitoring sites based on Case 5 model |               |                                                                                      |      |             |       |      |                |      |             |       |      |
| Case 5 Model                                                                                                                            |               | NSE                                                                                  | RMSE | Correlation | Bias  | KGE  | NSE            | RMSE | Correlation | Bias  | KGE  |
|                                                                                                                                         | USGS 05481000 | 0.91                                                                                 | 2.23 | 0.96        | 0.68  | 0.93 | 0.90           | 2.41 | 0.97        | 1.08  | 0.88 |
|                                                                                                                                         | USGS 05482500 | 0.92                                                                                 | 1.57 | 0.97        | 0.66  | 0.88 | 0.92           | 1.50 | 0.97        | 0.74  | 0.90 |
|                                                                                                                                         | WQS 0005      | 0.93                                                                                 | 0.78 | 0.98        | 0.16  | 0.91 | 0.55           | 2.20 | 0.78        | 0.62  | 0.68 |
|                                                                                                                                         | WQS 0012      | 0.49                                                                                 | 1.69 | 0.76        | -0.65 | 0.75 | 0.53           | 2.09 | 0.75        | 0.01  | 0.75 |

Table S2.6 Contributing Area and annual nitrate loads based on Case 2 and Case 5 models at the randomly selected four sensor-based continuous nitrate monitoring sites.

| Site ID               | Contributing Watershed Area (sq km.) | Annual Nitrate Load (Kg/ha/year) based on Case 2 Model | Annual Nitrate Load (Kg/ha/year) based on Case 5 Model |
|-----------------------|--------------------------------------|--------------------------------------------------------|--------------------------------------------------------|
| USGS 05481000         | 2127.5                               | 46.10                                                  | 48.20                                                  |
| USGS 05482500         | 4176.0                               | 35.70                                                  | 35.70                                                  |
| WQS 0005              | 1480.8                               | 19.10                                                  | 20.90                                                  |
| WQS 0012 <sup>1</sup> | 12.3                                 | 27.17                                                  | 26.78                                                  |

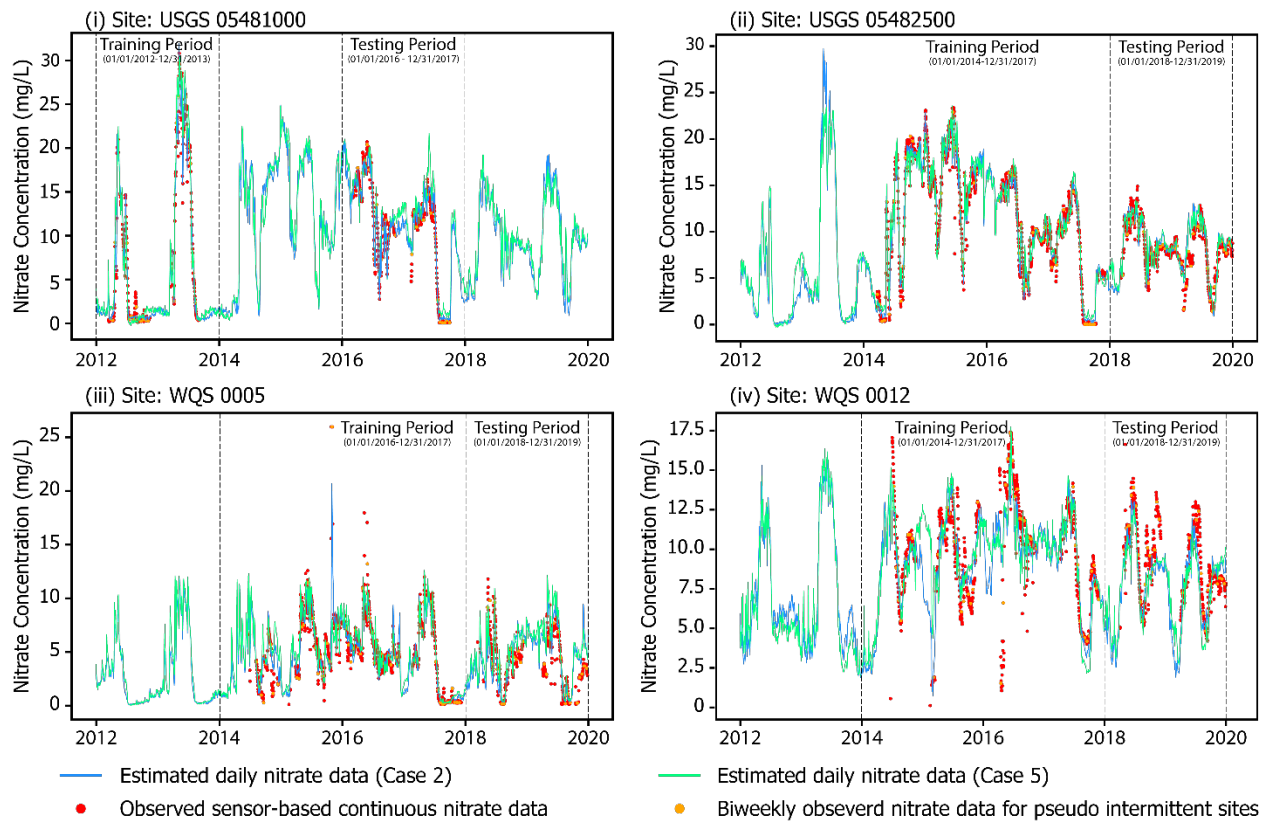


Figure S2.7 Comparison among time series plots of two sensor-based continuous nitrate monitoring sites based on the developed Case 2 and Case 5 model. Figures S2.8i, S2.8ii, S2.8iii and S2.8iv showed observed continuous nitrate time series, biweekly data used in the pseudo intermittent nitrate monitoring sites, estimated time series based on Case 2 and Case 5 model of sites USGS 05481000, USGS 05482500, WQS0005, and WQS 0012, respectively. The respective evaluation metrics for period 2012-2019 are calculated based on their respective observed nitrate data. The test-period model

performances for four sensor-based high-frequency sites: (S7i) Site USGS 05481000 with its continuous nitrate data using Case 2 model [RMSE: 0.96, NSE: 0.98, Bias: 0.11, Correlation: 0.99, and KGE: 0.98], and Site USGS 05481000 as pseudo intermittent nitrate monitoring site using Case 5 model [RMSE: 1.61, NSE: 0.96, Bias: 0.69, Correlation: 0.98, and KGE: 0.91]; (S7ii) Site USGS 05482500 with its continuous nitrate data using Case 2 model [RMSE: 1.11, NSE: 0.96, Bias: 0.19, Correlation: 0.98, and KGE: 0.96], and Site USGS 05482500 as pseudo intermittent nitrate monitoring site using Case 5 model [RMSE: 1.20, NSE: 0.95, Bias: 0.33, Correlation: 0.98, and KGE: 0.93]; (S7iii) Site WQS 0005 with its continuous nitrate data using Case 2 model [RMSE: 1.57, NSE: 0.78, Bias: 0.28, Correlation: 0.89, and KGE: 0.79], and Site WQS 0005 as pseudo intermittent nitrate monitoring site using Case 5 model [RMSE: 2.38, NSE: 0.58, Bias: 0.40, Correlation: 0.77, and KGE: 0.63]; and (S7iv) Site WQS 0012 with its continuous nitrate data using Case 2 model [RMSE: 1.66, NSE: 0.70, Bias: -0.22, Correlation: 0.84, and KGE: 0.79], and Site WQS 0012 as pseudo intermittent nitrate monitoring site using Case 5 model [RMSE: 1.90, NSE: 0.59, Bias: -0.43, Correlation: 0.79, and KGE: 0.78].

### Comparison among four seasons' stream nitrate concentration estimates

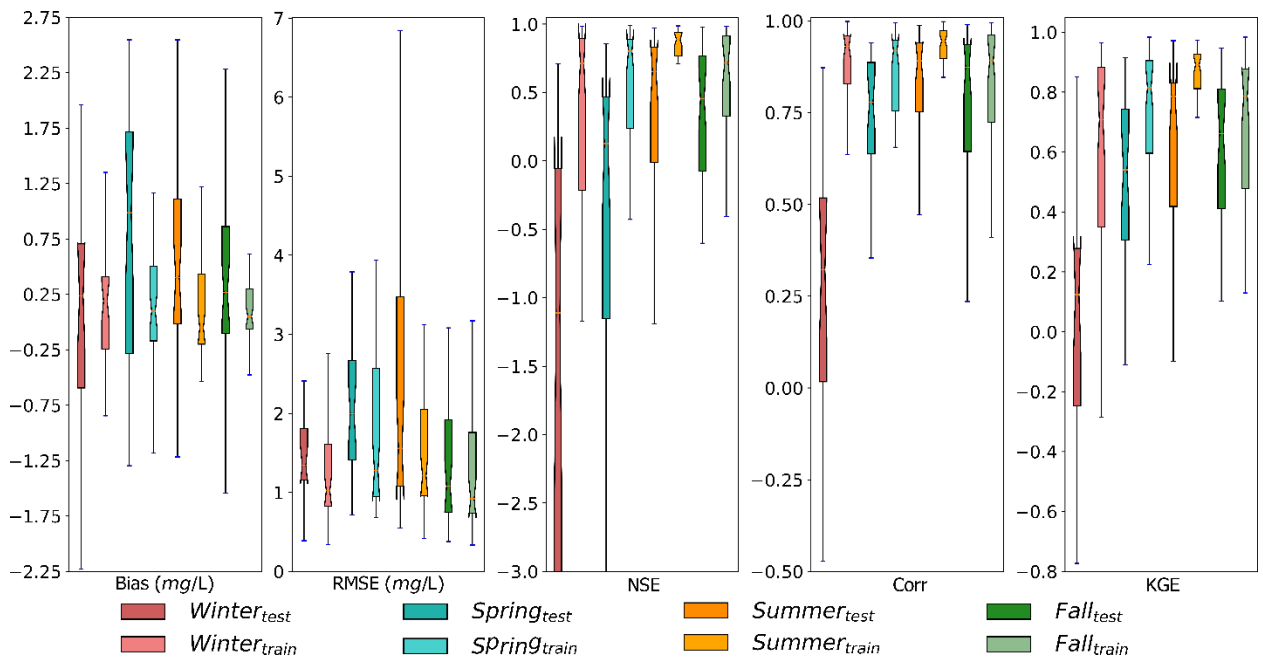


Figure S2.8 Boxplots of LSTM model outputs simulated with four different hyperparameters combinations for Case 5

Table S2.7 Scores of evaluation matrices based on the estimated daily stream nitrate concentration for four seasons' using Case 2 model.

| Simulation Periods | Seasons | Bias  | RMSE | NSE   | Correlation | KGE  |
|--------------------|---------|-------|------|-------|-------------|------|
| Testing            | Winter  | 0.24  | 1.35 | -1.11 | 0.32        | 0.12 |
|                    | Spring  | 0.99  | 1.99 | 0.13  | 0.78        | 0.54 |
|                    | Summer  | 0.4   | 1.56 | 0.65  | 0.89        | 0.79 |
|                    | Fall    | 0.27  | 1.08 | 0.45  | 0.87        | 0.66 |
| Training           | Winter  | 0.19  | 1.02 | 0.71  | 0.93        | 0.71 |
|                    | Spring  | 0.1   | 1.28 | 0.8   | 0.92        | 0.81 |
|                    | Summer  | -0.05 | 1.21 | 0.89  | 0.95        | 0.89 |
|                    | Fall    | 0.05  | 0.92 | 0.72  | 0.89        | 0.79 |

## Slope (b) and Correlation values between the simulated correlation and discharge

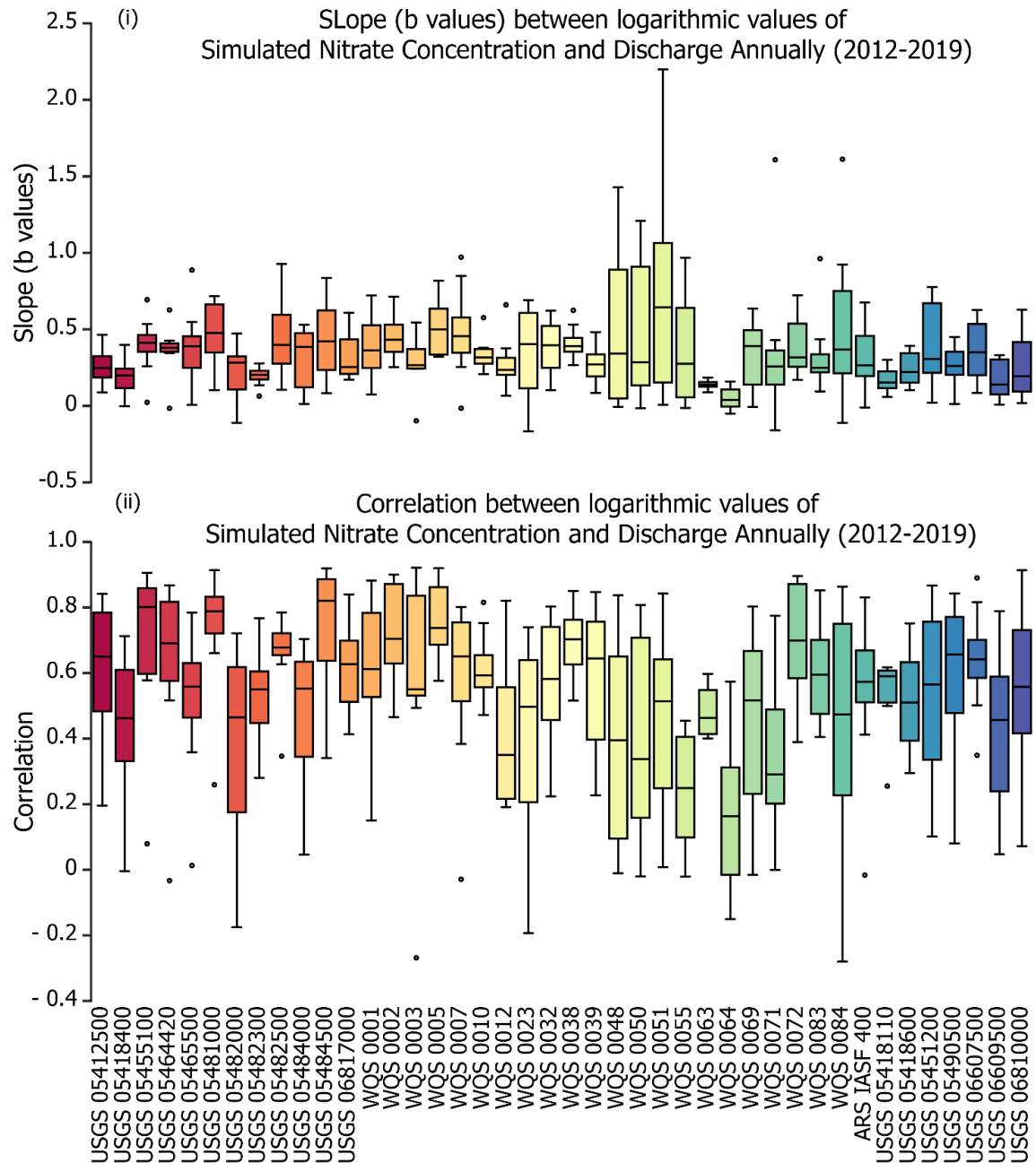


Figure S2.9 Boxplots of (i) slope (b) and (ii) correlation values calculated based on the logarithmic nitrate concentration (estimated) versus logarithmic stream discharge plots for all 42 nitrate monitoring sites. Slope (b) and correlation values were estimated annually. Estimated slope (b) value ranges from - 0.2 to 2.10. Calculated correlation values varied between - 0.25 and 0.90. The slope and correlation values indicated the interannual variability in the c-Q relationships of each site.

### Time series of plots of best and poor performed nitrate monitoring sites.

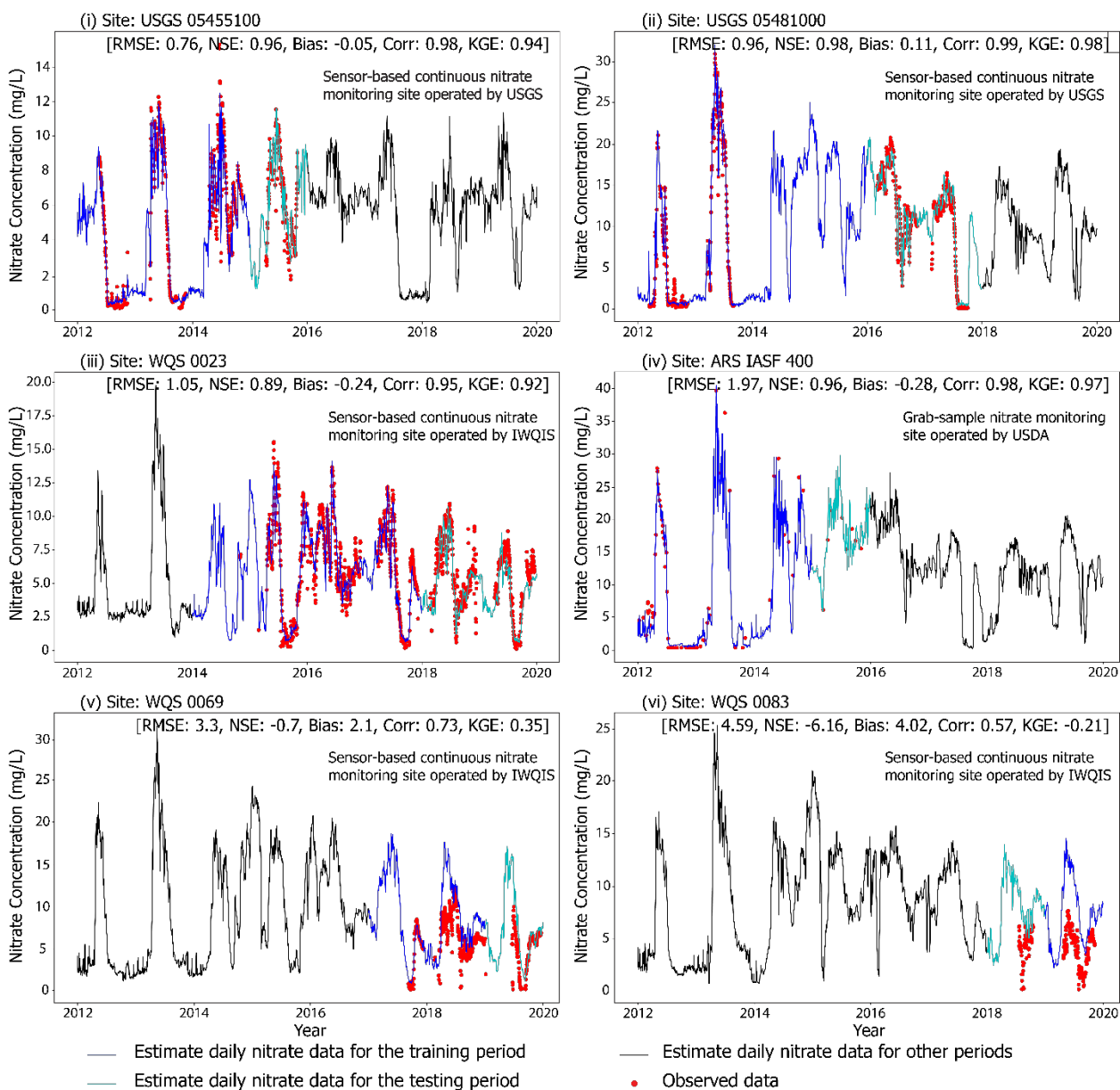


Figure S2.10 Time series plots of continuous daily nitrate concentration for sensor-based continuous nitrate monitoring sites for period 2012-2019. The test-period model performances for sensor-based nitrate monitoring sites: (4i) Site USGS 05455100 [RMSE: 0.76, NSE: 0.96, Bias: -0.05, Correlation: 0.98, and KGE: 0.94], (4ii) Site USGS 05481000 [RMSE: 0.96, NSE: 0.98, Bias: 0.11, Correlation: 0.99, and KGE: 0.98], (4iii) Site WQS 0023 [RMSE: 1.05, NSE: 0.89, Bias: -0.24, Correlation: 0.95,



and KGE: 0.92], (4iv) Site WQS 0010 [RMSE: 1.08, NSE: 0.87, Bias: 0.04, Correlation: 0.93, and KGE: 0.87], (4v) Site WQS 0069 [RMSE: 3.30, NSE: -0.70, Bias: 2.10, Correlation: 0.73, and KGE: 0.35], (4vi) Site WQS 0083 [RMSE: 4.59, NSE: -6.16, Bias: 4.02, Correlation: 0.57, and KGE: -0.21]. Figures S2.9i, S2.9ii, 2.9iii, and 2.9iv show remarkably good performance. Figures S2.9v and S2.9vi have not offered reasonably poor evaluation metrics scores.

### Spatial distribution of Case 2 metrics.

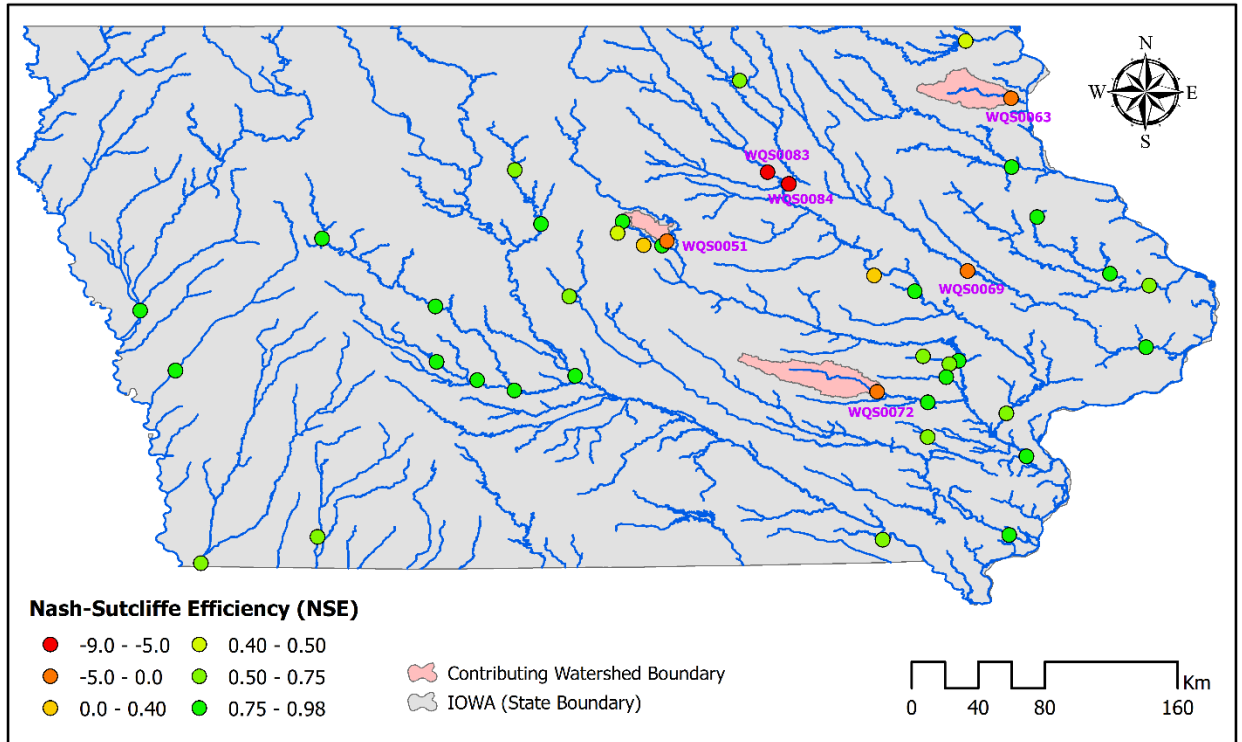


Figure S2.11 Spatial distribution of test-period Nash-Sutcliffe Efficiency (NSE) for all nitrate monitoring sites of Case 2 across Iowa. Sites WQS0083 and WQS0084 do not have enough nitrate information as well as neighboring data-rich sensor-based nitrate monitoring sites. Sites WQS0051, WQS0063, and WQS0072 have small contributing head-watersheds and therefore, downscaled Fertilizer N and Manure N data may not represent actual N application scenario for these watersheds.

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## Appendix B

### Site identity of the selected high-frequency nitrate monitoring sites

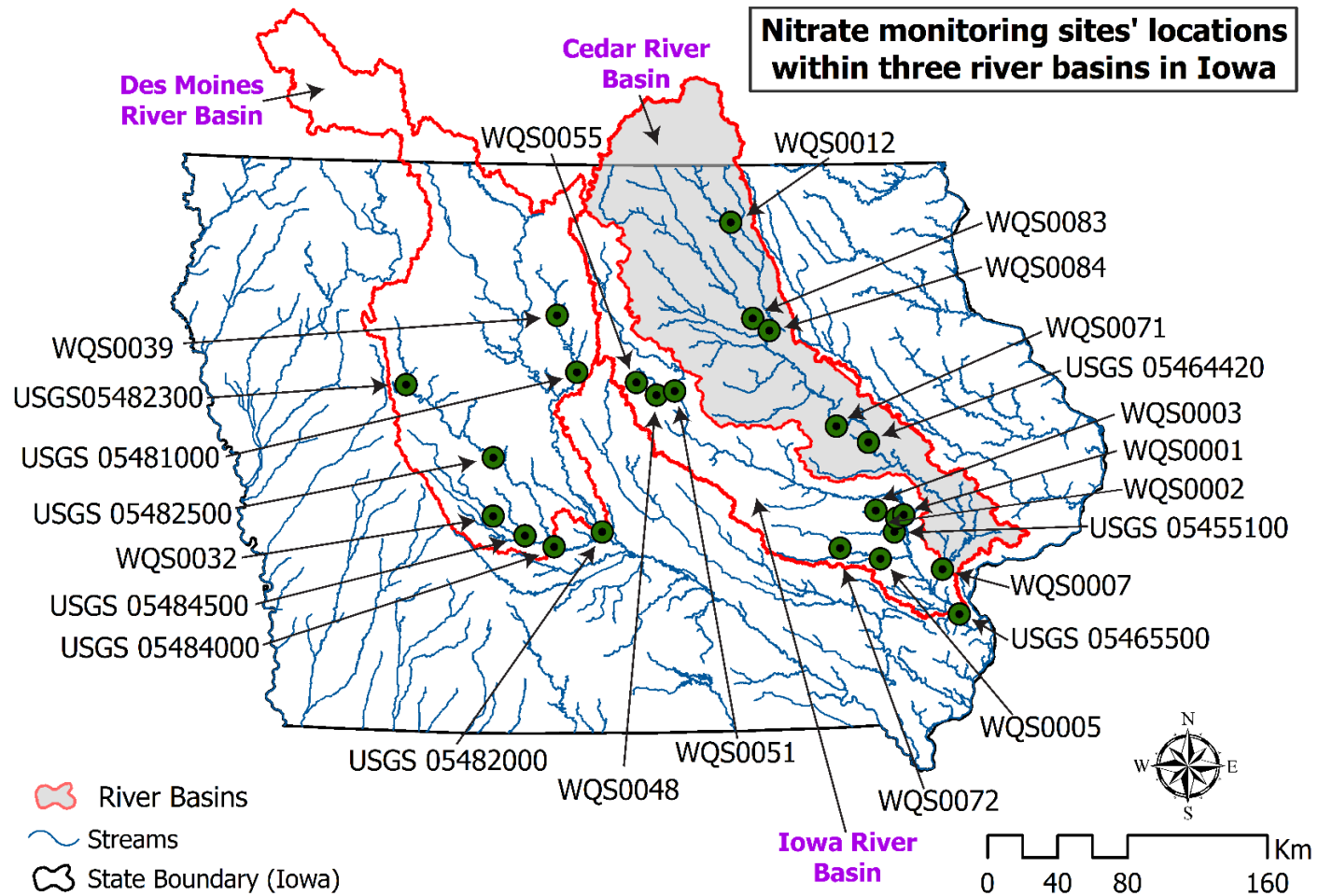


Figure S3.1 Site identities of the selected high-frequency nitrate monitoring sites within the selected three river basins in Iowa.

Table S3.1 A list of high-frequency nitrate monitoring sites available in the three selected region. Nitrate data of these monitoring sites was used to develop DL models.

| Basin No               | Site No       | Model 1                     |                          | Model 2        |             | Model 5                                                                                                                                                                                      |             | Model 6        |             |
|------------------------|---------------|-----------------------------|--------------------------|----------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|-------------|
|                        |               | Train and test <sup>1</sup> | Model input <sup>2</sup> | Train and test | Model input | Train and test                                                                                                                                                                               | Model input | Train and test | Model input |
| Des Moines River Basin | USGS 05481000 | x                           |                          | x              |             | x                                                                                                                                                                                            |             | x              |             |
|                        | USGS 05482000 | x                           |                          | x              |             | x                                                                                                                                                                                            |             |                | x           |
|                        | USGS 05482300 |                             | x                        | x              |             | x                                                                                                                                                                                            |             | x              |             |
|                        | USGS 05482500 | x                           |                          | x              |             |                                                                                                                                                                                              | x           | x              |             |
|                        | USGS 05484000 | x                           |                          | x              |             | x                                                                                                                                                                                            |             | x              |             |
|                        | USGS 05484500 | x                           |                          |                | x           | x                                                                                                                                                                                            |             | x              |             |
|                        | WQS 0032      | x                           |                          | x              |             | x                                                                                                                                                                                            |             | x              |             |
|                        | WQS 0039      | x                           |                          | x              |             | x                                                                                                                                                                                            |             | x              |             |
|                        |               |                             |                          |                |             |                                                                                                                                                                                              |             |                |             |
| Iowa River Basin       |               | Model 3                     |                          | Model 4        |             | Model 7                                                                                                                                                                                      |             |                |             |
|                        |               | Train and test              | Model input              | Train and test | Model input | Train and test                                                                                                                                                                               | Model input |                |             |
|                        | USGS 05455100 | x                           |                          | x              |             | x                                                                                                                                                                                            |             |                |             |
|                        | USGS 05465500 | x                           |                          |                | x           | x                                                                                                                                                                                            |             |                |             |
|                        | WQS 0001      |                             | x                        | x              |             | x                                                                                                                                                                                            |             |                |             |
|                        | WQS 0002      | x                           |                          | x              |             | x                                                                                                                                                                                            |             |                |             |
|                        | WQS 0003      | x                           |                          | x              |             |                                                                                                                                                                                              | x           |                |             |
|                        | WQS 0005      | x                           |                          | x              |             | x                                                                                                                                                                                            |             |                |             |
|                        | WQS 0048      | x                           |                          | x              |             | x                                                                                                                                                                                            |             |                |             |
|                        | WQS 0051      | x                           |                          | x              |             | x                                                                                                                                                                                            |             |                |             |
| WQS 0055               | x             |                             | x                        |                | x           |                                                                                                                                                                                              |             |                |             |
| WQS 0072               | x             |                             | x                        |                | x           |                                                                                                                                                                                              |             |                |             |
|                        |               |                             |                          |                |             |                                                                                                                                                                                              |             |                |             |
| Cedar River Basin      |               | Model 8                     |                          | Model 9        |             |                                                                                                                                                                                              |             |                |             |
|                        |               | Train and test              | Model input              | Train and test | Model input |                                                                                                                                                                                              |             |                |             |
|                        | USGS 05465500 | x                           |                          |                | x           | 1. The biweekly subsampled (randomly) nitrate data of all high-frequency nitrate monitoring sites were used for DL model training. Remaining daily nitrate data were used for model testing. |             |                |             |
|                        | USGS 05464420 | x                           |                          | x              |             |                                                                                                                                                                                              |             |                |             |
|                        | WQS 0007      |                             | x                        | x              |             | 2. Sites for model inputs were selected based on the quantity of daily nitrate concentration availability.                                                                                   |             |                |             |
|                        | WQS 0012      | x                           |                          | x              |             |                                                                                                                                                                                              |             |                |             |
|                        | WQS 0071      | x                           |                          | x              |             |                                                                                                                                                                                              |             |                |             |
|                        | WQS 0083      | x                           |                          | x              |             |                                                                                                                                                                                              |             |                |             |
| WQS 0084               | x             |                             | x                        |                |             |                                                                                                                                                                                              |             |                |             |

Table S3.2 Data availability, training, and testing periods of high-frequency nitrate monitoring sites for model development and combinations of hyper-parameters to identify the best models for daily nitrate prediction.

| Model No.                                       | Site No.      | Data availability | Training <sup>1</sup> and testing <sup>2</sup> period | Batch size | Hidden layer size | Time window      | No. of epoch              | Random seeds |
|-------------------------------------------------|---------------|-------------------|-------------------------------------------------------|------------|-------------------|------------------|---------------------------|--------------|
| Model 1,<br>Model 2,<br>Model 5, and<br>Model 6 | USGS 05481000 | 2012-2017         | 2012-2017                                             | 4, 7       | 7, 10, 12         | 120, 150,<br>180 | 2000, 2500,<br>3000, 3500 | 0, 1, 3, 5   |
|                                                 | USGS 05482000 | 2013-2019         | 2013-2019                                             |            |                   |                  |                           |              |
|                                                 | USGS 05482300 | 2012-2019         | 2012-2019                                             |            |                   |                  |                           |              |
|                                                 | USGS 05482500 | 2012-2019         | 2012-2019                                             |            |                   |                  |                           |              |
|                                                 | USGS 05484000 | 2016-2019         | 2016-2019                                             |            |                   |                  |                           |              |
|                                                 | USGS 05484500 | 2013-2019         | 2013-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0032      | 2014-2019         | 2014-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0039      | 2016-2019         | 2016-2019                                             |            |                   |                  |                           |              |
| Model 3,<br>Model 4, and<br>Model 7             | USGS 05455100 | 2012-2015         | 2012-2015                                             | 5, 9       | 9, 10, 12         | 120, 150,<br>180 | 2000, 2500,<br>3000, 3500 | 0, 1, 3, 5   |
|                                                 | USGS 05465500 | 2012-2019         | 2012-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0001      | 2012-2019         | 2012-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0002      | 2012-2019         | 2012-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0003      | 2012-2019         | 2012-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0005      | 2012-2019         | 2012-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0048      | 2016-2019         | 2016-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0051      | 2016-2019         | 2016-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0055      | 2017-2019         | 2017-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0072      | 2017-2019         | 2017-2019                                             |            |                   |                  |                           |              |
| Model 8 and<br>Model 9                          | USGS 05465500 | 2012-2019         | 2012-2019                                             | 3, 6       | 6, 10, 12         | 120, 150,<br>180 | 2000, 2500,<br>3000, 3500 | 0, 1, 3, 5   |
|                                                 | USGS 05464420 | 2012-2019         | 2012-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0007      | 2012-2019         | 2012-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0012      | 2014-2019         | 2014-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0071      | 2017-2019         | 2017-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0083      | 2018-2019         | 2018-2019                                             |            |                   |                  |                           |              |
|                                                 | WQS 0084      | 2018-2019         | 2018-2019                                             |            |                   |                  |                           |              |

<sup>1</sup> Training data for each site is the biweekly subsampled nitrate data used for model development.

<sup>2</sup> Testing data for each site is the remaining daily nitrate data used for model evaluation.

Table S3.3 Calibration parameters for WRTDS-based continuous daily nitrate estimation.

| Basin No.              | Site No.      | WRTDS Estimation Parameter (EGRETS) |                             |                |           |             |                         | Kalman (EGRETS) |       |      |
|------------------------|---------------|-------------------------------------|-----------------------------|----------------|-----------|-------------|-------------------------|-----------------|-------|------|
|                        |               | WindowY (Year)                      | WindowQ (m <sup>3</sup> /s) | WindowS (Year) | minNumObs | minNumUncen | edgeAdjust (True/False) | rho             | niter | seed |
| Des Moines River Basin | USGS 05481000 | 3.0                                 | 1.75                        | 0.40           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | USGS 05482000 | 3.5                                 | 2.00                        | 0.50           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | USGS 05482300 | 4.0                                 | 2.00                        | 0.50           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | USGS 05482500 | 4.0                                 | 1.50                        | 0.50           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | USGS 05484000 | 2.0                                 | 1.70                        | 0.55           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | USGS 05484500 | 2.0                                 | 1.50                        | 0.40           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0032      | 3.0                                 | 1.50                        | 0.40           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0039      | 3.5                                 | 1.50                        | 0.30           | 50        | 20          | True                    | 0.95            | 500   | 1    |
| Iowa River Basin       | USGS 05455100 | 2.0                                 | 1.50                        | 0.50           | 45        | 20          | True                    | 0.95            | 500   | 1    |
|                        | USGS 05465500 | 2.0                                 | 2.00                        | 0.30           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0001      | 2.5                                 | 2.00                        | 0.40           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0002      | 2.5                                 | 1.50                        | 0.40           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0003      | 3.5                                 | 2.00                        | 0.75           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0005      | 4.0                                 | 2.00                        | 0.45           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0048      | 2.5                                 | 1.70                        | 0.75           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0051      | 3.0                                 | 1.50                        | 0.30           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0055      | 2.0                                 | 2.00                        | 0.60           | 45        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0072      | 2.0                                 | 2.00                        | 0.30           | 39        | 20          | True                    | 0.95            | 500   | 1    |
| Cedar River Basin      | USGS 05464420 | 2.0                                 | 2.00                        | 0.40           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | USGS 05465500 | 2.0                                 | 2.00                        | 0.30           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0007      | 4.0                                 | 2.00                        | 0.40           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0012      | 4.0                                 | 2.00                        | 0.30           | 60        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0071      | 2.0                                 | 1.65                        | 0.30           | 40        | 20          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0083      | 2.0                                 | 1.50                        | 0.50           | 15        | 10          | True                    | 0.95            | 500   | 1    |
|                        | WQS 0084      | 2.0                                 | 1.50                        | 0.40           | 20        | 20          | True                    | 0.95            | 500   | 1    |

Table S3.4 Median evaluation metrics for simulated nine (9) data-driven models.

| Model No | Bias  | RMSE | Correlation | NSE   | KGE  |
|----------|-------|------|-------------|-------|------|
| 1        | 0.18  | 1.50 | 0.93        | 0.86  | 0.92 |
| 2        | -0.12 | 1.64 | 0.95        | 0.89  | 0.94 |
| 3        | 0.12  | 2.01 | 0.82        | 0.66  | 0.80 |
| 4        | 0.22  | 1.93 | 0.84        | 0.69  | 0.80 |
| 5        | 0.03  | 1.75 | 0.94        | 0.87  | 0.92 |
| 6        | 2.89  | 2.95 | 0.66        | -3.13 | 0.80 |
| 7        | -0.03 | 2.11 | 0.80        | 0.63  | 0.77 |
| 8        | -0.16 | 1.68 | 0.69        | 0.36  | 0.67 |
| 9        | 0.23  | 1.50 | 0.78        | 0.51  | 0.72 |

Table S3.5 The best evaluation metrics and corresponding model number based on testing data for each nitrate monitoring site within all three basins.

| Basin No                  | Site No       | Bias  | RMSE | Correlation | NSE  | KGE  | Model No |
|---------------------------|---------------|-------|------|-------------|------|------|----------|
| Des Moines River Basin    | USGS 05481000 | 0.10  | 1.75 | 0.97        | 0.94 | 0.97 | 5        |
|                           | USGS 05482000 | -0.18 | 1.32 | 0.95        | 0.89 | 0.94 | 2        |
|                           | USGS 05482300 | 0.08  | 1.64 | 0.95        | 0.89 | 0.94 | 2        |
|                           | USGS 05482500 | 0.20  | 1.37 | 0.98        | 0.95 | 0.96 | 1        |
|                           | USGS 05484000 | -0.12 | 0.93 | 0.93        | 0.86 | 0.93 | 2        |
|                           | USGS 05484500 | -0.03 | 1.21 | 0.96        | 0.92 | 0.95 | 1        |
|                           | WQS 0032      | 0.03  | 2.22 | 0.87        | 0.74 | 0.85 | 5        |
|                           | WQS 0039      | 0.22  | 3.20 | 0.72        | 0.51 | 0.66 | 5        |
| Iowa River Basin          | USGS 05455100 | -0.18 | 1.21 | 0.95        | 0.89 | 0.90 | 7        |
|                           | WQS 0001      | 0     | 1.53 | 0.89        | 0.78 | 0.84 | 4        |
|                           | WQS 0002      | -0.03 | 1.37 | 0.89        | 0.79 | 0.88 | 7        |
|                           | WQS 0003      | 0.22  | 1.93 | 0.84        | 0.69 | 0.80 | 4        |
|                           | WQS 0005      | -0.10 | 1.88 | 0.84        | 0.70 | 0.80 | 4        |
|                           | WQS 0048      | 1.58  | 4.24 | 0.76        | 0.48 | 0.70 | 3        |
|                           | WQS 0051      | 0.70  | 2.21 | 0.83        | 0.61 | 0.81 | 4        |
|                           | WQS 0055      | -0.76 | 4.92 | 0.70        | 0.46 | 0.64 | 7        |
| Iowa & Cedar River Basins | WQS 0072      | 0.64  | 2.12 | 0.69        | 0.19 | 0.61 | 4        |
|                           | USGS 05465500 | -0.32 | 1.19 | 0.91        | 0.80 | 0.83 | 3        |
|                           | USGS 05464420 | 0.10  | 1.42 | 0.88        | 0.77 | 0.87 | 9        |
|                           | WQS 0007      | 0.17  | 1.61 | 0.86        | 0.72 | 0.86 | 9        |
|                           | WQS 0012      | 0.38  | 2.09 | 0.74        | 0.53 | 0.65 | 9        |
|                           | WQS 0071      | 0.29  | 1.59 | 0.71        | 0.38 | 0.70 | 9        |
|                           | WQS 0083      | 0.48  | 1.31 | 0.81        | 0.40 | 0.68 | 9        |
|                           | WQS 0084      | 0.18  | 1.37 | 0.75        | 0.49 | 0.75 | 9        |

Table S3.6 Stream distances between the pseudo-low-frequency nitrate monitoring sites and their respective high-frequency nitrate monitoring sites that help to predict daily nitrate concentrations at the pseudo-low-frequency nitrate monitoring sites.

| Basin's Name              | Pseudo-low-frequency Nitrate Monitoring Site No. | Model No. | Input High-frequency Nitrate Monitoring Sites | Stream Distance (Km) |
|---------------------------|--------------------------------------------------|-----------|-----------------------------------------------|----------------------|
| Des Moines River Basin    | USGS 05481000                                    | 5         | USGS 05482500                                 | 315                  |
|                           | USGS 05482000                                    | 2         | USGS 05484500                                 | 52                   |
|                           | USGS 05482300                                    | 2         | USGS 05484500                                 | 215                  |
|                           | USGS 05482500                                    | 1         | USGS 05482300                                 | 118                  |
|                           | USGS 05484000                                    | 2         | USGS 05484500                                 | 25                   |
|                           | USGS 05484500                                    | 1         | USGS 05482300                                 | 215                  |
|                           | WQS 0032                                         | 5         | USGS 05482500                                 | 154                  |
|                           | WQS 0039                                         | 5         | USGS 05482500                                 | 118                  |
| Iowa River Basin          | USGS 05455100                                    | 7         | WQS 0003                                      | 61                   |
|                           | WQS 0001                                         | 4         | USGS 05465500                                 | 108                  |
|                           | WQS 0002                                         | 7         | WQS 0003                                      | 17                   |
|                           | WQS 0003                                         | 4         | USGS 05465500                                 | 119                  |
|                           | WQS 0005                                         | 4         | USGS 05465500                                 | 88                   |
|                           | WQS 0048                                         | 3         | WQS 0001                                      | 261                  |
|                           | WQS 0051                                         | 4         | USGS 05465500                                 | 349                  |
|                           | WQS 0055                                         | 7         | WQS 0003                                      | 310                  |
|                           | WQS 0072                                         | 4         | USGS 05465500                                 | 126                  |
| Iowa & Cedar River Basins | USGS 05465500                                    | 3         | WQS 0001                                      | 103                  |
| Cedar River Basin         | USGS 05464420                                    | 9         | USGS 05465500                                 | 181                  |
|                           | WQS 0007                                         | 9         | USGS 05465500                                 | 41                   |
|                           | WQS 0012                                         | 9         | USGS 05465500                                 | 433                  |
|                           | WQS 0071                                         | 9         | USGS 05465500                                 | 217                  |
|                           | WQS 0083                                         | 9         | USGS 05465500                                 | 328                  |
|                           | WQS 0084                                         | 9         | USGS 05465500                                 | 313                  |



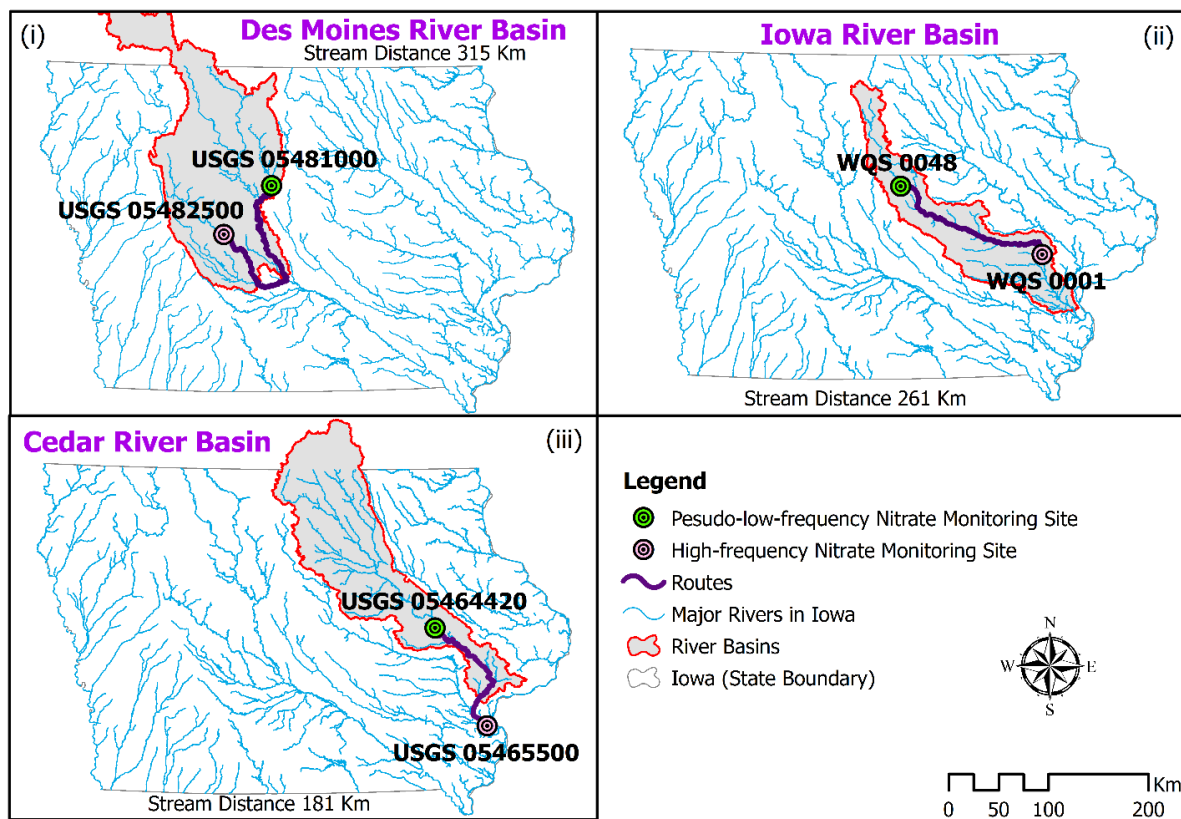


Figure S3.2 Calculated stream distances between the target pseudo-low-frequency nitrate monitoring sites and their respective high-frequency nitrate monitoring sites whose nitrate and stream discharge data were used as inputs to the developed DL models.

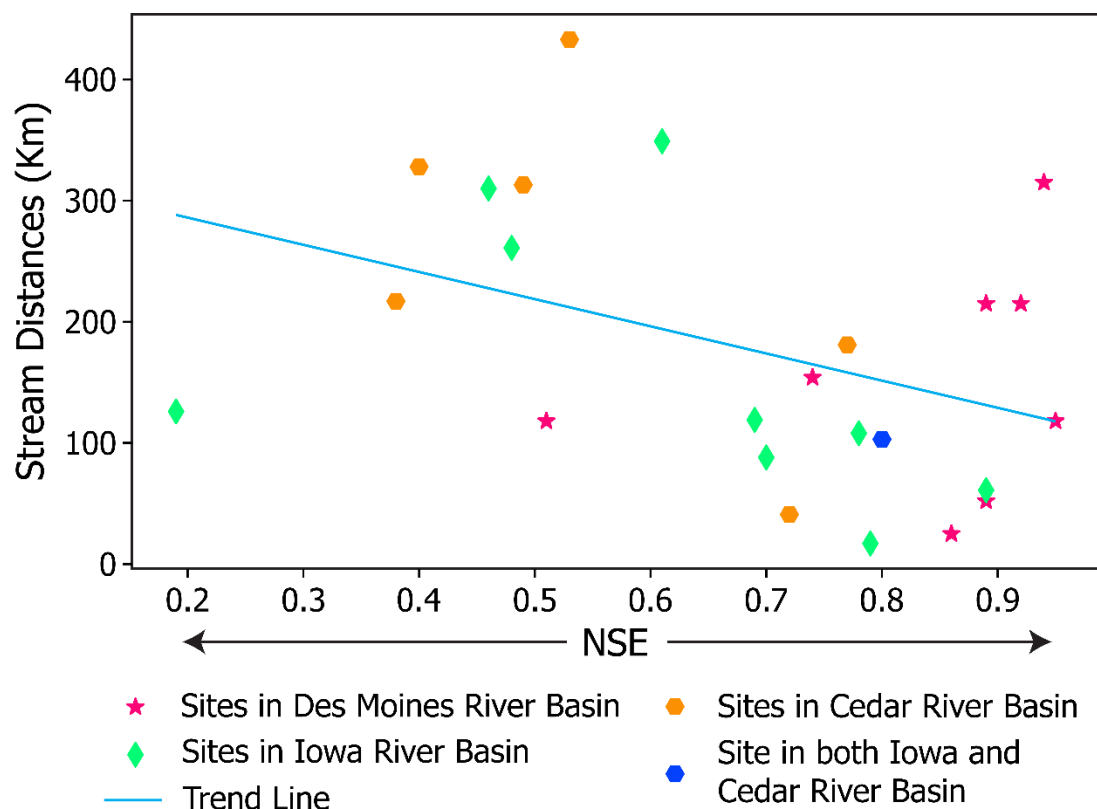


Figure S3.3 River basin-wise Nash Sutcliffe Efficiency (NSE) performance of each site and its respective stream distances from the input high-frequency nitrate monitoring site.

Table S3.7 Evaluation metrics comparison between the data-driven model-based and WRTDS model-based daily nitrate estimates.

| Basin No.                 | Site No.      | Bias  |       |          | RMSE |       |          | Correlation |       |          | NSE  |        |          | KGE  |       |          |
|---------------------------|---------------|-------|-------|----------|------|-------|----------|-------------|-------|----------|------|--------|----------|------|-------|----------|
|                           |               | DL    | WRTDS | LOADES T | DL   | WRTDS | LOADES T | DL          | WRTDS | LOADES T | DL   | WRTDS  | LOADES T | DL   | WRTDS | LOADES T |
| Des Moines River Basin    | USGS 05481000 | 0.10  | -0.07 | -1.66    | 1.75 | 1.97  | 3.77     | 0.97        | 0.96  | 0.88     | 0.94 | 0.93   | 0.73     | 0.97 | 0.96  | 0.76     |
|                           | USGS 05482000 | -0.18 | 0.12  | -1.39    | 1.32 | 0.90  | 3.47     | 0.95        | 0.97  | 0.6      | 0.89 | 0.95   | 0.23     | 0.94 | 0.97  | 0.36     |
|                           | USGS 05482300 | 0.08  | 0.12  | -2.02    | 1.64 | 1.44  | 4.05     | 0.95        | 0.96  | 0.73     | 0.89 | 0.91   | 0.33     | 0.94 | 0.94  | 0.43     |
|                           | USGS 05482500 | 0.20  | 0.03  | -1.65    | 1.37 | 1.91  | 3.75     | 0.98        | 0.95  | 0.83     | 0.95 | 0.9    | 0.61     | 0.96 | 0.95  | 0.65     |
|                           | USGS 05484000 | -0.12 | 0.13  | -1.38    | 0.93 | 0.94  | 2.3      | 0.93        | 0.93  | 0.71     | 0.86 | 0.86   | 0.16     | 0.93 | 0.93  | 0.4      |
|                           | USGS 05484500 | -0.03 | 0     | -1.39    | 1.21 | 1.59  | 2.91     | 0.96        | 0.93  | 0.82     | 0.92 | 0.86   | 0.54     | 0.95 | 0.90  | 0.57     |
|                           | WQS 0032      | 0.03  | -0.11 | -1.32    | 2.22 | 2.06  | 6.53     | 0.87        | 0.89  | -0.25    | 0.74 | 0.78   | -0.55    | 0.85 | 0.88  | -0.36    |
|                           | WQS 0039      | 0.22  | 1.17  | -1.14    | 3.20 | 4.51  | 4.05     | 0.72        | 0.71  | 0.51     | 0.51 | 0.02   | 0.19     | 0.66 | 0.52  | 0.34     |
| Iowa River Basin          | USGS 05455100 | -0.18 | -0.12 | -1.5     | 1.21 | 1.54  | 2.65     | 0.95        | 0.91  | 0.84     | 0.89 | 0.83   | 0.48     | 0.90 | 0.91  | 0.49     |
|                           | WQS 0001      | 0     | 0     | -0.78    | 1.53 | 1.44  | 2.59     | 0.89        | 0.90  | 0.66     | 0.78 | 0.81   | 0.38     | 0.84 | 0.90  | 0.53     |
|                           | WQS 0002      | -0.03 | -0.05 | -0.83    | 1.37 | 1.44  | 2.04     | 0.89        | 0.89  | 0.79     | 0.79 | 0.77   | 0.54     | 0.88 | 0.88  | 0.6      |
|                           | WQS 0003      | 0.22  | 0.05  | -1.04    | 1.93 | 1.74  | 2.35     | 0.84        | 0.87  | 0.81     | 0.69 | 0.75   | 0.55     | 0.80 | 0.87  | 0.57     |
|                           | WQS 0005      | -0.10 | 0.19  | -0.34    | 1.88 | 1.30  | 2.18     | 0.84        | 0.93  | 0.78     | 0.70 | 0.86   | 0.6      | 0.80 | 0.91  | 0.67     |
|                           | WQS 0048      | 1.58  | 0.38  | -2.36    | 4.24 | 2.86  | 4.15     | 0.76        | 0.90  | 0.82     | 0.48 | 0.76   | 0.51     | 0.70 | 0.86  | 0.72     |
|                           | WQS 0051      | 0.70  | 0.26  | -1.81    | 2.21 | 1.90  | 3.32     | 0.83        | 0.87  | 0.67     | 0.61 | 0.71   | 0.13     | 0.81 | 0.84  | 0.61     |
|                           | WQS 0055      | -0.76 | 2.02  | -6.88    | 4.92 | 5.20  | 9.4      | 0.70        | 0.78  | 0.35     | 0.46 | 0.40   | -0.9     | 0.64 | 0.71  | -0.05    |
|                           | WQS 0072      | 0.64  | 1.62  | 9.32     | 2.12 | 8.51  | 12.49    | 0.69        | 0.37  | 0.66     | 0.19 | -11.99 | -26.48   | 0.61 | -1.89 | -2.99    |
| Iowa & Cedar River Basins | USGS 05465500 | -0.32 | -0.07 | -0.82    | 1.19 | 0.84  | 2.08     | 0.91        | 0.95  | 0.71     | 0.80 | 0.90   | 0.41     | 0.83 | 0.94  | 0.57     |
| Cedar River Basin         | USGS 05464420 | 0.10  | -0.02 | -1.18    | 1.42 | 1.17  | 2.14     | 0.88        | 0.92  | 0.78     | 0.77 | 0.84   | 0.43     | 0.87 | 0.88  | 0.63     |
|                           | WQS 0007      | 0.17  | 0     | CBC*     | 1.61 | 1.48  | CBC*     | 0.86        | 0.88  | CBC*     | 0.72 | 0.77   | CBC*     | 0.86 | 0.88  | CBC*     |
|                           | WQS 0012      | 0.38  | 0.13  | -1.73    | 2.09 | 1.68  | 3.18     | 0.74        | 0.84  | 0.47     | 0.53 | 0.70   | -0.11    | 0.65 | 0.84  | 0.18     |
|                           | WQS 0071      | 0.29  | -0.31 | -1.88    | 1.59 | 1.44  | 2.62     | 0.71        | 0.76  | 0.53     | 0.38 | 0.49   | -0.62    | 0.70 | 0.76  | 0.42     |
|                           | WQS 0083      | 0.48  | 0.38  | -0.41    | 1.31 | 1.34  | 1.88     | 0.81        | 0.78  | 0.46     | 0.40 | 0.38   | -0.16    | 0.68 | 0.69  | 0.45     |
|                           | WQS 0084      | 0.18  | 0.16  | -1.03    | 1.37 | 1.19  | 1.95     | 0.75        | 0.80  | 0.58     | 0.49 | 0.62   | -0.01    | 0.75 | 0.78  | 0.53     |

\* CBC = Cannot be computed.

Table S3.8 Nitrate loads comparison for individual sites based on data-driven model, WRTDS, and  
LOADEST.

| Basin No.                          | Site No.      | Annual Average Nitrate Load (Metric Ton per<br>Year) |         |         | Standard Error<br>(LOADEST) |
|------------------------------------|---------------|------------------------------------------------------|---------|---------|-----------------------------|
|                                    |               | DL model                                             | WRTDS   | LOADEST |                             |
| Des<br>Moines<br>River<br>Basin    | USGS 05481000 | 299452                                               | 316517  | 527026  | 300343                      |
|                                    | USGS 05482000 | 1656071                                              | 1664057 | 1841724 | 354162                      |
|                                    | USGS 05482300 | 234107                                               | 220316  | 239612  | 40681                       |
|                                    | USGS 05482500 | 536228                                               | 527248  | 803855  | 301876                      |
|                                    | USGS 05484000 | 208709                                               | 227264  | 213457  | 55592                       |
|                                    | USGS 05484500 | 1015167                                              | 1014676 | 935657  | 129500                      |
|                                    | WQS 0032      | 143226                                               | 131052  | 147607  | 41127                       |
|                                    | WQS 0039      | 165795                                               | 189969  | 307447  | 559268                      |
| Iowa River<br>Basin                | USGS 05455100 | 38450                                                | 41767   | 32557   | 22276                       |
|                                    | WQS 0001      | 539772                                               | 583029  | 612257  | 163267                      |
|                                    | WQS 0002      | 19606                                                | 20376   | 21785   | 11793                       |
|                                    | WQS 0003      | 16341                                                | 14791   | 14482   | 5935                        |
|                                    | WQS 0005      | 111471                                               | 127989  | 121149  | 72177                       |
|                                    | WQS 0048      | 57771                                                | 59898   | 41456   | 11262                       |
|                                    | WQS 0051      | 19712                                                | 21234   | 16445   | 4163                        |
|                                    | WQS 0055      | 23030                                                | 36260   | 14408   | CBC*                        |
|                                    | WQS 0072      | 142465                                               | 332930  | 220645  | 745864                      |
| Iowa &<br>Cedar<br>River<br>Basins | USGS 05465500 | 2463886                                              | 2554863 | 2948497 | 540683                      |
| Cedar<br>River<br>Basin            | USGS 05464420 | 1987445                                              | 1978376 | 2004922 | 216540                      |
|                                    | WQS 0007      | 1919468                                              | 1725684 | CBC*    | CBC*                        |
|                                    | WQS 0012      | 53902                                                | 50287   | 47172   | 8941                        |
|                                    | WQS 0071      | 21502                                                | 17709   | 14397   | 2352                        |
|                                    | WQS 0083      | 41924                                                | 45106   | 58154   | CBC*                        |
|                                    | WQS 0084      | 739711                                               | 631895  | 632404  | 143589                      |

\* CBC = Cannot be computed.

Table S3.9 Comparing the site-wise day counts when observed and estimated nitrate concentration (by DL, WRTDS, and LOADEST) exceeded the USEPA Maximum contamination level (MCL) of 10 mg/L.

| Basin No.                       | Site No.      | Total<br>observed<br>NO <sub>3</sub><br>Count | DL Model                                    |       |                                     |       |                   | WRTDS Model                         |       |                   | LOADEST Model                       |       |                   |
|---------------------------------|---------------|-----------------------------------------------|---------------------------------------------|-------|-------------------------------------|-------|-------------------|-------------------------------------|-------|-------------------|-------------------------------------|-------|-------------------|
|                                 |               |                                               | Observed NO <sub>3</sub> > 10 mg/L<br>Count | %     | Predicted NO <sub>3</sub> > 10 mg/L |       | Percent<br>Diff.* | Predicted NO <sub>3</sub> > 10 mg/L |       | Percent<br>Diff.* | Predicted NO <sub>3</sub> > 10 mg/L |       | Percent<br>Diff.* |
|                                 |               |                                               |                                             |       | Count                               | %     |                   | Count                               | %     |                   | Count                               | %     |                   |
| Des Moines<br>River Basin       | USGS 05481000 | 1208                                          | 662                                         | 54.8  | 682                                 | 56.46 | 1.66              | 658                                 | 54.47 | -0.33             | 496                                 | 41.06 | -13.74            |
|                                 | USGS 05482000 | 2093                                          | 520                                         | 24.84 | 554                                 | 26.47 | 1.62              | 552                                 | 26.37 | 1.53              | 61                                  | 2.91  | -21.93            |
|                                 | USGS 05482300 | 2529                                          | 1214                                        | 48    | 1230                                | 48.64 | 0.63              | 1256                                | 49.66 | 1.66              | 895                                 | 35.39 | -12.61            |
|                                 | USGS 05482500 | 2143                                          | 958                                         | 44.7  | 954                                 | 44.52 | -0.19             | 980                                 | 45.73 | 1.03              | 716                                 | 33.41 | -11.29            |
|                                 | USGS 05484000 | 1342                                          | 120                                         | 8.94  | 122                                 | 9.09  | 0.15              | 146                                 | 10.88 | 1.94              | 0                                   | 0     | -8.94             |
|                                 | USGS 05484500 | 2465                                          | 740                                         | 30.02 | 744                                 | 30.18 | 0.16              | 772                                 | 31.32 | 1.3               | 245                                 | 9.94  | -20.08            |
|                                 | WQS 0032      | 1359                                          | 510                                         | 37.53 | 486                                 | 35.76 | -1.77             | 498                                 | 36.64 | -0.88             | 132                                 | 9.71  | -27.81            |
|                                 | WQS 0039      | 760                                           | 299                                         | 39.34 | 236                                 | 31.05 | -8.29             | 381                                 | 50.13 | 10.79             | 74                                  | 9.74  | -29.61            |
| Iowa River<br>Basin             | USGS 05455100 | 705                                           | 73                                          | 10.35 | 39                                  | 5.53  | -4.82             | 65                                  | 9.22  | -1.13             | 0                                   | 0     | -10.35            |
|                                 | WQS 0001      | 2222                                          | 240                                         | 10.8  | 144                                 | 6.48  | -4.32             | 211                                 | 9.50  | -1.31             | 378                                 | 41.81 | -22.12            |
|                                 | WQS 0002      | 1890                                          | 63                                          | 3.33  | 33                                  | 1.75  | -1.59             | 82                                  | 4.34  | 1.01              | 0                                   | 0     | -3.33             |
|                                 | WQS 0003      | 1910                                          | 245                                         | 12.83 | 181                                 | 9.48  | -3.35             | 240                                 | 12.57 | -0.26             | 0                                   | 0     | -12.83            |
|                                 | WQS 0005      | 1751                                          | 69                                          | 3.94  | 60                                  | 3.43  | -0.51             | 106                                 | 6.05  | 2.11              | 0                                   | 0     | -3.94             |
|                                 | WQS 0048      | 904                                           | 578                                         | 63.94 | 731                                 | 80.86 | 16.92             | 555                                 | 61.39 | -2.54             | 378                                 | 41.81 | -22.12            |
|                                 | WQS 0051      | 878                                           | 407                                         | 46.36 | 530                                 | 60.36 | 14.01             | 407                                 | 46.36 | 0                 | 215                                 | 24.49 | -21.87            |
|                                 | WQS 0055      | 725                                           | 583                                         | 80.41 | 568                                 | 78.34 | -2.07             | 594                                 | 81.93 | 1.52              | 40                                  | 5.52  | -74.9             |
|                                 | WQS 0072      | 548                                           | 6                                           | 1.09  | 18                                  | 3.28  | 2.19              | 34                                  | 6.20  | 5.11              | 284                                 | 51.82 | 50.73             |
| Iowa &<br>Cedar River<br>Basins | USGS 05465500 | 2057                                          | 80                                          | 3.89  | 27                                  | 1.31  | -2.58             | 67                                  | 3.26  | -0.63             | 0                                   | 0     | -3.89             |
| Cedar River<br>Basin            | USGS 05464420 | 1722                                          | 269                                         | 15.62 | 262                                 | 15.21 | -0.41             | 245                                 | 14.23 | -1.39             | 13                                  | 0.75  | -14.87            |
|                                 | WQS 0007      | 1856                                          | 118                                         | 6.36  | 109                                 | 5.87  | -0.48             | 82                                  | 4.42  | -1.94             | 0                                   | 0     | -6.36             |
|                                 | WQS 0012      | 1373                                          | 656                                         | 47.78 | 689                                 | 50.18 | 2.4               | 686                                 | 49.96 | 2.18              | 23                                  | 1.68  | -46.1             |
|                                 | WQS 0071      | 621                                           | 70                                          | 11.27 | 78                                  | 12.56 | 1.29              | 55                                  | 8.86  | -2.42             | 0                                   | 0     | -11.27            |
|                                 | WQS 0083      | 291                                           | 0                                           | 0     | 1                                   | 0.34  | 0.34              | 0                                   | 0     | 0                 | 4                                   | 1.37  | 1.37              |
|                                 | WQS 0084      | 360                                           | 662                                         | 1.94  | 3                                   | 0.83  | -1.11             | 9                                   | 2.5   | 0.56              | 0                                   | 0     | -1.94             |

\* Percent Diff = Percentage difference between Observed and Predicted NO<sub>3</sub> greater than USEPA MCL (10 mg/L)

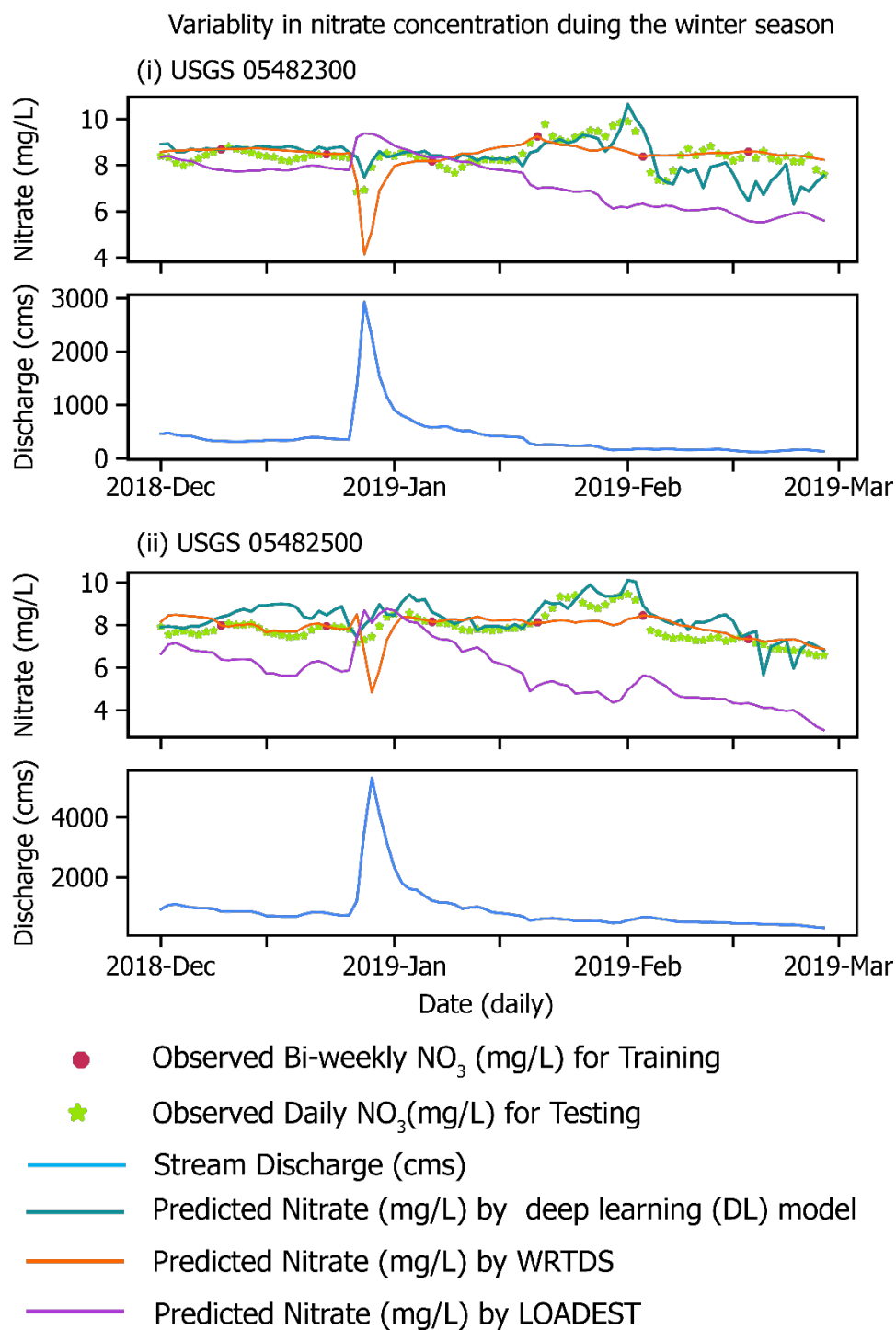


Figure S3.4 Nitrate concentration variabilities estimated based on DL, WRTDS, and LOADEST models and their fluctuations during significant stream discharge events.

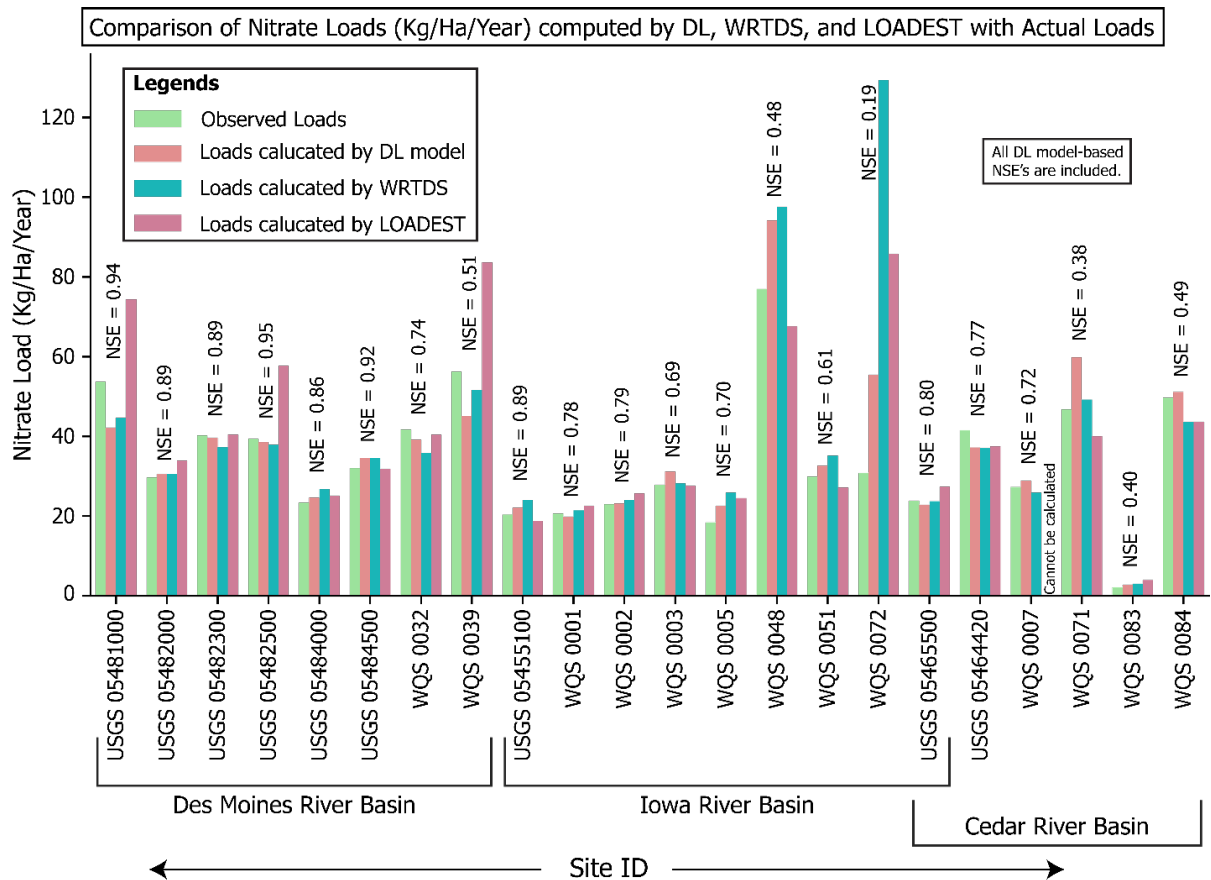


Figure S3.5 Comparing observed daily nitrate load with estimated daily nitrate loads (kg per ha per day) calculated based on DL, WRTDS, and LOADEST models for each nitrate monitoring sites within Des Moines, Iowa, and Cedar River basins.

## Appendix C

### Details on the nitrate monitoring site's location

Most of the nitrate monitoring sites are situated in Des Moines, Racoon, Iowa, and Cedar rivers.

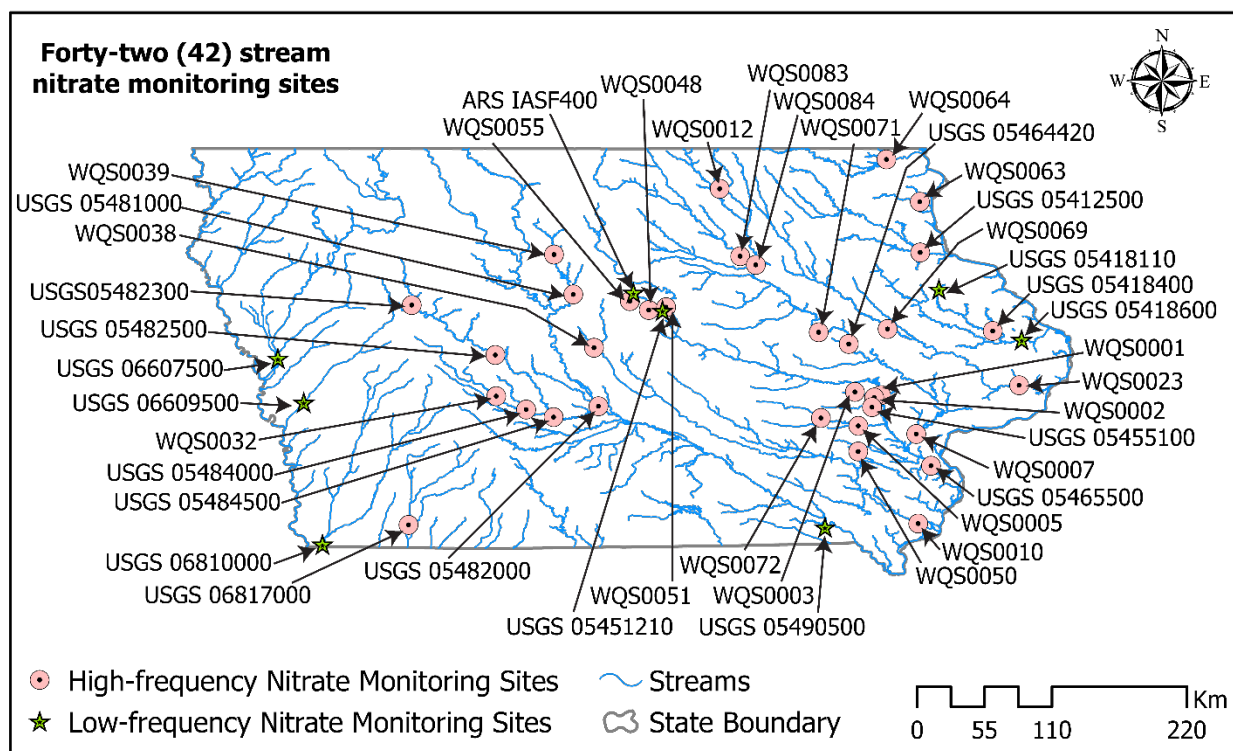


Figure S4.1 Site IDs and their respective stream locations of forty-two (42) nitrate monitoring sites in Iowa: thirty-four (34) high-frequency nitrate monitoring sites and eight (8) low-frequency nitrate monitoring sites.



Table S4.1 Data availability, training, and testing periods for Iowa's thirty-four high-frequency sensor-based nitrate monitoring sites and eight (8) low-frequency nitrate monitoring sites.

| Site No.      | Data Availability Period | Training Period for Scenario 1 and 2, and Testing period for Scenario 1* | Testing Periods for Scenario 2                                                                                                                  |
|---------------|--------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| USGS 05412500 | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| USGS 05418400 | 2012 - 2014              | 2012 - Jun 2014                                                          | Jun 2014 - Dec 2014                                                                                                                             |
| USGS 05455100 | 2012 - 2015              | 2012 - Jun 2015                                                          | Jun 2015 - Dec 2015                                                                                                                             |
| USGS 05464420 | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| USGS 05465500 | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| USGS 05481000 | 2012 - 2017              | 2012 - 2016                                                              | 2017                                                                                                                                            |
| USGS 05482000 | 2013 - 2019              | 2013 - 2018                                                              | 2019                                                                                                                                            |
| USGS 05482300 | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| USGS 05482500 | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| USGS 05484000 | 2016 - 2019              | 2016 - 2018                                                              | 2019                                                                                                                                            |
| USGS 05484500 | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| USGS 06817000 | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0001      | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0002      | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0003      | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0005      | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0007      | 2012 - 2019              | 2012 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0010      | 2013 - 2019              | 2013 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0012      | 2014 - 2019              | 2014 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0023      | 2014 - 2019              | 2014 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0032      | 2014 - 2019              | 2014 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0038      | 2016 - 2019              | 2016 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0039      | 2016 - 2019              | 2016 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0048      | 2016 - 2019              | 2016 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0050      | 2016 - 2019              | 2016 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0051      | 2016 - 2019              | 2017 - 2019                                                              | 2016                                                                                                                                            |
| WQS 0055      | 2017 - 2019              | 2018 - 2019                                                              | 2017                                                                                                                                            |
| WQS 0063      | 2017 - 2019              | 2018 - 2019                                                              | 2017                                                                                                                                            |
| WQS 0064      | 2017 - 2019              | 2018 - 2019                                                              | 2017                                                                                                                                            |
| WQS 0069      | 2017 - 2019              | 2017 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0071      | 2017 - 2019              | 2017 - 2018                                                              | 2019                                                                                                                                            |
| WQS 0072      | 2017 - 2019              | 2018 - 2019                                                              | 2017                                                                                                                                            |
| WQS 0083      | 2018 - 2019              | Aug 2018 – Dec 2019                                                      | Jan 2018 – Aug 2018                                                                                                                             |
| WQS 0084      | 2018 - 2019              | Aug 2018 – Dec 2019                                                      | Jan 2018 – Aug 2018                                                                                                                             |
| ARS_IASF400   | 2012 - 2015              | 2012 - 2014                                                              | Biweekly nitrate data of these low-frequency nitrate monitoring sites were used for model development. These sites were not used more analysis. |
| USGS_05418110 | 2012 - 2014              | 2012 – Mar 2014                                                          |                                                                                                                                                 |
| USGS_05418600 | 2012 - 2014              | 2012 – Mar 2014                                                          |                                                                                                                                                 |
| USGS_05451210 | 2012 - 2019              | 2012 - 2017                                                              |                                                                                                                                                 |
| USGS_05490500 | 2012 - 2019              | 2012 - 2017                                                              |                                                                                                                                                 |
| USGS_06607500 | 2012 - 2014              | 2012 - Mar 2014                                                          |                                                                                                                                                 |
| USGS_06609500 | 2012 - 2014              | 2012 - Mar 2014                                                          |                                                                                                                                                 |
| USGS_06810000 | 2012 - 2014              | 2012 - Mar 2014                                                          |                                                                                                                                                 |

\* In Scenario 1, biweekly subsampled nitrate data were used for a specific period to train the model. The remaining daily nitrate data for the same period were used for model performance evaluation.

### Dynamic Time Warping (DTW) Method

Dynamic time warping is a widely used time series analysis method that measures similarities between two-time sequences (Vintsyuk, 1968). The method generally calculated the optimal time match between two specific sequences by following a few rules: (i) each index of the first time sequence should be matched with one or more indices of the second time sequence, and vice versa; (ii) the first index of first time sequence should be matched with the first index of the second time sequence, (iii) the last index from the first time sequence should be matched with the last index of the second time sequence; (iv) the mapping of the indices of the first time sequence to indices of the second time sequence must be monotonically increasing. The method warped time sequences non-linearly in the time dimension to determine a measure of their similarity. Though DTW measures a distance-like quantity between two-time sequences, the method does not guarantee that the triangle inequality holds. The method is highly used in automatic speech recognition (Müller et al., 2006), online signature recognition, and shape matching (Dupas et al., 2015).

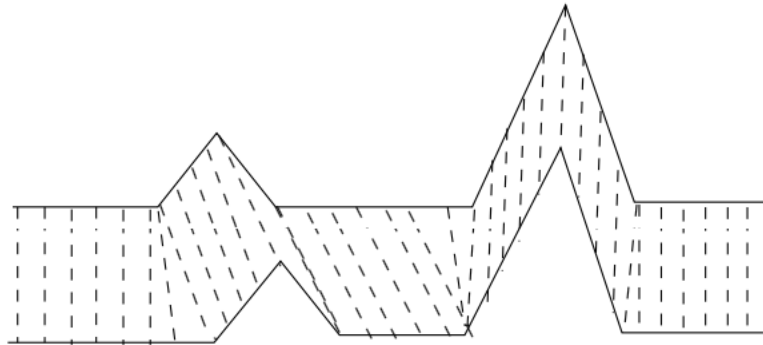


Figure S4.2 Dynamic Time Warping (DTW) between two-time sequence pieces. The dotted lines represented the time-warp relation. One point in the upper time sequence is connected to several points in the lower time sequence and vice versa.

Table S4.2 Target (33) and input (1) high-frequency nitrate monitoring sites used for the eight (8) deep learning (DL) models development.

| Site No.      | Model 1 | Model 2 | Model 3 | Model 4 | Model 5 | Model 6 | Model 7 | Model 8 |
|---------------|---------|---------|---------|---------|---------|---------|---------|---------|
| USGS 05412500 | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| USGS 05418400 | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| USGS 05455100 | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| USGS 05464420 | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| USGS 05465500 | 1       | 1       | 1       | 2       | 1       | 1       | 1       | 1       |
| USGS 05481000 | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| USGS 05482000 | 1       | 1       | 1       | 1       | 1       | 2       | 1       | 1       |
| USGS 05482300 | 2       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| USGS 05482500 | 1       | 1       | 1       | 1       | 2       | 1       | 1       | 1       |
| USGS 05484000 | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| USGS 05484500 | 1       | 2       | 1       | 1       | 1       | 1       | 1       | 1       |
| USGS 06817000 | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0001      | 1       | 1       | 2       | 1       | 1       | 1       | 1       | 1       |
| WQS 0002      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0003      | 1       | 1       | 1       | 1       | 1       | 1       | 2       | 1       |
| WQS 0005      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0007      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 2       |
| WQS 0010      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0012      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0023      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0032      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0038      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0039      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0048      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0050      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0051      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0055      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0063      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0064      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0069      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0071      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0072      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0083      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |
| WQS 0084      | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |

1 = Target nitrate monitoring sites with biweekly subsampled nitrate data

2 = Input nitrate monitoring sites with daily nitrate and stream discharge data

Table S4.3 Combinations of hyper-parameters used to develop DL models and identify the best hyper-parameters combinations for each model under Scenario 1 and 2

| Model No.                | No. of epoch | Batch size | Hidden layer size | Time window | Random seeds |
|--------------------------|--------------|------------|-------------------|-------------|--------------|
| Model 1<br>to<br>Model 8 | 2000, 2500   | 21, 41     | 45, 50, 60        | 150, 180    | 0, 1, 3, 5   |

Table S4.4 Median test-period evaluation metrics of eight (8) DL models under Scenario 1 and Scenario 2 with the combinations of hyper-parameters used to develop DL models.

| Scenario No | Model No | Epoch | Batch Size | Hidden Layer Size | Time window | Testing Period |      |       |             |      |
|-------------|----------|-------|------------|-------------------|-------------|----------------|------|-------|-------------|------|
|             |          |       |            |                   |             | Bias           | RMSE | NSE   | Correlation | KGE  |
| Scenario 1  | Model 1  | 2500  | 41         | 60                | 150         | 0.15           | 1.53 | 0.80  | 0.89        | 0.87 |
| Scenario 1  | Model 2  | 2500  | 41         | 60                | 150         | -0.01          | 1.46 | 0.80  | 0.9         | 0.86 |
| Scenario 1  | Model 3  | 2500  | 41         | 60                | 150         | 0.15           | 1.74 | 0.73  | 0.87        | 0.84 |
| Scenario 1  | Model 4  | 2500  | 41         | 50                | 150         | 0.20           | 1.70 | 0.79  | 0.89        | 0.85 |
| Scenario 1  | Model 5  | 2500  | 41         | 60                | 150         | 0.22           | 1.59 | 0.79  | 0.90        | 0.85 |
| Scenario 1  | Model 6  | 2500  | 41         | 50                | 150         | 0.44           | 2.66 | 0.46  | 0.76        | 0.65 |
| Scenario 1  | Model 7  | 2500  | 41         | 50                | 180         | 0.34           | 1.92 | 0.74  | 0.89        | 0.82 |
| Scenario 1  | Model 8  | 2500  | 41         | 50                | 150         | 0.26           | 1.73 | 0.74  | 0.87        | 0.83 |
| Scenario 2  | Model 1  | 2500  | 41         | 60                | 150         | 0.91           | 1.70 | 0.49  | 0.86        | 0.62 |
| Scenario 2  | Model 2  | 2500  | 41         | 60                | 150         | 0.44           | 1.70 | 0.46  | 0.85        | 0.67 |
| Scenario 2  | Model 3  | 2500  | 41         | 60                | 150         | 0.82           | 1.81 | 0.12  | 0.84        | 0.48 |
| Scenario 2  | Model 4  | 2500  | 41         | 50                | 150         | 0.85           | 2.02 | 0.24  | 0.82        | 0.49 |
| Scenario 2  | Model 5  | 2500  | 41         | 60                | 150         | 1.24           | 2.07 | 0.19  | 0.79        | 0.58 |
| Scenario 2  | Model 6  | 2500  | 41         | 50                | 150         | 1.93           | 2.85 | -0.69 | 0.69        | 0.22 |
| Scenario 2  | Model 7  | 2500  | 41         | 50                | 180         | 1.14           | 2.34 | -0.03 | 0.82        | 0.44 |
| Scenario 2  | Model 8  | 2500  | 41         | 50                | 150         | 1.19           | 2.88 | -0.45 | 0.76        | 0.28 |

Table S4.5 Individual best model performance (based on test-period evaluation metrics of eight (8) developed DL models) for each of thirty-four (34) nitrate monitoring sites under Scenario 1 and Scenario 2.

| Site No       | Evaluation Metrics for Scenario 1 |      |       |       |      | Model No | Input High-freq.<br>Nitrate Site | Evaluation Metrics for Scenario 2 |      |       |       |      | Model No | Input High-freq.<br>Nitrate Site |
|---------------|-----------------------------------|------|-------|-------|------|----------|----------------------------------|-----------------------------------|------|-------|-------|------|----------|----------------------------------|
|               | Bias                              | RMSE | Corr. | NSE   | KGE  |          |                                  | Bias                              | RMSE | Corr. | NSE   | KGE  |          |                                  |
| USGS 05412500 | 0.05                              | 1.00 | 0.93  | 0.87  | 0.90 | Model 7  | WQS 0003                         | 1.63                              | 2.44 | 0.72  | -1.81 | 0.14 | Model 4  | USGS 05465500                    |
| USGS 05418400 | -0.24                             | 1.25 | 0.91  | 0.78  | 0.69 | Model 5  | USGS 05482500                    | 0.02                              | 0.23 | 0.99  | 0.98  | 0.98 | Model 5  | USGS 05482500                    |
| USGS 05455100 | -0.11                             | 1.06 | 0.96  | 0.93  | 0.90 | Model 2  | USGS 05484500                    | -0.01                             | 0.40 | 0.98  | 0.97  | 0.94 | Model 5  | USGS 05482500                    |
| USGS 05464420 | 0                                 | 1.12 | 0.93  | 0.86  | 0.90 | Model 1  | USGS 05482300                    | 0.70                              | 1.65 | 0.88  | 0.28  | 0.47 | Model 4  | USGS 05465500                    |
| USGS 05465500 | 0.07                              | 1.00 | 0.94  | 0.88  | 0.88 | Model 8  | WQS 0007                         | 0.41                              | 1.24 | 0.79  | 0.55  | 0.77 | Model 6  | USGS 05482000                    |
| USGS 05481000 | 0.05                              | 1.71 | 0.97  | 0.95  | 0.97 | Model 8  | WQS 0007                         | 0.07                              | 0.40 | 0.99  | 0.99  | 0.99 | Model 1  | USGS 05482300                    |
| USGS 05482000 | 0.12                              | 1.34 | 0.94  | 0.89  | 0.93 | Model 2  | USGS 05484500                    | 1.42                              | 2.05 | 0.80  | 0.03  | 0.61 | Model 2  | USGS 05484500                    |
| USGS 05482300 | -0.09                             | 1.47 | 0.96  | 0.92  | 0.94 | Model 2  | USGS 05484500                    | 0.44                              | 2.03 | 0.73  | 0.30  | 0.67 | Model 2  | USGS 05484500                    |
| USGS 05482500 | -0.01                             | 1.36 | 0.98  | 0.96  | 0.96 | Model 2  | USGS 05484500                    | 0.67                              | 1.45 | 0.90  | 0.68  | 0.81 | Model 2  | USGS 05484500                    |
| USGS 05484000 | -0.30                             | 1.02 | 0.93  | 0.85  | 0.91 | Model 2  | USGS 05484500                    | 0.26                              | 1.04 | 0.84  | 0.68  | 0.83 | Model 2  | USGS 05484500                    |
| USGS 05484500 | 0.02                              | 1.06 | 0.97  | 0.94  | 0.96 | Model 1  | USGS 05482300                    | 0.73                              | 1.40 | 0.86  | 0.64  | 0.77 | Model 1  | USGS 05482300                    |
| USGS 06817000 | 0.04                              | 1.01 | 0.92  | 0.85  | 0.88 | Model 5  | USGS 05482500                    | 0.96                              | 1.17 | 0.89  | 0.20  | 0.62 | Model 1  | USGS 05482300                    |
| WQS 0001      | -0.11                             | 1.54 | 0.89  | 0.80  | 0.87 | Model 1  | USGS 05482300                    | 0.61                              | 1.35 | 0.88  | 0.53  | 0.68 | Model 4  | USGS 05465500                    |
| WQS 0002      | 0.16                              | 0.96 | 0.95  | 0.90  | 0.93 | Model 2  | USGS 05484500                    | 2.31                              | 3.01 | 0.69  | -0.43 | 0.32 | Model 7  | WQS 0003                         |
| WQS 0003      | -0.09                             | 1.45 | 0.91  | 0.83  | 0.90 | Model 2  | USGS 05484500                    | 2.48                              | 3.59 | 0.50  | -0.79 | 0.28 | Model 2  | USGS 05484500                    |
| WQS 0005      | 0.17                              | 1.38 | 0.92  | 0.85  | 0.91 | Model 5  | USGS 05482500                    | 1.27                              | 2.31 | 0.85  | 0.19  | 0.46 | Model 3  | WQS 0001                         |
| WQS 0007      | 0.10                              | 1.13 | 0.93  | 0.87  | 0.90 | Model 1  | USGS 05482300                    | 1.18                              | 2.77 | 0.41  | -0.27 | 0.33 | Model 4  | USGS 05465500                    |
| WQS 0010      | 0.03                              | 1.39 | 0.90  | 0.80  | 0.85 | Model 5  | USGS 05482500                    | 0.44                              | 0.95 | 0.92  | 0.81  | 0.80 | Model 2  | USGS 05484500                    |
| WQS 0012      | -0.03                             | 1.81 | 0.82  | 0.68  | 0.74 | Model 2  | USGS 05484500                    | 0.73                              | 1.70 | 0.72  | 0.29  | 0.70 | Model 2  | USGS 05484500                    |
| WQS 0023      | 0.11                              | 1.18 | 0.93  | 0.86  | 0.89 | Model 1  | USGS 05482300                    | 0.46                              | 1.14 | 0.91  | 0.79  | 0.84 | Model 6  | USGS 05482000                    |
| WQS 0032      | -0.11                             | 2.17 | 0.89  | 0.79  | 0.82 | Model 4  | USGS 05465500                    | 0.36                              | 2.53 | 0.46  | 0.04  | 0.43 | Model 2  | USGS 05484500                    |
| WQS 0038      | 0.17                              | 1.92 | 0.89  | 0.80  | 0.88 | Model 2  | USGS 05484500                    | 1.41                              | 2.28 | 0.89  | 0.62  | 0.74 | Model 1  | USGS 05482300                    |
| WQS 0039      | 1.03                              | 3.49 | 0.71  | 0.45  | 0.65 | Model 2  | USGS 05484500                    | 1.65                              | 3.30 | 0.69  | 0.27  | 0.56 | Model 2  | USGS 05484500                    |
| WQS 0048      | -0.21                             | 2.14 | 0.93  | 0.86  | 0.91 | Model 1  | USGS 05482300                    | 5.52                              | 7.69 | 0.47  | -2.18 | 0.07 | Model 3  | WQS 0001                         |
| WQS 0050      | -0.21                             | 2.41 | 0.88  | 0.76  | 0.72 | Model 2  | USGS 05484500                    | -0.48                             | 2.68 | 0.84  | 0.66  | 0.60 | Model 3  | WQS 0001                         |
| WQS 0051      | 2.53                              | 3.77 | 0.77  | -0.03 | 0.60 | Model 2  | USGS 05484500                    | -0.27                             | 0.50 | 0.92  | 0.79  | 0.91 | Model 2  | USGS 05484500                    |
| WQS 0055      | -1.23                             | 4.57 | 0.55  | 0.06  | 0.54 | Model 2  | USGS 05484500                    | -0.51                             | 1.61 | 0.99  | 0.97  | 0.93 | Model 2  | USGS 05484500                    |
| WQS 0063      | 1.27                              | 1.84 | 0.55  | -2.16 | 0.28 | Model 1  | USGS 05482300                    | 0.04                              | 0.84 | 0.91  | 0.83  | 0.82 | Model 5  | USGS 05482500                    |

| Site No  | Evaluation Metrics for Scenario 1 |      |       |       |      | Model No | Input High-freq.<br>Nitrate Site | Evaluation Metrics for Scenario 2 |      |       |      |       | Model No | Input High-freq.<br>Nitrate Site |
|----------|-----------------------------------|------|-------|-------|------|----------|----------------------------------|-----------------------------------|------|-------|------|-------|----------|----------------------------------|
|          | Bias                              | RMSE | Corr. | NSE   | KGE  |          |                                  | Bias                              | RMSE | Corr. | NSE  | KGE   |          |                                  |
| WQS 0064 | 1.00                              | 1.64 | 0.74  | 0.28  | 0.62 | Model 4  | USGS 05465500                    | 0.04                              | 0.33 | 0.98  | 0.96 | 0.94  | Model 2  | USGS 05484500                    |
| WQS 0069 | -0.05                             | 1.25 | 0.87  | 0.75  | 0.87 | Model 2  | USGS 05484500                    | 1.00                              | 1.95 | 0.77  | 0.28 | 0.67  | Model 2  | USGS 05484500                    |
| WQS 0071 | -0.13                             | 1.41 | 0.64  | 0.38  | 0.58 | Model 8  | WQS 0007                         | 0.76                              | 1.68 | 0.74  | 0.42 | 0.62  | Model 3  | WQS 0001                         |
| WQS 0072 | 1.80                              | 2.74 | 0.63  | -0.29 | 0.40 | Model 2  | USGS 05484500                    | -0.30                             | 2.30 | 0.49  | 0.06 | -0.08 | Model 3  | WQS 0001                         |
| WQS 0083 | 0.71                              | 1.90 | 0.79  | -0.24 | 0.32 | Model 1  | USGS 05482300                    | 0.26                              | 0.99 | 0.86  | 0.71 | 0.70  | Model 3  | WQS 0001                         |
| WQS 0084 | 1.17                              | 2.23 | 0.70  | -0.46 | 0.44 | Model 1  | USGS 05482300                    | -0.68                             | 1.28 | 0.85  | 0.61 | 0.68  | Model 5  | USGS 05482500                    |

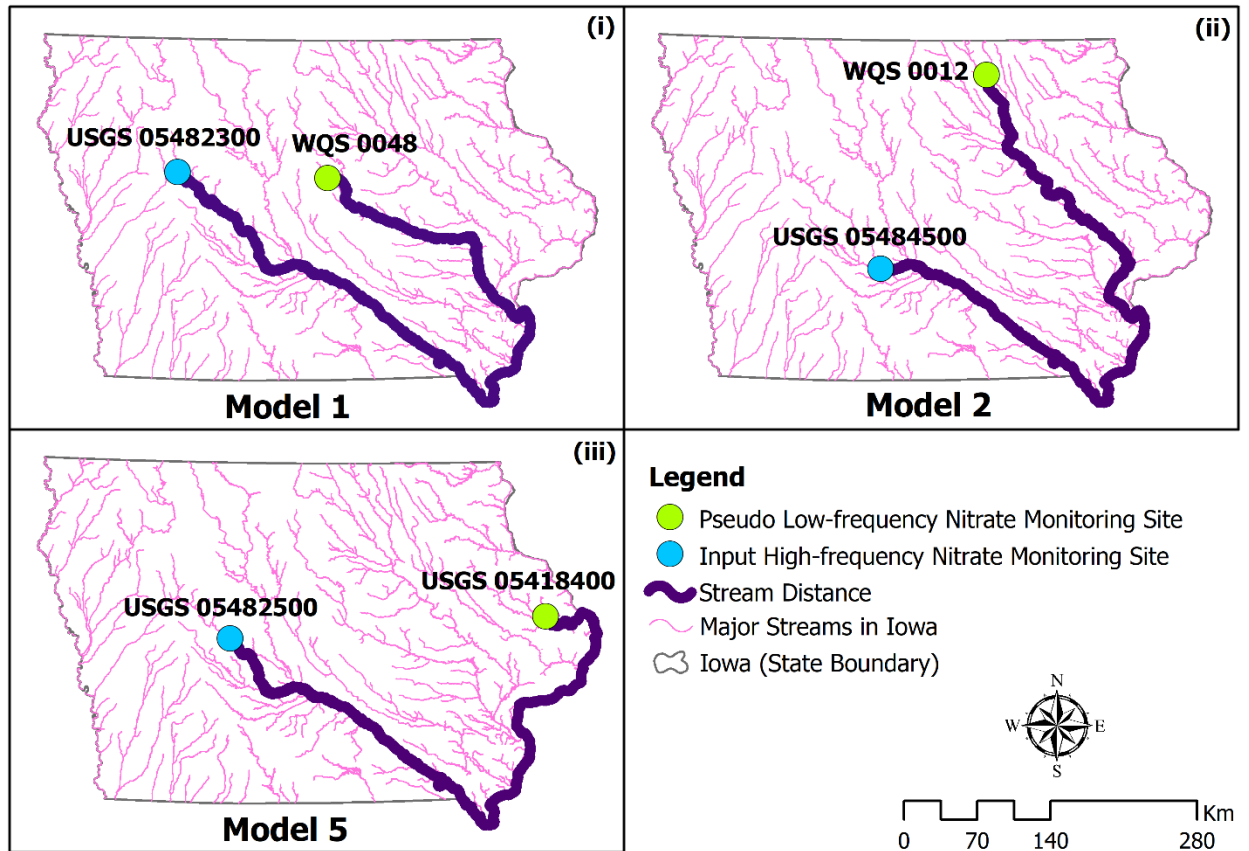


Figure S4.3 Stream distances between the pseudo-low-frequency nitrate monitoring sites and respective input high-frequency nitrate monitoring sites used for DL models development: (i) stream distance between pseudo-low-frequency site, WQS 0048 and input high-frequency site, USGS 05482300 (Model 1); (ii) stream distance between pseudo-low-frequency site, WQS 0012 and input high-frequency site, USGS 05484500 (Model 2); (iii) stream distance between pseudo-low-frequency site, USGS 05418400 and input high-frequency site, USGS 05482500 (Model 5).

Table S4.6 Euclidean distances between the target biweekly subsampled nitrate monitoring sites and respective input high-frequency nitrate monitoring site (based on the identified best models for each site) in Scenario 1

| Site No.      | Model No. | NSE values | Input High-Freq. Nitrate Sites | Euclidean Distance (Km) |
|---------------|-----------|------------|--------------------------------|-------------------------|
| USGS 05412500 | Model 7   | 0.87       | WQS0003                        | 120                     |
| USGS 05418400 | Model 5   | 0.78       | USGS 05482500                  | 302                     |
| USGS 05455100 | Model 2   | 0.93       | USGS 05484500                  | 195                     |
| USGS 05464420 | Model 1   | 0.86       | USGS 05482300                  | 266                     |
| USGS 05465500 | Model 8   | 0.88       | WQS 0007                       | 27                      |
| USGS 05481000 | Model 8   | 0.95       | WQS 0007                       | 237                     |
| USGS 05482000 | Model 2   | 0.89       | USGS 05484500                  | 29                      |
| USGS 05482300 | Model 2   | 0.92       | USGS 05484500                  | 126                     |
| USGS 05482500 | Model 2   | 0.96       | USGS 05484500                  | 62                      |
| USGS 05484000 | Model 2   | 0.85       | USGS 05484500                  | 18                      |
| USGS 05484500 | Model 1   | 0.94       | USGS 05482300                  | 126                     |
| USGS 06817000 | Model 5   | 0.85       | USGS 05482500                  | 148                     |
| WQS 0001      | Model 1   | 0.80       | USGS 05482300                  | 294                     |
| WQS 0002      | Model 2   | 0.90       | USGS 05484500                  | 197                     |
| WQS 0003      | Model 2   | 0.83       | USGS 05484500                  | 185                     |
| WQS 0005      | Model 5   | 0.85       | USGS 05482500                  | 229                     |
| WQS 0007      | Model 1   | 0.87       | USGS 05482300                  | 324                     |
| WQS 0010      | Model 5   | 0.80       | USGS 05482500                  | 293                     |
| WQS 0012      | Model 2   | 0.68       | USGS 05484500                  | 211                     |
| WQS 0023      | Model 1   | 0.86       | USGS 05482300                  | 374                     |
| WQS 0032      | Model 4   | 0.79       | USGS 05465500                  | 272                     |
| WQS 0038      | Model 2   | 0.80       | USGS 05484500                  | 61                      |
| WQS 0039      | Model 2   | 0.45       | USGS 05484500                  | 132                     |
| WQS 0048      | Model 1   | 0.86       | USGS 05482300                  | 143                     |
| WQS 0050      | Model 2   | 0.76       | USGS 05484500                  | 189                     |
| WQS 0051      | Model 2   | -0.03      | USGS 05484500                  | 113                     |
| WQS 0055      | Model 2   | 0.06       | USGS 05484500                  | 105                     |
| WQS 0063      | Model 1   | -2.16      | USGS 05482300                  | 316                     |
| WQS 0064      | Model 4   | 0.28       | USGS 05465500                  | 251                     |
| WQS 0069      | Model 2   | 0.75       | USGS 05484500                  | 216                     |
| WQS 0071      | Model 8   | 0.38       | WQS 0007                       | 102                     |
| WQS 0072      | Model 2   | -0.29      | USGS 05484500                  | 164                     |
| WQS 0083      | Model 1   | -0.24      | USGS 05482300                  | 202                     |
| WQS 0084      | Model 1   | -0.46      | USGS 05482300                  | 210                     |



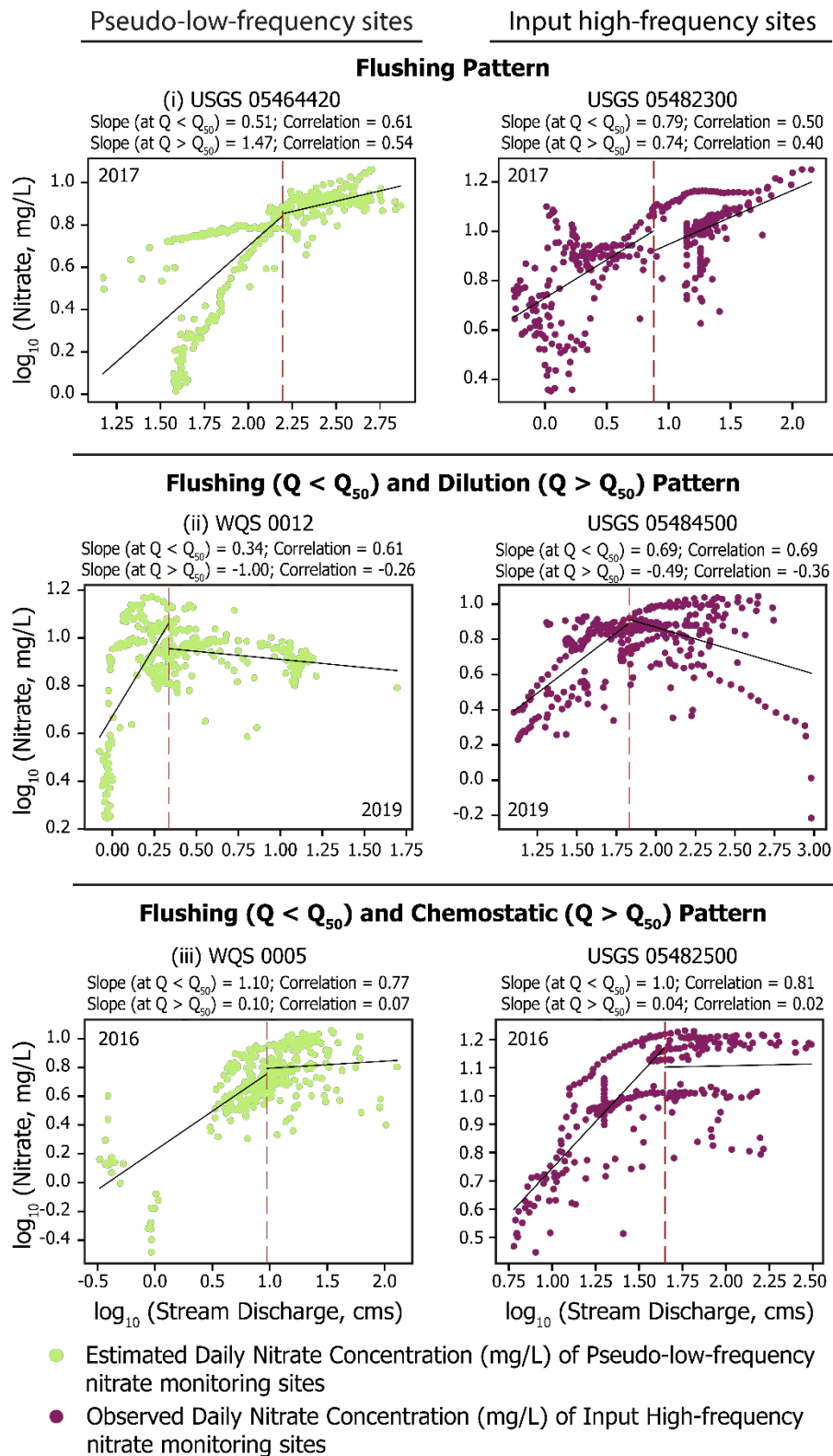


Figure S4.4 Dominant nitrate concentration ( $c$ ) and stream discharge ( $Q$ ) relationship patterns were determined using estimated nitrate concentrations and stream discharge of pseudo-low-frequency sites (left-side plots). Observed nitrate  $c$ - $Q$  relationship patterns of input high-frequency nitrate monitoring sites, which contributed to the best prediction performance for respective pseudo-low-frequency sites, were also shown in the right-side plots. Three dominant patterns were identified from the analyses (i)

flushing pattern (where pseudo-low-frequency and high-frequency input sites were USGS 05464420 and USGS 05482300), (ii) flushing before  $Q$  less than median  $Q$  ( $Q < Q_{50}$ ) and dilution after  $Q$  greater than median  $Q$  ( $Q > Q_{50}$ ) (pseudo-low-frequency and high-frequency input sites: WQS 0012 and USGS 05484500), and (iii) flushing ( $Q < Q_{50}$ ) and chemostatic ( $Q > Q_{50}$ ) (pseudo-low-frequency and high-frequency input sites: WQS 0005 and USGS 05482500).

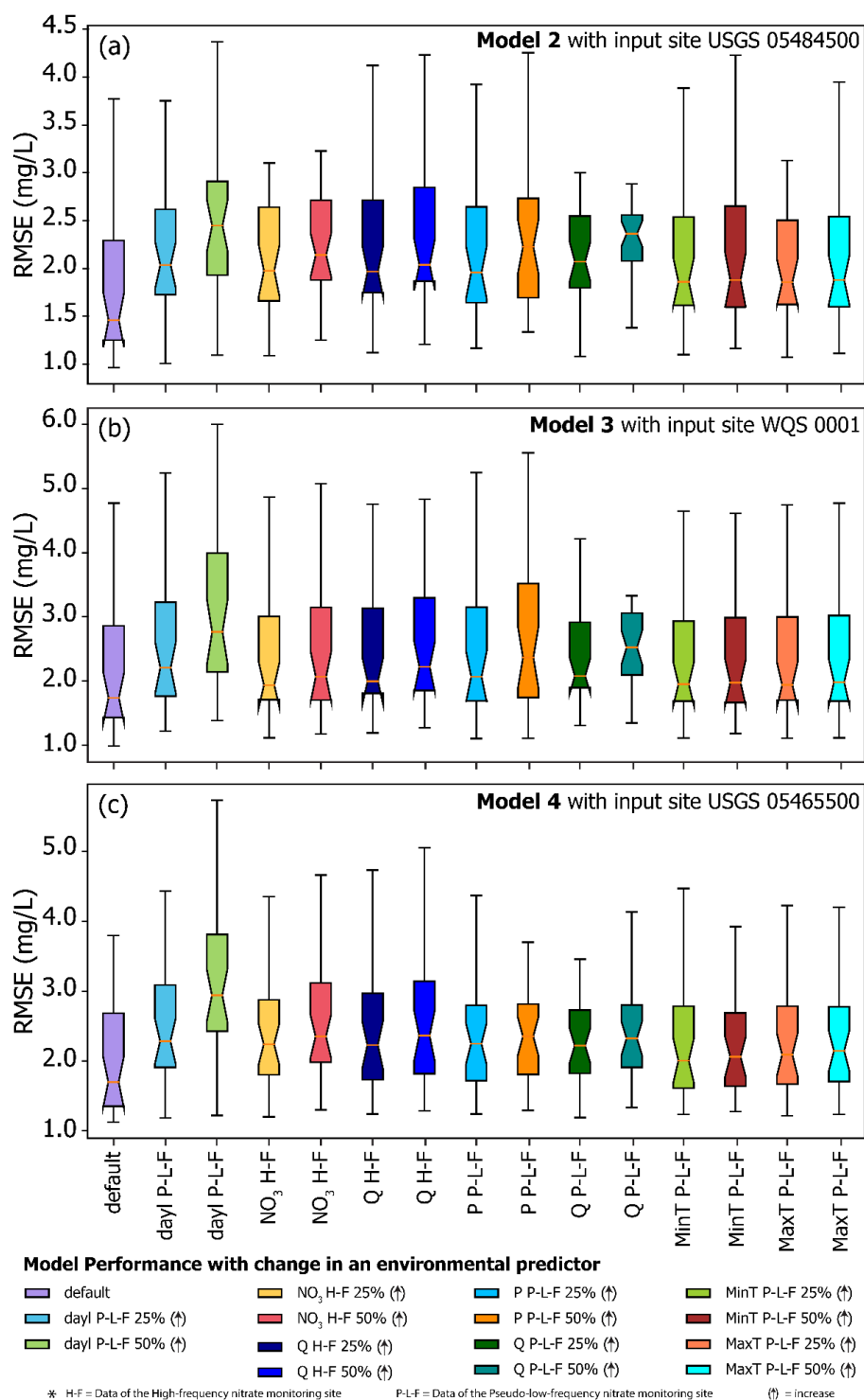


Figure S4.5 Boxplots of RMSEs calculated for Model 2, Model 3, and Model 4 after increasing (directional sign) 25% and 50% of the magnitude of each daily varying environmental characteristics, including day length (dayl), stream discharge (Q), precipitation (P), maximum temperature (MaxT), minimum temperature (MinT) of pseudo-low-frequency (P-L-F) nitrate monitoring sites, and daily nitrate concentrations (NO<sub>3</sub>) and stream discharge (Q) of respective input high-frequency nitrate monitoring (H-F) site.

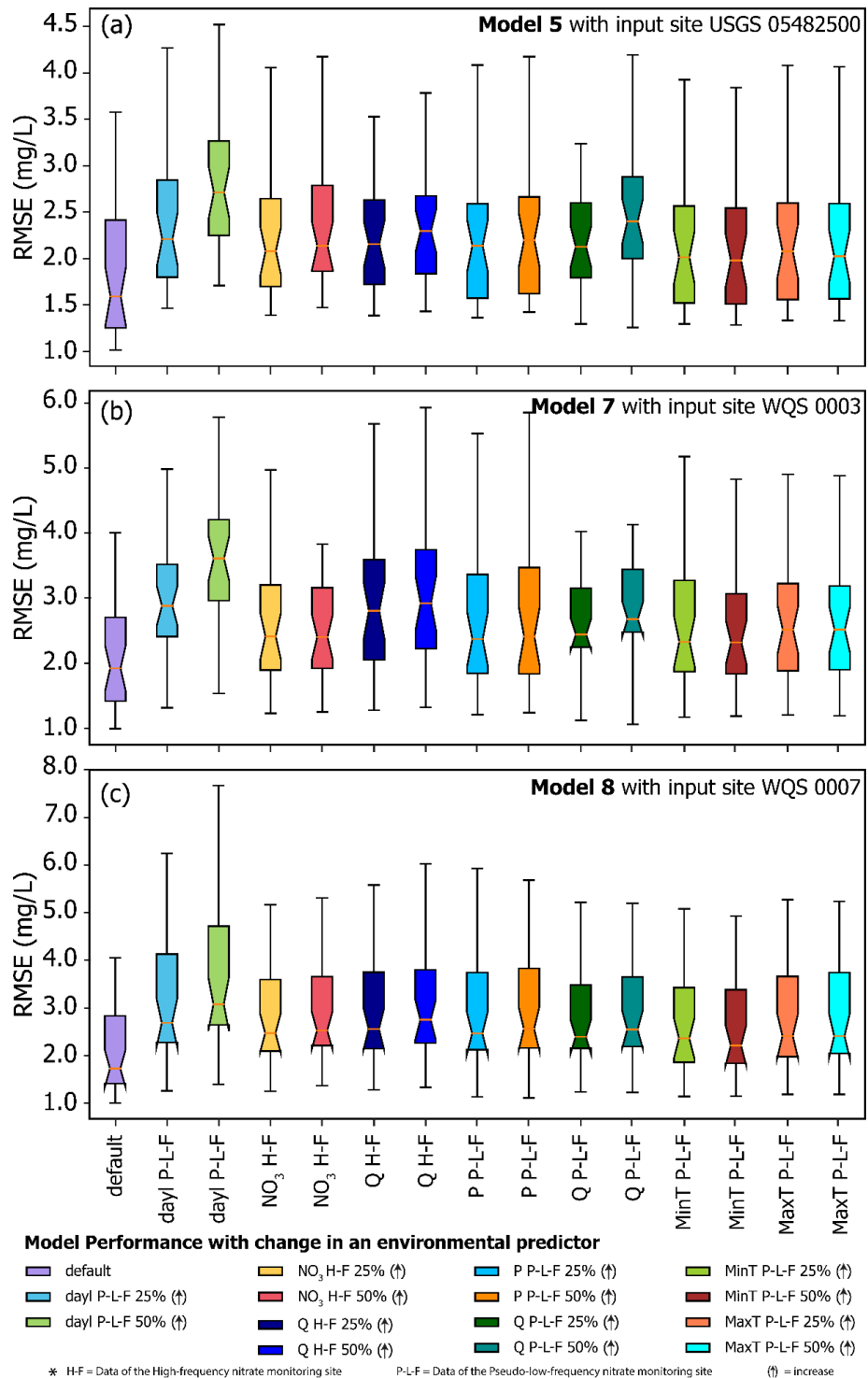


Figure S4.6 Boxplots of RMSEs calculated for Model 5, Model 7, and Model 8 after increasing (directional sign) 25% and 50% of the magnitude of each daily varying environmental characteristics, including day length (dayl), stream discharge (Q), precipitation (P), maximum temperature (MaxT), minimum temperature (MinT) of pseudo-low-frequency (P-L-F) nitrate monitoring sites, and daily nitrate concentrations (NO<sub>3</sub>) and stream discharge (Q) of respective input high-frequency nitrate monitoring (H-F) site.

Table S4.7 Site-wise similarity indices for Day length and Stream Discharge and their normalized values.

| Sites ID      | Similarity Index based on Day Length | Similarity Index based on Stream Discharge | Similarity Index based on Day Length (Normalized) | Similarity Index based on Stream Discharge (Normalized) | Model No | Input high-frequency nitrate monitoring sites |
|---------------|--------------------------------------|--------------------------------------------|---------------------------------------------------|---------------------------------------------------------|----------|-----------------------------------------------|
| USGS 05412500 | 2637.19                              | 531725.2                                   | 0.86                                              | 0.15                                                    | Model 7  | WQS0003                                       |
| USGS 05418400 | 9.5                                  | 323428.47                                  | 0                                                 | 0.09                                                    | 5        | USGS 05482500                                 |
| USGS 05455100 | 122.1                                | 759334.1                                   | 0.04                                              | 0.22                                                    | 2        | USGS 05484500                                 |
| USGS 05464420 | 59.02                                | 2060777.14                                 | 0.02                                              | 0.6                                                     | 1        | USGS 05482300                                 |
| USGS 05465500 | 58.52                                | 2867506.2                                  | 0.02                                              | 0.83                                                    | 8        | WQS 0007                                      |
| USGS 05481000 | 5.07                                 | 2257426.33                                 | 0                                                 | 0.66                                                    | 8        | WQS 0007                                      |
| USGS 05482000 | 2331.92                              | 1677358.33                                 | 0.76                                              | 0.49                                                    | 2        | USGS 05484500                                 |
| USGS 05482300 | 124.49                               | 642032.09                                  | 0.04                                              | 0.19                                                    | 2        | USGS 05484500                                 |
| USGS 05482500 | 80.49                                | 577824.58                                  | 0.03                                              | 0.17                                                    | 2        | USGS 05484500                                 |
| USGS 05484000 | 57.28                                | 718964.02                                  | 0.02                                              | 0.21                                                    | 2        | USGS 05484500                                 |
| USGS 05484500 | 124.55                               | 645257.19                                  | 0.04                                              | 0.19                                                    | 1        | USGS 05482300                                 |
| USGS 06817000 | 3066.67                              | 405280.44                                  | 1                                                 | 0.12                                                    | 5        | USGS 05482500                                 |
| WQS 0001      | 99.81                                | 529331                                     | 0.03                                              | 0.15                                                    | 1        | USGS 05482300                                 |
| WQS 0002      | 83.96                                | 775919.49                                  | 0.03                                              | 0.23                                                    | 2        | USGS 05484500                                 |
| WQS 0003      | 83.96                                | 781269.25                                  | 0.03                                              | 0.23                                                    | 2        | USGS 05484500                                 |
| WQS 0005      | 1158.73                              | 447816.29                                  | 0.38                                              | 0.13                                                    | 5        | USGS 05482500                                 |
| WQS 0007      | 5.06                                 | 2298323.55                                 | 0                                                 | 0.67                                                    | 1        | USGS 05482300                                 |
| WQS 0010      | 1834.32                              | 894829.15                                  | 0.6                                               | 0.26                                                    | 5        | USGS 05482500                                 |
| WQS 0012      | 2509.88                              | 773064.37                                  | 0.82                                              | 0.22                                                    | 2        | USGS 05484500                                 |
| WQS 0023      | 50.74                                | 765566.22                                  | 0.02                                              | 0.22                                                    | 1        | USGS 05482300                                 |
| WQS 0032      | 98.94                                | 3444491.25                                 | 0.03                                              | 1                                                       | 4        | USGS 05465500                                 |
| WQS 0038      | 0                                    | 741357.83                                  | 0                                                 | 0.22                                                    | 2        | USGS 05484500                                 |
| WQS 0039      | 403.23                               | 778173.92                                  | 0.13                                              | 0.23                                                    | 2        | USGS 05484500                                 |
| WQS 0048      | 58.39                                | 155793.07                                  | 0.02                                              | 0.05                                                    | 1        | USGS 05482300                                 |
| WQS 0050      | 226.91                               | 786686.4                                   | 0.07                                              | 0.23                                                    | 2        | USGS 05484500                                 |
| WQS 0051      | 57.87                                | 777683.83                                  | 0.02                                              | 0.23                                                    | 2        | USGS 05484500                                 |
| WQS 0055      | 57.87                                | 783820.74                                  | 0.02                                              | 0.23                                                    | 2        | USGS 05484500                                 |
| WQS 0063      | 126.99                               | 146567.28                                  | 0.04                                              | 0.04                                                    | 1        | USGS 05482300                                 |
| WQS 0064      | 1737.55                              | 3260057.77                                 | 0.57                                              | 0.95                                                    | 4        | USGS 05465500                                 |
| WQS 0069      | 211.38                               | 790837.39                                  | 0.07                                              | 0.23                                                    | 2        | USGS 05484500                                 |
| WQS 0071      | 184.01                               | 2423469.44                                 | 0.06                                              | 0.7                                                     | 8        | WQS 0007                                      |
| WQS 0072      | 122.1                                | 749797.49                                  | 0.04                                              | 0.22                                                    | 2        | USGS 05484500                                 |
| WQS 0083      | 137.65                               | 151610.19                                  | 0.04                                              | 0.04                                                    | 1        | USGS 05482300                                 |
| WQS 0084      | 236.57                               | 493982.63                                  | 0.08                                              | 0.14                                                    | 1        | USGS 05482300                                 |

Table S4.8 Evaluation metrics and statistics of four seasons for each nitrate monitoring site

| Four Seasons  |          |                            | Winter |      |       |         |       | Spring |      |       |        |       | Summer |       |       |       |       | Fall  |       |       |       |       |
|---------------|----------|----------------------------|--------|------|-------|---------|-------|--------|------|-------|--------|-------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Site No.      | Model No | Input high frequency sites | Bias   | RMSE | Corr. | NSE     | KGE   | Bias   | RMSE | Corr. | NSE    | KGE   | Bias   | RMSE  | Corr. | NSE   | KGE   | Bias  | RMSE  | Corr. | NSE   | KGE   |
| USGS 05412500 | 7        | WQS 0003                   | -      | -    | -     | -       | -     | 0.06   | 1.19 | 0.85  | 0.72   | 0.82  | 0.05   | 1.08  | 0.92  | 0.85  | 0.89  | 0.08  | 0.72  | 0.93  | 0.86  | 0.88  |
| USGS 05418400 | 5        | USGS 05482500              | -      | -    | -     | -       | -     | -0.39  | 1.3  | 0.85  | 0.69   | 0.72  | -0.45  | 1.56  | 0.92  | 0.78  | 0.67  | 0.21  | 0.41  | 0.73  | 0.38  | 0.61  |
| USGS 05455100 | 2        | USGS 05484500              | -      | -    | -     | -       | -     | -0.07  | 1.1  | 0.86  | 0.74   | 0.84  | -0.18  | 1.25  | 0.96  | 0.91  | 0.88  | -0.03 | 0.65  | 0.98  | 0.95  | 0.87  |
| USGS 05464420 | 1        | USGS 05482300              | -0.29  | 1.03 | 0.96  | 0.84    | 0.68  | 0.11   | 1.26 | 0.89  | 0.79   | 0.88  | -0.2   | 1.18  | 0.93  | 0.86  | 0.86  | 0.1   | 1.02  | 0.86  | 0.75  | 0.81  |
| USGS 05465500 | 8        | WQS 0007                   | 0.51   | 1.09 | 0.85  | 0.65    | 0.8   | -0.15  | 1.22 | 0.88  | 0.77   | 0.83  | -0.04  | 1.05  | 0.94  | 0.89  | 0.86  | 0.21  | 0.72  | 0.95  | 0.89  | 0.92  |
| USGS 05481000 | 8        | WQS 0007                   | -      | -    | -     | -       | -     | -0.12  | 1.84 | 0.96  | 0.92   | 0.96  | -0.22  | 1.67  | 0.98  | 0.95  | 0.96  | 0.55  | 1.64  | 0.97  | 0.92  | 0.93  |
| USGS 05482000 | 2        | USGS 05484500              | -0.06  | 1.31 | 0.94  | 0.89    | 0.89  | -0.1   | 1.34 | 0.95  | 0.9    | 0.93  | 0.11   | 1.17  | 0.96  | 0.92  | 0.96  | 0.46  | 1.49  | 0.92  | 0.82  | 0.89  |
| USGS 05482300 | 2        | USGS 05484500              | 0.04   | 1.28 | 0.94  | 0.88    | 0.94  | 0.01   | 1.46 | 0.96  | 0.92   | 0.94  | -0.02  | 1.52  | 0.97  | 0.94  | 0.93  | -0.37 | 1.53  | 0.95  | 0.9   | 0.92  |
| USGS 05482500 | 2        | USGS 05484500              | 0.11   | 1.05 | 0.98  | 0.95    | 0.92  | -0.04  | 1.54 | 0.97  | 0.94   | 0.97  | -0.03  | 1.46  | 0.98  | 0.95  | 0.94  | 0     | 1.08  | 0.99  | 0.97  | 0.96  |
| USGS 05484000 | 2        | USGS 05484500              | -0.47  | 0.79 | 0.92  | 0.75    | 0.88  | -0.01  | 1.19 | 0.89  | 0.79   | 0.86  | -0.4   | 0.99  | 0.94  | 0.85  | 0.92  | -0.35 | 1.04  | 0.94  | 0.85  | 0.81  |
| USGS 05484500 | 1        | USGS 05482300              | -0.47  | 0.79 | 0.92  | 0.75    | 0.88  | -0.01  | 1.19 | 0.89  | 0.79   | 0.86  | -0.4   | 0.99  | 0.94  | 0.85  | 0.92  | -0.35 | 1.04  | 0.94  | 0.85  | 0.81  |
| USGS 06817000 | 5        | USGS 05482500              | 0.41   | 0.6  | 0.64  | -0.52   | 0.6   | -0.18  | 1.18 | 0.9   | 0.81   | 0.81  | -0.03  | 1.1   | 0.93  | 0.87  | 0.88  | 0.31  | 0.73  | 0.93  | 0.7   | 0.7   |
| WQS 0001      | 1        | USGS 05482300              | -0.44  | 1.28 | 0.89  | 0.76    | 0.85  | -0.63  | 2.35 | 0.74  | 0.5    | 0.66  | -0.14  | 1.54  | 0.86  | 0.73  | 0.81  | 0.19  | 1.17  | 0.91  | 0.82  | 0.9   |
| WQS 0002      | 2        | USGS 05484500              | 0.5    | 0.91 | 0.97  | 0.91    | 0.85  | 0.25   | 1.04 | 0.93  | 0.86   | 0.88  | 0.29   | 1.13  | 0.94  | 0.87  | 0.91  | -0.06 | 0.72  | 0.96  | 0.93  | 0.93  |
| WQS 0003      | 2        | USGS 05484500              | 0.99   | 2.41 | 0.76  | 0.47    | 0.68  | -0.23  | 1.51 | 0.88  | 0.77   | 0.78  | -0.16  | 1.41  | 0.92  | 0.85  | 0.92  | -0.03 | 1.32  | 0.92  | 0.83  | 0.91  |
| WQS 0005      | 5        | USGS 05482500              | 0.48   | 1.01 | 0.94  | 0.83    | 0.87  | -0.24  | 1.51 | 0.83  | 0.67   | 0.8   | 0      | 0.97  | 0.96  | 0.92  | 0.96  | 0.46  | 1.66  | 0.9   | 0.77  | 0.8   |
| WQS 0007      | 1        | USGS 05482300              | -0.3   | 0.78 | 0.96  | 0.89    | 0.8   | 0.45   | 1.7  | 0.85  | 0.7    | 0.73  | 0.03   | 1.04  | 0.95  | 0.9   | 0.88  | 0.04  | 0.93  | 0.92  | 0.85  | 0.87  |
| WQS 0010      | 5        | USGS 05482500              | 0.23   | 1.16 | 0.88  | 0.74    | 0.72  | 0.02   | 1.74 | 0.72  | 0.52   | 0.56  | 0.01   | 1.5   | 0.91  | 0.83  | 0.84  | 0.02  | 1.07  | 0.89  | 0.78  | 0.88  |
| WQS 0012      | 2        | USGS 05484500              | -1.03  | 1.22 | 0.99  | 0.92    | 0.89  | 0.27   | 2.72 | 0.7   | 0.48   | 0.6   | -0.06  | 1.67  | 0.88  | 0.76  | 0.78  | -0.11 | 1.28  | 0.81  | 0.65  | 0.67  |
| WQS 0023      | 1        | USGS 05482300              | 0.3    | 1.74 | 0.77  | 0.57    | 0.65  | 0.04   | 1.25 | 0.78  | 0.57   | 0.77  | -0.21  | 1.37  | 0.94  | 0.87  | 0.84  | 0.37  | 1.1   | 0.88  | 0.74  | 0.82  |
| WQS 0032      | 4        | USGS 05465500              | -0.64  | 1.01 | 0.98  | 0.91    | 0.81  | 0.34   | 2.02 | 0.93  | 0.86   | 0.85  | -0.2   | 2.43  | 0.74  | 0.52  | 0.71  | -0.19 | 2.15  | 0.9   | 0.79  | 0.74  |
| WQS 0038      | 2        | USGS 05484500              | -      | -    | -     | -       | -     | -0.26  | 2.44 | 0.71  | 0.42   | 0.71  | 0.65   | 1.51  | 0.96  | 0.89  | 0.89  | -0.09 | 2     | 0.79  | 0.61  | 0.76  |
| WQS 0039      | 2        | USGS 05484500              | -      | -    | -     | -       | -     | 2.82   | 6.15 | 0.4   | -0.23  | 0.26  | 0.58   | 1.89  | 0.94  | 0.86  | 0.84  | 0.28  | 1.99  | 0.8   | 0.63  | 0.75  |
| WQS 0048      | 1        | USGS 05482300              | -      | -    | -     | -       | -     | -1.62  | 2.64 | 0.69  | -0.43  | 0.56  | -0.28  | 2.04  | 0.95  | 0.9   | 0.9   | 0.35  | 2.34  | 0.88  | 0.77  | 0.86  |
| WQS 0050      | 2        | USGS 05484500              | -1.45  | 2.17 | 0.99  | 0.79    | 0.63  | -2.35  | 3.3  | 0.8   | 0.23   | 0.57  | 1.04   | 2.73  | 0.88  | 0.72  | 0.7   | -0.41 | 1.77  | 0.93  | 0.84  | 0.77  |
| WQS 0051      | 2        | USGS 05484500              | 2.96   | 3.06 | 0.95  | -4.9    | 0.35  | 1.71   | 3.84 | 0.16  | -1.33  | 0.14  | 2.98   | 4.28  | 0.81  | -0.04 | 0.56  | 2.71  | 3.14  | 0.92  | -0.08 | 0.53  |
| WQS 0055      | 2        | USGS 05484500              | 0.74   | 1.22 | 0.84  | 0.3     | 0.72  | 0.67   | 4.25 | 0.22  | -0.21  | 0.16  | -3.75  | 5.76  | 0.57  | -0.72 | 0.5   | -0.21 | 3.67  | 0.56  | 0.05  | 0.56  |
| WQS 0063      | 1        | USGS 05482300              | 1.22   | 1.33 | -0.42 | -7.76   | -0.61 | 3.23   | 3.38 | 0.87  | -6.6   | 0.16  | 0.95   | 1.89  | 0.81  | -4.4  | -0.75 | 0.42  | 0.91  | 0.47  | -0.05 | 0.38  |
| WQS 0064      | 4        | USGS 05465500              | 0.59   | 0.64 | -0.03 | -16.46  | -0.1  | 2.08   | 2.33 | 0.84  | -1.03  | 0.57  | 1.32   | 2.01  | 0.82  | 0.39  | 0.55  | 0.55  | 1.14  | 0.72  | 0.35  | 0.69  |
| WQS 0069      | 2        | USGS 05484500              | 0.23   | 0.53 | 0.9   | 0.4     | 0.53  | 0.46   | 1.65 | 0.72  | 0.34   | 0.7   | -0.33  | 1.39  | 0.89  | 0.79  | 0.86  | -0.2  | 1.08  | 0.86  | 0.73  | 0.82  |
| WQS 0071      | 8        | WQS 0007                   | 0.38   | 0.95 | 0.9   | 0.58    | 0.48  | -0.13  | 1.89 | 0.42  | 0.02   | 0.39  | 0.01   | 1.58  | 0.79  | 0.61  | 0.62  | -0.32 | 1.06  | 0.52  | -0.11 | 0.52  |
| WQS 0072      | 2        | USGS 05484500              | 1.99   | 2.39 | -0.77 | -33.81  | -1.35 | 2.29   | 3.04 | 0.54  | -1.26  | 0.27  | 1.74   | 2.32  | 0.86  | 0.14  | 0.49  | 1.45  | 2.92  | 0.4   | -0.15 | 0.15  |
| WQS 0083      | 1        | USGS 05482300              | 1.22   | 1.33 | -0.42 | -7.76   | -0.61 | 4.57   | 4.99 | 0.86  | -10.44 | -0.43 | 1.41   | 2.78  | 0.79  | -1.21 | -0.01 | 1.1   | 2.21  | 0.78  | -1.56 | -0.07 |
| WQS 0084      | 1        | USGS 05482300              | 3.12   | 3.16 | 0.02  | -470.44 | -1.39 | 3.18   | 3.45 | 0.5   | -7.98  | 0.27  | 0.74   | 2.86  | 0.76  | -0.1  | 0.4   | 1.56  | 2.41  | 0.74  | -1.93 | 0.07  |
| Statistics    | Min      |                            | -1.45  | 0.53 | -0.77 | -470.44 | -1.39 | -2.35  | 1.04 | 0.16  | -10.44 | -0.43 | -3.75  | 0.97  | 0.57  | -4.4  | -0.75 | -0.41 | 0.41  | 0.4   | -1.93 | -0.07 |
|               | Max      |                            | 3.12   | 3.16 | 0.99  | 0.95    | 0.94  | 4.57   | 6.15 | 0.97  | 0.94   | 0.97  | 2.98   | 5.76  | 0.98  | 0.95  | 0.96  | 2.71  | 3.67  | 0.99  | 0.97  | 0.96  |
|               | Median   |                            | 0.3    | 1.16 | 0.9   | 0.74    | 0.72  | 0.015  | 1.72 | 0.85  | 0.62   | 0.725 | -0.025 | 1.515 | 0.925 | 0.85  | 0.86  | 0.09  | 1.155 | 0.9   | 0.77  | 0.81  |

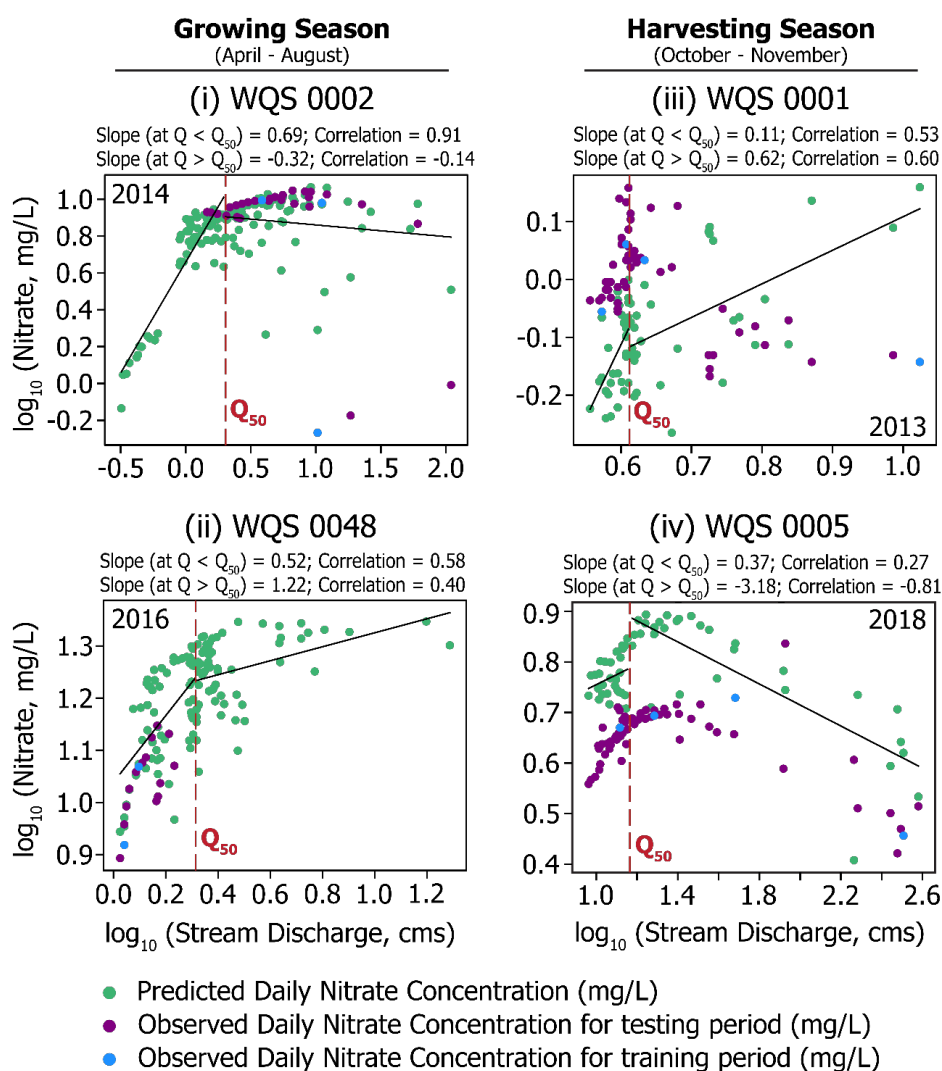


Figure S4.7 Daily nitrate concentration – stream discharge (c-Q) relationships of nitrate monitoring sites for planting and harvesting seasons that did not follow the c-Q relationships of observed daily nitrate concentration data: (i) Site WQS 0002 has observed nitrate concentration only at high Q ( $Q >$  median Q) during planting season of the year 2014; (ii) Site WQS 0048 has observed nitrate concentration only at low Q ( $Q <$  median Q) during planting season of the year 2016; (iii) Site WQS 0001 showed different estimated c-Q relationship pattern than the observed c-Q relationship during harvesting season of the year 2013; (iv) Site WQS 0005 showed same estimated c-Q relationship pattern as the observed, but with overestimated daily nitrate concentration during harvesting season of the year 2013.

## Appendix D

### Sites selected for DL modeling.

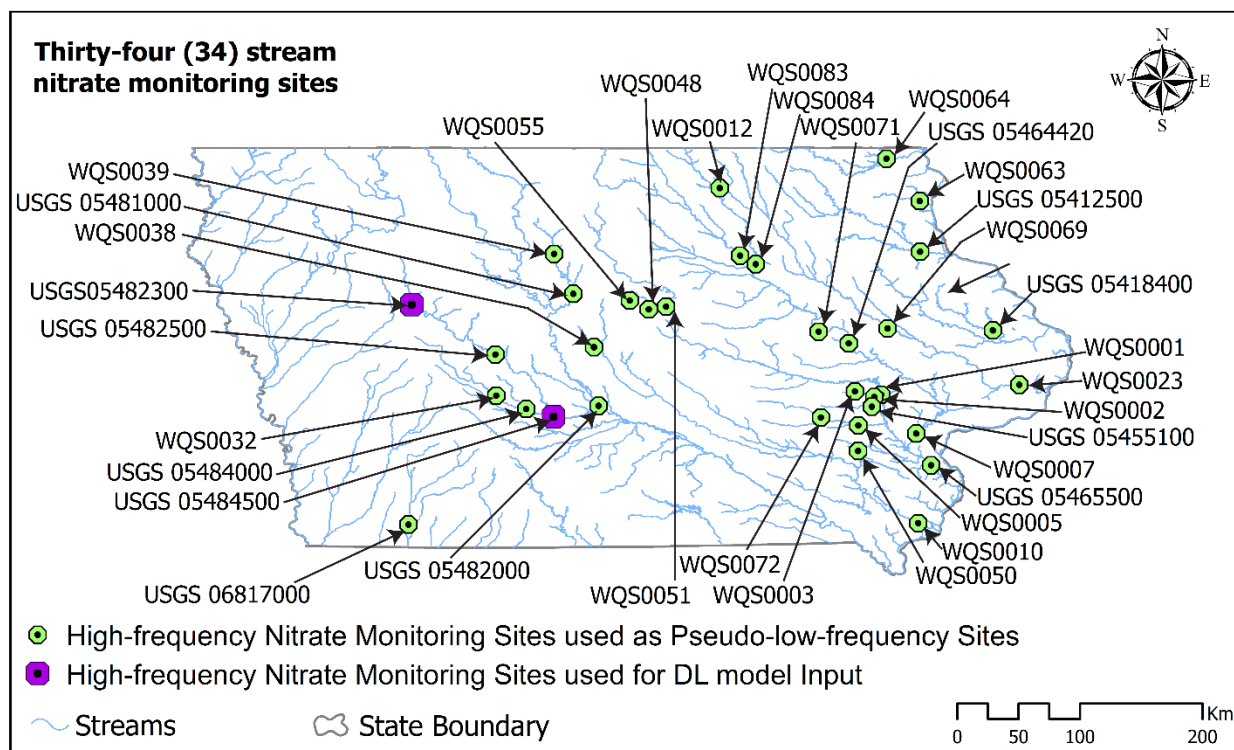


Figure S5.1 Site identities of the selected high-frequency nitrate monitoring sites within Iowa where thirty-two (32) high-frequency nitrate monitoring sites used as pseudo-low-frequency sites and the high-frequency nitrate data of remaining two (2) high-frequency sites used as DL model input.



## Data Availability, training, and testing periods

Table S5.1 Data availability, and training and testing periods for sensor-based high-frequency and periodically collected low-frequency nitrate monitoring sites in Iowa.

| Data Type                                                                                                                         | Site No.      | Data Availability Period | Training and Testing Periods <sup>1</sup> |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|-------------------------------------------|
| High-frequency nitrate monitoring sites. Nitrate concentration data of these sites was used as input during DL model development. | USGS 05482300 | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | USGS 05484500 | 2012-2019                | 2012-2019                                 |
| High-frequency nitrate monitoring sites used as Pseudo-low-frequency sites for the DL model training.                             | USGS 05412500 | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | USGS 05418400 | 2012-2014                | 2012-2014                                 |
|                                                                                                                                   | USGS 05455100 | 2012-2015                | 2012-2015                                 |
|                                                                                                                                   | USGS 05464420 | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | USGS 05465500 | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | USGS 05481000 | 2014-2017                | 2014-2017                                 |
|                                                                                                                                   | USGS 05482000 | 2013-2019                | 2013-2019                                 |
|                                                                                                                                   | USGS 05482500 | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | USGS 05484000 | 2016-2019                | 2016-2019                                 |
|                                                                                                                                   | USGS 06817000 | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | WQS 0001      | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | WQS 0002      | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | WQS 0003      | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | WQS 0005      | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | WQS 0007      | 2012-2019                | 2012-2019                                 |
|                                                                                                                                   | WQS 0010      | 2013-2019                | 2013-2019                                 |
|                                                                                                                                   | WQS 0012      | 2014-2019                | 2014-2019                                 |
|                                                                                                                                   | WQS 0023      | 2014-2019                | 2014-2019                                 |
|                                                                                                                                   | WQS 0032      | 2014-2019                | 2014-2019                                 |
|                                                                                                                                   | WQS 0038      | 2016-2019                | 2016-2019                                 |
|                                                                                                                                   | WQS 0039      | 2016-2019                | 2016-2019                                 |
|                                                                                                                                   | WQS 0048      | 2016-2019                | 2016-2019                                 |
|                                                                                                                                   | WQS 0050      | 2016-2019                | 2016-2019                                 |
|                                                                                                                                   | WQS 0051      | 2016-2019                | 2016-2019                                 |
|                                                                                                                                   | WQS 0055      | 2017-2019                | 2017-2019                                 |
|                                                                                                                                   | WQS 0063      | 2017-2019                | 2017-2019                                 |
|                                                                                                                                   | WQS 0064      | 2017-2019                | 2017-2019                                 |
|                                                                                                                                   | WQS 0069      | 2017-2019                | 2017-2019                                 |
|                                                                                                                                   | WQS 0071      | 2017-2019                | 2017-2019                                 |
|                                                                                                                                   | WQS 0072      | 2017-2019                | 2017-2019                                 |
|                                                                                                                                   | WQS 0083      | 2018-2019                | 2018-2019                                 |
|                                                                                                                                   | WQS 0084      | 2018-2019                | 2018-2019                                 |

<sup>1</sup> DL model training and testing periods were the same because the DL models were trained with randomly subsampled biweekly nitrate data of pseudo-low-frequency sites, and the remaining daily nitrate data of the same period was used for model testing.

Table S5.2 Five (5) developed cases focusing on major environmental variables that influence stream nitrate concentration and hyper-parameters.

| Case No. | Nitrate Monitoring Sites | Major variables | Batch size | Hidden layer size | Time window | No. of epoch | Random seeds |
|----------|--------------------------|-----------------|------------|-------------------|-------------|--------------|--------------|
| 1        | 32 pseudo-low-           | P               | 16, 32     | 32, 35, 40        | 150, 180    | 2500, 3000   | 0, 1, 3, 5   |
| 2        | frequency and 2          | Q               |            |                   |             |              |              |
| 3        | high-frequency           | SimQ            |            |                   |             |              |              |
| 4        | nitrate                  | P and SimQ      |            |                   |             |              |              |
| 5        | monitoring sites         | P and Q         |            |                   |             |              |              |

P = Precipitation, Q = Observed Stream Discharge, SimQ = Simulated Stream Discharge.

### **Combination of hyperparameters offered the best performance.**

The identified best DL models for all five cases could successfully fill the temporal nitrate data gaps for pseudo-low-frequency monitoring sites (Table 1). The best model for Case 2 (with stream discharge as the significant variable) was identified with batch size 32, a time window of 150 days, and a hidden layer size 40 (Table S3 in the SI document). In the remaining cases, hyperparameter combinations of batch size 32, a time window of 180 days, and a hidden layer size of 35 (Case 4 and Case 5) or 40 (Case 1 and Case 3) provided the best results for the model testing periods.

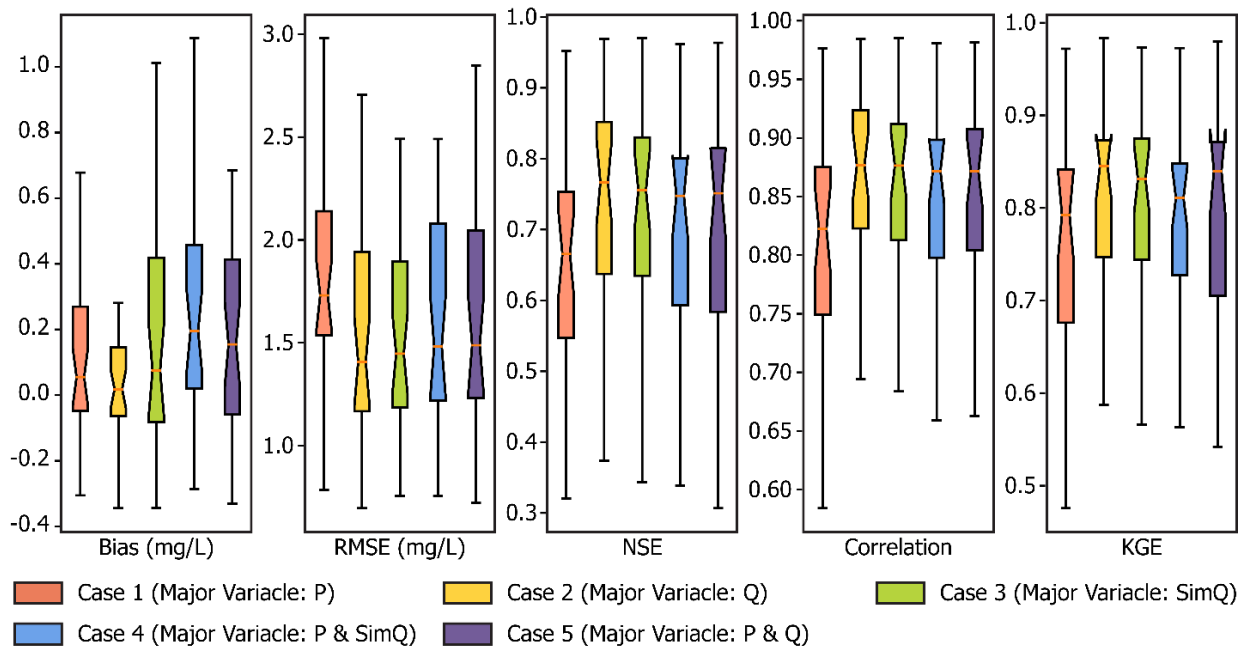


Figure S5.2 Comparison of the DL model performances for the testing period among five (5) simulation cases. Precipitation, observed stream discharge, and simulated stream discharge were significant variables for Case 1, Case 2, and Case 3, respectively, to fill the stream nitrate data gaps in the pseudo-low-frequency nitrate data. For Case 4 and Case 5, combinations of precipitation and simulated stream discharge and precipitation and observed stream discharge were considered critical variables for DL model development.

Table S5.3 Median evaluation metric values for each combination of hyperparameters for all five cases considering the test periods of all the nitrate monitoring sites.

| Case No. | Epoch | Batch size | Hidden Layer Size | Time Window | Evaluation Metrics for Testing Periods |      |      |             |      |
|----------|-------|------------|-------------------|-------------|----------------------------------------|------|------|-------------|------|
|          |       |            |                   |             | Bias                                   | RMSE | NSE  | Correlation | KGE  |
| Case1    | 2500  | 32         | 35                | 150         | 0.05                                   | 1.74 | 0.66 | 0.82        | 0.78 |
|          |       |            | 40                |             | 0.07                                   | 1.77 | 0.64 | 0.81        | 0.76 |
|          |       |            | 35                | 180         | 0.05                                   | 1.79 | 0.65 | 0.81        | 0.76 |
|          |       |            | 40                |             | 0.06                                   | 1.73 | 0.67 | 0.82        | 0.79 |
| Case2    | 2500  | 32         | 35                | 150         | 0.12                                   | 1.49 | 0.75 | 0.87        | 0.82 |
|          |       |            | 40                |             | 0.02                                   | 1.41 | 0.77 | 0.88        | 0.85 |
|          |       |            | 35                | 180         | 0.13                                   | 1.47 | 0.75 | 0.87        | 0.84 |
|          |       |            | 40                |             | 0.15                                   | 1.49 | 0.75 | 0.87        | 0.84 |
| Case3    | 2500  | 32         | 35                | 150         | 0.16                                   | 1.52 | 0.75 | 0.87        | 0.82 |
|          |       |            | 40                |             | 0.2                                    | 1.44 | 0.75 | 0.88        | 0.82 |
|          |       |            | 35                | 180         | 0.16                                   | 1.47 | 0.75 | 0.87        | 0.82 |
|          |       |            | 40                |             | 0.08                                   | 1.45 | 0.76 | 0.88        | 0.83 |
| Case4    | 2500  | 32         | 35                | 150         | 0.06                                   | 1.53 | 0.74 | 0.87        | 0.82 |
|          |       |            | 40                |             | 0.14                                   | 1.47 | 0.74 | 0.87        | 0.82 |
|          |       |            | 35                | 180         | 0.20                                   | 1.48 | 0.75 | 0.87        | 0.81 |
|          |       |            | 40                |             | 0.18                                   | 1.50 | 0.75 | 0.87        | 0.83 |
| Case5    | 2500  | 32         | 35                | 150         | 0.16                                   | 1.46 | 0.92 | 0.96        | 0.83 |
|          |       |            | 40                |             | 0.07                                   | 1.49 | 0.93 | 0.96        | 0.84 |
|          |       |            | 35                | 180         | 0.15                                   | 1.49 | 0.90 | 0.95        | 0.84 |
|          |       |            | 40                |             | 0.08                                   | 1.56 | 0.92 | 0.96        | 0.83 |

Table S5.4 Site-wise DL model performance metrics for all five (5) cases

| Site No.      | Bias   |        |        |        |        | RMSE   |        |        |        |        | Correlation |        |        |        |        | NSE    |        |        |        |        | KGE    |        |        |        |        |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
|               | Case 1 | Case 2 | Case 3 | Case 4 | Case 5 | Case 1 | Case 2 | Case 3 | Case 4 | Case 5 | Case 1      | Case 2 | Case 3 | Case 4 | Case 5 | Case 1 | Case 2 | Case 3 | Case 4 | Case 5 | Case 1 | Case 2 | Case 3 | Case 4 | Case 5 |
| USGS 05412500 | 0.2    | 0.08   | 0.06   | 0.21   | 0.15   | 1.64   | 1.03   | 1.12   | 1.21   | 1.15   | 0.80        | 0.93   | 0.91   | 0.9    | 0.91   | 0.63   | 0.85   | 0.83   | 0.8    | 0.82   | 0.77   | 0.85   | 0.85   | 0.82   | 0.83   |
| USGS 05418400 | 0      | -0.03  | 0.05   | 0.02   | 0.21   | 1.52   | 1.20   | 1.16   | 1.18   | 1.20   | 0.79        | 0.87   | 0.88   | 0.87   | 0.87   | 0.60   | 0.75   | 0.77   | 0.76   | 0.75   | 0.77   | 0.85   | 0.83   | 0.83   | 0.85   |
| USGS 05455100 | -0.31  | -0.34  | -0.09  | 0.01   | -0.07  | 1.72   | 1.29   | 1.19   | 1.18   | 1.24   | 0.89        | 0.94   | 0.95   | 0.95   | 0.94   | 0.78   | 0.88   | 0.89   | 0.90   | 0.89   | 0.83   | 0.87   | 0.88   | 0.89   | 0.93   |
| USGS 05464420 | -0.06  | 0.01   | -0.21  | -0.07  | -0.19  | 1.47   | 1.11   | 1.14   | 1.28   | 1.24   | 0.87        | 0.93   | 0.92   | 0.90   | 0.91   | 0.75   | 0.86   | 0.85   | 0.81   | 0.83   | 0.84   | 0.89   | 0.89   | 0.85   | 0.87   |
| USGS 05465500 | -0.05  | -0.02  | -0.34  | -0.29  | -0.33  | 1.23   | 1.07   | 1.28   | 1.37   | 1.23   | 0.90        | 0.92   | 0.90   | 0.88   | 0.90   | 0.81   | 0.85   | 0.79   | 0.76   | 0.80   | 0.86   | 0.88   | 0.82   | 0.80   | 0.84   |
| USGS 05481000 | -0.24  | -0.19  | -0.18  | 0      | -0.06  | 2.57   | 2.05   | 1.95   | 2.10   | 2.05   | 0.94        | 0.96   | 0.96   | 0.96   | 0.96   | 0.87   | 0.92   | 0.93   | 0.92   | 0.92   | 0.92   | 0.95   | 0.95   | 0.95   | 0.96   |
| USGS 05482000 | 0.11   | 0.13   | -0.12  | 0.07   | 0.07   | 1.20   | 1.19   | 1.12   | 1.22   | 1.25   | 0.95        | 0.96   | 0.96   | 0.95   | 0.95   | 0.91   | 0.91   | 0.92   | 0.91   | 0.90   | 0.95   | 0.94   | 0.96   | 0.95   | 0.95   |
| USGS 05482500 | -0.15  | -0.05  | -0.12  | 0.13   | 0.05   | 1.32   | 1.07   | 1.05   | 1.18   | 1.15   | 0.98        | 0.98   | 0.99   | 0.98   | 0.98   | 0.95   | 0.97   | 0.97   | 0.96   | 0.96   | 0.97   | 0.98   | 0.97   | 0.97   | 0.98   |
| USGS 05484000 | -0.11  | -0.10  | 0.04   | 0.17   | 0.08   | 0.78   | 0.70   | 0.75   | 0.76   | 0.72   | 0.95        | 0.96   | 0.95   | 0.96   | 0.96   | 0.90   | 0.92   | 0.91   | 0.91   | 0.92   | 0.95   | 0.96   | 0.94   | 0.95   | 0.96   |
| USGS 06817000 | -0.03  | 0      | 0.09   | 0.18   | 0.17   | 1.25   | 0.99   | 1.03   | 1.06   | 1.10   | 0.87        | 0.92   | 0.92   | 0.91   | 0.91   | 0.76   | 0.85   | 0.84   | 0.83   | 0.81   | 0.87   | 0.91   | 0.91   | 0.90   | 0.90   |
| WQS0001       | 0.07   | 0.03   | -0.05  | 0.10   | 0      | 1.70   | 1.55   | 1.68   | 1.71   | 1.65   | 0.86        | 0.88   | 0.86   | 0.85   | 0.87   | 0.73   | 0.78   | 0.74   | 0.73   | 0.75   | 0.85   | 0.85   | 0.83   | 0.81   | 0.84   |
| WQS0002       | 0.32   | 0.12   | 0.30   | 0.34   | 0.32   | 1.70   | 1.48   | 1.46   | 1.50   | 1.49   | 0.83        | 0.87   | 0.88   | 0.87   | 0.88   | 0.67   | 0.75   | 0.76   | 0.75   | 0.75   | 0.80   | 0.86   | 0.85   | 0.85   | 0.86   |
| WQS0003       | 0.17   | 0.14   | 0.24   | 0.29   | 0.18   | 2.02   | 1.68   | 1.72   | 1.96   | 1.76   | 0.82        | 0.87   | 0.87   | 0.83   | 0.87   | 0.66   | 0.76   | 0.75   | 0.68   | 0.74   | 0.80   | 0.84   | 0.84   | 0.81   | 0.85   |
| WQS0005       | 0.31   | 0.15   | 0.30   | 0.30   | 0.32   | 2.09   | 1.72   | 1.79   | 1.65   | 1.81   | 0.81        | 0.87   | 0.86   | 0.88   | 0.86   | 0.63   | 0.75   | 0.73   | 0.77   | 0.73   | 0.78   | 0.86   | 0.84   | 0.84   | 0.83   |
| WQS0007       | 0.21   | -0.01  | -0.08  | 0.08   | -0.08  | 1.75   | 1.47   | 1.49   | 1.51   | 1.52   | 0.83        | 0.88   | 0.88   | 0.87   | 0.87   | 0.67   | 0.77   | 0.77   | 0.76   | 0.75   | 0.80   | 0.85   | 0.83   | 0.82   | 0.84   |
| WQS0010       | 0.02   | -0.04  | -0.06  | 0      | -0.10  | 1.55   | 1.33   | 1.34   | 1.41   | 1.42   | 0.87        | 0.90   | 0.90   | 0.89   | 0.89   | 0.75   | 0.82   | 0.81   | 0.79   | 0.79   | 0.82   | 0.84   | 0.86   | 0.82   | 0.85   |
| WQS0012       | 0.38   | 0.10   | 0.57   | 0.68   | 0.42   | 2.15   | 1.82   | 1.89   | 2.03   | 2.02   | 0.73        | 0.80   | 0.81   | 0.78   | 0.77   | 0.51   | 0.64   | 0.62   | 0.56   | 0.56   | 0.68   | 0.71   | 0.74   | 0.69   | 0.71   |
| WQS0023       | -0.09  | -0.30  | -0.18  | -0.14  | -0.13  | 1.61   | 1.19   | 1.24   | 1.32   | 1.23   | 0.86        | 0.93   | 0.92   | 0.91   | 0.92   | 0.73   | 0.85   | 0.84   | 0.82   | 0.84   | 0.79   | 0.87   | 0.88   | 0.85   | 0.89   |
| WQS0032       | -0.09  | -0.15  | 0.10   | 0.22   | 0.16   | 2.14   | 1.93   | 1.97   | 2.20   | 2.06   | 0.88        | 0.90   | 0.90   | 0.87   | 0.89   | 0.76   | 0.81   | 0.80   | 0.75   | 0.78   | 0.87   | 0.88   | 0.87   | 0.86   | 0.87   |
| WQS0038       | 0.17   | 0.24   | 0.41   | 0.52   | 0.41   | 2.17   | 1.98   | 1.96   | 2.08   | 2.05   | 0.85        | 0.88   | 0.89   | 0.88   | 0.88   | 0.73   | 0.77   | 0.78   | 0.75   | 0.76   | 0.80   | 0.84   | 0.81   | 0.79   | 0.86   |
| WQS0039       | 0.26   | 0.22   | 0.19   | 0.40   | 0.67   | 3.66   | 3.41   | 3.45   | 3.69   | 3.60   | 0.63        | 0.69   | 0.68   | 0.63   | 0.66   | 0.35   | 0.44   | 0.43   | 0.34   | 0.38   | 0.60   | 0.68   | 0.66   | 0.60   | 0.63   |
| WQS0048       | 1.26   | 1.62   | 1.01   | 1.09   | 1.54   | 3.77   | 3.65   | 3.27   | 3.46   | 3.77   | 0.80        | 0.83   | 0.85   | 0.83   | 0.81   | 0.59   | 0.62   | 0.69   | 0.66   | 0.59   | 0.74   | 0.74   | 0.78   | 0.73   | 0.71   |
| WQS0050       | 0.04   | 0.02   | 0.30   | 0.49   | 0      | 2.98   | 2.71   | 2.49   | 2.49   | 2.85   | 0.79        | 0.83   | 0.86   | 0.86   | 0.81   | 0.62   | 0.69   | 0.74   | 0.74   | 0.65   | 0.66   | 0.76   | 0.81   | 0.81   | 0.69   |
| WQS0051       | 0.68   | 0.62   | 0.76   | 1.01   | 0.69   | 2.01   | 2.06   | 1.84   | 2.18   | 2.24   | 0.88        | 0.84   | 0.90   | 0.87   | 0.82   | 0.68   | 0.66   | 0.73   | 0.62   | 0.60   | 0.81   | 0.82   | 0.84   | 0.79   | 0.80   |
| WQS0055       | 0.15   | -0.60  | 0.68   | 0.98   | -0.90  | 4.46   | 4.18   | 4.76   | 4.59   | 4.88   | 0.75        | 0.79   | 0.74   | 0.75   | 0.70   | 0.56   | 0.61   | 0.50   | 0.53   | 0.47   | 0.70   | 0.73   | 0.73   | 0.71   | 0.62   |
| WQS0063       | 0.53   | 0.28   | 0.50   | 0.45   | 0.52   | 1.64   | 1.29   | 1.43   | 1.46   | 1.49   | 0.27        | 0.57   | 0.46   | 0.41   | 0.47   | -0.24  | 0.23   | 0.05   | 0.02   | -0.02  | 0.21   | 0.52   | 0.37   | 0.32   | 0.43   |
| WQS0064       | 0      | 0.02   | 0.06   | -0.12  | 0.16   | 1.33   | 0.99   | 1.11   | 1.16   | 1.11   | 0.70        | 0.85   | 0.81   | 0.80   | 0.81   | 0.48   | 0.71   | 0.64   | 0.60   | 0.64   | 0.55   | 0.75   | 0.62   | 0.56   | 0.67   |
| WQS0069       | -0.03  | -0.10  | 0      | -0.04  | 0.08   | 1.54   | 1.10   | 1.29   | 1.45   | 1.20   | 0.80        | 0.90   | 0.86   | 0.82   | 0.88   | 0.62   | 0.81   | 0.74   | 0.67   | 0.77   | 0.76   | 0.82   | 0.79   | 0.76   | 0.82   |
| WQS0071       | -0.23  | 0.18   | 0.52   | 0.84   | 0.48   | 2.22   | 1.60   | 1.82   | 2.14   | 1.68   | 0.58        | 0.67   | 0.64   | 0.66   | 0.64   | -0.21  | 0.37   | 0.19   | -0.12  | 0.31   | 0.48   | 0.66   | 0.63   | 0.55   | 0.61   |
| WQS0072       | 0.64   | 0.91   | 0.92   | 0.93   | 1.22   | 1.95   | 1.97   | 1.91   | 1.92   | 2.19   | 0.66        | 0.70   | 0.71   | 0.71   | 0.69   | 0.32   | 0.31   | 0.34   | 0.34   | 0.14   | 0.60   | 0.59   | 0.57   | 0.58   | 0.54   |
| WQS0083       | 0.87   | 0.04   | 0.44   | 0.43   | 0.43   | 1.74   | 1.21   | 1.23   | 1.18   | 1.41   | 0.74        | 0.76   | 0.80   | 0.78   | 0.79   | -0.04  | 0.49   | 0.48   | 0.52   | 0.31   | 0.54   | 0.75   | 0.75   | 0.74   | 0.63   |
| WQS0084       | -0.04  | -0.30  | -0.26  | 0.05   | -0.06  | 1.76   | 1.35   | 1.43   | 1.44   | 1.45   | 0.66        | 0.75   | 0.70   | 0.70   | 0.72   | 0.16   | 0.50   | 0.44   | 0.44   | 0.43   | 0.61   | 0.74   | 0.67   | 0.70   | 0.72   |

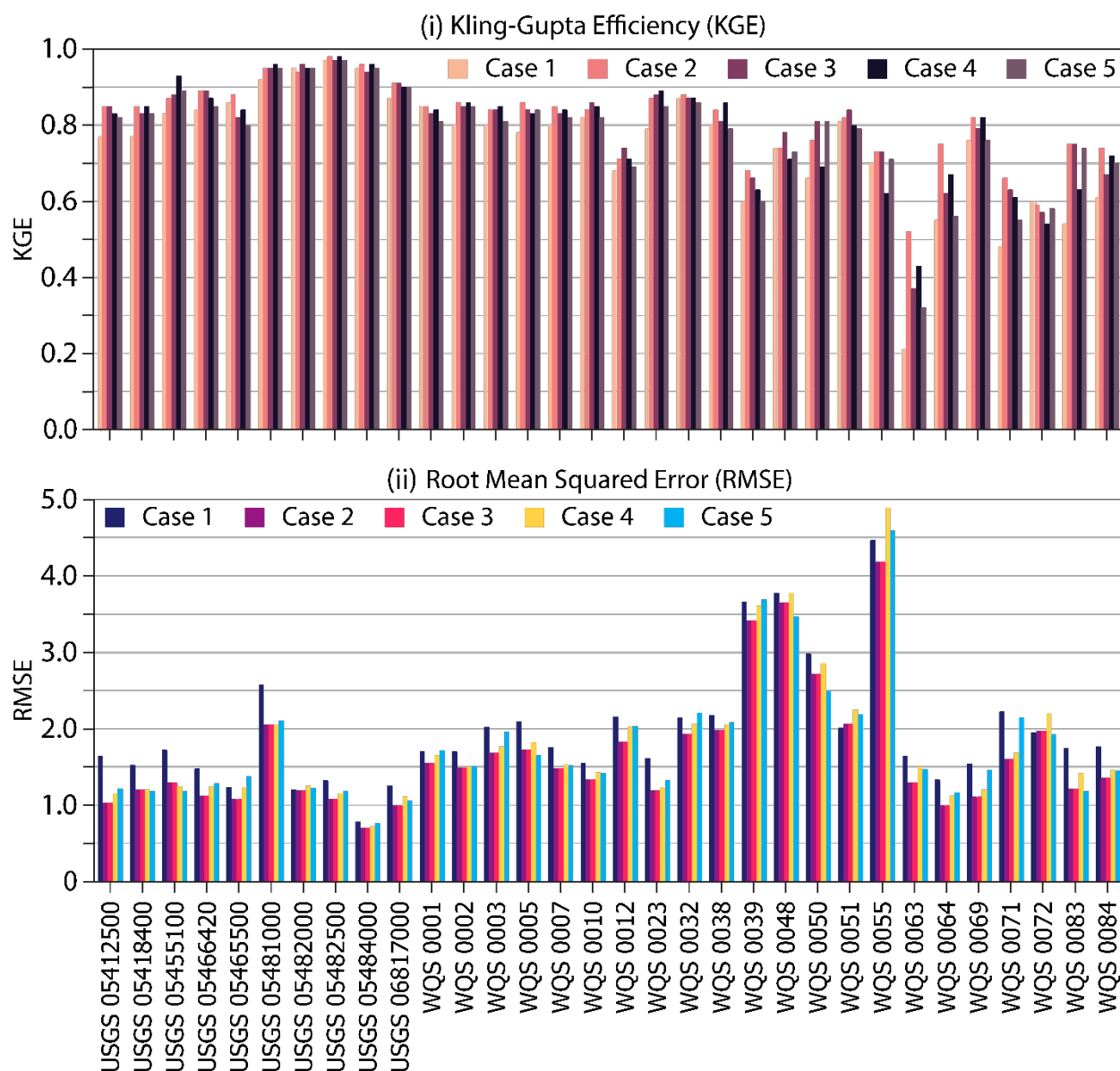


Figure S5.3 Site-wise evaluation metrics for all thirty-two (32) pseudo-low-frequency nitrate monitoring sites in Iowa (i) Kling-Gupta Efficiency (KGE) and (ii) Root Mean Squared Error (RMSE).

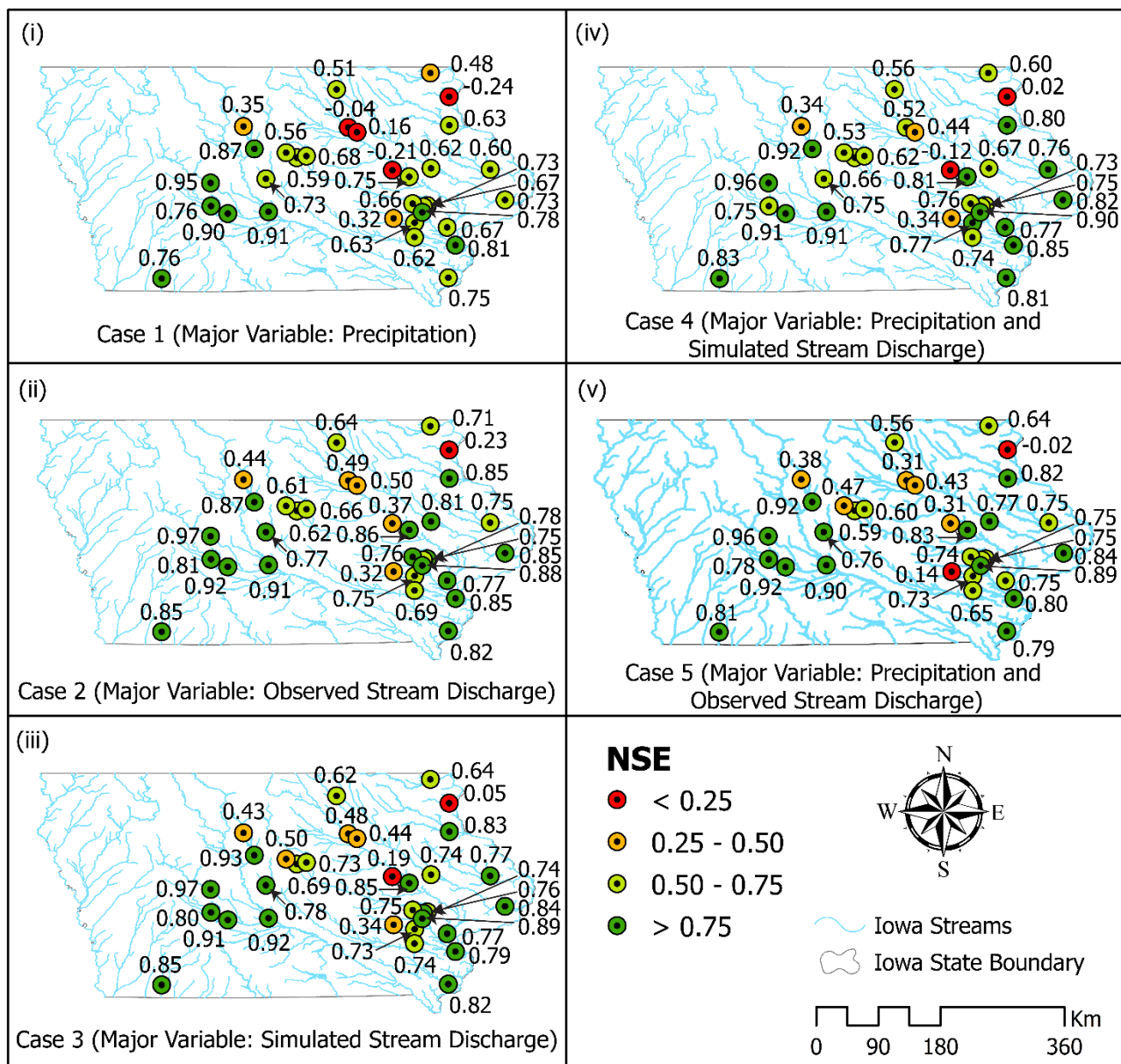


Figure S5.4 Site-wise evaluation metrics for all thirty-two (32) pseudo-low-frequency nitrate monitoring sites in Iowa (i) Kling-Gupta Efficiency (KGE) and (ii) Root Mean Squared Error (RMSE).

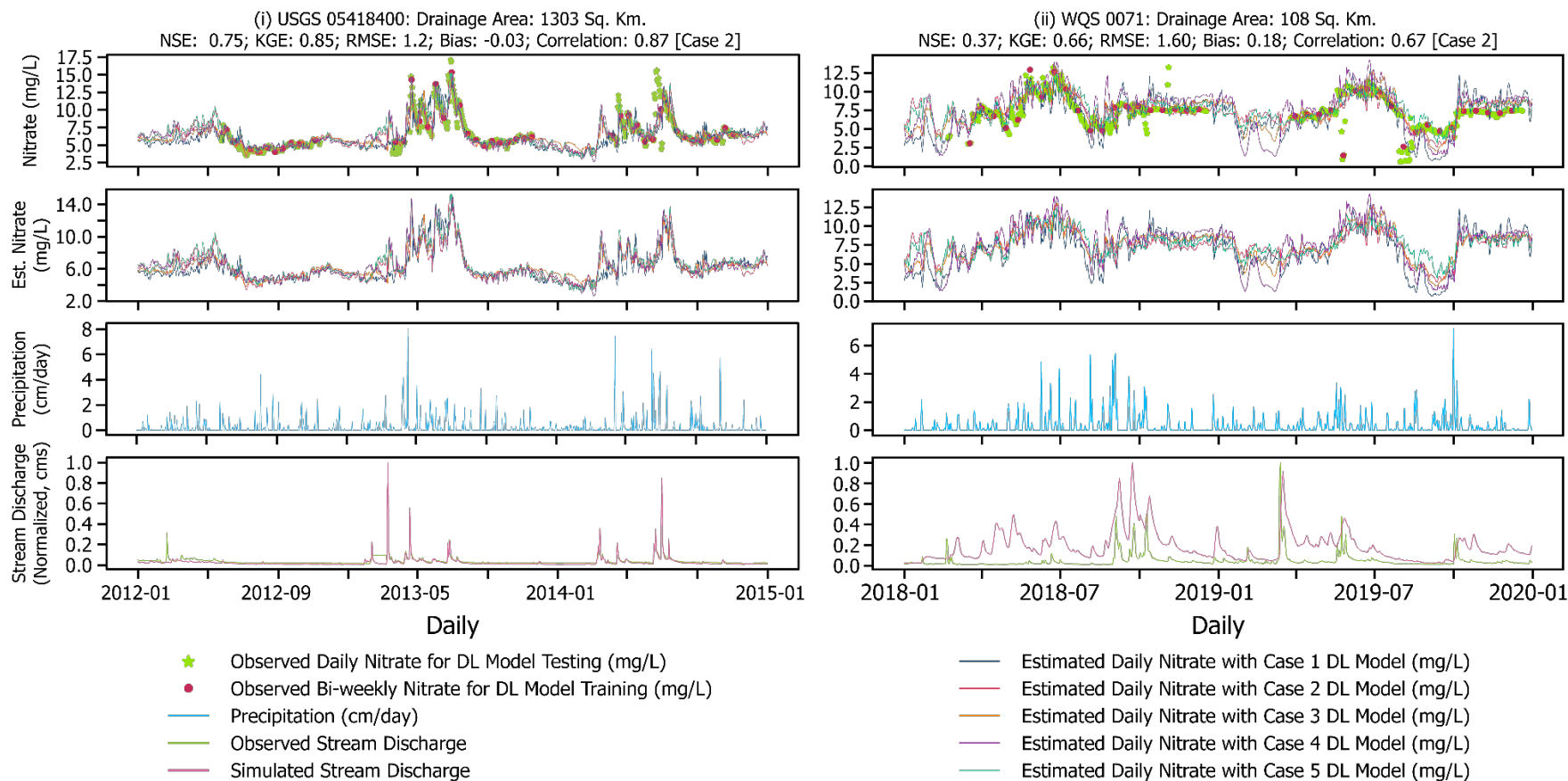


Figure S5.5 Time series plots of observed and estimated daily nitrate concentrations for five cases, precipitation, observed and simulated stream discharge of pseudo-low-frequency for training and testing periods: (i) USGS 05418400 (a well-performed site), and (ii) WQS 0071 (a comparatively poor-performed site).



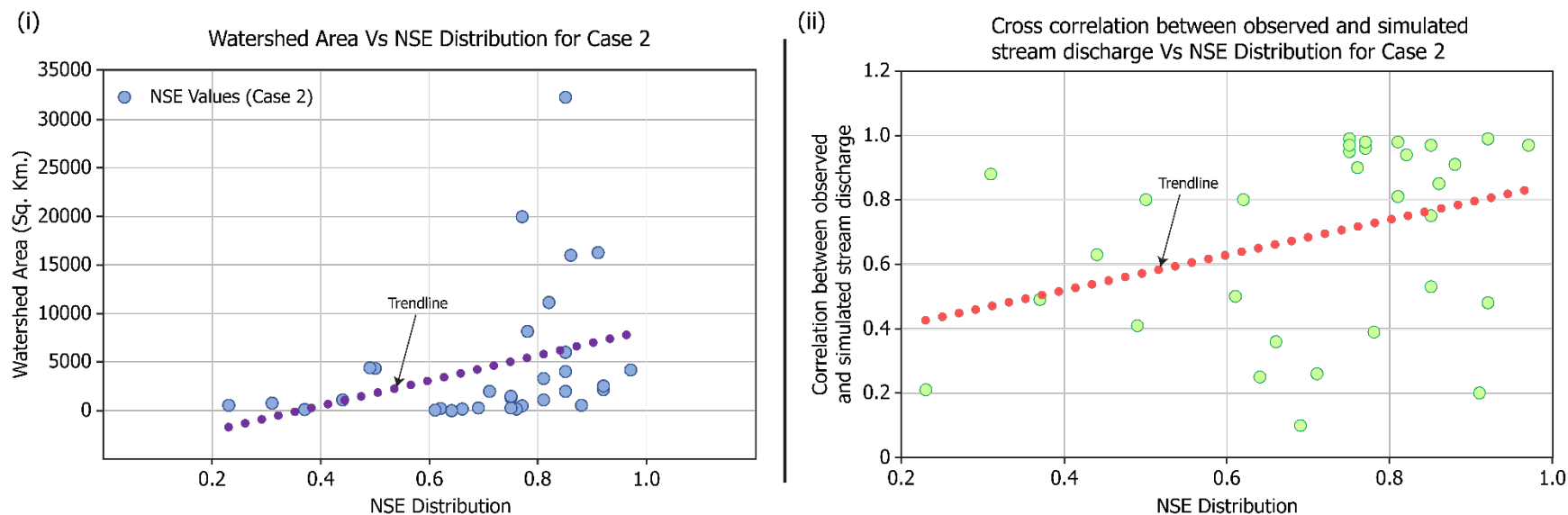


Figure S5.6 Two plots of (i) contributing watershed areas versus site-wise evaluation metrics (NSE for Case 2) and (ii) cross-correlation between observed and simulated stream discharges versus site-wise evaluation metrics (NSE for Case 2) all thirty-two (32) pseudo-low-frequency nitrate monitoring sites in Iowa. Trendlines represented increasing watershed areas and correlation between observed and simulated stream discharge with higher NSE values.

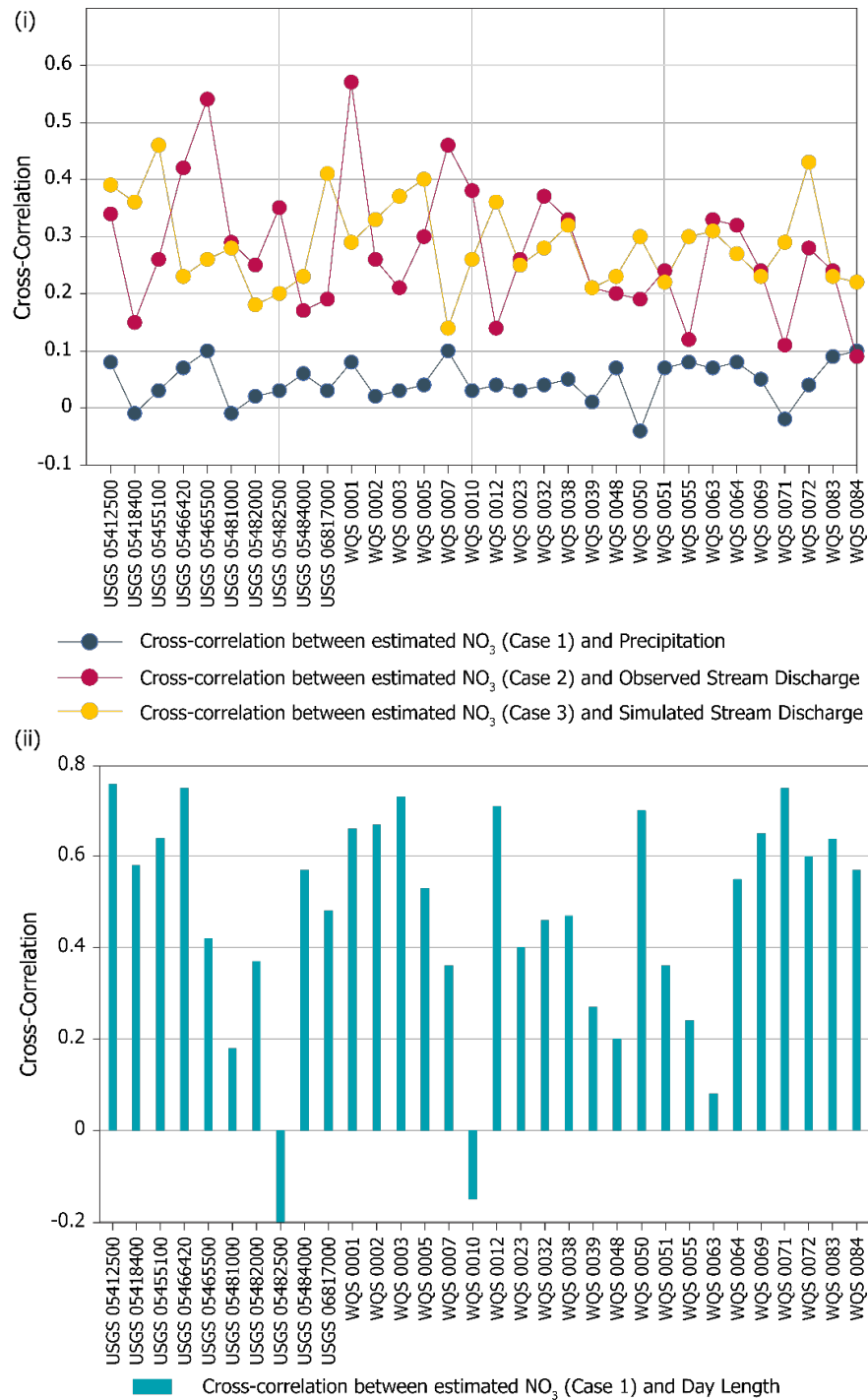


Figure S5.7 Cross-correlations between estimated nitrate concentrations calculated in three cases and precipitation, observed stream discharge, simulated stream discharge, and day length. (i) Cross-correlation between estimate nitrate concentration calculated in Case 1, Case 2, and Case 3 and precipitation, observed stream discharge, and simulated stream discharge, respectively; (ii) cross-correlation between estimated nitrate concentrations (Case 1) and day length.

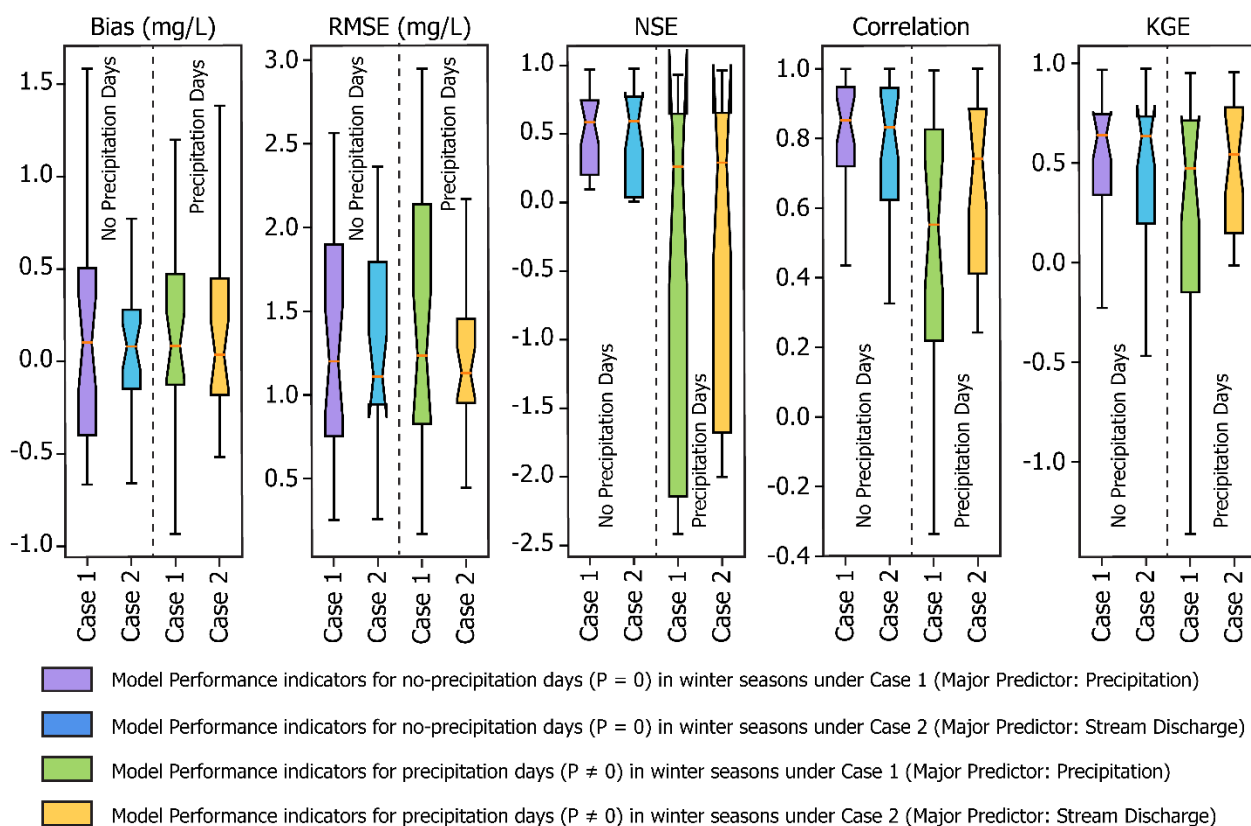


Figure S5.8 Boxplots comparison between the Case 1 and Case 2 DL models' performance for no-precipitation and precipitation days during winter seasons.

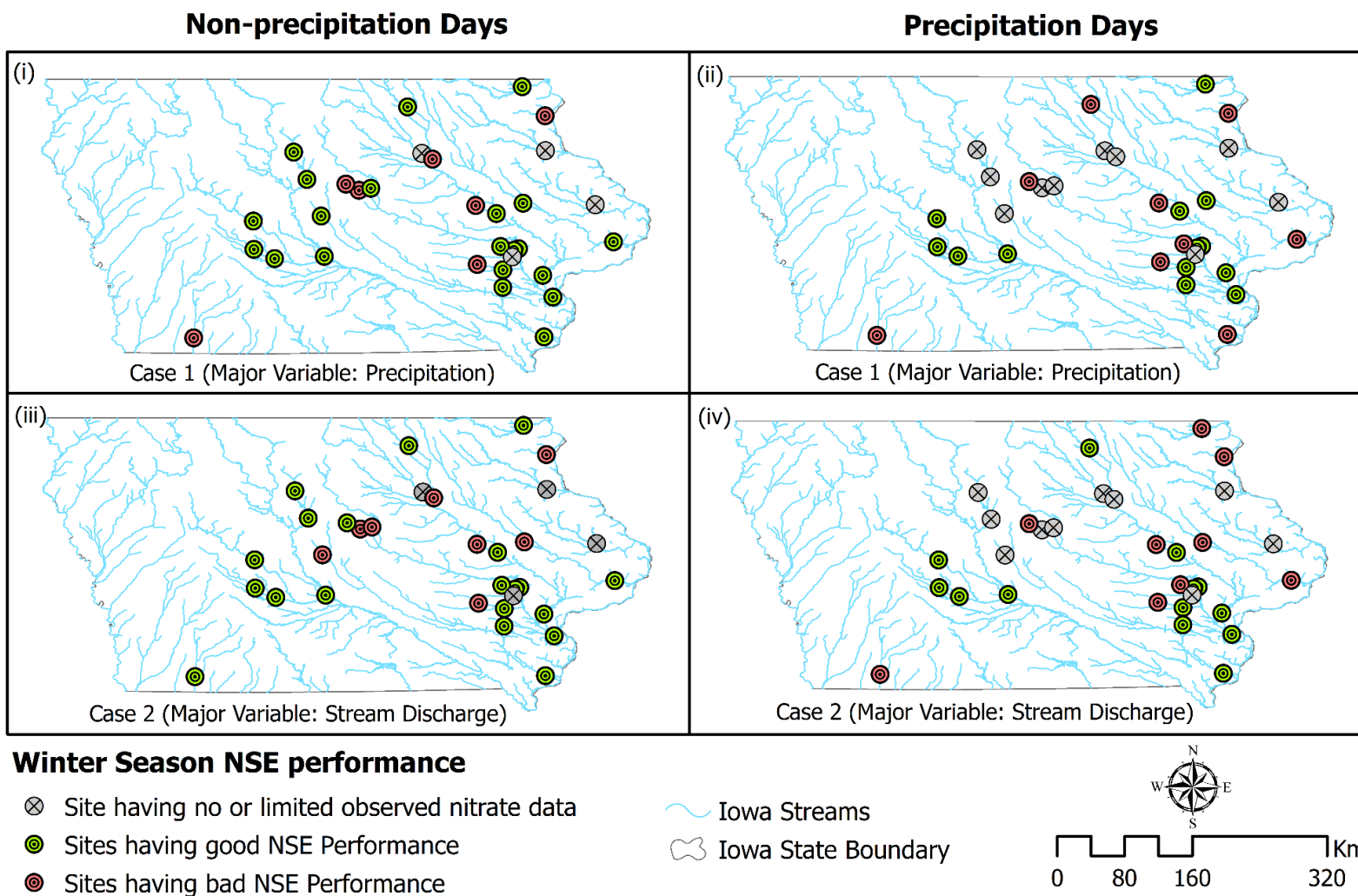


Figure S5.9 Regional distribution of NSE performance of all thirty-four (34) nitrate monitoring sites for no-precipitation and precipitation days during winter seasons based on Case 1 and Case 2 DL model performance.

# Gourab Saha

Ph.D.: Fall 2023

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## Education

**The Pennsylvania State University, University Park, USA.** Ph.D. in ABE.  
CGPA 3.85/4.00. Advisor: Prof. Cibir Raj.

*August 2019 – December 30*

**The Pennsylvania State University, University Park, USA.** M.Sc. in ABE.  
CGPA 3.89/4.00. Advisor: Prof. Cibir Raj.

*January 2017 - July 2019*

**Bangladesh University of Engineering & Tech. (BUET), Dhaka, Bangladesh.** B.Sc. in WRE.  
CGPA 3.50/4.00. Advisor: Prof. Md. Sabbir Mostafa Khan.

*May 2007 - April 2012*

## Work Experience

**Penn State University, University Park, PA, USA.**  
Graduate Assistant, Agricultural and Biological Engineering.

*January 2017 – July 2023*

**Bangladesh Water Development Board, Dhaka, Bangladesh.**  
Assistant Engineer (Civil), Design Circle 6.

*October 2014 - December 2016*

**Institute of Water Modelling, Dhaka, Bangladesh.**  
Junior Engineer, Irrigation and Groundwater Modeling Division.

*August 2012 - October 2014*

## Publication(s)

### Journal and Conference Paper (s)

[J1] **G. K. Saha**, et. al., "Towards a Robust Land Suitability Framework for Manure Management: Modeling Impacts and Evaluating Biophysical Characteristics, Journal of American Water Resources Association (**JAWRA**), 2022.

[J2] **G. K. Saha**, et. al., "Development of a Land Suitability Framework for Sustainable Manure Utilization", (**Transactions of the ASABE**), 2020.

[J3] A. Saha, **G. K. Saha**, et. al., "Evaluating Water Quality Benefits of Manure Management in the Susquehanna River Basin", Journal of Environmental Quality (**JEQ**), 2022.

[J4] **G. K. Saha**, et. al., "A deep learning-based novel approach to generate continuous daily stream nitrate concentration for nitrate data-sparse watersheds", Science of the Total Environment (**STOTEN**), 2023.

[C1] **G. K. Saha**, et. al., "Geospatial Landscape Analysis for Livestock Manure Management in Western Pennsylvania", (**ASABE Annual International Meeting**) 2018.

### Report and Blog

[R1] **G. K. Saha** et. al., "Manurefest Destiny: Opportunities for Animal Agriculture in Western Pennsylvania", submitted to Pennsylvania Department of Agriculture (**PDA**), 2018.

[B1] **G. K. Saha**, "Manure can be a resource when applied in the right places", **Medium** and **Penn State Institute of Energy and Environment**, 2022.

### Developed Tool(s)

[T1] **G. K. Saha**, "Land Suitability Analysis Tools for Sustainable Manure Utilization", Available at **ScholarSphere** and **GitHub**, 2019.

[T2] **G. K. Saha**, "LSTM Nitrate (an LSTM-based daily stream nitrate prediction model)", Available at **GitHub**, 2023.

## Awards and Honors

- 2023 **ABE Outstanding Graduate Student Award** from ABE Department at Penn State.
- 2022 **ASABE Superior Paper Award** from ASABE for land suitability analysis research.
- 2<sup>nd</sup> position in **State College Borough Sustainable Environment Meeting** poster presentation, 2019.
- 1<sup>st</sup> position in **NABEC** poster presentation, 2018.
- **Two Travel Awards** for ASABE and AGU conferences.
- **University Merit Scholarship** for 4 years during bachelor tenure due to outstanding results (2007 – 2011).

## Outreach Activities

**Reviewer** of (i) Journal of Environmental Quality (JEQ); (ii) Agrosystems, Geosciences, and Environment. **Member** of (iii) American Geophysical Union (AGU); (iv) American Society of Civil Engineers (ASCE); (v) American Society of Agricultural and Biological Engineers (ASABE).